

Fermi Notes on Physics

Volume Five

Introductory Undergraduate
Lectures on Electricity and
Magnetism

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Magnetism

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Electrostatic Fields and Potential

1.1 Electricity at School

The electrostatics taught at school describes the forces between charges at rest, the electric field that carries those forces, and the potential that measures the work they do. Since you have met this material before, this section states the school results briefly, in the notation that the rest of the course will use. The three sections that follow take these results to university level. Section 1.2 writes Coulomb's law and the field in vector form and works out the fields of charge spread along lines and over surfaces, Section 1.3 introduces the flux of a field and Gauss's law, and Section 1.4 develops the potential and the energy stored in a system of charges.

Charge

Electric charge is a property of the particles that make up atoms. The proton carries a charge $+e$, the electron a charge $-e$, and the neutron carries no charge, where e is called the elementary charge. Every charge found on an isolated particle is a whole multiple of e , an experimental fact first established by Millikan, who measured the charges on small oil drops held between charged plates and found that they were all multiples of a single value. The SI unit of

charge is the coulomb. An atom has as many electrons as protons and is therefore neutral, and a body becomes charged when it gains or loses electrons.

Charge is found by experiment to be conserved: the total charge of an isolated system does not change. Charging by friction is one example of this principle. When a polythene rod is rubbed with a woollen cloth, electrons are transferred from the cloth to the rod, and the rod acquires a negative charge while the cloth acquires an equal positive charge, so the total remains zero. In a conductor, such as a metal, some of the electrons are free to move through the material, while in an insulator, such as polythene or glass, the electrons are bound to their atoms and a charge placed on the surface stays where it was put.

Coulomb's Law and Field Lines

Point charges q_1 and q_2 placed a distance r apart are found by experiment to exert on each other forces given by

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}, \quad (1.1)$$

directed along the line joining them, where ϵ_0 is called the electric constant, or the permittivity of free space. This relation is called Coulomb's law. Like charges repel and unlike charges attract, so a positive value of F in this formula means a repulsion. The force obeys an inverse-square law, as Newton's law of gravitation does, although the electric force between two electrons is enormously larger than their gravitational attraction. When several charges act on a single charge, the force on it is observed to be the vector sum of the forces that the charges would exert separately, so the force on a charge placed between two others that repel it is the difference of the two repulsions.

The electric field at a point is the force per unit charge on a small positive test charge q_0 placed there, $\mathbf{E} = \mathbf{F}/q_0$, and its unit is the newton per coulomb. The field of a point charge q is given by

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}, \quad (1.2)$$

and points directly away from the charge when q is positive and towards it when q is negative. The force on a charge q placed in a

field $\mathbf{E}$ is $q\mathbf{E}$, which depends only on the field at the position of the charge and not on the arrangement of the charges that produce the field.

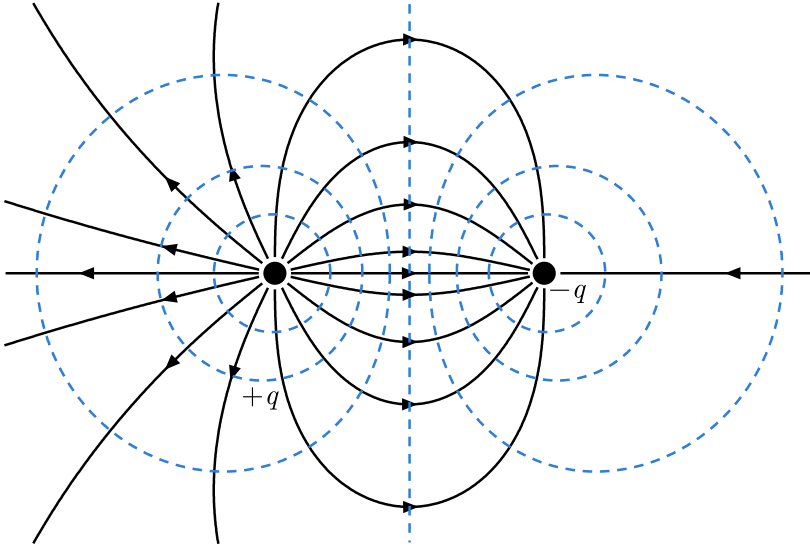


Figure 1. The field lines (solid) and the equipotential surfaces (accent, dashed) of a positive charge $+q$ and a negative charge $-q$. The field lines leave the positive charge and end on the negative one, and they cross every equipotential surface at right angles.

A field can be pictured with field lines, drawn so that the tangent to a line at each point gives the direction of the field there. Field lines begin on positive charges and end on negative charges, they are crowded together where the field is strong and spread apart where it is weak, and they never cross, since the field at a point has only one direction. The field lines of an isolated point charge are straight lines running radially outwards from the charge, or inwards to it if the charge is negative, and they spread apart as the field weakens with distance. The field lines of a pair of equal and opposite charges are drawn in Figure 1.

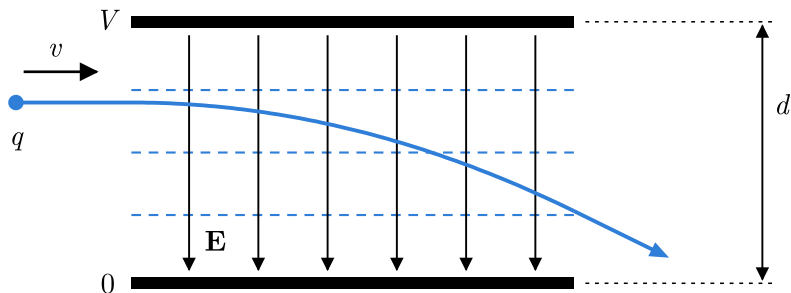


Figure 2. A particle of charge q entering the uniform field between two parallel plates, a distance d apart and held at the potentials V and 0 . The field lines run from the plate at the higher potential to the other plate, the equipotential surfaces (dashed) are planes parallel to the plates, and the path of the particle (accent) is a parabola between the plates and a straight line beyond them.

The Uniform Field between Parallel Plates

When two parallel metal plates a distance d apart are connected to a supply that holds them at a potential difference V , the field between them is uniform, except near the edges of the plates. It points from the plate at the higher potential to the other plate, and its magnitude is

$$E = \frac{V}{d}. \quad (1.3)$$

A particle of charge q and mass m between the plates experiences a constant force qE and therefore a constant acceleration qE/m perpendicular to the plates. A particle that enters the field moving parallel to the plates, as in Figure 2, keeps its velocity parallel to the plates and accelerates uniformly across them, just as a projectile keeps its horizontal velocity while it falls, so its path between the plates is a parabola. Once it has left the plates, no force acts on it, and it moves in a straight line. A particle that is instead released

from rest at one plate and accelerated across to the other gains a kinetic energy qV , which is how the electrons in an electron gun are given their speed, and a beam of such electrons can then be deflected by a second pair of plates.

Potential and Potential Difference

The potential at a point is the work done per unit positive charge in bringing a small test charge from infinity to that point. Its unit is the volt, defined as one joule per coulomb, and at a distance r from a point charge q it is

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}. \quad (1.4)$$

The potential is a scalar, so the potential of several charges is the sum of their separate potentials. The work needed to move a charge q from a point at potential V_1 to a point at potential V_2 is $q(V_2 - V_1)$, and the potential energy of two point charges a distance r apart is $U = q_1q_2/4\pi\epsilon_0r$. The field points in the direction in which the potential decreases most rapidly, and its component along a direction x is minus the potential gradient, $E = -\Delta V/\Delta x$, which gives $E = V/d$ for the parallel plates.

A surface on which the potential has the same value everywhere is called an equipotential surface. No work is done in moving a charge along an equipotential surface, so the field has no component along it, and the field lines cross the equipotential surfaces at right angles, as Figure 1 shows. The equipotential surfaces of a point charge are spheres centred on the charge, and those of the parallel plates are planes parallel to the plates, as Figure 2 shows. Near each charge of the pair in Figure 1 the equipotential surfaces are very nearly spheres, and the plane midway between the two charges is the equipotential surface on which the potential is zero, since every point of it is equidistant from charges of equal magnitude and opposite sign.

Energies on the atomic scale are often quoted in electronvolts. An electronvolt is defined as the energy gained by an electron, of charge $-e$, when it moves through a potential difference of one volt, so that $1 \text{ eV} = e \times 1 \text{ V}$.

1.2 Coulomb's Law and the Electric Field

This section first writes Coulomb's law and the electric field in vector form, then shows how the field of charge spread continuously along a line, over a surface, or through a volume is found by integration, and finally works out the fields of a straight segment, a ring, and a disc. The fields of an infinite line and an infinite plane follow as limits.

Coulomb's Law in Vector Form

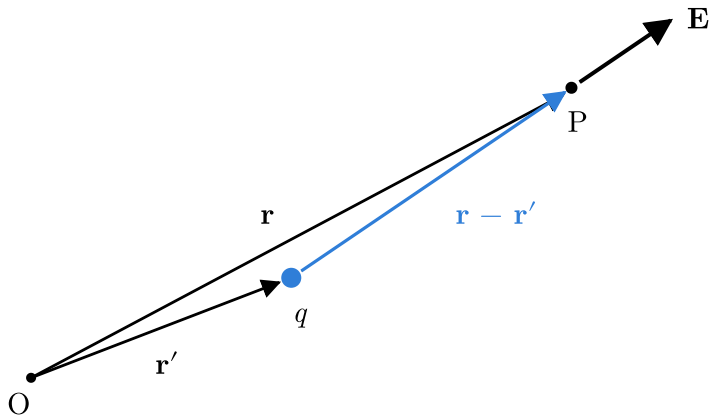


Figure 3. A point charge q at the source point $\mathbf{r}'$ and a field point P at $\mathbf{r}$, both measured from an origin O . When q is positive, the field at P points along the separation $\mathbf{r} - \mathbf{r}'$ (accent).

Let a point charge q sit at the position $\mathbf{r}'$, which is called the source point, and let a second charge q_0 sit at the position $\mathbf{r}$, which is called the field point, both measured from an origin O , as in Figure 3. The vector $\mathbf{r} - \mathbf{r}'$ points from the source point to the field point, and dividing it by its magnitude gives a unit vector in that direction. Coulomb's law gives the force on q_0 as

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{qq_0(\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3}, \quad (1.5)$$

since the magnitude of this vector is $|qq_0|/4\pi\epsilon_0|\mathbf{r} - \mathbf{r}'|^2$, as the school

form of the law requires. The direction of the force follows from the signs of the charges. When the charges have the same sign the product qq_0 is positive and the force on q_0 points away from q , and when they have opposite signs it points towards q . Exchanging the two charges reverses the sign of $\mathbf{r} - \mathbf{r}'$, so the force on q is equal and opposite to the force on q_0 , in agreement with Newton's third law.

The electric field at $\mathbf{r}$ is the force per unit charge on the test charge, $\mathbf{E} = \mathbf{F}/q_0$. The test charge has to be small enough that it does not disturb the charges producing the field, since the charges on nearby bodies would otherwise move in response to it. The field of the point charge is therefore

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q(\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3}, \quad (1.6)$$

and for a charge at the origin it reduces to $\mathbf{E} = q\hat{\mathbf{r}}/4\pi\epsilon_0 r^2$, where $\hat{\mathbf{r}}$ is the unit vector pointing away from the origin. This has the same form as the gravitational field of a point mass m in Section 5.1 of Vol. 1 of the Fermi Notes on Physics, with $-Gm$ replaced by $q/4\pi\epsilon_0$. The change of sign reflects the fact that two masses attract each other, whereas two like charges repel.

Superposition

When several charges are present, experiment shows that the force on a test charge is the vector sum of the forces that each charge would exert on it alone, and that the force between any two of the charges is unaffected by the presence of the others. This experimental fact is called the principle of superposition. The field of point charges $q_1, q_2, \dots$ at the positions $\mathbf{r}_1, \mathbf{r}_2, \dots$ is therefore

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_i q_i \frac{\mathbf{r} - \mathbf{r}_i}{|\mathbf{r} - \mathbf{r}_i|^3}. \quad (1.7)$$

At the point midway between two equal charges, for example, the two terms are equal in magnitude and opposite in direction, and the field there is zero.

Continuous Distributions of Charge

A charged body contains an enormous number of electrons and nuclei, and on the scale of a charged rod or a charged sheet the graininess of the charge is far too fine to notice. It is then a valid approximation to treat the charge as spread continuously through the body, and to describe it by a density. Charge spread through a volume, for example through a charged plastic bead, is described by its charge per unit volume at the source point $\mathbf{r}'$, which is called the volume charge density $\rho(\mathbf{r}')$, so that a small volume $d\mathcal{V}'$ carries a charge $dq = \rho d\mathcal{V}'$. Charge spread over a surface, for example over a thin charged sheet, is described by its charge per unit area, which is called the surface charge density σ , with $dq = \sigma da'$ on a patch of area da' . Charge spread along a thin wire or thread is described by its charge per unit length, which is called the line charge density λ , with $dq = \lambda d\ell'$ on a length $d\ell'$.

Each small piece of the distribution acts as a point charge, and superposition turns the sum over point charges into an integral. For a volume distribution the field is

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \rho(\mathbf{r}') \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3} d\mathcal{V}', \quad (1.8)$$

and the fields of surface and line distributions are the same integral with $\rho d\mathcal{V}'$ replaced by $\sigma da'$ or $\lambda d\ell'$. The primed coordinates of the source point are the variables of integration, while the field point $\mathbf{r}$ is held fixed. The integrand is a vector, and the integral is done one component at a time, as Section 5.2 of Vol. 1 noted of the corresponding gravitational integral.

A point charge can be written as a density with the three-dimensional delta function $\delta^3(\mathbf{r}) = \delta(x)\delta(y)\delta(z)$, built from the delta function of Section 2.3 of Vol. 3, which Section 6.3 of Vol. 4 of the Fermi Notes on Physics treats more fully. The integral of $\delta^3(\mathbf{r})$ over any volume containing the origin is one, and its integral over a volume that does not contain the origin is zero. A charge q at $\mathbf{r}_0$ has the density $\rho(\mathbf{r}') = q\delta^3(\mathbf{r}' - \mathbf{r}_0)$, and applying the property $\int f(x)\delta(x - x_0)dx = f(x_0)$ of Section 2.3 of Vol. 3 to each coordinate in turn reduces the integral above to the field of a point charge at $\mathbf{r}_0$.

The integrals over a surface or a volume are defined, as in Sections

2.3 and 3.2 of Vol. 4, as limits of sums over small patches or small volumes. In the cylindrical coordinates (s, ϕ, z) , with $x = s \cos \phi$ and $y = s \sin \phi$, the element of volume is $dV = s ds d\phi dz$, and in the spherical coordinates (r, θ, ϕ) it is $r^2 \sin \theta dr d\theta d\phi$, as Section 2.3 of Vol. 4 shows. The element of area of a flat surface perpendicular to the z -axis is $s ds d\phi$, that of a cylinder of radius s about the axis is $s d\phi dz$, and that of a sphere of radius r is $r^2 d\Omega$, where $d\Omega = \sin \theta d\theta d\phi$ is the element of solid angle of Section 4.1 of Vol. 3. As an example, the total charge on a disc of radius a carrying a uniform density σ is

$$Q = \int_0^{2\pi} \int_0^a \sigma s ds d\phi = \pi a^2 \sigma, \quad (1.9)$$

and doing the integral over ϕ first amounts to slicing the disc into rings of area $2\pi s ds$, as the disc was sliced to find its moment of inertia in Section 3.3 of Vol. 1.

The Field of a Straight Segment

As a first example, consider a thin straight rod carrying a uniform line charge density λ . We set up coordinates such that the rod lies along the z -axis between z_1 and z_2 , and such that the field point P lies in the plane $z = 0$, a distance s from the axis, as in Figure 4. Any field point can be brought to this position by moving the origin along the rod to the foot O of the perpendicular from P. Let $\hat{\mathbf{s}}$ be the unit vector pointing from O towards P. The element of the rod between z' and $z' + dz'$ carries a charge $\lambda dz'$, the separation is $\mathbf{r} - \mathbf{r}' = s\hat{\mathbf{s}} - z'\hat{\mathbf{z}}$, and its magnitude is $(s^2 + z'^2)^{1/2}$, so the element contributes

$$d\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{\lambda (s\hat{\mathbf{s}} - z'\hat{\mathbf{z}})}{(s^2 + z'^2)^{3/2}} dz'. \quad (1.10)$$

The integral is simplest in terms of the angle β at which the element is seen from P, measured from the perpendicular PO and taken as positive for elements with $z' > 0$. From Figure 4, $z' = s \tan \beta$, so that $dz' = s \sec^2 \beta d\beta$ and $s^2 + z'^2 = s^2 \sec^2 \beta$. When these expressions are substituted, the powers of s and $\sec \beta$ cancel to leave

$$d\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{\lambda}{s} (\cos \beta \hat{\mathbf{s}} - \sin \beta \hat{\mathbf{z}}) d\beta. \quad (1.11)$$

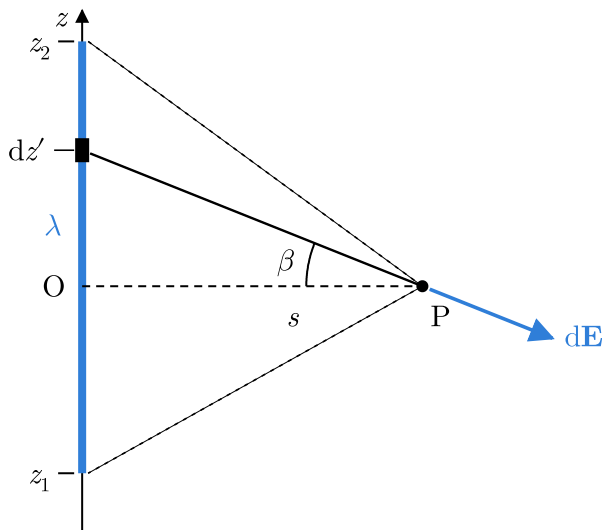


Figure 4. The field of a uniformly charged straight segment, which lies on the z -axis between z_1 and z_2 , at a point P a perpendicular distance s from the axis. The element dz' is seen from P at the angle β to the perpendicular PO , and its field $d\mathbf{E}$ (accent) points along the line from the element to P .

The ends of the rod are seen at the angles β_1 and β_2 , with $\tan \beta_1 = z_1/s$ and $\tan \beta_2 = z_2/s$. Integrating between these limits gives the component of the field away from the rod,

$$E_s = \frac{1}{4\pi\epsilon_0} \frac{\lambda}{s} (\sin \beta_2 - \sin \beta_1) , \quad (1.12)$$

and the component along the rod,

$$E_z = \frac{1}{4\pi\epsilon_0} \frac{\lambda}{s} (\cos \beta_2 - \cos \beta_1) . \quad (1.13)$$

When P lies opposite the midpoint of the rod, $\beta_1 = -\beta_2$ and E_z vanishes, as it must, since the elements at z' and $-z'$ then give equal and opposite components along the rod.

For an infinitely long line the angles tend to $\beta_1 = -\pi/2$ and $\beta_2 =$

$\pi/2$, and the field becomes

$$\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 s} \hat{\mathbf{s}}. \quad (1.14)$$

The same limit applies to a finite rod at points much closer to the rod than to either of its ends. The field of the line goes as $1/s$, which falls more slowly with distance than the field of a point charge. The reason is that most of the field at P comes from the part of the line within a distance of order s of the foot O, and the charge on that part grows in proportion to s , which partly compensates for the inverse-square fall of the field of each element.

The Field of a Ring on Its Axis

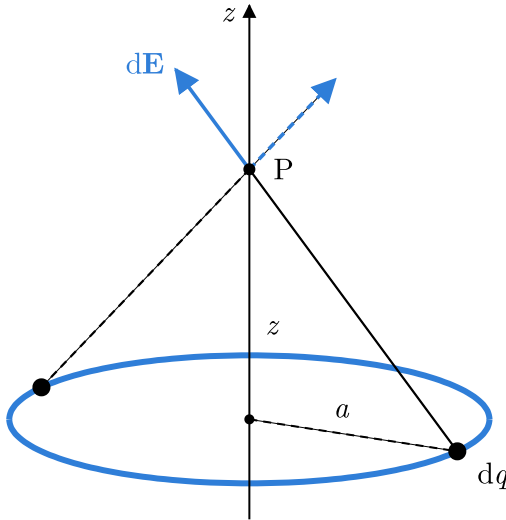


Figure 5. A uniformly charged ring of radius a and a point P on its axis at a height z above its centre. The fields $d\mathbf{E}$ of the element dq and of the element diametrically opposite it (dashed) have equal components along the axis, and their components perpendicular to the axis cancel.

As another example, consider a thin ring of radius a carrying a total charge Q spread uniformly around it. Let the ring lie in the xy -plane with its centre at the origin, and let P be a point on the

z -axis at height z , as in Figure 5. Every element dq of the ring is at the same distance $(a^2 + z^2)^{1/2}$ from P, and its field has a component along the axis equal to the magnitude of its field times $z/(a^2 + z^2)^{1/2}$, the cosine of the angle between the axis and the line from the element to P. The component perpendicular to the axis is cancelled by the element on the opposite side of the ring, whose field has the same magnitude and the same component along the axis. The field on the axis therefore lies along the axis, and adding the axial components of all the elements gives

$$E_z = \frac{1}{4\pi\epsilon_0} \frac{z}{(a^2 + z^2)^{3/2}} \int dq. \quad (1.15)$$

The integral is the total charge Q , and so

$$E_z = \frac{1}{4\pi\epsilon_0} \frac{Qz}{(a^2 + z^2)^{3/2}}. \quad (1.16)$$

At the centre of the ring the field is zero, since the fields of diametrically opposite elements cancel there. Below the ring z is negative and the field points downwards, away from the ring when Q is positive. Far from the ring, where $z \gg a$, the term a^2 can be neglected, and the field becomes $Q/4\pi\epsilon_0 z^2$, the field of a point charge Q . The gravitational field on the axis of a ring, which has the same form, was found in Problem 2 of Section 5.2 of Vol. 1, and you will find in Problem 4 the height at which the field on the axis is strongest.

The Field of a Disc on Its Axis

Let us consider one last example, a thin disc of radius a carrying a uniform surface charge density σ , with the field point P on its axis at a height $z > 0$. We slice the disc into thin rings, as in Figure 6, since every part of a ring is at the same distance from P and the result for a ring can then be used directly. The ring of radius s' and width ds' has area $2\pi s' ds'$ and charge $dq = 2\pi\sigma s' ds'$. Replacing Q by dq and a by s' in the field of the ring gives its contribution

$$dE_z = \frac{\sigma z}{2\epsilon_0} \frac{s' ds'}{(s'^2 + z^2)^{3/2}}. \quad (1.17)$$

With the substitution $w = s'^2 + z^2$, for which $dw = 2s' ds'$, the

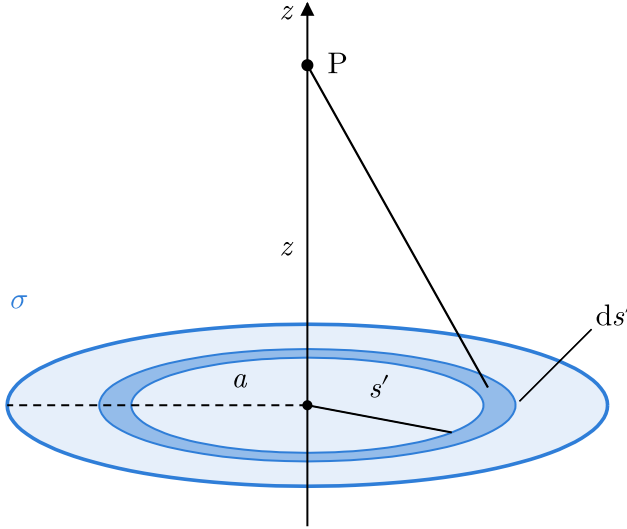


Figure 6. A uniformly charged disc of radius a , sliced into rings. Every part of the shaded ring, of radius s' and width ds' , lies at the same distance from the point P on the axis.

integral over s' becomes

$$\int_0^a \frac{s' ds'}{(s'^2 + z^2)^{3/2}} = \frac{1}{2} \int_{z^2}^{a^2+z^2} w^{-3/2} dw = \frac{1}{z} - \frac{1}{\sqrt{a^2 + z^2}}, \quad (1.18)$$

and the field on the axis is therefore

$$E_z = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{a^2 + z^2}} \right). \quad (1.19)$$

The disc looks the same from below as from above, so at the point a distance z below the disc the field has the same magnitude and points downwards.

We can check the result by looking at limiting cases. Far from the disc, where $z \gg a$, the binomial expansion gives

$$\frac{z}{\sqrt{a^2 + z^2}} = \left(1 + \frac{a^2}{z^2} \right)^{-1/2} \approx 1 - \frac{a^2}{2z^2}, \quad (1.20)$$

so that $E_z \approx \sigma a^2 / 4\epsilon_0 z^2 = Q / 4\pi\epsilon_0 z^2$, where $Q = \pi a^2 \sigma$ is the charge of the disc. From far away the disc looks like a point charge, as it

should. In the opposite limit, where $a \rightarrow \infty$ or, for a finite disc, where $z \ll a$, the second term in the bracket is negligible, and the field of an infinite plane of charge is

$$\mathbf{E} = \frac{\sigma}{2\epsilon_0} \hat{\mathbf{n}}, \quad (1.21)$$

where $\hat{\mathbf{n}}$ is the unit vector perpendicular to the plane, pointing away from it on the side where the field is measured. This field does not depend on the distance from the plane. As P moves away from the plane the field of each element weakens as the inverse square of its distance, but the area of the plane lying within a given angle of the perpendicular from P grows as the square of that distance, and the weakening of the field and the growth of the area cancel. On crossing the plane the field reverses direction, so its component along a fixed normal jumps by σ/ϵ_0 .

1.3 Electric Flux and Gauss's Law

The fields of Section 1.2 were found by adding up the contributions of all the elements of a distribution, one component at a time. When the distribution has enough symmetry there is a much shorter route to the field, which rests on a relation between the field on a closed surface and the charge inside that surface. We first define the flux of a field through a surface and the solid angle, then prove Gauss's law and apply it to a sphere, a line, and a plane, and finally turn it into a differential equation that holds at every point.

Flux through a Surface

A small flat patch of a surface is described by the vector $d\mathbf{a} = \hat{\mathbf{n}} da$, whose magnitude is the area of the patch and whose direction is the unit normal $\hat{\mathbf{n}}$ to the patch, and which is called the vector area of the patch. For a surface with an edge, such as a disc, either of the two normals may be chosen, as long as the normal is taken on the same side of the surface everywhere, while for a closed surface, such as a sphere, the normal is taken to point outwards.

The quantity $\mathbf{E} \cdot d\mathbf{a} = E \cos \eta da$, where η is the angle between the field and the normal, is called the flux of the field $\mathbf{E}$ through the

patch, and the flux through a whole surface S is the surface integral

$$\Phi_E = \int_S \mathbf{E} \cdot d\mathbf{a}, \quad (1.22)$$

defined, as in Section 1.2, as the limit of a sum over small patches, and treated at length in Section 3.2 of Vol. 4. An integral over a closed surface is marked by a circle on the integral sign, $\oint_S$. For example, a uniform field $\mathbf{E}$ passing through a flat area A whose normal makes an angle η with the field gives a flux $EA \cos \eta$, which is largest when the normal is parallel to the field and zero when the field lies in the plane of the area. If the number of field lines drawn through a unit area is proportional to the strength of the field, the flux is proportional to the number of lines crossing the surface.

The Solid Angle

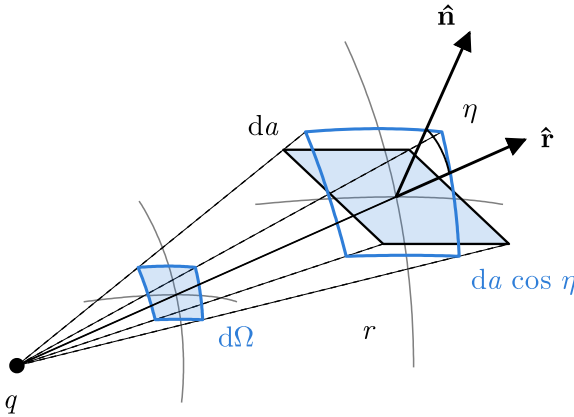


Figure 7. A small patch da (lightly shaded) at a distance r from a point charge q , with its unit normal $\hat{n}$ at an angle η to the radial direction $\hat{r}$, which is the direction of the field. The rays from q through the corners of the patch (dashed) cut the sphere of radius r centred on q in a patch (accent outline) of area $da \cos \eta$, and they cut a smaller sphere centred on q in a smaller patch that subtends the same solid angle $d\Omega$. A faint arc of each sphere runs through its patch.

Just as a plane angle is the length of the arc it cuts from a circle centred on its vertex divided by the radius, a solid angle is the area it cuts from a sphere centred on its vertex divided by the square of the radius, and the full solid angle is 4π . The element of solid angle $d\Omega = \sin\theta d\theta d\phi$ of Section 4.1 of Vol. 3 is the area of a small patch of a sphere of unit radius, in agreement with this definition.

Let us consider a small patch da at a distance r from the origin, whose normal makes an angle η with the radial direction $\hat{\mathbf{r}}$, as in Figure 7. The rays from the origin through the edges of the patch cut out a small patch of the sphere of radius r whose area is the projection of da perpendicular to $\hat{\mathbf{r}}$, namely $da \cos\eta$, since for a small patch the rays are very nearly parallel and the sphere is very nearly flat. The patch therefore subtends the solid angle

$$d\Omega = \frac{\hat{\mathbf{r}} \cdot d\mathbf{a}}{r^2}, \quad (1.23)$$

which is negative when the normal points back towards the origin.

The flux of the field of a point charge q at the origin through the patch is then

$$\mathbf{E} \cdot d\mathbf{a} = \frac{1}{4\pi\epsilon_0} \frac{q \hat{\mathbf{r}} \cdot d\mathbf{a}}{r^2} = \frac{1}{4\pi\epsilon_0} q d\Omega. \quad (1.24)$$

The flux depends on the distance and the orientation of the patch only through the solid angle that the patch subtends. This result depends on the inverse-square law, since the area cut from a cone of given solid angle grows as r^2 while the field falls as $1/r^2$.

Gauss's Law

Let a closed surface S enclose a point charge q , and divide the space around the charge into narrow cones with their vertices at the charge, as in Figure 8. A cone of solid angle $d\Omega$ that crosses the surface once contributes a flux $(q/4\pi\epsilon_0) d\Omega$. If the surface is folded, a cone may cross it three or more times, but the crossings after the first come in pairs, one inwards and one outwards. At an inward crossing the outward normal points back towards the charge, so $\hat{\mathbf{r}} \cdot d\mathbf{a}$ is negative, and the crossings of a pair cancel. Adding the cones in all directions gives a total solid angle 4π and a total flux q/ϵ_0 . When the charge lies outside the surface, every cone that

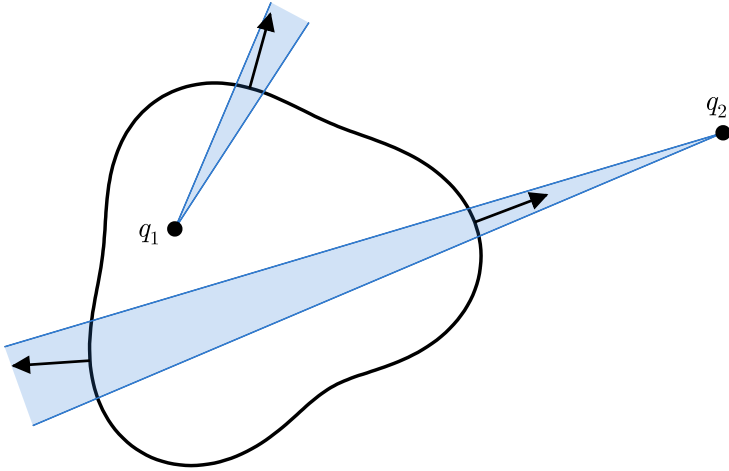


Figure 8. A closed surface, with a narrow cone drawn from a charge q_1 inside it and another from a charge q_2 outside it. The cone from q_1 crosses the surface once, going outwards. The cone from q_2 crosses it twice, once inwards and once outwards, and the arrows show the outward normal at each crossing.

meets the surface crosses it an even number of times, in pairs of inward and outward crossings, and the total flux is zero.

A general distribution of charge is a superposition of point charges, and the flux of the total field is the sum of the fluxes of the separate fields. The charges inside S contribute their charge divided by ϵ_0 , and the charges outside contribute nothing, so that

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0}, \quad (1.25)$$

where Q_{enc} is the total charge enclosed by S , equal to $\int \rho dV$ over the volume inside S for a continuous distribution. This result is called Gauss's law, which Section 3.3 of Vol. 4 derives in another way, from the divergence theorem. A surface to which the law is applied is called a Gaussian surface. Gauss's law holds for any distribution of charges at rest and for any closed surface. A charge outside the surface changes the field on the surface but not the total flux through it. A charge placed just outside a sphere, for example,

strengthens the field on the near side of the sphere, but the flux it sends in through that side is equal to the flux it sends out through the far side.

To find the field itself from Gauss's law, we need to know from the symmetry of the distribution which way the field points and that its magnitude is the same over parts of the surface, as it is in the examples that follow.

The Field of a Spherical Distribution

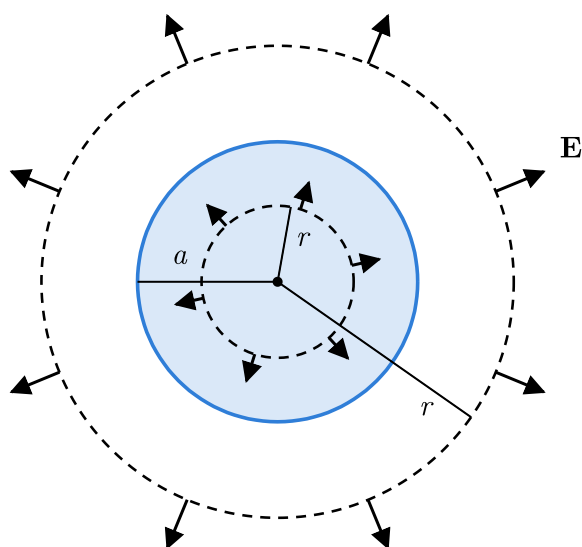


Figure 9. A spherically symmetric distribution of charge of radius a (accent), with Gaussian spheres of radius r (dashed) inside and outside it. On each Gaussian sphere the field is radial and has the same magnitude everywhere.

We begin with a distribution of charge with spherical symmetry, whose density $\rho(r)$ depends only on the distance r from a centre. A rotation about any axis through the centre leaves the distribution unchanged, so it must leave the field unchanged as well. A rotation through half a turn about the line from the centre through a point P leaves P where it is, and it would reverse any component of the field at P perpendicular to that line, so no such component can exist,

and the field at P is radial. A rotation carrying P to any other point at the same distance from the centre shows that the magnitude of the field is the same at both points. The field therefore has the form $\mathbf{E} = E(r) \hat{\mathbf{r}}$.

On a Gaussian sphere of radius r centred on the distribution, as in Figure 9, the flux is $E(r) 4\pi r^2$, and Gauss's law gives

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q_{\text{enc}}(r)}{r^2} \hat{\mathbf{r}}, \quad Q_{\text{enc}}(r) = \int_0^r \rho(r') 4\pi r'^2 dr', \quad (1.26)$$

where the enclosed charge is found with the spherical element of volume of Section 1.2, integrated over the angles. At any point outside the distribution the enclosed charge is the total charge Q , and the field is that of a point charge Q at the centre. For a thin spherical shell of charge the enclosed charge is zero at every point inside the shell, and the field there is zero. These two results make up the shell theorem of Section 5.2 of Vol. 1, proved there for gravitation by slicing the shell into rings. Gauss's law gives the theorem without any integration, and the same argument applies to gravitation, which also obeys an inverse-square law.

For a sphere of radius a with a total charge Q spread uniformly through it, the charge enclosed by a sphere of radius $r < a$ is the fraction r^3/a^3 of the whole, and the field is

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Qr}{a^3} \hat{\mathbf{r}} \quad (r < a), \quad \mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{\mathbf{r}} \quad (r > a). \quad (1.27)$$

At $r = a$ the expressions for the field inside and outside the sphere agree.

The Field of an Infinite Line

The next example is an infinite straight line carrying a uniform line charge density λ . The line is unchanged by a shift along itself and by a rotation about itself, so the field depends only on the distance s from the line. A reflection in the plane through a point P perpendicular to the line, or in the plane containing the line and P, also leaves the line unchanged, and it would reverse a component of the field along the line or along the direction $\hat{\phi}$ in which ϕ increases, so $\mathbf{E} = E(s) \hat{\mathbf{s}}$.

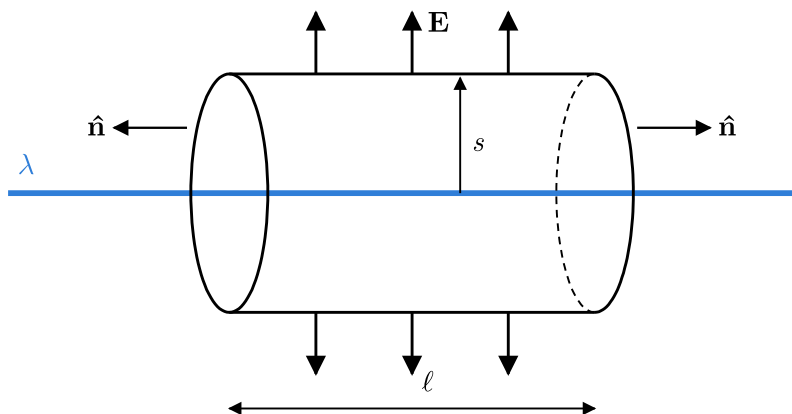


Figure 10. A Gaussian cylinder of radius s and length ℓ around an infinite line of charge. The field crosses the curved surface at right angles and lies along the end faces, whose outward normals $\hat{\mathbf{n}}$ point along the line.

The Gaussian surface is a closed cylinder of radius s and length ℓ with the line as its axis, drawn in Figure 10. The field lies along the flat end faces, so they contribute no flux, while on the curved surface the field is perpendicular to the surface and has the same magnitude everywhere, giving a flux $E(s) 2\pi s\ell$. The enclosed charge is $\lambda\ell$, and Gauss's law gives

$$E(s) = \frac{\lambda}{2\pi\epsilon_0 s}, \quad (1.28)$$

which is the field found in Section 1.2 as the limit of a long segment.

The Field of an Infinite Plane

The last example is an infinite plane carrying a uniform surface charge density σ . By arguments like those for the line, the field is perpendicular to the plane, depends only on the distance from it, and, by reflection in the plane, has the same magnitude at equal distances on the two sides, pointing away from the plane when σ is positive.

The Gaussian surface is a short cylinder, often called a pillbox, whose axis is perpendicular to the plane and whose end faces, each

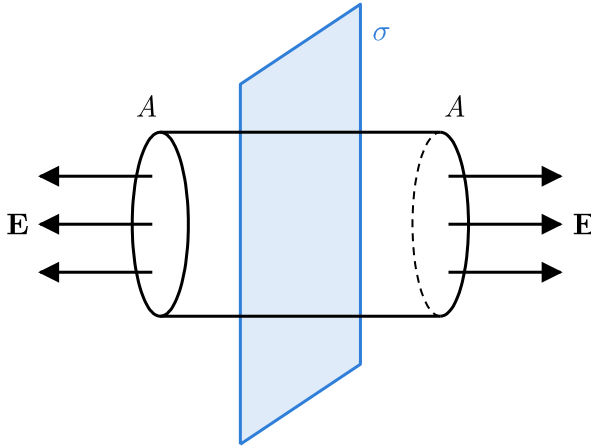


Figure 11. A Gaussian pillbox crossing an infinite plane of charge (accent), with end faces of area A parallel to the plane. The field leaves through both end faces and lies along the curved surface.

of area A , lie at equal distances on the two sides, as in Figure 11. The curved surface contributes no flux, since the field lies along it, and each end face contributes EA . The enclosed charge is σA , so $2EA = \sigma A/\epsilon_0$ and

$$E = \frac{\sigma}{2\epsilon_0}, \quad (1.29)$$

in agreement with the limit of the disc in Section 1.2. The field does not depend on the distance from the plane, since moving the end faces further out does not change the enclosed charge.

The Divergence

To turn Gauss's law into a statement about the field at a single point, we apply it to a very small volume, and for this we need the divergence of the field, $\nabla \cdot \mathbf{E} = \partial E_x/\partial x + \partial E_y/\partial y + \partial E_z/\partial z$, which is defined in Section 3.1 of Vol. 4. Section 3.3 of Vol. 4 shows, by adding the fluxes through the faces of a small box, that the divergence at a point is the outward flux per unit volume from a small region around the point, in the limit in which the region

shrinks to the point. The field $\kappa\mathbf{r}$, with κ a positive constant, has divergence 3κ , and its field lines spread out from every point, while a uniform field has zero divergence.

The Divergence Theorem

Adding the fluxes out of the small boxes into which a volume $\mathcal{V}$ bounded by a closed surface S can be divided gives the divergence theorem, or Gauss's theorem, which is proved in Section 3.3 of Vol. 4 and states that, for a field whose components and their first partial derivatives are continuous throughout the volume,

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \int_{\mathcal{V}} \nabla \cdot \mathbf{E} d\mathcal{V}. \quad (1.30)$$

Gauss's Law in Differential Form

Applying the divergence theorem to the left-hand side of Gauss's law, and writing the enclosed charge as the integral of ρ , gives Gauss's law in differential form,

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad (1.31)$$

as Section 3.3 of Vol. 4 shows. This equation relates the field near a point to the charge density at that point.

As an example, consider again the uniformly charged sphere. Inside it the field is $\mathbf{E} = Q\mathbf{r}/4\pi\epsilon_0 a^3 = \rho\mathbf{r}/3\epsilon_0$, where $\rho = 3Q/4\pi a^3$, and since $\nabla \cdot \mathbf{r} = \partial x/\partial x + \partial y/\partial y + \partial z/\partial z = 3$, its divergence is ρ/ϵ_0 , as it should be. Outside, the field is a multiple of $\mathbf{r}/r^3$, whose divergence is zero wherever $r \neq 0$, as Section 3.1 of Vol. 4 shows, and so the divergence is zero outside the sphere, where there is no charge.

The field of a point charge at the origin is also a multiple of $\mathbf{r}/r^3$, so it has zero divergence everywhere except at the origin, where the field is not defined, and yet its flux through any sphere centred on the origin is q/ϵ_0 . The vanishing divergence away from the origin and the flux q/ϵ_0 are consistent if the divergence of $\hat{\mathbf{r}}/r^2$ is written with the delta function of Section 1.2,

$$\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 4\pi \delta^3(\mathbf{r}), \quad (1.32)$$

which is zero away from the origin and whose integral over any volume containing the origin is 4π . With this expression the field of a point charge satisfies $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ with $\rho = q\delta^3(\mathbf{r})$, and Gauss's law in differential form holds for point charges as well as for continuous distributions.

1.4 The Electrostatic Potential and Field Energy

At school the potential was defined as the work done per unit charge in bringing a test charge from infinity, which assumes that this work does not depend on the path taken. In this section we first show that the line integral of an electrostatic field is independent of the path, and then define the potential, relate it to the field, and use it to find the fields of the distributions of Section 1.2 more easily. Following that, we combine the potential with Gauss's law to obtain Poisson's and Laplace's equations, and finally we find the energy of a system of charges and show that it can be written as an integral over the field.

The Line Integral of the Field

The line integral of the field along a path from A to B is defined, as the work done by a force along a path was in Section 2.3 of Vol. 1, by dividing the path into short steps $d\boldsymbol{\ell}$, forming the scalar product $\mathbf{E} \cdot d\boldsymbol{\ell}$ on each step, adding up the results, and taking the limit of shorter steps. The result is written $\int_A^B \mathbf{E} \cdot d\boldsymbol{\ell}$, and in general its value depends on the path, as the examples of Section 3.2 of Vol. 4 show.

Let us take first the field of a point charge q at the origin, and a path from a point A at distance r_A to a point B at distance r_B , as in Figure 12. Each step $d\boldsymbol{\ell}$ can be split into a part along $\hat{\mathbf{r}}$, whose length is the change dr in the distance from the charge, and a part perpendicular to $\hat{\mathbf{r}}$. The field is radial, so only the first part contributes to the scalar product, and $\mathbf{E} \cdot d\boldsymbol{\ell} = q dr/4\pi\epsilon_0 r^2$. The line integral is therefore

$$\int_A^B \mathbf{E} \cdot d\boldsymbol{\ell} = \frac{1}{4\pi\epsilon_0} q \int_{r_A}^{r_B} \frac{dr}{r^2} = \frac{1}{4\pi\epsilon_0} q \left(\frac{1}{r_A} - \frac{1}{r_B} \right), \quad (1.33)$$

which depends only on the distances of the two ends from the charge

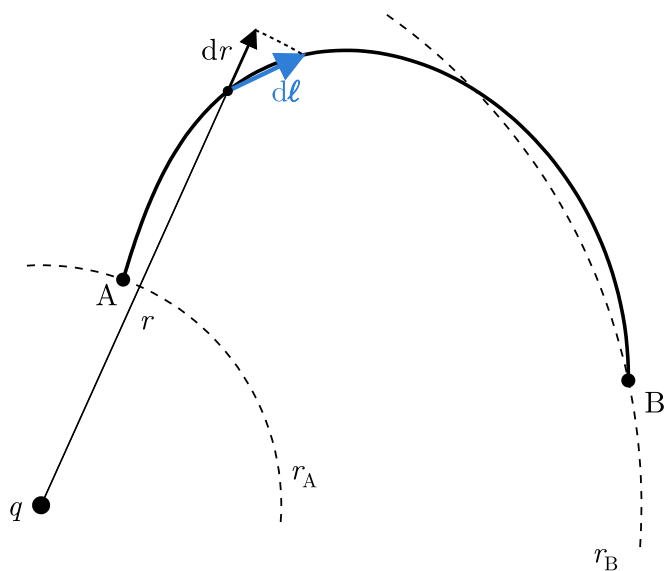


Figure 12. A path from A to B near a point charge q , with the dashed arcs of radius r_A and r_B centred on the charge. A step $d\ell$ along the path (accent) is split into a radial part, whose length dr is the change in the distance from q , and a part along the dotted arc, perpendicular to the field.

and not on the path between them. The field of any distribution of charges at rest is a sum of fields of point charges, and its line integral is the sum of their line integrals, each of which is independent of the path. The line integral of an electrostatic field from A to B is therefore the same along every path, and the integral of the field around any closed path is zero. The force $q\mathbf{E}$ on a charge is conservative in the sense of Section 2.3 of Vol. 1, and a charge carried round a closed loop and back to its starting point has had no net work done on it by the field.

The Potential

Since the line integral depends only on its end points, we can choose a reference point $\mathbf{r}_{\text{ref}}$ and define the potential at $\mathbf{r}$ as

$$V(\mathbf{r}) = - \int_{\mathbf{r}_{\text{ref}}}^{\mathbf{r}} \mathbf{E} \cdot d\boldsymbol{\ell}. \quad (1.34)$$

The potential difference between two points is then $V(B) - V(A) = - \int_A^B \mathbf{E} \cdot d\boldsymbol{\ell}$, and moving the reference point only adds a constant to V , which does not change any potential difference. For charge confined to a finite region, the reference point is usually taken at infinity. An external agent that moves a test charge q_0 slowly from A to B must supply a force $-q_0\mathbf{E}$ and does work $q_0[V(B) - V(A)]$, so q_0V is the potential energy of the test charge, and V is the work per unit charge of the school definition. For a point charge at $\mathbf{r}'$, the line integral above, taken from infinity, gives

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{|\mathbf{r} - \mathbf{r}'|}, \quad (1.35)$$

and superposition gives the potential of a continuous distribution as

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathcal{V}'. \quad (1.36)$$

The integrand is a scalar, so there are no components to resolve.

Since the line integral of the field does not depend on the path, Section 3.2 of Vol. 4 shows that the field is minus the gradient of the potential,

$$\mathbf{E} = -\nabla V. \quad (1.37)$$

The field is perpendicular to the equipotential surfaces and points towards lower potential, as Figure 1 showed, and its unit can equally be written as the volt per metre. Since $E_x = -\partial V/\partial x$ and $E_y = -\partial V/\partial y$, and the order of two partial differentiations makes no difference, an electrostatic field satisfies $\partial E_x/\partial y = \partial E_y/\partial x$ and the two corresponding equalities for the other pairs of coordinates. This is the test for a conservative force of Section 2.3 of Vol. 1, and a field that fails it, such as $\kappa(0, x, 0)$ with κ a constant, cannot be produced by charges at rest.

The Potentials of the Distributions of Section 1.2

We return first to the ring of Section 1.2, with charge Q and radius a . Every element of the ring is at the distance $(a^2 + z^2)^{1/2}$ from the point on the axis at height z , so the potential there is

$$V(z) = \frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{a^2 + z^2}}, \quad (1.38)$$

and differentiating gives $E_z = -dV/dz = Qz/4\pi\epsilon_0(a^2 + z^2)^{3/2}$, as in Section 1.2. The potential on the axis alone gives only the component of the field along the axis, since the other components need the potential off the axis. On the axis those components vanish by symmetry, as Section 1.2 showed.

The disc of Section 1.2, with radius a and density σ , is treated in the same way. The ring of radius s' and width ds' carries a charge $2\pi\sigma s' ds'$, all at the distance $(s'^2 + z^2)^{1/2}$, and adding the rings gives

$$V(z) = \frac{\sigma}{2\epsilon_0} \int_0^a \frac{s' ds'}{\sqrt{s'^2 + z^2}} = \frac{\sigma}{2\epsilon_0} \left(\sqrt{a^2 + z^2} - |z| \right), \quad (1.39)$$

where the integral follows from the substitution $w = s'^2 + z^2$ used in Section 1.2. For $z > 0$, minus the derivative with respect to z is $(\sigma/2\epsilon_0)[1 - z/(a^2 + z^2)^{1/2}]$, the field found in Section 1.2. Far from the disc the binomial expansion gives $(a^2 + z^2)^{1/2} - z \approx a^2/2z$, and the potential becomes $Q/4\pi\epsilon_0 z$, with $Q = \pi a^2 \sigma$.

For the infinite line the integral diverges, because the line carries an infinite charge, and the reference point cannot be placed at infinity. We integrate the field from a reference distance s_0 instead, which gives

$$V(s) = - \int_{s_0}^s \frac{\lambda}{2\pi\epsilon_0 s'} ds' = - \frac{\lambda}{2\pi\epsilon_0} \ln \frac{s}{s_0}. \quad (1.40)$$

The potential of a straight segment of finite length can be referred to infinity, and you will find it on the perpendicular bisector of the segment in Problem 1.

Finally, we return to the uniformly charged sphere of Section 1.3, with charge Q and radius a . Outside the sphere the field is that of a point charge, so $V = Q/4\pi\epsilon_0 r$ for $r \geq a$. Inside, the field is

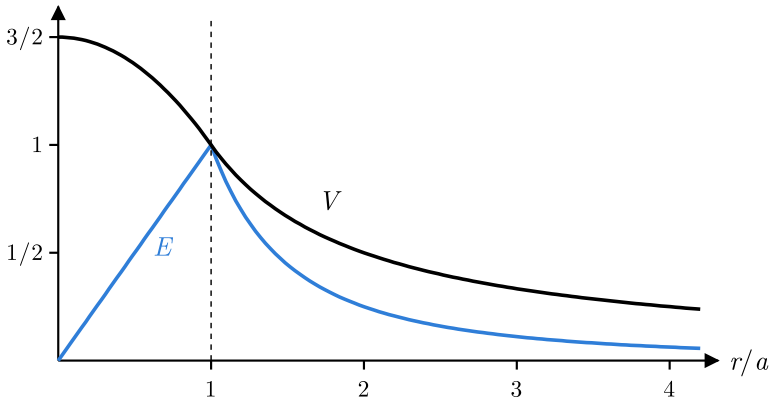


Figure 13. The field E (accent) and the potential V (solid) of a uniformly charged sphere of radius a , with r in units of a , E in units of $Q/4\pi\epsilon_0 a^2$, and V in units of $Q/4\pi\epsilon_0 a$. Inside the sphere the field grows linearly and the potential is a parabola with its maximum at the centre, and outside they go as $1/r^2$ and $1/r$.

$Qr/4\pi\epsilon_0 a^3$, and integrating it inwards from the surface gives

$$V(r) = \frac{1}{4\pi\epsilon_0} \frac{Q}{a} - \frac{1}{4\pi\epsilon_0} \int_a^r \frac{Qr'}{a^3} dr' = \frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \left(3 - \frac{r^2}{a^2} \right), \quad (1.41)$$

which has the same form as the gravitational potential of a uniform solid sphere found in Problem 1 of Section 5.2 of Vol. 1. The field and the potential are drawn in Figure 13. The potential is continuous at the surface, and it has its largest value, $3Q/8\pi\epsilon_0 a$, at the centre, where its slope and the field are both zero.

Poisson's and Laplace's Equations

Substituting $\mathbf{E} = -\nabla V$ into Gauss's law in differential form gives, as in Section 7.1 of Vol. 4, Poisson's equation

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}, \quad (1.42)$$

where $\nabla^2 V = \partial^2 V/\partial x^2 + \partial^2 V/\partial y^2 + \partial^2 V/\partial z^2$ is the Laplacian of V , the divergence of its gradient. In a region where there is no charge it becomes $\nabla^2 V = 0$, which is called Laplace's equation. Poisson's

equation relates the second derivatives of the potential at a point to the charge density at that point, while the integral for V above gives the potential from the charge everywhere. The methods of solving Laplace's equation for a region with given conditions on its boundary are not treated in this course, and Sections 7.1 to 7.3 of Vol. 4 treat them at length.

We can check Poisson's equation inside the uniformly charged sphere. There $r^2 = x^2 + y^2 + z^2$, so $\nabla^2(r^2) = 2 + 2 + 2 = 6$, and the potential found above gives $\nabla^2 V = -6Q/8\pi\epsilon_0 a^3 = -\rho/\epsilon_0$, with $\rho = 3Q/4\pi a^3$.

For a point charge at the origin, $V = q/4\pi\epsilon_0 r$. Section 3.1 of Vol. 4 shows that $\nabla(1/r) = -\mathbf{r}/r^3 = -\hat{\mathbf{r}}/r^2$, and the divergence of this gradient was stated in Section 1.3 to be zero away from the origin and to be $-4\pi\delta^3(\mathbf{r})$ when the origin is included. Therefore

$$\nabla^2\left(\frac{1}{r}\right) = -4\pi\delta^3(\mathbf{r}), \quad (1.43)$$

and the potential of a point charge satisfies Poisson's equation with $\rho = q\delta^3(\mathbf{r})$. This is the behaviour of the Laplacian of $1/r$ at the origin that Section 5.1 of Vol. 3 quoted as beyond that course.

The Energy of a System of Charges

The energy of a set of point charges $q_1, q_2, \dots$ at the positions $\mathbf{r}_1, \mathbf{r}_2, \dots$ is the work needed to assemble them, bringing them slowly in from infinity one at a time. No work is needed for the first charge, since no field acts on it. The second charge is brought in against the field of the first, which takes work $q_2 V_1(\mathbf{r}_2) = q_1 q_2 / 4\pi\epsilon_0 |\mathbf{r}_2 - \mathbf{r}_1|$, where V_1 is the potential of the first charge. The third is brought in against the fields of the first two, and so on. Each pair of charges contributes once, and the total is

$$U = \frac{1}{4\pi\epsilon_0} \sum_{i < j} \frac{q_i q_j}{|\mathbf{r}_i - \mathbf{r}_j|}, \quad (1.44)$$

which does not depend on the order in which the charges were brought in, since it is a sum over pairs. For two opposite charges the energy is negative, and work must be supplied to pull them apart. Summing over all i and j with $i \neq j$ counts each pair twice,

so the energy can also be written

$$U = \frac{1}{2} \sum_i q_i V_i, \quad (1.45)$$

where V_i is the potential at $\mathbf{r}_i$ due to all the charges other than q_i . For a continuous distribution the sum becomes an integral,

$$U = \frac{1}{2} \int \rho V d\mathcal{V}, \quad (1.46)$$

where V is now the potential of the whole distribution. Including the charge in the element $d\mathcal{V}$ itself makes no difference in the limit of small elements, since the potential that a small sphere of radius b and density ρ produces at its centre goes as ρb^2 and vanishes as $b \rightarrow 0$.

The uniformly charged sphere serves once more as an example. With the potential found above, and the spherical element of volume integrated over the angles,

$$U = \frac{\rho}{2} \frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \int_0^a \left(3 - \frac{r^2}{a^2}\right) 4\pi r^2 dr = \frac{\rho Q a^2}{5\epsilon_0}, \quad (1.47)$$

and with $\rho = 3Q/4\pi a^3$ the energy of the sphere is

$$U = \frac{1}{4\pi\epsilon_0} \frac{3Q^2}{5a}. \quad (1.48)$$

The Energy Density of the Field

The energy can be expressed in terms of the field alone. Replacing ρ by $\epsilon_0 \nabla \cdot \mathbf{E}$ gives $U = \frac{1}{2} \epsilon_0 \int V \nabla \cdot \mathbf{E} d\mathcal{V}$, and to rearrange the integrand we need the following identity

$$\nabla \cdot (V\mathbf{E}) = V \nabla \cdot \mathbf{E} + \mathbf{E} \cdot \nabla V. \quad (1.49)$$

It is one of the identities proved in Section 3.1 of Vol. 4. With $\nabla V = -\mathbf{E}$, the identity gives $V \nabla \cdot \mathbf{E} = \nabla \cdot (V\mathbf{E}) + E^2$.

The integral $\frac{1}{2} \int \rho V d\mathcal{V}$ receives contributions only where ρ is non-zero, so it can be taken over any volume $\mathcal{V}$ that contains all the charge. Using the divergence theorem on the first term, we obtain

$$U = \frac{\epsilon_0}{2} \oint_S V \mathbf{E} \cdot d\mathbf{a} + \frac{\epsilon_0}{2} \int_{\mathcal{V}} E^2 d\mathcal{V}, \quad (1.50)$$

where S is the surface of $\mathcal{V}$. Let $\mathcal{V}$ be a sphere of radius r centred on the distribution, and let r grow without limit. Far from a distribution confined to a finite region, the potential falls at least as fast as $1/r$ and the field at least as fast as $1/r^2$, while the area of the sphere grows as r^2 , so the surface integral falls at least as fast as $1/r$ and vanishes as $r \rightarrow \infty$. The volume integral then extends over all space, and

$$U = \frac{\epsilon_0}{2} \int E^2 d\mathcal{V}. \quad (1.51)$$

The energy can therefore be regarded as spread through space with the density

$$u = \frac{1}{2} \epsilon_0 E^2, \quad (1.52)$$

which is called the energy density of the electric field. The two expressions for U give the same total. The first assigns the energy to the places where there is charge, and the second to the places where there is field, including empty space, and it shows that the energy of a continuous distribution is never negative.

Once again, we take the uniformly charged sphere as an example. Using the field of Section 1.3, the energy in the field outside the sphere is

$$\frac{\epsilon_0}{2} \int_a^\infty \left(\frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \right)^2 4\pi r^2 dr = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{2a}, \quad (1.53)$$

and the energy in the field inside it is

$$\frac{\epsilon_0}{2} \int_0^a \left(\frac{1}{4\pi\epsilon_0} \frac{Qr}{a^3} \right)^2 4\pi r^2 dr = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{10a}. \quad (1.54)$$

The sum is $(1/4\pi\epsilon_0)(3Q^2/5a)$, in agreement with the calculation from ρV , and five sixths of the energy lies in the field outside the sphere.

For point charges the sum over pairs and the integral of the energy density differ. The field energy of a single point charge q , integrated over the region outside a small sphere of radius b about it, is $q^2/8\pi\epsilon_0 b$ by the calculation above, and it grows without limit as $b \rightarrow 0$. When the field of a set of point charges is squared, the square of the field of each charge alone gives such an infinite term, called the self-energy of the charge, and the products of the fields

of different charges give the energy of the pairs. The sum over pairs leaves out the self-energies, which do not change when the charges are moved, so the change in the energy of a set of point charges when they are rearranged is correctly given by the sum over pairs.

Problems

§1.1 Electricity at School

1. Positive point charges q_1 and q_2 are placed a distance ℓ apart. Find the point on the line between them at which the electric field is zero.
2. An electron starts from rest and is accelerated through a potential difference V . Find its final speed, and show that its kinetic energy, expressed in electronvolts, is equal to V expressed in volts.
3. A particle of charge q and mass m enters the region between the plates of Figure 2 with speed v parallel to the plates. The plates have length ℓ , separation d , and potential difference V . Show that when the particle leaves the plates it has been deflected by $qV\ell^2/2mdv^2$ towards one plate, and find the angle between its final velocity and the plates.
4. An oil drop of mass m carrying a negative charge is held at rest between two horizontal plates a distance d apart, with a potential difference V between them. Find the magnitude of the charge on the drop, and say which plate is at the higher potential.

§1.2 Coulomb's Law and the Electric Field

1. A thin rod bent into a semicircle of radius a carries a charge Q spread uniformly along it. Find the electric field at the centre of the semicircle.
2. A thin rod of length ℓ lies along the z -axis between $z = 0$ and $z = \ell$ and carries a uniform line charge density λ . Find the field at the point $z = -z_0$ on the axis, where $z_0 > 0$, and check that it reduces to the field of a point charge when $z_0 \gg \ell$.

3. A square loop of side ℓ carries a uniform line charge density λ . Use the field of a straight segment to find the field at a point on the axis of the loop, a height z above its centre, and check the limit $z \gg \ell$.
4. Find the height z at which the field on the axis of the ring of Figure 5 is strongest, and find the field there.
5. A flat annulus, bounded by circles of radii a and $b > a$, carries a uniform surface charge density σ . Find the field on its axis, and check that it reduces to the field of a disc when $a \rightarrow 0$.

§1.3 Electric Flux and Gauss's Law

1. A point charge q sits at the centre of a cube. Find the flux of its field through one face of the cube. The charge is then moved to one corner of the cube. Find the flux through the whole surface of the cube, and through each of the three faces that do not meet at that corner.
2. A long cylinder of radius a carries a uniform charge density ρ . Find the field inside and outside the cylinder, and check that the field inside satisfies $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$.
3. An infinite slab of thickness 2ℓ , lying between the planes $x = -\ell$ and $x = \ell$, carries a uniform charge density ρ . Use a pillbox placed symmetrically about the plane $x = 0$ to find the field inside and outside the slab.
4. Concentric thin spherical shells, of radii a and $b > a$, carry charges Q_1 and Q_2 . Find the field in the three regions $r < a$, $a < r < b$, and $r > b$. What must Q_2 be for the field outside both shells to vanish?
5. Inside a sphere of radius a the field is $\mathbf{E} = E_0(r/a)^2 \hat{\mathbf{r}}$, where E_0 is a constant. Find the charge density $\rho(r)$ in two ways, first from Gauss's law in integral form and then from $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$.
6. A uniform field $\mathbf{E}$ is parallel to the axis of a hemisphere of radius a . Find the flux of the field through the curved surface of the hemisphere, by closing the surface with a flat disc.

§1.4 The Electrostatic Potential and Field Energy

1. A thin rod of length 2ℓ carries a uniform line charge density λ . Show by differentiation that $\ln(z' + \sqrt{z'^2 + s^2})$ has the derivative $1/\sqrt{z'^2 + s^2}$ with respect to z' , and use this result to find the potential at a point a distance s from the midpoint of the rod, on the perpendicular bisector. Find the field there from the potential, and check it against the field of a straight segment in Section 1.2.
2. A thin spherical shell of radius a carries a charge Q spread uniformly over it. Find the potential inside and outside the shell, and find the energy of the shell in two ways, first from $\frac{1}{2} \int \sigma V da$ and then from the energy density of the field.
3. Equal charges q are placed at the four corners of a square of side ℓ . Find the work needed to assemble them from infinity.
4. Take the potential on the axis of the charged cylinder of Problem 2 of Section 1.3 to be zero. Find the potential inside the cylinder, and check that it satisfies Poisson's equation.
5. Which of the fields $\mathbf{E}_1 = \kappa(y, x, 0)$ and $\mathbf{E}_2 = \kappa(y, -x, 0)$, where κ is a constant, could be produced by charges at rest? For the one that could, find the potential and the charge density.
6. Show that the energy of the uniformly charged sphere can be found by building it up in thin shells, each brought in from infinity and added to the surface of the sphere already assembled, and that the result agrees with the one found in the text.
7. Find the work needed to bring a point charge q from infinity to the centre of the uniformly charged sphere, through a narrow tunnel drilled along a radius.

Conductors and Capacitance

2.1 Conductors and Capacitance

The charge distributions of the previous chapter were given in advance, as though every charge were held in place. In a conductor some of the charges are free to move, and they move until the field inside the material vanishes. In this section we first review the capacitor as it is treated at school, then work out the properties of a conductor in electrostatic equilibrium and use them to find the capacitance of parallel plates, concentric spheres, and coaxial cylinders, and finally find the energy stored in a capacitor and the force between its plates.

Capacitors at School

At school a capacitor is introduced as a pair of conducting plates separated by an insulator, such as air or a thin plastic film. Connected to a battery, it charges until one plate carries a charge Q and the other $-Q$, with the potential difference V of the battery between them. The charge is proportional to the potential difference, and the constant of proportionality,

$$C = \frac{Q}{V}, \quad (2.1)$$

is called the capacitance, whose unit, the farad, is defined as one coulomb per volt. For parallel plates of area A a distance d apart,

$C = \epsilon_0 A/d$, and filling the gap with an insulating material raises C by a factor ϵ_r , which is called the relative permittivity of the material. Capacitors in parallel share a potential difference, and capacitors in series carry the same charge, so that the combined capacitance of C_1 and C_2 is given by

$$C = C_1 + C_2 \quad (\text{in parallel}), \quad \frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} \quad (\text{in series}). \quad (2.2)$$

A charged capacitor stores the energy

$$U = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{Q^2}{2C}, \quad (2.3)$$

which it can release quickly, as it does, for example, in the flash of a camera, where the charge on a capacitor is sent through a lamp in a small fraction of a second to give a brief, bright burst of light. The results for a capacitor with nothing between its plates rest on the way in which charge arranges itself on a conductor, so we begin by working out this arrangement.

Conductors in Electrostatic Equilibrium

A conductor is a material in which some of the charges are free to move through the whole body. The conduction electrons in a metal such as copper, for example, are free to move in this way. In an electric field the free charges move under the force $q\mathbf{E}$ until no force acts on them, and the conductor is then said to be in electrostatic equilibrium, a state with the following properties:

1. the electric field is zero at every point inside the material;
2. the charge density is zero inside the material, so any net charge sits on the surface;
3. the whole conductor, including its surface, is at a single potential;
4. just outside the surface, the field is perpendicular to the surface and has magnitude σ/ϵ_0 , where σ is the surface charge density there.

The first property holds because a field inside the material would push the free charges, which would then not be in equilibrium. For

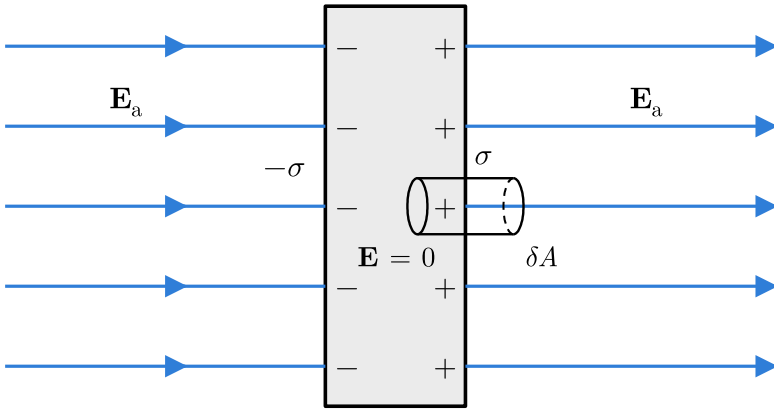


Figure 14. A conducting slab in a uniform external field $\mathbf{E}_a$. Charges of density $-\sigma$ and σ gather on the two faces, and their field cancels $\mathbf{E}_a$ inside the slab. The pillbox, whose flat faces have area δA , straddles the face on the right.

instance, let a conducting slab be placed in a uniform field $\mathbf{E}_a$ perpendicular to its faces, as in Figure 14. The field pushes electrons to the face on the left, leaving an equal positive charge on the face on the right. Each face acts as one of the infinite charged planes of Section 1.3, with a field $\sigma/2\epsilon_0$ on either side, so the fields of the two faces add to σ/ϵ_0 inside the slab, pointing against $\mathbf{E}_a$, and cancel outside it. The electrons stop moving when $\sigma = \epsilon_0 E_a$, and the field is then zero inside the slab and equal to $\mathbf{E}_a$ outside it.

The second property follows from Gauss's law in the differential form of Section 1.3, since $\mathbf{E} = 0$ inside the material gives $\rho = \epsilon_0 \nabla \cdot \mathbf{E} = 0$ there. Any net charge therefore sits in a layer at the surface, thin enough to be described by a surface charge density σ . The third property follows from the line integral of Section 1.4, since the potential difference between two points of the conductor is $-\int \mathbf{E} \cdot d\mathbf{l}$ along any path joining them, and a path that runs through the material, where $\mathbf{E} = 0$, gives zero.

For the fourth property, a component of $\mathbf{E}$ along the surface would

push the surface charges along it, so in equilibrium the field just outside is perpendicular to the surface. To find its magnitude, we use the pillbox of Figure 14, a short cylinder whose flat faces, each of area δA , lie parallel to the surface, one just inside the conductor and one just outside it. There is no flux through the inner face, where the field is zero, nor through the curved side, where the field is either zero or parallel to the side. The flux through the outer face is $E_n \delta A$, where E_n is the component of the field along the unit normal $\hat{\mathbf{n}}$ pointing out of the conductor, and the charge enclosed is $\sigma \delta A$, so Gauss's law gives

$$\mathbf{E} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}. \quad (2.4)$$

For the slab, with $\sigma = \epsilon_0 E_a$ on the face on the right in Figure 14, this is the field $\mathbf{E}_a$ found above.

Cavities and Electrostatic Screening

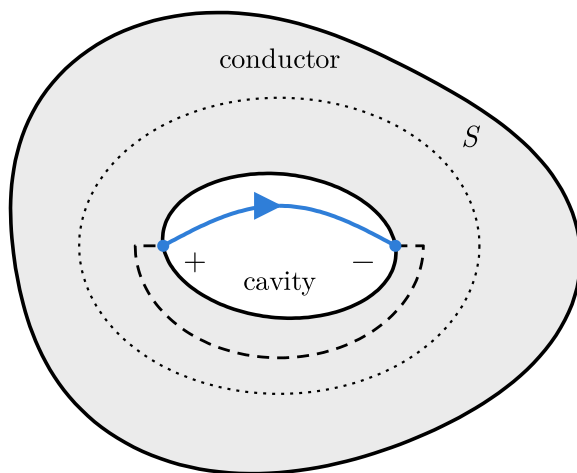


Figure 15. A conductor with an empty cavity. The dotted surface S lies in the material and surrounds the cavity. The accent curve is the field line that would cross the cavity if its wall carried charges of both signs, and the dashed curve closes the path through the material.

We now consider a conductor with an empty cavity inside it, as in Figure 15. The field is zero everywhere on the surface S , which lies

in the material and surrounds the cavity, so by Gauss's law the total charge on the wall of the cavity is zero. Could the wall carry positive charge in one place and an equal negative charge in another? If it did, a field line would cross the cavity from the positive charge to the negative charge, like the accent curve, and the line integral of $\mathbf{E}$ along it would be positive. Closing the path through the material along the dashed curve, where the field is zero, would give a closed path with a positive line integral, whereas the independence of the path found in Section 1.4 requires this integral to be zero. There is therefore no charge on the wall and no field in the cavity, for any arrangement of charges outside the conductor. A closed conducting shell thus shields the space inside it from the fields of charges outside, and this effect is called electrostatic screening. A sensitive instrument is often screened in this way, by a metal box or by an enclosure of fine wire mesh called a Faraday cage.

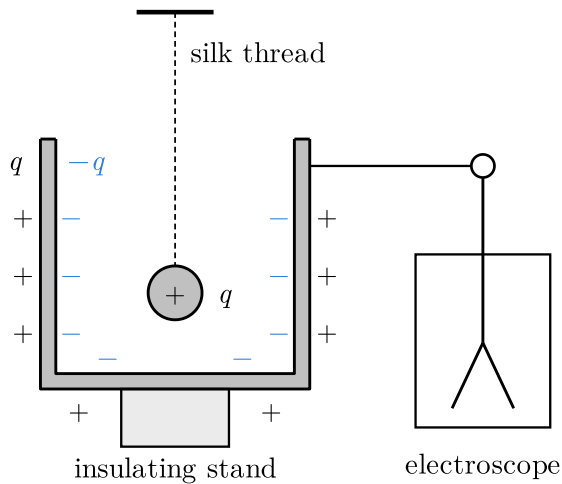


Figure 16. Faraday's ice-pail experiment. A metal ball carrying a charge q hangs on a silk thread inside a metal pail on an insulating stand. The accent minus signs on the inner wall of the pail show the induced charge $-q$, and the plus signs on its outer surface show the charge q . The pail is joined by a wire to an electroscope, whose leaves stand apart.

If a charge q is placed in the cavity, the flux through S is still zero, so the wall of the cavity carries a total charge $-q$, and if the

conductor was uncharged, a charge q appears on its outer surface, by the conservation of charge. Faraday demonstrated this by lowering a charged metal ball on a silk thread into an insulated metal ice pail connected to an electroscope, as in Figure 16. An electroscope is a metal rod with a pair of thin metal leaves, usually of gold foil, hanging side by side from its lower end inside a glass case. When charge flows onto the rod, the two leaves carry charge of the same sign and repel each other, so they spread apart, and the angle between them grows with the charge. Joined by a wire to the outside of the pail, the electroscope therefore shows the charge on the outer surface. Once the ball was well inside, the leaves stayed the same distance apart as the ball was moved about, and also when it touched the bottom, where its charge cancelled the charge $-q$ on the inner wall and left the charge q outside.

Capacitance

A capacitor, in general, is any pair of conductors carrying charges Q and $-Q$. The potential difference V between them is proportional to Q , because multiplying every charge by the same factor multiplies the field everywhere by that factor, by the principle of superposition, so the field still vanishes inside both conductors, the new arrangement is again in equilibrium, and the line integral that gives V is multiplied by the same factor. The ratio $C = Q/V$ therefore depends only on the shapes, the sizes, and the arrangement of the conductors. To find it, we first place charges Q and $-Q$ on the conductors and find the field between them, then integrate the field from one conductor to the other to find V , and finally divide Q by V .

The Parallel-Plate Capacitor

The first capacitor we treat is made of two parallel conducting plates of area A , a distance d apart, carrying charges Q and $-Q$, as in Figure 17. If d is small compared with the width of the plates, the plates can be treated as infinite planes, and the non-uniform field near their edges can be neglected. The fields of the two plates, each of magnitude $Q/2\epsilon_0 A$ like the field of the infinite plane of Section 1.3, then cancel outside the pair, so by the fourth property the outer faces carry no charge, and the charges sit on the inner faces with

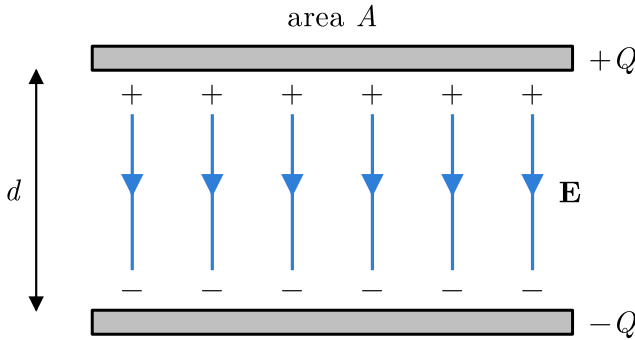


Figure 17. A parallel-plate capacitor with plates of area A a distance d apart. The charges Q and $-Q$ sit on the inner faces, and the field between the plates is uniform. The non-uniform field near the edges is not drawn.

densities $\pm Q/A$. The fourth property also gives the field $Q/\epsilon_0 A$ just outside the inner face of the positive plate, and since the field between the plates is uniform,

$$E = \frac{Q}{\epsilon_0 A}, \quad V = Ed = \frac{Qd}{\epsilon_0 A}. \quad (2.5)$$

Dividing Q by V gives $C = \epsilon_0 A/d$, the formula quoted in the school review.

Concentric Spheres

The next arrangement is a conducting sphere of radius a carrying a charge Q , inside a concentric conducting shell of inner radius b carrying $-Q$, as in Figure 18. By the spherical symmetry the field between them is radial, and Gauss's law applied to the dashed sphere of radius r gives $4\pi r^2 E = Q/\epsilon_0$ for $a < r < b$. The potential difference between the conductors is

$$V = - \int_b^a \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} dr = \frac{1}{4\pi\epsilon_0} Q \left(\frac{1}{a} - \frac{1}{b} \right), \quad (2.6)$$

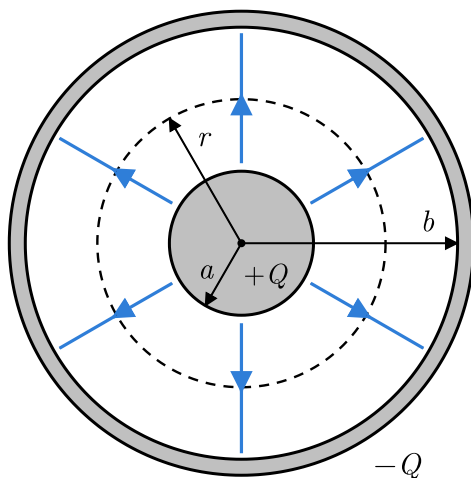


Figure 18. A spherical capacitor in cross-section, made of a conducting sphere of radius a with charge Q inside a concentric conducting shell of inner radius b with charge $-Q$. The dashed sphere of radius r is the Gaussian surface.

and dividing Q by this potential difference gives

$$C = \frac{4\pi\epsilon_0 ab}{b - a}. \quad (2.7)$$

When the gap $b - a$ is small compared with a , C is close to $\epsilon_0(4\pi a^2)/(b - a)$, the parallel-plate result for plates whose area is that of the sphere. When b is made very large, C approaches $4\pi\epsilon_0 a$, which is called the capacitance of an isolated sphere and relates the charge on the sphere to its potential, measured from zero at infinity.

Coaxial Cylinders

As a last example, consider a long conducting cylinder of radius a inside a coaxial conducting tube of inner radius b , which is the arrangement of the conductors in a coaxial cable, such as the lead that joins a television to its aerial. Let the conductors carry charges λ and $-\lambda$ per unit length, and let them be long enough for the fields near the ends to be neglected. Gauss's law applied to the dashed cylinder of Figure 19, through which the radial field passes only

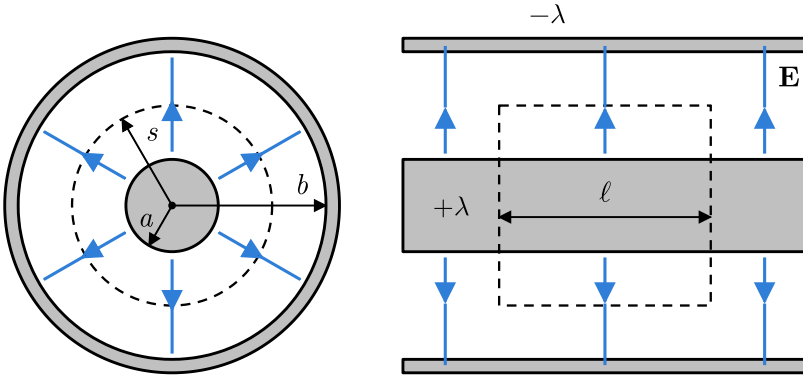


Figure 19. A coaxial capacitor, seen end-on on the left and in a section through its axis on the right. The inner conductor has radius a and charge λ per unit length, and the outer conductor has inner radius b and charge $-\lambda$ per unit length. The dashed Gaussian cylinder has radius s and length ℓ .

at the curved surface, gives $2\pi s\ell E = \lambda\ell/\epsilon_0$ for $a < s < b$, so that $E = \lambda/2\pi\epsilon_0 s$, the field of the infinite line of Section 1.3. The potential difference is

$$V = - \int_b^a \frac{\lambda}{2\pi\epsilon_0 s} ds = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{b}{a}, \quad (2.8)$$

and a length ℓ of the cable, whose inner conductor carries $Q = \lambda\ell$, has the capacitance

$$C = \frac{2\pi\epsilon_0\ell}{\ln(b/a)}. \quad (2.9)$$

The capacitance per unit length, $2\pi\epsilon_0/\ln(b/a)$, depends on the radii only through their ratio.

Capacitors in Series and in Parallel

Plates joined by a wire form a single conductor and are therefore at the same potential. In the parallel combination of Figure 20(a), both capacitors have the potential difference V , so they carry

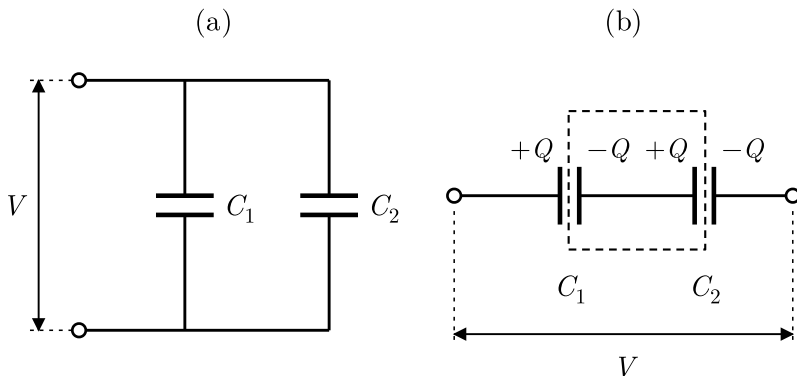


Figure 20. Capacitors connected (a) in parallel and (b) in series, with a potential difference V across each combination. In (b) the dashed box encloses the isolated conductor formed by the inner plates and the wire between them.

charges C_1V and C_2V , and the combination behaves as a single capacitor of capacitance $C_1 + C_2$.

In the series combination of Figure 20(b), the right-hand plate of C_1 , the left-hand plate of C_2 , and the wire between them form an isolated conductor, shown by the dashed box, whose total charge stays zero by the conservation of charge. The facing plates of each capacitor carry equal and opposite charges, since a closed surface that runs through the material of both plates and encloses the gap has no flux through it, the field outside the gap being neglected as in the parallel-plate capacitor. A charge Q on the left-hand plate of C_1 is therefore matched by $-Q$ on its right-hand plate, which leaves Q on the left-hand plate of C_2 and $-Q$ on its right-hand plate. Both capacitors carry the same charge, and their potential differences add to give

$$V = \frac{Q}{C_1} + \frac{Q}{C_2}, \quad (2.10)$$

so the combination behaves as a single capacitor with $1/C = 1/C_1 + 1/C_2$. The same arguments give $C = C_1 + C_2 + \dots$ for any number of capacitors in parallel and $1/C = 1/C_1 + 1/C_2 + \dots$ for any number in series.

The Energy Stored in a Capacitor

To find the energy stored in a capacitor, let us charge it by carrying small charges dq from the negative plate to the positive plate. When the plates carry q and $-q$, the potential difference is q/C , and carrying dq across requires the work $(q/C) dq$, so

$$U = \int_0^Q \frac{q}{C} dq = \frac{Q^2}{2C}, \quad (2.11)$$

which, with $Q = CV$, gives the three expressions of the school review.

The same energy is found by integrating the energy density $u = \frac{1}{2}\epsilon_0 E^2$ of Section 1.4 over the space in which the field is non-zero. For parallel plates, the field $Q/\epsilon_0 A$ fills a volume Ad , which gives $U = Q^2 d/2\epsilon_0 A = Q^2/2C$. For the concentric spheres, the thin shell between the radii r and $r + dr$ has volume $d\mathcal{V} = 4\pi r^2 dr$ and holds the energy $u d\mathcal{V} = Q^2 dr/8\pi\epsilon_0 r^2$, so

$$U = \frac{Q^2}{8\pi\epsilon_0} \int_a^b \frac{dr}{r^2} = \frac{Q^2}{8\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right), \quad (2.12)$$

which is again $Q^2/2C$, and you will check in Problem 4 that the charging integral and the integral of the energy density also agree for the coaxial capacitor.

The Force between the Plates

The plates of a charged capacitor attract one another, and the force between them can be found from the stored energy. Let the plates of a parallel-plate capacitor be isolated, so that their charges Q and $-Q$ stay fixed, and let x be their separation. The stored energy is $U = Q^2 x/2\epsilon_0 A$, and since no charge can reach or leave either plate, the work done by the force between the plates comes entirely from U . The force on one plate is therefore minus the derivative of U with respect to its position, as for the conservative forces of Section 2.3 of Vol. 1 of the Fermi Notes on Physics, and with x increasing as that plate moves away,

$$F = -\frac{dU}{dx} = -\frac{Q^2}{2\epsilon_0 A}. \quad (2.13)$$

The negative sign means that the force acts to decrease x , so the

plates attract, with a force that does not depend on their separation while x remains small compared with the width of the plates.

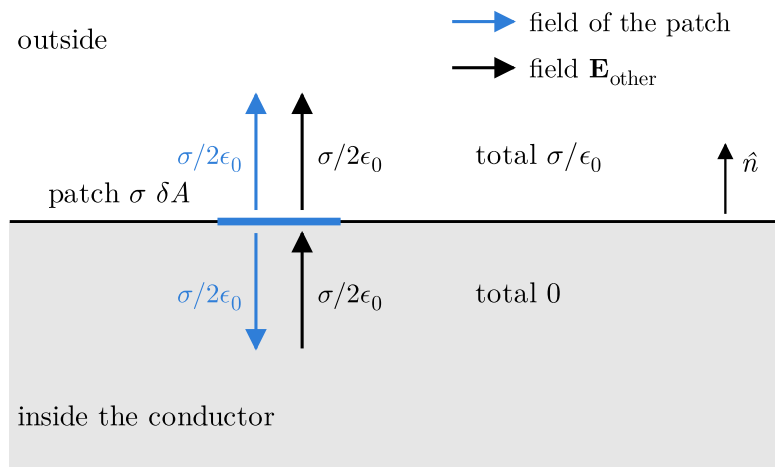


Figure 21. The field near a small patch of charge $\sigma \delta A$ on the surface of a conductor, whose interior (shaded) lies below the surface. The accent arrows show the field of the patch, of magnitude $\sigma/2\epsilon_0$, pointing away from the patch on both sides of the surface, and the plain arrows show the field $\mathbf{E}_{\text{other}}$ of all the other charges, which has the same magnitude and direction on both sides. The total field is zero just inside the conductor and σ/ϵ_0 along $\hat{n}$ just outside.

The force can also be found from the field, although multiplying the charge Q by the field σ/ϵ_0 between the plates gives twice the force above. The discrepancy arises because a patch of charge $\sigma \delta A$ on the surface exerts no net force on itself, since the forces between its parts cancel in pairs by Newton's third law, so only the field $\mathbf{E}_{\text{other}}$ of all the other charges acts on it. At points very close to the patch, its field is that of an infinite plane, $(\sigma/2\epsilon_0) \hat{n}$ outside and $-(\sigma/2\epsilon_0) \hat{n}$ inside, while $\mathbf{E}_{\text{other}}$ is the same on both sides, since every charge that produces it lies at a finite distance from the patch and its field is therefore continuous across the patch, as in Figure 21. The total field is zero just inside, so $\mathbf{E}_{\text{other}} = (\sigma/2\epsilon_0) \hat{n}$, half the field just outside. The patch therefore feels the outward force per

unit area

$$\frac{\sigma^2}{2\epsilon_0} = \frac{1}{2}\epsilon_0 E^2, \quad (2.14)$$

where $E = \sigma/\epsilon_0$ is the field just outside, and this force per unit area is called the electrostatic pressure. The electrostatic pressure is equal to the energy density of the field just outside the surface, and the force on the surface points out of the conductor for either sign of σ , since reversing σ reverses both the charge of the patch and $\mathbf{E}_{\text{other}}$. On the inner face of the positive plate the force points towards the other plate, and multiplying the pressure by A gives the force $Q^2/2\epsilon_0 A$ found from the energy. You will see in Problem 5 how the energy is divided between the capacitor, a battery, and the agent that moves the plates, when the capacitor stays connected to the battery.

Problems

§2.1 Conductors and Capacitance

1. Conducting spheres of radii a and b are placed far apart and joined by a long thin wire, and a total charge Q is shared between them. Neglecting the charge on the wire and the effect of each sphere on the other, find the charge on each sphere, and show that the surface charge densities satisfy $\sigma_a/\sigma_b = b/a$. Which sphere has the larger field just outside its surface?
2. A conducting sphere of outer radius b carries a net charge Q and has a concentric spherical cavity of radius a , at whose centre a point charge q is held. Find the charges on the wall of the cavity and on the outer surface, and find the field at every distance r from the centre. What is the potential of the conductor, taking the potential to be zero at infinity?
3. An uncharged conducting slab of thickness h , with faces of area A , is slid between the plates of a parallel-plate capacitor of plate area A and separation d , parallel to the plates and without touching them. Show that the capacitance becomes $\epsilon_0 A/(d - h)$ for any position of the slab between the plates.
4. By integrating the energy density between the conductors, find the energy stored in a length ℓ of the coaxial capacitor of Fig-

ure 19, and check that it equals $Q^2/2C$, with $Q = \lambda\ell$ and the capacitance found in the text.

5. A parallel-plate capacitor with plates of area A is kept connected to a battery that holds the potential difference between its plates at V , and the plates are pulled slowly apart from separation d_1 to separation d_2 . Find the change in the stored energy, the change in the charge on the positive plate, and the work done by the battery. Using the force $Q^2/2\epsilon_0 A$ found in the text, find the work done by the agent that pulls the plates apart, and show that the energy is conserved. Explain why $-dU/dx$, evaluated with V held fixed, gives a force of the wrong sign.
6. A capacitor C_1 is charged by a battery to a potential difference V_i and then disconnected from it. Its plates are then connected by wires to the plates of an uncharged capacitor C_2 , so that the two capacitors are in parallel. Find the final potential difference, and show that the stored energy falls by the fraction $C_2/(C_1 + C_2)$ of its initial value.

Electric Fields in Matter

3.1 Dipoles and Polarisation

In Section 2.1 we saw that the free electrons of a conductor move until the field inside it vanishes. An insulating material is called a dielectric; common examples include glass, plastics, and dry air. In a dielectric every electron is bound to its atom or molecule, so that a field can move it only a very small distance. These small displacements still change the field, as the school result that a dielectric raises the capacitance of a capacitor shows. This section first treats the electric dipole, then shows how atoms and molecules come to carry dipole moments in a field, and finally describes a polarised material by its dipole moment per unit volume and finds the charge that this polarisation produces.

The Electric Dipole

A pair of charges q and $-q$ held a small distance apart is called an electric dipole. The positive and negative charges of a molecule of hydrogen chloride form a dipole of this kind, for example, since its electrons spend more of their time near the chlorine atom than near the hydrogen atom. Let ℓ be the displacement from the negative charge to the positive one. The electric dipole moment of the pair is defined as

$$\mathbf{p} = q\boldsymbol{\ell}, \quad (3.1)$$

a vector pointing from $-q$ to q , measured in coulomb metres.

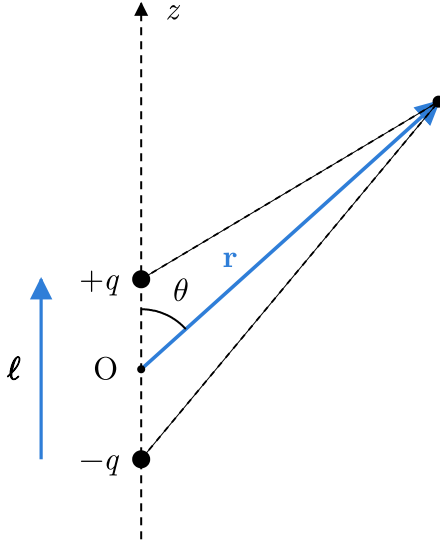


Figure 22. A dipole formed by charges q and $-q$ at $\ell/2$ and $-\ell/2$ on the z -axis, on either side of the origin O . The field point at $\mathbf{r}$ (accent) makes an angle θ with the z -axis, and the dashed lines show its separations $|\mathbf{r} - \ell/2|$ and $|\mathbf{r} + \ell/2|$ from the two charges.

We set up coordinates such that the origin lies midway between the charges and the z -axis points along ℓ , as shown in Figure 22. Adding the potentials of the two point charges, found in Section 1.4, gives the potential at the field point $\mathbf{r}$,

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} q \left(\frac{1}{|\mathbf{r} - \ell/2|} - \frac{1}{|\mathbf{r} + \ell/2|} \right). \quad (3.2)$$

We want this potential at distances r much larger than ℓ . Multiplying out $(\mathbf{r} - \ell/2) \cdot (\mathbf{r} - \ell/2)$ gives $|\mathbf{r} - \ell/2|^2 = r^2 - \mathbf{r} \cdot \ell + \ell^2/4$, and therefore

$$\frac{1}{|\mathbf{r} - \ell/2|} = \frac{1}{r} \left(1 - \frac{\hat{\mathbf{r}} \cdot \ell}{r} + \frac{\ell^2}{4r^2} \right)^{-1/2} \approx \frac{1}{r} \left(1 + \frac{\hat{\mathbf{r}} \cdot \ell}{2r} \right), \quad (3.3)$$

where the binomial expansion $(1 + \zeta)^{-1/2} \approx 1 - \zeta/2$ has been used and terms of order $(\ell/r)^2$ have been dropped. Changing the sign of

ℓ gives $1/|\mathbf{r} + \ell/2|$. The terms $1/r$ cancel in the difference, and the potential of the dipole is

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}, \quad (3.4)$$

in which θ is the polar angle of the field point, measured from the direction of $\mathbf{p}$.

The potential goes as $1/r^2$, falling off faster than that of either charge alone, because far away the potentials of the two charges nearly cancel. The dipole obtained in the limit in which ℓ goes to zero with p held fixed is called a point dipole, and its potential is given by this formula at every $r > 0$.

For a neutral set of point charges q_i at positions $\mathbf{r}_i$, the dipole moment is defined as

$$\mathbf{p} = \sum_i q_i \mathbf{r}_i, \quad (3.5)$$

which for the pair above is $q(\ell/2) - q(-\ell/2) = q\ell$, and you will show in Problem 1 that it does not depend on the choice of origin. The expansion above, with $\ell/2$ replaced by the position $\mathbf{r}_i$ of each charge, gives $1/|\mathbf{r} - \mathbf{r}_i| \approx 1/r + \hat{\mathbf{r}} \cdot \mathbf{r}_i/r^2$, and since the terms q_i/r add up to zero, the potential far from the set of charges is that of a point dipole with the moment $\mathbf{p}$. The terms of higher order in r_i/r , which make up what is called the multipole expansion, are not treated in this course.

The Field of a Dipole

The field follows from $\mathbf{E} = -\nabla V$, found in Section 1.4, with the gradient in the spherical coordinates of Section 3.4 of Vol. 4 of the Fermi Notes on Physics, whose components along $\hat{\mathbf{r}}$ and along the unit vector $\hat{\boldsymbol{\theta}}$, which points in the direction of increasing θ , are $\partial V/\partial r$ and $(1/r) \partial V/\partial \theta$. The components of the field are therefore

$$E_r = -\frac{\partial V}{\partial r}, \quad E_\theta = -\frac{1}{r} \frac{\partial V}{\partial \theta}, \quad (3.6)$$

and since the potential does not depend on ϕ , the field has no component in the direction of increasing ϕ . Differentiating the potential of the dipole gives

$$E_r = \frac{1}{4\pi\epsilon_0} \frac{2p \cos \theta}{r^3}, \quad E_\theta = \frac{1}{4\pi\epsilon_0} \frac{p \sin \theta}{r^3}. \quad (3.7)$$

On the axis of the dipole the field points along $\mathbf{p}$, and in the plane through its centre perpendicular to the axis the field points opposite to $\mathbf{p}$ and has half the magnitude it has on the axis at the same distance. Resolving the unit vector $\hat{\mathbf{z}}$ into its components along $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$ gives $\mathbf{p} = p(\cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}})$, and with this the two components combine into

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{3(\mathbf{p} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \mathbf{p}}{r^3}, \quad (3.8)$$

a form that makes no reference to the coordinates.

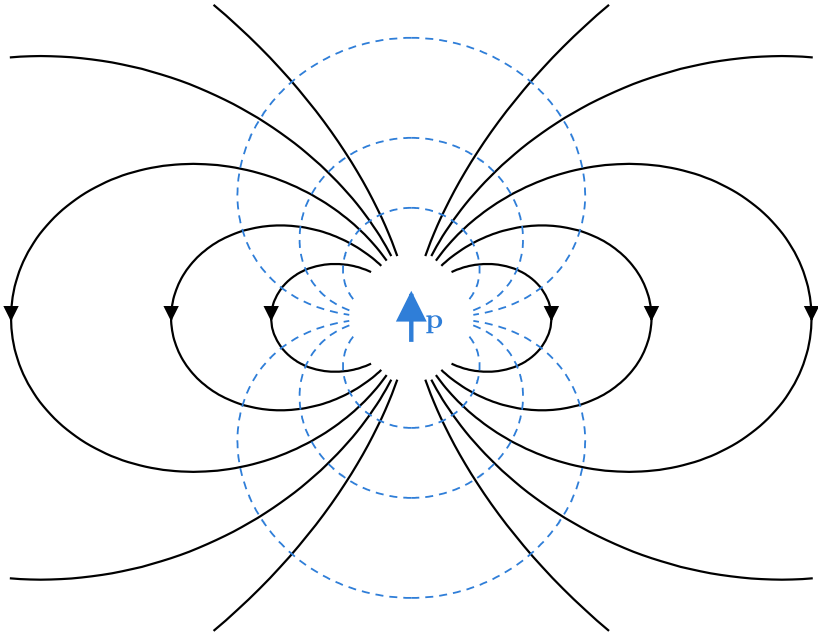


Figure 23. Field lines (solid) and equipotential surfaces (dashed, accent) of a point dipole $\mathbf{p}$ at the centre, pointing up the page. The field lines are the curves $r = r_0 \sin^2 \theta$, and the equipotential surfaces meet the plane of the figure in the curves on which $\cos \theta / r^2$ is constant. The arrows show the direction of the field where each field line crosses the plane perpendicular to $\mathbf{p}$, and the region close to the dipole is left blank.

The field lines of a point dipole are shown in Figure 23, and you will show in Problem 2 that they are the curves $r = r_0 \sin^2 \theta$, with a different constant r_0 on each line. The field lines cross the equipotential

tential surfaces, on which $\cos \theta/r^2$ is constant, at right angles, since a gradient is perpendicular to the contours of its function, as shown in Section 2.3 of Vol. 1 of the Fermi Notes on Physics.

Torque and Energy in a Uniform Field

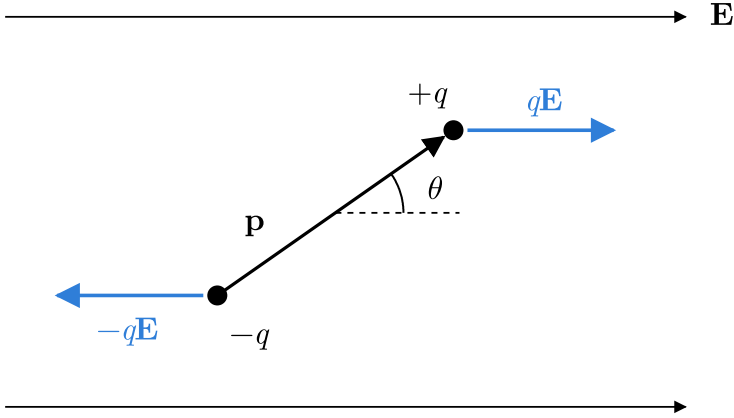


Figure 24. A dipole in a uniform field $\mathbf{E}$, with its moment $\mathbf{p}$ pointing from $-q$ to q at an angle θ to the field. The forces $q\mathbf{E}$ and $-q\mathbf{E}$ (accent) add up to zero, and together they exert the torque $\mathbf{p} \times \mathbf{E}$, which turns the dipole towards the direction of the field.

Let us place a dipole in a uniform field $\mathbf{E}$, as shown in Figure 24. The forces $q\mathbf{E}$ and $-q\mathbf{E}$ on its charges add up to zero, so a uniform field exerts no net force on a dipole. The two forces act at different points, however, and together they exert a torque. We write the torque $\mathbf{r} \times \mathbf{F}$ of Section 3.2 of Vol. 1 as $\mathbf{\Gamma}$, and take moments about the centre of the dipole,

$$\mathbf{\Gamma} = \frac{\ell}{2} \times q\mathbf{E} + \left(-\frac{\ell}{2}\right) \times (-q\mathbf{E}) = \mathbf{p} \times \mathbf{E}. \quad (3.9)$$

Since the forces add up to zero, this torque is the same about every point, as shown in Section 3.2 of Vol. 1. Let θ now be the angle between $\mathbf{p}$ and $\mathbf{E}$, the polar angle of $\mathbf{p}$ measured from the direction of the field. The torque has magnitude $pE \sin \theta$ and turns the dipole towards the field, as a magnetic field turns a compass needle.

The potential energy of the dipole is the sum of the potential energies qV of its charges, found in Section 1.4, where V is the potential of the applied field and $\mathbf{r}$ is now the centre of the dipole,

$$U = qV(\mathbf{r} + \boldsymbol{\ell}/2) - qV(\mathbf{r} - \boldsymbol{\ell}/2). \quad (3.10)$$

For a small dipole this difference is $q\boldsymbol{\ell} \cdot \nabla V$, by the relation $dV = \nabla V \cdot d\mathbf{r}$ of Section 1.4, and with $\mathbf{E} = -\nabla V$,

$$U = -\mathbf{p} \cdot \mathbf{E} = -pE \cos \theta. \quad (3.11)$$

The energy is lowest when the dipole points along the field. Its derivative $dU/d\theta = pE \sin \theta$ is the magnitude of the torque, which acts to reduce θ and so to lower the energy. The expression $U = -\mathbf{p} \cdot \mathbf{E}$ also holds in a field that varies from place to place, provided the field changes little over the length of the dipole.

The Force in a Non-Uniform Field

Where the field varies, the forces on the two charges no longer cancel. By the relation $dE_x = \nabla E_x \cdot d\mathbf{r}$, the difference between the values of E_x at the two charges of a small dipole is $\boldsymbol{\ell} \cdot \nabla E_x$, and similarly for E_y and E_z , so the net force $q\mathbf{E}(\mathbf{r} + \boldsymbol{\ell}/2) - q\mathbf{E}(\mathbf{r} - \boldsymbol{\ell}/2)$ is

$$\mathbf{F} = (\mathbf{p} \cdot \nabla) \mathbf{E}, \quad (3.12)$$

where $\mathbf{p} \cdot \nabla$ stands for $p_x \partial/\partial x + p_y \partial/\partial y + p_z \partial/\partial z$, applied to each component of $\mathbf{E}$ in turn.

As an example, consider a dipole at a distance r from a point charge $Q > 0$, pointing directly away from it and so lined up with its field, as shown in Figure 25. Along this line the field is radial with magnitude $Q/4\pi\epsilon_0 r^2$, and $\mathbf{p} \cdot \nabla$ is $p \partial/\partial r$, so

$$\mathbf{F} = p \frac{\partial}{\partial r} \left(\frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \right) \hat{\mathbf{r}} = -\frac{1}{4\pi\epsilon_0} \frac{2pQ}{r^3} \hat{\mathbf{r}}, \quad (3.13)$$

which points towards the charge, because the negative end of the dipole is the nearer one and is attracted more strongly than the positive end is repelled. A dipole lined up with a non-uniform field is in general drawn towards the region where the field is stronger. You will check in Problem 3 that the force the dipole exerts on the charge is equal and opposite to this force.

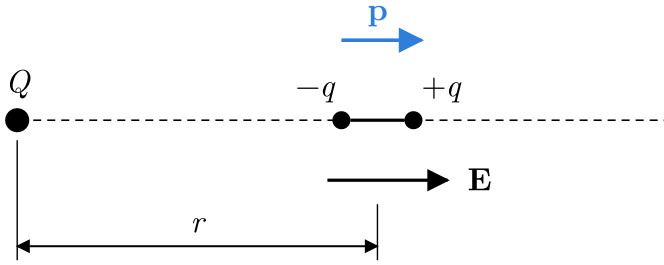


Figure 25. A dipole at a distance r from a point charge $Q > 0$, with $-q$ nearer the charge and q further from it, so that its moment $\mathbf{p}$ (accent) points directly away from the charge, along the field $\mathbf{E}$ of the charge at the position of the dipole.

Induced Dipoles

An atom has a positive nucleus surrounded by a cloud of negative charge formed by its electrons, and with no field applied the centre of the cloud coincides with the nucleus. In a field $\mathbf{E}$ the nucleus is pushed along the field and the cloud the opposite way, until the attraction between them balances the forces of the field. The atom then has a dipole moment along the field, which is said to be induced.

To estimate the induced moment, consider the model of an atom shown in Figure 26, in which the nucleus is a point charge q and the electron cloud is a sphere of radius a with charge $-q$ spread uniformly through it, which keeps its shape. The field inside a uniformly charged sphere was found from Gauss's law in Section 1.3, and at a distance $r < a$ from the centre of the cloud the field of the cloud has magnitude $qr/4\pi\epsilon_0 a^3$ and points towards the centre. A nucleus displaced by ℓ from the centre is therefore pulled back with the force $-q^2\ell/4\pi\epsilon_0 a^3$, and in equilibrium this force balances $q\mathbf{E}$. The induced moment $\mathbf{p} = q\ell$ is then

$$\mathbf{p} = 4\pi\epsilon_0 a^3 \mathbf{E}. \quad (3.14)$$

The same balance gives $\ell/a = E/(q/4\pi\epsilon_0 a^2)$, and since laboratory fields are very much weaker than the field $q/4\pi\epsilon_0 a^2$ of the nucleus at

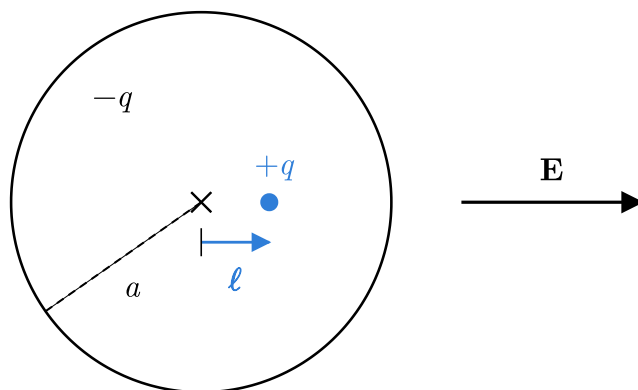


Figure 26. A model of an atom in a field $\mathbf{E}$. The nucleus, of charge q , is displaced by ℓ along the field from the centre (cross) of a uniform spherical electron cloud of radius a and charge $-q$. The displacement is drawn greatly enlarged.

the edge of the atom, the displacement is a very small fraction of the radius. The cloud of a real atom is more complicated than a uniform sphere, but for so small a displacement the restoring force is still proportional to the displacement, as Section 4.2 of Vol. 1 showed for a small displacement from a stable equilibrium. The induced moment of any atom is therefore proportional to the field, and the constant of proportionality is called the atomic polarisability α ,

$$\mathbf{p} = \alpha \mathbf{E}. \quad (3.15)$$

In the model $\alpha = 4\pi\epsilon_0 a^3$, so $\alpha/4\pi\epsilon_0$ is comparable with the volume of the atom. The induced moment of a molecule can depend on the direction of the field relative to the molecule, and in this course we consider only materials in which it is parallel to the field.

Permanent Dipoles

Some molecules have a dipole moment with no field applied, and these are called polar molecules; examples include hydrogen chloride, which appeared in Section 5.1 of Vol. 3 as a molecule with an

electric dipole moment, and water.

In a gas or a liquid of polar molecules the dipoles point in random directions. A field exerts on each the torque $\mathbf{p} \times \mathbf{E}$, which tends to line it up, while collisions keep changing its direction at random, and the outcome is a small mean moment along the field. Let $\langle p_E \rangle$ be the mean component of the moment of a molecule along the field, and let $\mathcal{T}$ be the temperature. In Problem 5 you will use the Boltzmann factor of Section 4.4 of Vol. 2 to show that, provided pE is much smaller than $k_B\mathcal{T}$, as it is at ordinary temperatures in laboratory fields,

$$\langle p_E \rangle = \frac{p^2 E}{3k_B \mathcal{T}}. \quad (3.16)$$

This mean moment is proportional to the field, and it goes as $1/\mathcal{T}$, in the same way as the magnetisation of the paramagnet of Section 5.3 of Vol. 2, which obeys Curie's law. A polar molecule is also polarised by induction, and the induced moment and the mean moment from the alignment add.

Polarisation

At distances large compared with the spacing between its molecules, the field of a dielectric depends only on how much dipole moment there is in each small region of it. The polarisation $\mathbf{P}$ of a material is defined as its dipole moment per unit volume. If a small volume $d\mathcal{V}$ contains the dipole moment $d\mathbf{p}$, then

$$\mathbf{P} = \frac{d\mathbf{p}}{d\mathcal{V}}, \quad (3.17)$$

where $d\mathcal{V}$ is small compared with the material but contains very many molecules. With n_d dipoles per unit volume, of mean moment $\langle \mathbf{p} \rangle$, the polarisation is $n_d \langle \mathbf{p} \rangle$, which for a dilute gas of atoms of polarisability α is $n_d \alpha \mathbf{E}$. The polarisation is measured in coulombs per square metre, the unit of a surface charge density. A material in which $\mathbf{P}$ is not zero is said to be polarised, and one that stays polarised with no field applied is called an electret; examples include some waxes that have been allowed to solidify in a strong field.

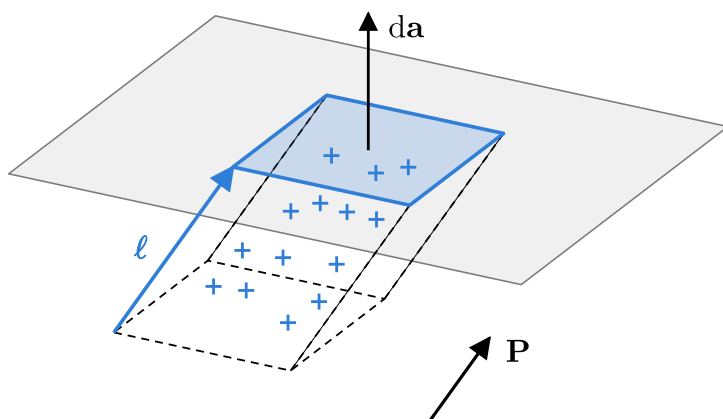


Figure 27. A small element da (lightly shaded, with an accent outline) of a surface (faintly shaded) drawn inside a polarised material, with the vector da drawn along its normal. When the positive charges are displaced by ℓ relative to the negative charges, the positive charges in the slanted prism (dashed) of base da and edge ℓ , which lies behind the element, cross the surface. The polarisation $\mathbf{P}$ is parallel to ℓ .

Bound Charge

A polarised dielectric can carry a net charge in places, because polarising it moves charge across any surface drawn in the material. To find how much, we model each molecule as a charge q displaced by ℓ from a charge $-q$, so that $\mathbf{P} = n_d q \ell$, with only the positive charges moving. The positive charges that cross a small element of surface da , shown in Figure 27, are those that start in the slanted prism of base da and edge ℓ behind it, whose volume is $\ell \cdot da$. The charge that crosses in the direction of da is therefore

$$dQ = n_d q \ell \cdot da = \mathbf{P} \cdot da, \quad (3.18)$$

which depends only on the polarisation, although the calculation used a particular model of the molecules.

At the outer surface of the material, with $\hat{\mathbf{n}}$ the unit normal pointing outwards, the charge that crosses cannot leave, and it collects in a

thin layer. The surface therefore carries the charge per unit area

$$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}, \quad (3.19)$$

which is called the bound surface charge density. A volume $\mathcal{V}$ inside the material was neutral before the polarisation was set up, and it loses the charge $\oint \mathbf{P} \cdot d\mathbf{a}$ through its surface, which by the divergence theorem of Section 1.3, treated at length in Section 3.3 of Vol. 4, is $\int \nabla \cdot \mathbf{P} d\mathcal{V}$. Since this holds for every volume, however small, the charge per unit volume left behind is

$$\rho_b = -\nabla \cdot \mathbf{P}, \quad (3.20)$$

which is called the bound volume charge density.

These charges are called bound because they stay attached to their molecules. They produce a field as any other charge does, so the field of a polarised material, averaged over many molecules, is the field of σ_b and ρ_b , and you will confirm this in Problem 7 by adding up the potentials of the dipoles. The total bound charge of a piece of dielectric is zero, since by the divergence theorem the volume charge $-\int \nabla \cdot \mathbf{P} d\mathcal{V}$ is minus the surface charge $\oint \mathbf{P} \cdot d\mathbf{a}$. Where the polarisation is uniform there is no bound volume charge, because the positive end of each dipole sits next to the negative end of its neighbour and their charges cancel.

A Uniformly Polarised Slab

As a further example, consider a slab of thickness w , with faces much larger than w , uniformly polarised with $\mathbf{P}$ perpendicular to its faces, as shown in Figure 28. Since $\mathbf{P}$ is uniform, $\rho_b = 0$. On the face towards which $\mathbf{P}$ points, $\sigma_b = P$, and on the opposite face, where $\hat{\mathbf{n}}$ points the other way, $\sigma_b = -P$. The slab therefore has the field of a pair of large parallel sheets with charge densities $\pm P$, the arrangement of the parallel-plate capacitor of Section 2.1, and the field is zero outside the slab, while inside it is

$$\mathbf{E} = -\frac{\mathbf{P}}{\epsilon_0}, \quad (3.21)$$

so that the face on which $\sigma_b = P$ is at a potential Pw/ϵ_0 above the other. The field of the bound charges points against the polarisation, so where the polarisation has been produced by an applied field, the bound charges weaken the field inside the material.

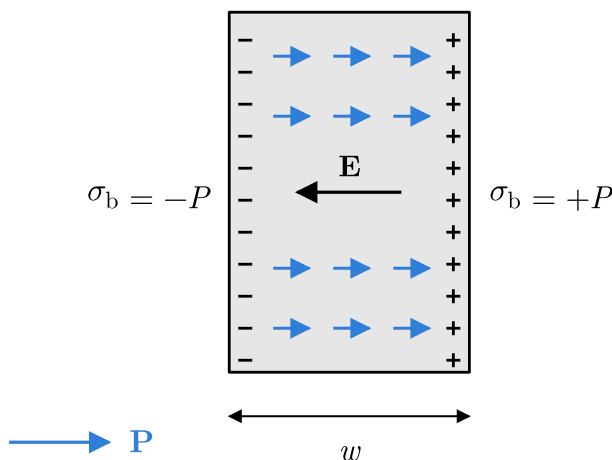


Figure 28. A slab of thickness w , uniformly polarised with $\mathbf{P}$ perpendicular to its faces. Inside the slab the ends of neighbouring dipoles (accent arrows) cancel, leaving bound surface charge densities $-P$ and P on the faces. The field of these charges is $\mathbf{E} = -\mathbf{P}/\epsilon_0$ inside the slab and zero outside it.

3.2 The Electric Displacement and Linear Dielectrics

The charge in and around a piece of matter can be divided into two kinds. The bound charge of Section 3.1 appears when the molecules of a dielectric are polarised, and all other charge is called free charge; examples include the charge on the plates of a capacitor and any charge rubbed onto the surface of a dielectric by friction. In this section we first use this division to write Gauss's law in a form that refers only to the free charge, then define a linear dielectric, in which the polarisation is proportional to the field, and finally treat a capacitor filled with a dielectric and a point charge embedded in one.

Free and Bound Charge

At any point in a dielectric, the charge density is the sum of the free and the bound charge densities,

$$\rho = \rho_f + \rho_b, \quad (3.22)$$

where ρ_f is the density of free charge. The free charge is the charge we control, by placing it on conductors or putting it into the material, and the bound charge is the response of the material to the field. The difficulty in finding the field in a dielectric is that the bound charge cannot be written down in advance. It is fixed by the polarisation, the polarisation is fixed by the field, and the field is produced by all the charge, bound as well as free. Gauss's law in the form found in Section 1.3, $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$, holds with ρ the total charge density, since bound charge produces a field in the same way as any other charge.

The Electric Displacement

Substituting $\rho_b = -\nabla \cdot \mathbf{P}$ from Section 3.1 into Gauss's law gives

$$\epsilon_0 \nabla \cdot \mathbf{E} = \rho_f - \nabla \cdot \mathbf{P}, \quad (3.23)$$

and moving the divergence of the polarisation to the left-hand side gives

$$\nabla \cdot (\epsilon_0 \mathbf{E} + \mathbf{P}) = \rho_f. \quad (3.24)$$

The combination in brackets is called the electric displacement,

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}, \quad (3.25)$$

and in terms of it Gauss's law takes the form

$$\nabla \cdot \mathbf{D} = \rho_f. \quad (3.26)$$

This equation is called Gauss's law in matter. Integrating both sides over a volume and applying the divergence theorem to the left-hand side gives the integral form of the law,

$$\oint \mathbf{D} \cdot d\mathbf{a} = Q_f, \quad (3.27)$$

in which Q_f is the free charge enclosed by the closed surface. The bound surface charge is included in this result, because at the surface of a dielectric the polarisation falls to zero across a very thin

layer, and the bound volume charge $-\nabla \cdot \mathbf{P}$ in that layer is the charge σ_b spread through its thickness. In a vacuum $\mathbf{P} = 0$ and $\mathbf{D} = \epsilon_0 \mathbf{E}$. The displacement is measured in coulombs per square metre, as the polarisation is.

Gauss's law in matter has the same form as the law of Section 1.3, with $\epsilon_0 \mathbf{E}$ replaced by $\mathbf{D}$ and the total charge by the free charge, and this raises the following question: is $\mathbf{D}$ equal to ϵ_0 times the field that the free charge would produce if no dielectric were present? The answer is that it is for some arrangements and not for others. The uniformly polarised slab of Section 3.1 carries no free charge, and since the field inside it is $-\mathbf{P}/\epsilon_0$, the displacement is zero inside the slab as well as outside it, which is what the absence of free charge would suggest. As you showed in Problem 6 of Section 3.1, however, a uniformly polarised sphere has the uniform field $-\mathbf{P}/3\epsilon_0$ inside it, so that $\mathbf{D} = \frac{2}{3}\mathbf{P}$ inside the sphere, although there is no free charge anywhere. The reason for the difference is that $\mathbf{E}$ is the sum of the Coulomb fields of all the charges, bound as well as free, as in Section 1.2, and so is fixed once those charges are known, whereas there is no corresponding formula giving $\mathbf{D}$ from the free charge alone; $\mathbf{D}$ contains the polarisation, which is fixed by the way the material has been polarised, and $\mathbf{D}$ need not be the gradient of any function. Gauss's law in matter is therefore most useful where the symmetry of the whole arrangement, dielectric included, fixes the direction of $\mathbf{D}$ and shows that its magnitude is constant over a suitable closed surface, as in the plane, cylindrical, and spherical arrangements of Section 1.3.

Linear Dielectrics

The polarisation of most dielectrics is produced by the field, and so it depends on the field. The induced moment of an atom, found in Section 3.1, is proportional to the field, and the mean moment of a polar molecule is proportional to the field provided pE is much smaller than $k_B \mathcal{T}$. In many materials, as long as the field is much weaker than the fields inside atoms, the polarisation is therefore proportional to the field and parallel to it, and we write

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}, \quad (3.28)$$

where $\mathbf{E}$ is the total field in the material, including the field of the bound charges. The constant χ_e is called the electric susceptibility of the material, and the factor ϵ_0 is included so that χ_e has no dimensions. A material that obeys this relation is called a linear dielectric. For a dilute gas of atoms of polarisability α , the result $\mathbf{P} = n_d \alpha \mathbf{E}$ of Section 3.1 gives $\chi_e = n_d \alpha / \epsilon_0$. For a dilute gas of polar molecules, the result you found in Problem 5 of Section 3.1 gives the contribution $n_d p^2 / 3 \epsilon_0 k_B \mathcal{T}$ from the alignment of the dipoles, to which the contribution from induction is added. Since the polarisation of these materials points along the field, χ_e is positive.

In a dilute gas the field that acts on each molecule is, to a good approximation, the field $\mathbf{E}$ averaged over the material. In a liquid or a solid the neighbouring molecules are close enough to change the field at each molecule, and the relation between the polarisability of a molecule and the susceptibility of a dense material, known as the Clausius–Mossotti relation, is not treated in this course. Some materials are not linear dielectrics, among them the electrets of Section 3.1, which are polarised with no field applied.

In a linear dielectric the displacement is

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \epsilon_0 \chi_e \mathbf{E} = \epsilon_r \epsilon_0 \mathbf{E}, \quad \epsilon_r = 1 + \chi_e. \quad (3.29)$$

The number ϵ_r is called the relative permittivity of the material, and it is also known as the dielectric constant. It is the quantity quoted in your school textbooks for the material between the plates of a capacitor, and the product $\epsilon_r \epsilon_0$ is called the permittivity of the material. In a vacuum $\epsilon_r = 1$. For a gas at ordinary pressures ϵ_r is very close to one, because n_d is small, and for water it is much larger than for most other liquids, largely because its molecules are polar.

Suppose now that a region is filled with a single linear dielectric, so that χ_e has the same value everywhere in it. Then $\mathbf{P} = \epsilon_0 \chi_e \mathbf{E} = (\chi_e / \epsilon_r) \mathbf{D}$, and Gauss's law in matter gives the bound volume charge density in the region,

$$\rho_b = -\nabla \cdot \mathbf{P} = -\frac{\chi_e}{\epsilon_r} \rho_f. \quad (3.30)$$

Inside a uniform linear dielectric, bound volume charge appears only where there is free charge, and it has the opposite sign and a smaller

magnitude, so that it cancels part of the free charge. Bound surface charge can still appear on the surfaces of the dielectric. Boundary-value problems in linear dielectrics, the forces on dielectrics, and the energy stored in them are not treated in this course. The following two examples have enough symmetry for Gauss's law in matter to give $\mathbf{D}$ directly.

The Filled Parallel-Plate Capacitor

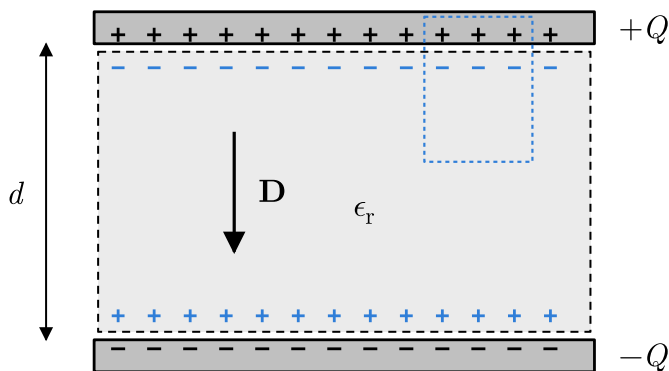


Figure 29. A parallel-plate capacitor with plates a distance d apart, filled with a linear dielectric of relative permittivity ϵ_r . The free charges Q and $-Q$ sit on the inner faces of the plates, and the bound charges (accent) on the faces of the dielectric have the opposite sign to the free charge on the neighbouring plate. The pillbox (dotted, accent) has one flat face inside the upper plate and the other in the dielectric, and the displacement $\mathbf{D}$ points from the positive plate towards the negative one.

Our first example is a parallel-plate capacitor whose plates, each of area A , are a distance d apart, with the gap between them filled by a linear dielectric of relative permittivity ϵ_r , as shown in Figure 29. The plates carry free charges Q and $-Q$, spread over their inner faces with surface densities $\sigma_f = Q/A$ and $-\sigma_f$. As in Section 2.1, the plates are taken to be large compared with d , so that the effects at their edges can be neglected, and by symmetry $\mathbf{D}$ is then perpendicular to the plates and has the same value at every point

of any plane parallel to them.

Let A_p be the area of each flat face of the pillbox drawn in Figure 29. Its upper face lies inside the metal of the upper plate, where $\mathbf{E} = 0$ by the result of Section 2.1 and $\mathbf{P} = 0$ since the metal is not a dielectric, so that $\mathbf{D} = 0$ there. No displacement crosses its curved side, since $\mathbf{D}$ is perpendicular to the plates and so parallel to that side. Its lower face lies in the dielectric, and the free charge it encloses is $\sigma_f A_p$. The pillbox also encloses the bound charge on the upper face of the dielectric, but Gauss's law in matter counts only the free charge. It follows that $DA_p = \sigma_f A_p$, so the displacement in the dielectric has magnitude σ_f and points from the positive plate towards the negative one, and a pillbox drawn at the lower plate gives the same result. The field in the dielectric has magnitude

$$E = \frac{D}{\epsilon_r \epsilon_0} = \frac{\sigma_f}{\epsilon_r \epsilon_0}, \quad (3.31)$$

which is smaller by the factor ϵ_r than the field σ_f/ϵ_0 found in Section 2.1 for an empty capacitor carrying the same charge. The potential difference between the plates is $V = Ed = Qd/\epsilon_r \epsilon_0 A$, and the capacitance is

$$C = \frac{Q}{V} = \frac{\epsilon_r \epsilon_0 A}{d}, \quad (3.32)$$

which is larger by the factor ϵ_r than the capacitance $\epsilon_0 A/d$ of the empty capacitor, a result you should be familiar with from school.

The reduction of the field is explained by the bound charges. The polarisation of the dielectric has magnitude

$$P = \epsilon_0 \chi_e E = \frac{\chi_e}{\epsilon_r} \sigma_f, \quad (3.33)$$

and it points in the same direction as the field, from the positive plate towards the negative one. On the face of the dielectric next to the positive plate, the outward normal points towards that plate, against $\mathbf{P}$, so $\sigma_b = -P$ there, and on the face next to the negative plate $\sigma_b = P$. The polarisation is uniform, so there is no bound volume charge. Each plate therefore faces a layer of bound charge of the opposite sign, as drawn in Figure 29, because the molecules of the dielectric are lined up with their negative ends towards the positive plate. The total charge per unit area at the upper surface

of the gap is

$$\sigma_f - \frac{\chi_e}{\epsilon_r} \sigma_f = \frac{\epsilon_r - \chi_e}{\epsilon_r} \sigma_f = \frac{\sigma_f}{\epsilon_r}, \quad (3.34)$$

using $\epsilon_r - \chi_e = 1$, and at the lower surface it is $-\sigma_f/\epsilon_r$. By the result of Section 2.1, two large parallel sheets carrying these total charge densities produce between them a field of magnitude $\sigma_f/\epsilon_r\epsilon_0$, in agreement with the result found from **D**. The bound charge next to each plate cancels part of the free charge on it, and the field in the gap is the field of the charge that remains.

If the plates are isolated while the dielectric is inserted, Q stays the same and the field and the potential difference both fall by the factor ϵ_r . If the plates stay connected to a battery, V stays the same, and charge flows onto the plates until their free charge has risen by the factor ϵ_r , as you will find in Problem 3. The larger charge stored at a given potential difference is one reason why practical capacitors usually contain a dielectric, another being that a thin solid film holds the plates a small, fixed distance apart.

A Point Charge in a Dielectric

As a second example, consider a point charge q at the centre of a spherical shell of linear dielectric, with inner radius a and outer radius b , as shown in Figure 30. The charge sits in a small empty cavity, which serves as a model of an ion in a liquid, since the molecules of the liquid cannot come closer to the centre of the ion than its radius. The arrangement is spherically symmetric, so **D** points radially and its magnitude depends only on the distance r from the charge. Let us apply Gauss's law in matter to a sphere of radius r centred on the charge. The dielectric carries no free charge, so the free charge enclosed is q for every r , and

$$\mathbf{D} = \frac{q}{4\pi r^2} \hat{\mathbf{r}} \quad (3.35)$$

at every $r > 0$. In the cavity and outside the shell $\mathbf{E} = \mathbf{D}/\epsilon_0$, which is the Coulomb field of q alone. Inside the dielectric the field is smaller by the factor ϵ_r ,

$$\mathbf{E} = \frac{1}{4\pi\epsilon_r\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}. \quad (3.36)$$

The polarisation of the dielectric is $\mathbf{P} = (\chi_e/\epsilon_r)\mathbf{D} = \chi_e q \hat{\mathbf{r}}/4\pi\epsilon_r r^2$.

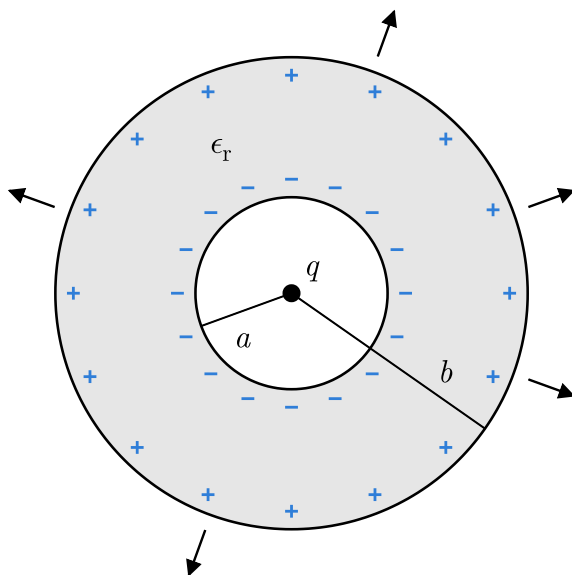


Figure 30. A positive point charge q in a cavity of radius a at the centre of a spherical shell of linear dielectric of outer radius b . The polarised dielectric carries negative bound charge (accent) on its inner surface and an equal amount of positive bound charge on its outer surface. Outside the shell the field (arrows) is the Coulomb field of q alone.

There is no bound volume charge in the dielectric, in agreement with the relation between ρ_b and ρ_f found above, and this can be checked directly. A thin spherical shell between the radii r and $r+dr$ loses the charge $4\pi(r+dr)^2P(r+dr)$ through its outer surface and gains the charge $4\pi r^2P(r)$ through its inner surface, and since r^2P does not depend on r , these charges are equal and the shell stays neutral. On the inner surface of the dielectric the outward normal points towards the charge, $\hat{\mathbf{n}} = -\hat{\mathbf{r}}$, and so

$$\sigma_b = -\frac{\chi_e q}{4\pi\epsilon_r a^2}. \quad (3.37)$$

Multiplying by the area $4\pi a^2$ gives the total bound charge on the inner surface, $-\chi_e q/\epsilon_r$, which does not depend on a . On the outer surface $\hat{\mathbf{n}} = \hat{\mathbf{r}}$, and a similar calculation gives the total bound charge $\chi_e q/\epsilon_r$, so that the bound charges on the two surfaces add up to zero.

Gauss's law of Section 1.3 gives the same field. A sphere of radius r inside the dielectric encloses the free charge q and the bound charge on the inner surface, whose sum is $q - \chi_e q / \epsilon_r = q / \epsilon_r$, and this total charge produces the field found above. The molecules near the charge are polarised with their ends of opposite sign facing it, and the layer of bound charge that they form around the charge cancels part of it, as seen from any point in the dielectric beyond.

Let the outer radius b now become very large. The outer bound charge is then spread over a very large sphere, and by the shell theorem of Section 1.3 its field is zero everywhere inside that sphere. A point charge in a large body of linear dielectric therefore has the field and the potential

$$\mathbf{E} = \frac{1}{4\pi\epsilon_r\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}, \quad V = \frac{1}{4\pi\epsilon_r\epsilon_0} \frac{q}{r}, \quad (3.38)$$

the potential being found by integrating the field inwards from infinity, as in Section 1.4. These are the results of Coulomb's law with ϵ_0 replaced by the permittivity $\epsilon_r\epsilon_0$. The electric forces between ions dissolved in water are weakened in a similar way, which is one reason why many salts dissolve readily in water, whose relative permittivity is large.

Problems

§3.1 Dipoles and Polarisation

1. Show that the dipole moment $\sum_i q_i \mathbf{r}_i$ of a set of charges whose total charge is zero is unchanged when the origin is moved. Find the dipole moment of a charge $-2q$ at the origin together with charges q at $(a, 0, 0)$ and at $(0, a, 0)$.
2. Along a field line of a point dipole, in a plane containing its axis, a small step has components dr and $r d\theta$ in the ratio $E_r : E_\theta$. Use this to show that the field lines are the curves $r = r_0 \sin^2 \theta$, and check that they cross the equipotential surfaces, on which $\cos \theta / r^2$ is constant, at right angles, by showing that a small step along a field line is perpendicular to a small step along an equipotential surface.
3. A point charge Q sits at the origin, and a small dipole $\mathbf{p} = p \hat{\mathbf{r}}$ sits at the position $r \hat{\mathbf{r}}$, pointing directly away from the charge,

as in Figure 25. Using the field of the dipole, find the force that the dipole exerts on the charge, and show that it is equal and opposite to the force $-2pQ\hat{\mathbf{r}}/4\pi\epsilon_0r^3$ found in the text.

4. An atom of polarisability α is at a distance r from a point charge Q . Show that the atom is attracted towards the charge for either sign of Q , with a force of magnitude $2\alpha Q^2/(4\pi\epsilon_0)^2r^5$.
5. A polar molecule of moment p is in a field $\mathbf{E}$ at temperature $\mathcal{T}$. The probability that its moment lies within a small solid angle $d\Omega$ about a direction at an angle θ to the field is proportional to $d\Omega$ multiplied by the Boltzmann factor $e^{-U/k_B\mathcal{T}}$ of Section 4.4 of Vol. 2, with $U = -pE \cos \theta$, and the element of solid angle is $d\Omega = \sin \theta d\theta d\phi$, as in Section 4.1 of Vol. 3. Let $\xi = pE/k_B\mathcal{T}$. Show that the mean component of the moment along the field is $\langle p_E \rangle = p (\coth \xi - 1/\xi)$, where $\coth \xi = \cosh \xi / \sinh \xi$ is formed from the hyperbolic functions of Section 1.2 of Vol. 4, and that this is approximately $p^2E/3k_B\mathcal{T}$ when ξ is much smaller than one.
6. A sphere of radius a is uniformly polarised, with polarisation $\mathbf{P}$ along the z -axis. First show that its bound surface charge density is $P \cos \theta$. Then treat the sphere as two balls of radius a , one with uniform charge density ρ and one with uniform charge density $-\rho$, whose centres are separated by a small displacement ℓ , so that $\mathbf{P} = \rho\ell$. Using the field inside a uniformly charged ball, found in Section 1.3, show that the field inside the sphere is uniform and equal to $-\mathbf{P}/3\epsilon_0$. Finally, show that outside the sphere the field is that of a point dipole of moment $\frac{4}{3}\pi a^3\mathbf{P}$ at its centre.
7. The potential at $\mathbf{r}$ of a point dipole $\mathbf{p}$ at the source point $\mathbf{r}'$ is $\mathbf{p} \cdot (\mathbf{r} - \mathbf{r}') / 4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^3$. Let ∇' denote the gradient with respect to $\mathbf{r}'$. Show in components that $\nabla'(1/|\mathbf{r} - \mathbf{r}'|) = (\mathbf{r} - \mathbf{r}') / |\mathbf{r} - \mathbf{r}'|^3$. Use this result, the product rule $\nabla' \cdot (g\mathbf{P}) = g \nabla' \cdot \mathbf{P} + \mathbf{P} \cdot \nabla' g$ of Section 3.1 of Vol. 4, and the divergence theorem to show that the potential of a polarised body occupying a volume $\mathcal{V}$, which is $\int_{\mathcal{V}} \mathbf{P}(\mathbf{r}') \cdot (\mathbf{r} - \mathbf{r}') d\mathcal{V}' / 4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^3$, equals the potential of the bound surface charge $\mathbf{P} \cdot \hat{\mathbf{n}}$ on its surface together with the bound volume charge $-\nabla' \cdot \mathbf{P}$ throughout it.

§3.2 The Electric Displacement and Linear Dielectrics

1. A parallel-plate capacitor with plates of area A a distance d apart contains a slab of linear dielectric of relative permittivity ϵ_r and thickness $w < d$, parallel to the plates, with the rest of the gap empty. Show that the capacitance is $\epsilon_0 A / (d - w + w/\epsilon_r)$, and that it does not depend on where the slab sits in the gap. Find the bound surface charge densities on the faces of the slab when the plates carry charges $\pm Q$.
2. The gap of a parallel-plate capacitor, with plates of area A a distance d apart, is filled by two slabs of linear dielectric side by side, each covering half the area of the plates and filling the whole gap, with relative permittivities ϵ_{r1} and ϵ_{r2} . Neglecting the effects of the edges, show that the capacitance is $\epsilon_0 A (\epsilon_{r1} + \epsilon_{r2}) / 2d$, and find the free surface charge density on each half of the positive plate when the potential difference is V .
3. An empty parallel-plate capacitor, with plates of area A a distance d apart, is connected to a battery that holds the potential difference between the plates at V . A slab of linear dielectric of susceptibility χ_e is then slid in so that it fills the gap. Find the charge that flows from the battery onto the positive plate, and show that the free charge added per unit area of that plate is equal in magnitude to the bound surface charge density on the face of the slab next to it.
4. A coaxial cable has an inner conductor of radius a and an outer conductor of inner radius b , and the space between them is filled with a linear dielectric of relative permittivity ϵ_r . The conductors carry free charges λ and $-\lambda$ per unit length. Find $\mathbf{D}$ and $\mathbf{E}$ between the conductors, show that the capacitance per unit length is $2\pi\epsilon_r\epsilon_0 / \ln(b/a)$, and find the total bound charge per unit length on the inner surface of the dielectric.
5. A sphere of radius a made of a linear dielectric of relative permittivity ϵ_r carries free charge spread uniformly through its volume with density ρ_f . Find $\mathbf{D}$ and $\mathbf{E}$ inside and outside the sphere. Find the bound volume charge density, both from the relation between ρ_b and ρ_f in the text and directly from $-\nabla \cdot \mathbf{P}$, and the bound surface charge density, and check that the total bound charge is zero.

6. For the point charge at the centre of the dielectric shell of Figure 30, find the potential at a distance r from the charge, with $a < r < b$, taking the potential to be zero far away. Check the result by adding the potentials of the charge q and of the bound charges on the two surfaces of the shell.

Electric Current and Circuits

4.1 Circuits at School

The circuits introduced at school are built from sources, resistors, capacitors, and coils joined by wires, and their behaviour can be described by a small set of relations between current, potential difference, and resistance. This section restates those relations in the notation that the rest of the chapter uses, together with the relation for an inductor. Section 4.2 then describes the flow of charge at each point inside a conductor and derives Ohm's law from a model of conduction in a metal, and Section 4.3 applies Kirchhoff's laws to circuits whose currents change with time.

Current, Potential Difference, and Resistance

The current I through a cross-section of a wire is the rate at which charge passes through it,

$$I = \frac{dQ}{dt}, \quad (4.1)$$

and its unit, the ampere, is defined as one coulomb per second. By convention the direction of the current is the direction in which positive charge would flow. In a metal the charge is carried by conduction electrons, which move in the opposite direction to the conventional current because their charge is $-e$.

The potential difference V between two points of a circuit is the energy transferred per unit charge as charge passes from one point to the other, and its unit is the volt, defined as one joule per coulomb. The resistance of a circuit component is the ratio of the potential difference across it to the current through it, and its unit, the ohm, is defined as one volt per ampere. A metal wire held at a constant temperature is observed to carry a current proportional to the potential difference across it, so that its resistance does not depend on V , and this result is known as Ohm's law. A circuit component that obeys Ohm's law is called ohmic. A filament lamp is not ohmic, since its filament warms up and its resistance rises as the current grows, and neither is a diode, which conducts far more easily in one direction than in the other. For a uniform wire of length ℓ and cross-sectional area A ,

$$R = \frac{\rho_r \ell}{A}, \quad (4.2)$$

where the resistivity ρ_r is a property of the material that depends on its temperature. A current I flowing through a potential difference V transfers energy at the rate $\mathcal{P} = IV$, and in a resistor this energy heats the resistor at the rate

$$\mathcal{P} = IV = I^2 R = \frac{V^2}{R}. \quad (4.3)$$

Resistors in Series and in Parallel

Resistors joined end to end, as in the top left of Figure 31, are said to be in series. The same current passes through each of them, since charge does not accumulate at the joins, and the potential differences across them add. Resistors joined between the same pair of points, as in the top right of the figure, are said to be in parallel. The potential difference across each of them is the same, and the currents through them add. The combined resistances are therefore

$$R_{\text{series}} = R_1 + R_2 + R_3 + \cdots, \quad \frac{1}{R_{\text{parallel}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots. \quad (4.4)$$

The resistance of a parallel combination is smaller than the smallest of the resistances in it, because each added resistor opens a further path for the current.

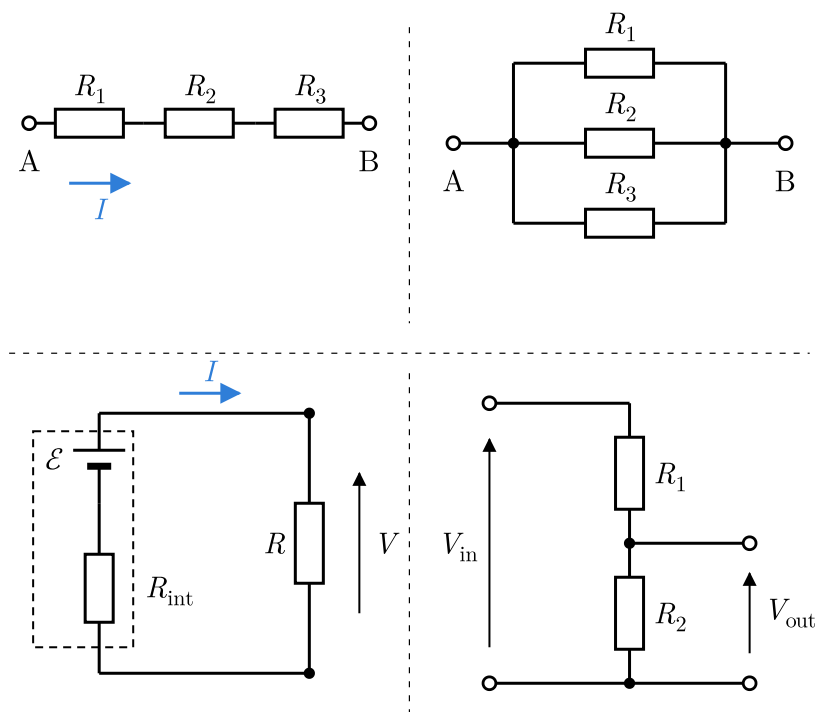


Figure 31. Top left, resistors in series, which carry the same current I . Top right, resistors in parallel, which have the same potential difference across them. Bottom left, a source of electromotive force $\mathcal{E}$ and internal resistance R_{int} , enclosed by the dashed box, driving a current through a load R , with the terminal potential difference V across the load. Bottom right, a potential divider, whose output V_{out} is taken across R_2 .

Electromotive Force and Internal Resistance

A source such as a cell or a battery gives energy to the charge that passes through it, and the electromotive force $\mathcal{E}$ of the source is the energy it gives per unit charge. In a cell, for example, this energy comes from chemical reactions, and the electromotive force is measured in volts. A real source has an internal resistance R_{int} , which can be drawn as a resistor in series with an ideal source, as in the bottom left of Figure 31. When the source drives a current I through a load of resistance R , part of the energy given to each unit

of charge is dissipated in the internal resistance, and the potential difference between its terminals is

$$V = \mathcal{E} - IR_{\text{int}}, \quad I = \frac{\mathcal{E}}{R + R_{\text{int}}}. \quad (4.5)$$

The terminal potential difference equals the electromotive force only when no current flows, which is why a voltmeter of very high resistance connected across a source measures its electromotive force closely.

Kirchhoff's Laws

Kirchhoff's first law, sometimes called the junction law, states that the total current flowing into a junction equals the total current flowing out of it. It expresses the conservation of charge, since in a steady state no charge accumulates at a junction. Kirchhoff's second law, sometimes called the loop law, states that around any closed loop of a circuit the sum of the electromotive forces equals the sum of the products IR for the resistors in the loop. It expresses the conservation of energy, since a charge carried once round the loop returns to the potential it started from.

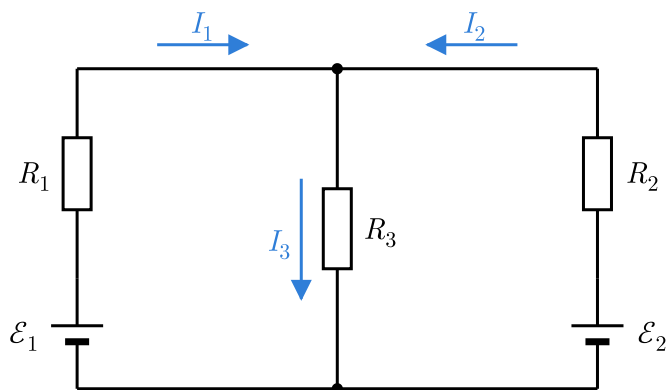


Figure 32. A network with two sources and three resistors. The arrows give the directions chosen for the currents I_1 , I_2 , and I_3 , and the long plate of each cell symbol is its positive terminal.

We apply Kirchhoff's laws to the network in Figure 32, which has two sources and three resistors. Let us choose a direction for the current in each branch, as the arrows show. A current that comes out negative when the equations are solved flows opposite to its arrow. The first law applied to the upper junction, and the second law applied to the left-hand and right-hand loops, give

$$I_1 + I_2 = I_3, \quad \mathcal{E}_1 = I_1 R_1 + I_3 R_3, \quad \mathcal{E}_2 = I_2 R_2 + I_3 R_3. \quad (4.6)$$

These are three equations for the three currents, and you will solve them in Problem 3. The outer loop, which runs through both sources, gives a fourth equation, $\mathcal{E}_1 - \mathcal{E}_2 = I_1 R_1 - I_2 R_2$, which is the difference of the two loop equations above and so contains nothing new.

The Potential Divider

A potential difference V_{in} applied across two resistors in series is divided between them in proportion to their resistances. Taking the output across R_2 , as in the bottom right of Figure 31, gives

$$V_{\text{out}} = \frac{R_2}{R_1 + R_2} V_{\text{in}}, \quad (4.7)$$

which holds provided that the output terminals draw a current small compared with the current through the resistors, that is, provided that anything connected to them has a resistance much larger than R_2 . An arrangement of this kind is called a potential divider. If R_2 is replaced by a circuit component whose resistance depends on its surroundings, such as a thermistor or a light-dependent resistor, the output voltage changes with the temperature or with the illumination, which is one way of building a sensing circuit.

Capacitors and Inductors in Circuits

The capacitance of a capacitor is $C = Q/V$, where Q is the charge on its positive plate and V the potential difference between its plates. A capacitor charged to Q_0 and then connected across a resistor R discharges, and its charge falls exponentially with time,

$$Q = Q_0 e^{-t/RC}, \quad (4.8)$$

with the current and the potential difference across the capacitor falling in the same way. A capacitor charged through a resistor R

from a source of electromotive force $\mathcal{E}$, starting uncharged, has the charge $Q = C\mathcal{E}(1 - e^{-t/RC})$. The product $\tau = RC$ is called the time constant of the circuit. It is the time in which the charge on a discharging capacitor falls to $1/e$ of its initial value, and the charge halves in the time $\tau \ln 2$.

A coil of wire is called an inductor. When the current through a coil changes, an electromotive force is observed to appear in the coil that opposes the change, and it is proportional to the rate of change of the current,

$$\mathcal{E} = -L \frac{dI}{dt}, \quad (4.9)$$

where L is called the self-inductance of the coil. Its unit, the henry, is defined as one volt second per ampere. The minus sign says that the induced electromotive force acts against the current when the current is increasing and with it when the current is decreasing. An ideal inductor has no resistance, so a steady current passes through it with no potential difference across it, and an inductor with a large self-inductance can be used in a circuit to smooth out rapid changes in the current.

4.2 Current Density, Conductivity, and Ohm's Law

The relations of Section 4.1 describe a resistor as a whole, through the current in it and the potential difference across it. We turn now to the flow of charge at each point inside a conductor. We first define the current density, then write Ohm's law as a relation between the current density and the electric field and explain it with a model of conduction in a metal, and finally use that relation to find resistances, to express the electromotive force and the rate of heating in terms of fields, and to find how quickly a charge placed inside a conductor leaves it.

Current Density

The current density $\mathbf{J}$ at a point is the vector that points in the direction in which positive charge flows there and has the magnitude dI/da , where dI is the current through a small area da placed perpendicular to the flow. Its unit is the ampere per square metre.

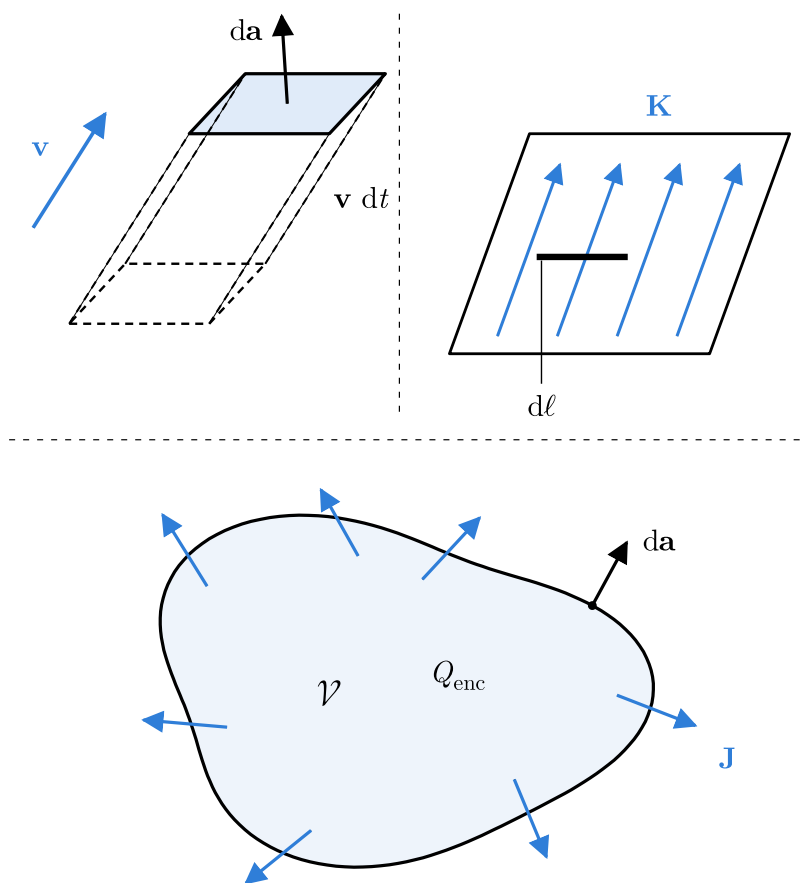


Figure 33. Top left, the charges that cross a surface element $d\mathbf{a}$ in a time dt are those inside the slanted prism whose sides are $\mathbf{v} dt$. Top right, a surface current density $\mathbf{K}$, with a line segment $d\ell$ drawn across the flow. Bottom, a fixed volume $\mathcal{V}$ containing the charge Q_{enc} , with current density $\mathbf{J}$ crossing its surface and the outward surface element $d\mathbf{a}$.

To relate $\mathbf{J}$ to the motion of the charges, suppose that charge of density ρ moves with velocity $\mathbf{v}$, and consider a small surface element $d\mathbf{a}$ of any orientation, drawn in the top left of Figure 33. In a time dt every charge moves through $\mathbf{v} dt$, so the charges that cross the element in that time are those that start inside the slanted prism whose base is the element and whose sides are parallel to $\mathbf{v} dt$. The volume of the prism is its base area da times its perpendicular height, which is the component $\hat{\mathbf{n}} \cdot \mathbf{v} dt$ of $\mathbf{v} dt$ along the unit normal $\hat{\mathbf{n}}$ of the element, so the volume is $\mathbf{v} \cdot d\mathbf{a} dt$, and the charge crossing the element per unit time is therefore

$$dI = \rho \mathbf{v} \cdot d\mathbf{a} = \mathbf{J} \cdot d\mathbf{a}, \quad \mathbf{J} = \rho \mathbf{v}. \quad (4.10)$$

The current through a surface of any shape is the sum of the currents through its elements,

$$I = \int \mathbf{J} \cdot d\mathbf{a}, \quad (4.11)$$

which is the flux of $\mathbf{J}$ through the surface, in the sense of Section 1.3. In a metal, where conduction electrons of number density n_e and charge $-e$ move with mean velocity $\mathbf{v}_d$, called the drift velocity, the current density is $\mathbf{J} = -n_e e \mathbf{v}_d$, and it points opposite to the motion of the electrons.

Where charge flows in a layer thin enough to be treated as a surface, such as a thin metal foil, the surface current density $\mathbf{K}$ is defined as the current per unit width measured across the flow, so that a line segment of length $d\ell$ drawn across the flow, as in the top right of Figure 33, is crossed by the current $K d\ell$. The argument above, with the prism replaced by a parallelogram, gives $\mathbf{K} = \sigma \mathbf{v}$ for a surface charge of density σ moving with velocity $\mathbf{v}$, and the unit of $\mathbf{K}$ is the ampere per metre.

The Continuity Equation

Let $\mathcal{V}$ be a fixed volume bounded by a closed surface, with the vector surface element $d\mathbf{a}$ pointing outwards, as in the bottom of Figure 33. Charge is conserved, so the charge Q_{enc} inside the volume can change only by flowing through the surface, and $\oint \mathbf{J} \cdot d\mathbf{a} = -dQ_{\text{enc}}/dt$. The argument that Section 3.3 of Vol. 4 of the Fermi Notes on Physics applies to the mass of a flowing fluid, with the

charge density ρ in place of the mass density and $\mathbf{J}$ in place of $\rho\mathbf{v}$, then gives

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}. \quad (4.12)$$

This relation is called the continuity equation, and it is the local form of the conservation of charge. A current is called steady when the charge density at every point stays constant, and for a steady current $\nabla \cdot \mathbf{J} = 0$, so the flux of $\mathbf{J}$ through every closed surface is zero. Applied to a small surface drawn round a junction of wires, this is Kirchhoff's first law of Section 4.1.

Ohm's Law at a Point

Experiment shows that in many materials held at a fixed temperature, including metals, the current density at a point is proportional to the electric field at that point,

$$\mathbf{J} = \sigma_c \mathbf{E}. \quad (4.13)$$

The constant σ_c is called the conductivity of the material, and its reciprocal, $\rho_r = 1/\sigma_c$, is the resistivity of Section 4.1. Like the relative permittivity of Section 3.2, the conductivity is a property of the material, and it is very large for a metal such as copper and very small for an insulator such as glass. The relation above is the form of Ohm's law that applies at each point, and a material that obeys it is called ohmic.

A Model of Conduction in a Metal

A metal consists of a lattice of positive ions through which the conduction electrons move rapidly in random directions, and each collision of an electron with the lattice sends it off in a new random direction. The mean distance between collisions is called the mean free path, which Section 1.3 of Vol. 2 of the Fermi Notes on Physics worked out for the molecules of a gas and called λ . Here we let ℓ_f be the mean free path, since λ is the line charge density. As in that section, if $\bar{v}$ is the mean speed of the electrons, the mean time between collisions is

$$\tau_c = \frac{\ell_f}{\bar{v}}. \quad (4.14)$$

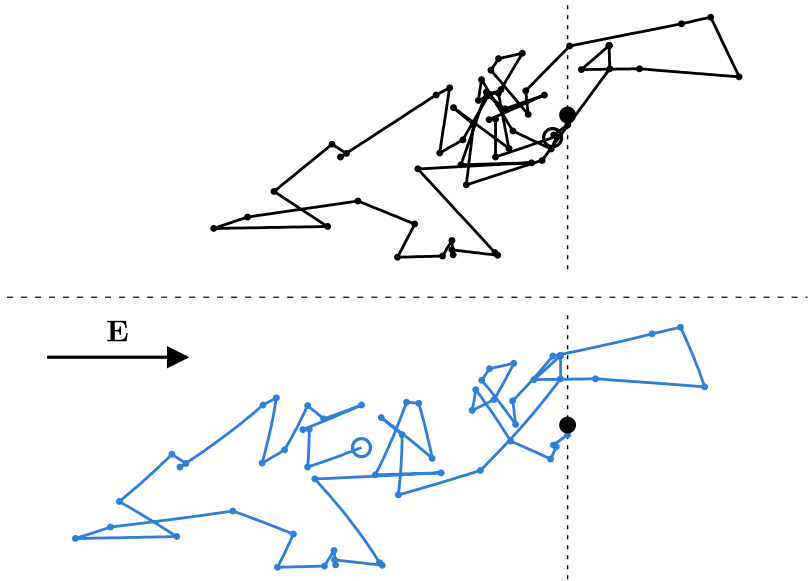


Figure 34. The path of a conduction electron in a metal without a field (top) and with a field $\mathbf{E}$ (bottom), computed with the same sequence of random velocities and times between collisions. The dots mark the collisions, the filled circle the starting point, and the open circle the end of the path. With the field, the path between collisions is a parabolic arc and the electron drifts opposite to $\mathbf{E}$; the drift is greatly exaggerated.

Without a field the velocities point equally in all directions and their mean is zero, so no current flows. In a field $\mathbf{E}$, each electron is accelerated at $-e\mathbf{E}/m_e$ between collisions, and its path between collisions becomes a parabolic arc, as in Figure 34. We assume that a collision leaves an electron with a velocity whose mean is zero, independently of the velocity it had before, and that every electron has the same chance dt/τ_c of colliding in a short interval dt , independently of its earlier history.

With these assumptions we can find how the drift velocity $\mathbf{v}_d$, the mean velocity of the electrons, changes with time. In the interval dt the field changes the momentum of every electron by $-e\mathbf{E} dt$. In the same interval a fraction dt/τ_c of the electrons collide, and since

these electrons are a random sample, their mean momentum before the collisions is $m_e \mathbf{v}_d$ and after the collisions is zero. The mean momentum of all the electrons therefore changes according to

$$m_e \frac{d\mathbf{v}_d}{dt} = -e\mathbf{E} - \frac{m_e}{\tau_c} \mathbf{v}_d. \quad (4.15)$$

This equation has the form of the equation of motion of a body falling with linear drag, which was solved in Section 1.3 of Vol. 1 of the Fermi Notes on Physics, and again as an example of a linear first-order equation in Section 5.1 of Vol. 4. In that equation the mass times the acceleration equals the weight minus a drag force proportional to the velocity, with a constant of proportionality that is called the drag constant, and the equation above has $-e\mathbf{E}$ in place of the weight and m_e/τ_c in place of the drag constant. The drift velocity therefore approaches a terminal value in a time of order τ_c , as you will show in Problem 5, and once it has reached that value the left-hand side vanishes, giving

$$\mathbf{v}_d = -\frac{e\tau_c}{m_e} \mathbf{E}. \quad (4.16)$$

Substituting this into $\mathbf{J} = -n_e e \mathbf{v}_d$ gives $\mathbf{J} = \sigma_c \mathbf{E}$, with

$$\sigma_c = \frac{n_e e^2 \tau_c}{m_e}. \quad (4.17)$$

The current density is proportional to the field because τ_c does not depend on $\mathbf{E}$. The drift speed in a metal carrying an ordinary current is very much smaller than $\bar{v}$, because the number density of conduction electrons is so large that a small drift speed carries a large current, as you will see in Problem 5, so the field hardly changes the speeds of the electrons between collisions, and for the same reason the drift in Figure 34 is greatly exaggerated.

The model has its limitations, and one of them shows in the temperature dependence of the resistivity. With $\bar{v}$ taken as the mean speed $\sqrt{8k_B \mathcal{T} / \pi m_e}$ of Section 1.2 of Vol. 2, where $\mathcal{T}$ is the temperature, written as in Section 3.1, and with ℓ_f fixed by the spacing of the ions, it gives a resistivity that goes as $\mathcal{T}^{1/2}$, whereas that of a pure metal near room temperature is close to proportional to $\mathcal{T}$. The motion of conduction electrons is governed by quantum mechanics, and the quantum theory of electrons in metals is not treated in this

course. The model is used here for the relation $\mathbf{J} = \sigma_c \mathbf{E}$ and for the expression of σ_c in terms of τ_c .

The Resistance of a Wire

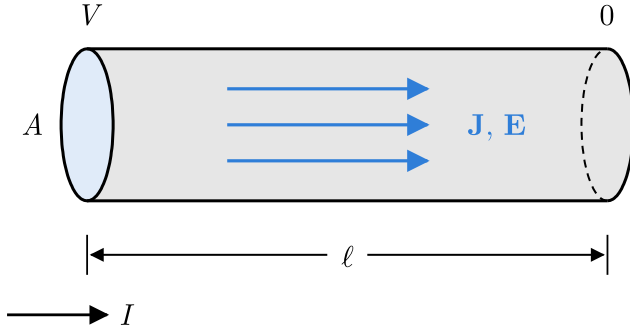


Figure 35. A straight wire of length ℓ and cross-sectional area A , with its ends held at the potentials V and 0 . The current density $\mathbf{J}$ and the field $\mathbf{E}$ are uniform and parallel to the axis, and the current I flows towards the end at the lower potential.

We can now recover the school relation for the resistance of a wire. For this, we take a straight wire of uniform conductivity σ_c , length ℓ , and cross-sectional area A , whose ends are held at the potentials V and 0 , as in Figure 35, carrying a steady current. Then $\nabla \cdot \mathbf{J} = 0$, so $\nabla \cdot \mathbf{E} = 0$ inside the wire, and Gauss's law in differential form, $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$, from Section 1.3, shows that there is no charge inside it. Any charge on a uniform ohmic conductor carrying a steady current lies on its surface, and the charges that gather on the surface of a wire, where it bends for example, keep the field inside it directed along the wire.

Inside a long straight wire, away from its ends, the field is uniform and parallel to the axis, so $V = E\ell$ by Section 1.4, and the current is $I = JA = \sigma_c EA$. The resistance is therefore

$$R = \frac{V}{I} = \frac{\ell}{\sigma_c A} = \frac{\rho_r \ell}{A}, \quad (4.18)$$

which is the relation quoted in Section 4.1, and $V = IR$ is Ohm's

law for the wire as a whole. You will find the resistance of a wire whose cross-section changes along its length in Problem 3.

The Resistance of a Spherical Shell

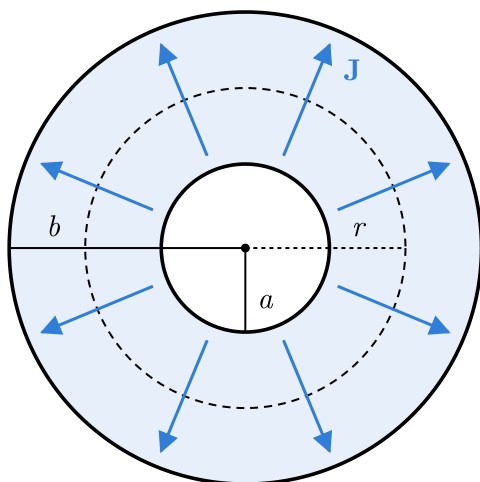


Figure 36. A cross-section through a shell of conducting material between concentric spheres of radii a and b . The current density $\mathbf{J}$ points radially outwards, and the same current crosses every sphere of radius r , such as the dashed one.

As an example in which the current density is not uniform, consider a material of conductivity σ_c that fills the space between two concentric conducting spheres of radii a and b , with $a < b$. The inner sphere is held at a potential V above the outer one, and a steady current I flows outwards through the material, as in Figure 36. By symmetry the current density points along $\hat{\mathbf{r}}$ and depends only on r . Since the current is steady, the flux of $\mathbf{J}$ out of the region between the inner sphere and a sphere of radius r is zero, so the same current I crosses every sphere of radius r between a and b , and

$$I = 4\pi r^2 J(r), \quad \mathbf{E} = \frac{\mathbf{J}}{\sigma_c} = \frac{I}{4\pi\sigma_c r^2} \hat{\mathbf{r}}. \quad (4.19)$$

The potential difference is the line integral of the field along a radius

from a to b ,

$$V = \frac{I}{4\pi\sigma_c} \int_a^b \frac{dr}{r^2} = \frac{I}{4\pi\sigma_c} \left(\frac{1}{a} - \frac{1}{b} \right), \quad (4.20)$$

and the resistance of the shell is

$$R = \frac{1}{4\pi\sigma_c} \left(\frac{1}{a} - \frac{1}{b} \right). \quad (4.21)$$

For a thin shell, with $b = a + \ell$ and $\ell \ll a$, the bracket is $\ell/a(a + \ell) \approx \ell/a^2$, and the resistance becomes $\ell/(4\pi a^2 \sigma_c)$, which is the resistance of a slab of thickness ℓ and area $4\pi a^2$, as it should be. The field in the shell has the same dependence on r as the field between the concentric spheres of the capacitor of Section 2.1, and you will show in Problems 1 and 2 that the product of the resistance and the capacitance of a pair of conductors of any shape, with the conducting material filling all the space in which the field is present, is ϵ_0/σ_c .

Electromotive Force

In a circuit carrying a steady current, the electric field is produced by charges whose distribution does not change, so by Section 1.4 the work it does on a charge q carried once round the circuit, $q \oint \mathbf{E} \cdot d\mathbf{l}$, is zero. Since the charges lose energy in the resistors, the electrostatic field alone cannot keep a current flowing round a closed circuit. A source supplies a further force $\mathbf{F}_s$ on the charges inside it, of chemical origin in a battery, for example, so that the total force on a charge q is $\mathbf{F} = q\mathbf{E} + \mathbf{F}_s$. The electromotive force of the circuit is defined as the work done per unit charge by this total force around the circuit,

$$\mathcal{E} = \frac{1}{q} \oint \mathbf{F} \cdot d\mathbf{l} = \frac{1}{q} \oint \mathbf{F}_s \cdot d\mathbf{l}, \quad (4.22)$$

where the second form follows from the vanishing of the line integral of $\mathbf{E}$. The force $\mathbf{F}_s$ acts only inside the source, so $\mathcal{E}$ is the integral of $\mathbf{F}_s/q$ through the source from its negative terminal to its positive terminal, taken in the direction drawn in Figure 37.

A charge carrier drifts in response to the total force on it, so inside the source Ohm's law becomes $\mathbf{J} = \sigma_c \mathbf{F}/q$. The integral of $\mathbf{F}/q$ through the source from the negative terminal to the positive one

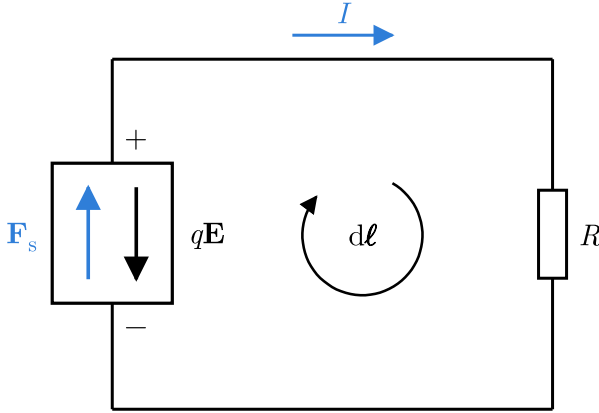


Figure 37. A circuit with a source and a resistor R . Inside the source the force $\mathbf{F}_s$ on a positive charge q points from the negative terminal to the positive one, and the electric force $q\mathbf{E}$ points the other way. The curved arrow shows the direction in which the line integral round the circuit is taken.

is therefore IR_{int} , by the argument used for the wire, where R_{int} is the internal resistance. The same integral is also $\mathcal{E} - (V_+ - V_-)$, where V_+ and V_- are the potentials of the terminals, so

$$V_+ - V_- = \mathcal{E} - IR_{\text{int}}, \quad (4.23)$$

which is the school relation of Section 4.1. For an ideal source, with no internal resistance, the force $\mathbf{F}_s$ exactly balances $q\mathbf{E}$ inside the source, and the electromotive force equals the potential difference between the terminals. The same reasoning applied to the whole circuit, with a resistor R outside the source, gives $\mathcal{E} = \oint (\mathbf{J}/\sigma_c) \cdot d\boldsymbol{\ell} = I(R + R_{\text{int}})$, which is Kirchhoff's second law for a single loop.

The Rate of Heating

A field $\mathbf{E}$ does work on a charge q moving with velocity $\mathbf{v}$ at the rate $q\mathbf{E} \cdot \mathbf{v}$. When this rate is added up over the charges in a small volume, the random parts of their velocities average to zero, so the rate per unit volume is $\rho \mathbf{v} \cdot \mathbf{E}$, with $\mathbf{v}$ the mean velocity, which can

be written as

$$\mathbf{J} \cdot \mathbf{E} = \sigma_c E^2 = \frac{J^2}{\sigma_c}, \quad (4.24)$$

where the last two forms hold in an ohmic material. When the drift velocity has reached its steady value, the kinetic energy of the electrons no longer increases, so the energy supplied by the field is passed to the lattice in collisions and raises the temperature of the conductor. The effect is called Joule heating. For the wire of Figure 35, $\mathbf{J} \cdot \mathbf{E}$ is the same throughout the volume $A\ell$, and the total rate of heating is $JE A\ell = (JA)(E\ell) = IV$, which is the school result $\mathcal{P} = IV = I^2 R$. A similar calculation with $\mathbf{F}_s$ in place of $q\mathbf{E}$ shows that an ideal source does work on the charges at the rate $\mathcal{E}I$.

The Decay of a Charge inside a Conductor

Section 2.1 showed that a conductor in electrostatic equilibrium has no charge inside it, and we can now find how quickly this equilibrium is reached. Suppose that a charge density $\rho(\mathbf{r}, 0)$ is placed inside a uniform ohmic conductor at $t = 0$. The continuity equation, Ohm's law, and Gauss's law in differential form give

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot \mathbf{J} = -\sigma_c \nabla \cdot \mathbf{E} = -\frac{\sigma_c}{\epsilon_0} \rho. \quad (4.25)$$

At each point this is a first-order differential equation in t , whose solution is

$$\rho(\mathbf{r}, t) = \rho(\mathbf{r}, 0) e^{-\sigma_c t / \epsilon_0}. \quad (4.26)$$

The charge density falls at the same rate at every point, for any initial distribution, and it falls to $1/e$ of its initial value in the time ϵ_0 / σ_c . Since charge is conserved, the charge that leaves the interior flows to the surface of the conductor, where the conductivity drops to zero and the argument above no longer applies. For a metal, $\epsilon_0 / \sigma_c = \epsilon_0 m_e / (n_e e^2 \tau_c)$ turns out to be much shorter than τ_c , because the number density n_e in its denominator is so large, and over so short a time the relation $\mathbf{J} = \sigma_c \mathbf{E}$ does not hold, since it assumes that the drift velocity has reached its steady value. The way in which the charge reaches the surface of a metal is not treated in this course, and we conclude only that it leaves the interior within a time of the order of τ_c .

4.3 Transients in RC and RL Circuits

A circuit containing a capacitor or an inductor does not respond instantly when a switch is flipped. The current and the charges change over a time set by the circuit components, and the changing currents that flow in this time are called transients. Section 4.1 quoted the exponential charging and discharging of a capacitor, and here we derive those results from Kirchhoff's laws. The equations that arise are linear differential equations of the first order, which are treated at length in Section 5.1 of Vol. 4. This section first extends the laws to circuits containing capacitors and inductors, then solves the circuits in which a capacitor discharges or charges through a resistor and follows the energy in them, and finally treats the growth and decay of the current in a circuit containing a resistor and an inductor.

Kirchhoff's Laws with Capacitors and Inductors

Section 4.2 derived Kirchhoff's laws for steady currents from the continuity equation and from the vanishing of the line integral of the electrostatic field. The currents we now treat change with time, and we assume that they change slowly enough that the laws still hold at each instant. More precisely, we assume that charge accumulates only on the plates of capacitors, so that the current at any instant is the same at every point along a wire without junctions, and that the potential differences around any closed loop of the circuit add up to the electromotive forces in it.

For each kind of circuit component we need the potential difference between its terminals. For a resistor it is IR , with the current flowing from the higher potential to the lower. For a capacitor with charge Q on one plate and $-Q$ on the other it is Q/C , with the plate carrying Q at the higher potential, as in Section 2.1. No charge crosses the gap between the plates; charge arrives on one plate and an equal charge leaves the other, so the current is the same in the wires on the two sides. We take the current I to be positive when it flows onto the plate that carries Q , and then

$$I = \frac{dQ}{dt}. \quad (4.27)$$

For an inductor we use the school relation of Section 4.1, $\mathcal{E} =$

$-L dI/dt$, and we take the inductor to be ideal, so that its wire has no resistance. For the current through it to be finite, the net force on the charges in its wire must then be zero, as it was inside the ideal source of Section 4.2, and the potential difference between its terminals balances the induced electromotive force. The potential of the terminal at which the current enters therefore exceeds that of the other terminal by

$$V_L = L \frac{dI}{dt}, \quad (4.28)$$

which is positive when the current is increasing and negative when it is decreasing. The potential differences across the three circuit components are therefore IR , Q/C , and $L dI/dt$, each measured in the direction of the current.

Discharging a Capacitor

Let us consider the circuit of Figure 38, in which a two-way switch connects a capacitor and a resistor either to a source, in position 1, or to each other, in position 2. Suppose that the capacitor carries the charge Q_0 and that the switch is moved to position 2 at $t = 0$. The loop now contains only the resistor and the capacitor, and the loop law gives

$$IR + \frac{Q}{C} = 0, \quad \frac{dQ}{dt} = -\frac{Q}{RC}. \quad (4.29)$$

This is the equation of exponential decay solved by separating the variables in Section 5.1 of Vol. 4, and its solution with the charge Q_0 at $t = 0$ is

$$Q = Q_0 e^{-t/\tau}, \quad \tau = RC, \quad (4.30)$$

which is the result quoted in Section 4.1. The time constant has the dimensions of a time, since an ohm is defined as a volt per ampere and a farad as a coulomb per volt, so that their product is a coulomb per ampere, which is a second. The current is

$$I = \frac{dQ}{dt} = -\frac{Q_0}{RC} e^{-t/\tau}. \quad (4.31)$$

It is negative because charge flows off the positive plate, round the circuit against the arrow of Figure 38. At $t = 0$ its magnitude is Q_0/RC , so at the instant the switch is moved the capacitor, across

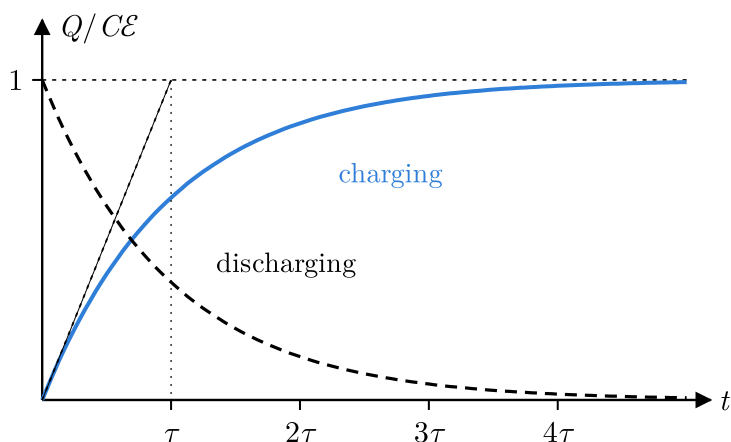
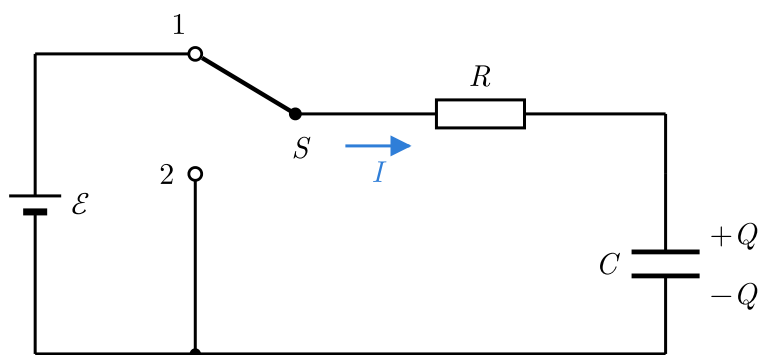


Figure 38. Top, a circuit in which the switch S connects a capacitor and a resistor to a source in position 1 and to each other in position 2. Bottom, the charge on the capacitor as a fraction of $C\mathcal{E}$, while it charges from zero (solid) and while it discharges from $C\mathcal{E}$ (dashed), against time, with $\tau = RC$. The dotted line through the origin is the tangent to the charging curve at $t = 0$, which reaches the final charge, marked by the horizontal dotted line, at $t = \tau$. While the capacitor charges, the current has the shape of the dashed curve.

which the potential difference is then Q_0/C , drives current through the resistor as a source of electromotive force Q_0/C would.

The discharge is drawn as the dashed curve in the lower part of Figure 38. The charge falls by the same factor e in every interval of length τ , from any starting value. The equation $dQ/dt = -Q/\tau$ also says that the tangent to the curve at any instant, continued as a straight line, reaches zero after a further time τ , which gives a way of finding the time constant from a measured curve.

Charging a Capacitor

Suppose now that the capacitor is uncharged and that the switch in Figure 38 is moved to position 1 at $t = 0$, connecting the source of electromotive force $\mathcal{E}$. The loop law gives

$$\mathcal{E} = IR + \frac{Q}{C}, \quad R \frac{dQ}{dt} = \mathcal{E} - \frac{Q}{C}. \quad (4.32)$$

The second equation has the same form as the equation of motion of a body of mass m falling from rest with linear drag, solved in Section 1.3 of Vol. 1, in which $m dv/dt$ equals the weight minus a drag force proportional to the speed v , the constant of proportionality being the drag constant of Section 4.2. The charge Q takes the place of the speed v , the resistance R that of the mass m , the electromotive force $\mathcal{E}$ that of the weight, and $1/C$ that of the drag constant. The terminal speed, which is the weight divided by the drag constant, then corresponds to the final charge $C\mathcal{E}$, and the time constant of the fall, which is the mass divided by the drag constant, to RC . The solution found there therefore gives

$$Q = C\mathcal{E} (1 - e^{-t/\tau}), \quad I = \frac{\mathcal{E}}{R} e^{-t/\tau}, \quad (4.33)$$

and it can be checked by substituting it into the differential equation, since $R dQ/dt = \mathcal{E}e^{-t/\tau}$ and $\mathcal{E} - Q/C = \mathcal{E}e^{-t/\tau}$. At $t = 0$ the uncharged capacitor has no potential difference across it, so the current starts at $\mathcal{E}/R$, the value it would have if the capacitor were replaced by a wire. As the charge grows, the potential difference Q/C opposes the source more and more strongly, and the current dies away. The charge approaches $C\mathcal{E}$, at which the capacitor has the potential difference $\mathcal{E}$ and no current flows. The charging curve

is the solid curve in Figure 38, and its tangent at $t = 0$ reaches the final charge at $t = \tau$.

Charging through a Potential Divider

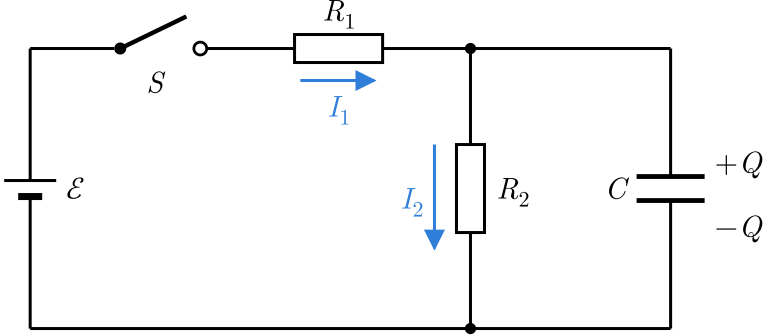


Figure 39. A capacitor charged through R_1 , with a resistor R_2 connected across it, and the currents I_1 and I_2 in the two resistors.

As an example of a circuit with more than one loop, consider the circuit of Figure 39, in which the capacitor has a resistor R_2 connected across it and is charged from the source through a resistor R_1 . The capacitor is uncharged when the switch is closed at $t = 0$. With the currents labelled as in the figure, the first law at the upper junction, the loop law for the loop containing the source, R_1 , and the capacitor, and the loop law for the loop containing the capacitor and R_2 give

$$I_1 = I_2 + \frac{dQ}{dt}, \quad \mathcal{E} = I_1 R_1 + \frac{Q}{C}, \quad I_2 R_2 = \frac{Q}{C}. \quad (4.34)$$

Solving the last two equations for I_1 and I_2 and substituting both currents into the first gives

$$\frac{dQ}{dt} = \frac{\mathcal{E}}{R_1} - \frac{Q}{C} \left(\frac{1}{R_1} + \frac{1}{R_2} \right). \quad (4.35)$$

Let $R_{\parallel} = R_1 R_2 / (R_1 + R_2)$ be the resistance of R_1 and R_2 in parallel.

Multiplying the equation by $R_{\parallel}$ puts it in the form

$$R_{\parallel} \frac{dQ}{dt} = \frac{R_2}{R_1 + R_2} \mathcal{E} - \frac{Q}{C}, \quad (4.36)$$

which is the charging equation with $R_{\parallel}$ in place of R and $\mathcal{E}R_2/(R_1 + R_2)$ in place of $\mathcal{E}$. The charge therefore grows as $1 - e^{-t/\tau}$, with the time constant $\tau = R_{\parallel}C$, towards a final potential difference across the capacitor equal to $\mathcal{E}R_2/(R_1 + R_2)$. This final value is the output of the potential divider of Section 4.1, which is to be expected, since no current flows into the capacitor once it is charged. The time constant is the one with which the capacitor would discharge if the source were replaced by a wire, since the capacitor would then be connected across R_1 and R_2 in parallel.

Energy in the Charging Circuit

To see what happens to the work done by the source while the capacitor charges, we multiply the loop equation $\mathcal{E} = IR + Q/C$ by $I = dQ/dt$, which gives

$$\mathcal{E}I = I^2R + \frac{Q}{C} \frac{dQ}{dt} = I^2R + \frac{d}{dt} \left(\frac{Q^2}{2C} \right). \quad (4.37)$$

The left-hand side is the rate at which the source does work, found in Section 4.2. The first term on the right is the rate of heating of the resistor, and the second is the rate of increase of the energy $Q^2/2C$ stored in the capacitor, found in Section 2.1. Over the whole charging process the charge $C\mathcal{E}$ passes through the source, which therefore does the work $C\mathcal{E}^2$, the product of the electromotive force and the charge, while the energy stored at the end is $\frac{1}{2}C\mathcal{E}^2$. The heat produced in the resistor is found by integrating the rate of heating, $I^2R = (\mathcal{E}^2/R)e^{-2t/\tau}$, over the whole process, which gives

$$\int_0^{\infty} I^2R dt = \frac{\mathcal{E}^2}{R} \frac{\tau}{2} = \frac{1}{2}C\mathcal{E}^2, \quad (4.38)$$

a result that agrees with the difference between the work done by the source and the energy stored. Half of the work done by the source is converted into heat, and this fraction does not depend on the resistance. This may seem surprising, since one might expect a smaller resistance to waste less energy. A smaller resistance does give a smaller rate of heating for a given current, but it also lets a

larger current flow, and the heating rate at the start, $\mathcal{E}^2/R$, grows as R falls while the time over which the heating lasts, of order RC , shrinks in proportion, so that their product does not depend on R . You will see in Problem 1 that the heat produced can be reduced by raising the electromotive force of the source in steps. When the capacitor discharges through the resistor, a similar calculation shows that all of the stored energy $Q_0^2/2C$ is converted into heat.

Growth of the Current in an RL Circuit

Let us now replace the capacitor by an inductor, as in Figure 40, and suppose that no current flows until the switch is moved to position 1 at $t = 0$. The loop law gives

$$\mathcal{E} = IR + L \frac{dI}{dt}. \quad (4.39)$$

This equation has the same form as the charging equation $\mathcal{E} = R dQ/dt + Q/C$, with the current I in place of the charge Q , the self-inductance L in place of R , and the resistance R in place of $1/C$. Its solution, with $I = 0$ at $t = 0$, is therefore

$$I = \frac{\mathcal{E}}{R} (1 - e^{-t/\tau}), \quad \tau = \frac{L}{R}. \quad (4.40)$$

The ratio L/R has the dimensions of a time, since a henry is defined as a volt second per ampere and an ohm as a volt per ampere. At $t = 0$ the current is zero, so there is no potential difference across the resistor, and the whole electromotive force of the source appears across the inductor, which gives $dI/dt = \mathcal{E}/L$. The current cannot jump suddenly to $\mathcal{E}/R$, since a sudden change in the current would need an infinite potential difference across the inductor. As the current grows, the potential difference across the resistor grows and the rate of growth falls, and the current approaches $\mathcal{E}/R$, the value it would have without the inductor. An ideal inductor therefore has no effect on a steady current, but it slows down every change in the current, over a time of order L/R . The growth of the current is drawn in the lower part of Figure 40.

Decay of the Current

Suppose next that at the time t_1 the switch in Figure 40 is moved to position 2, quickly compared with the time constant, when the

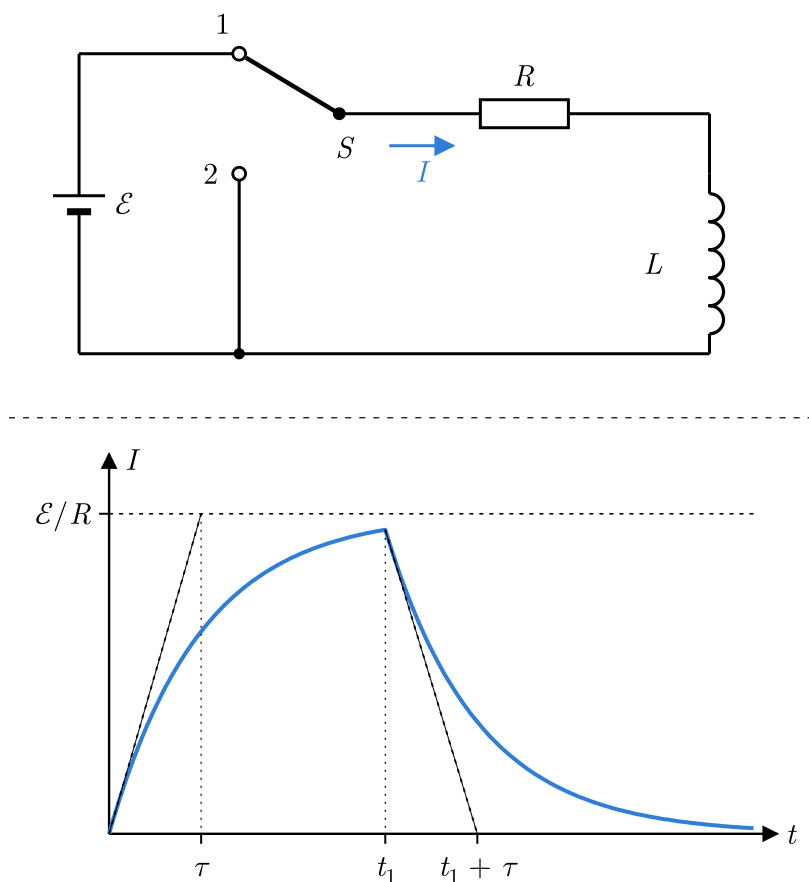


Figure 40. Top, the circuit of Figure 38 with the capacitor replaced by an ideal inductor L . Bottom, the current when the switch is moved to position 1 at $t = 0$ and to position 2 at $t = t_1$, with $\tau = L/R$. The two sloping dotted lines are the tangents to the curve at $t = 0$ and at $t = t_1$, which reach $\mathcal{E}/R$ and zero after a time τ .

current has the value I_1 . The source is now out of the circuit, and the loop law for the loop containing the resistor and the inductor gives

$$0 = IR + L \frac{dI}{dt}, \quad I = I_1 e^{-(t-t_1)/\tau}, \quad (4.41)$$

where the solution is that of the same equation of exponential decay as the discharge of the capacitor, solved in Section 5.1 of Vol. 4. The current continues to flow in the same direction after the source has been removed, and it decays with the same time constant L/R with which it grew, as the right-hand part of the curve in Figure 40 shows. While the current decays, dI/dt is negative, so the electromotive force $-L dI/dt$ of the inductor acts in the direction of the current and keeps it flowing through the resistor.

Multiplying the loop equation for the growing current by I , as we did for the capacitor, gives

$$\mathcal{E}I = I^2 R + \frac{d}{dt} \left(\frac{1}{2} LI^2 \right). \quad (4.42)$$

While the current grows, the source therefore does work at a rate greater than the rate of heating of the resistor, by the rate of increase of $\frac{1}{2}LI^2$, just as the source charging a capacitor did work faster than the resistor was heated, by the rate of increase of $Q^2/2C$. You will show in Problem 3 that the heat produced in the resistor while the current decays from I_1 to zero is $\frac{1}{2}LI_1^2$, which equals the work the source did, in excess of the heat produced, while the current grew from zero to I_1 .

Problems

§4.1 Circuits at School

1. A network is made of four identical resistors, each of resistance R , connected so that two of them in series are in parallel with the other two in series. Find the combined resistance. Then find the combined resistance when a fifth resistor R is connected between the midpoints of the two series pairs.
2. A source of electromotive force $\mathcal{E}$ and internal resistance R_{int} drives a current through a load R . Show that the power delivered

- to the load is greatest when $R = R_{\text{int}}$, and find this greatest power.
3. Solve the equations for the network of Figure 32, and show that the current through R_3 is $I_3 = (\mathcal{E}_1 R_2 + \mathcal{E}_2 R_1)/(R_1 R_2 + R_2 R_3 + R_3 R_1)$.
 4. A potential divider made of resistors R_1 and R_2 , with an input V_{in} , has a load of resistance R_L connected across R_2 . Find the output voltage, and show that it approaches the unloaded value when $R_L \gg R_2$.
 5. A capacitor discharges through a resistor. Show that the time in which its stored energy halves is $\frac{1}{2}\tau \ln 2$.
 6. The current through an inductor L rises steadily from zero to I_f in a time t_f . Find the magnitude of the electromotive force in the inductor during this time.

§4.2 Current Density, Conductivity, and Ohm's Law

1. The space between two coaxial conducting cylinders of radii a and b , with $a < b$, and length ℓ is filled with a material of conductivity σ_c . Neglecting the effects of the ends, find the resistance between the cylinders for a current flowing radially. Using the capacitances found in Section 2.1, show that the product of the resistance and the capacitance is ϵ_0/σ_c both for this arrangement and for the spherical shell of Figure 36.
2. Let two conductors be embedded in a material of conductivity σ_c that fills all of space. Assume that the field in the material is the same as the field that would be present in a vacuum if the conductors carried charges $\pm Q$ and had the same potential difference V between them. Use Gauss's law to show that the resistance between the conductors is $R = \epsilon_0/(\sigma_c C)$, where C is their capacitance in a vacuum.
3. A resistor has the shape of a truncated cone of length ℓ whose radius increases steadily from a at one end to b at the other, and it is made of a material of conductivity σ_c . Treating each thin slice across the cone as a short wire, which is a valid approximation when $b - a$ is small compared with ℓ , show that its resistance

- is $\ell/(\pi\sigma_c ab)$, and check that this reduces to the resistance of a uniform wire when $a = b$.
4. A thin disc of radius a carries a uniform surface charge density σ and rotates about its axis with angular velocity ω . Find the surface current density at a distance s from the axis, and find the total current crossing a radius of the disc. Check the result by counting the charge that crosses the radius in one revolution.
 5. The field $\mathbf{E}$ is switched on at $t = 0$ in a metal in which the drift velocity is initially zero. Solve the equation for $\mathbf{v}_d$ in the model of conduction, and show that the drift velocity reaches $1 - 1/e$ of its terminal value in the time τ_c . Show also that the drift speed in a wire of cross-sectional area A carrying a current I is $I/(n_e e A)$.
 6. By integrating $\mathbf{J} \cdot \mathbf{E}$ over the volume of the spherical shell of Figure 36, show that the total rate of heating is $I^2 R$, with R the resistance found in the text.
 7. A parallel-plate capacitor with plates of area A a distance d apart is filled with a linear dielectric of relative permittivity ϵ_r that also has a small conductivity σ_c . The capacitor is given a charge Q_0 and left isolated. Show that the charge leaks through the dielectric so that $Q = Q_0 e^{-\sigma_c t/\epsilon_r \epsilon_0}$, a result that does not depend on A or d .

§4.3 Transients in RC and RL Circuits

1. An uncharged capacitor C is charged through a resistor R by a source whose electromotive force is first set to $\mathcal{E}/2$ until the capacitor is fully charged, and then raised to $\mathcal{E}$. Using the energy balance of the text, show that the total heat produced in the resistor is $\frac{1}{4}C\mathcal{E}^2$, and that if the electromotive force is raised in N equal steps, with the capacitor fully charged at each step, the heat produced is $C\mathcal{E}^2/2N$.
2. A capacitor C with charge Q_0 is connected at $t = 0$ through a resistor R to an identical capacitor that is uncharged. Find the charges on the two capacitors as functions of time, and show that the time constant is $RC/2$. Show that the energy stored in the two capacitors at the end is half the initial stored energy, as

follows from the fall in stored energy that you found in Problem 6 of Section 2.1 when the two capacitances are equal, and that the heat produced in the resistor accounts for the difference.

3. For the circuit of Figure 40, show that the heat produced in the resistor while the current decays from I_1 to zero is $\frac{1}{2}LI_1^2$. Show also, from the energy equation of the text, that while the current grows from zero to I_1 the work done by the source exceeds the heat produced by $\frac{1}{2}LI_1^2$.
4. A source of electromotive force $\mathcal{E}$ and negligible internal resistance is connected through a switch and a resistor R_1 to an ideal inductor L , and a resistor R_2 is connected directly across the terminals of the inductor. The switch has been closed for a long time and is opened at $t = 0$. Find the current through R_2 before and just after the switch is opened, and show that just after it is opened the potential difference across R_2 is $\mathcal{E}R_2/R_1$. Find how this potential difference decays, and explain why a spark can jump across the contacts of a switch that breaks a circuit containing an inductor.
5. From $t = 0$ the electromotive force of a source rises steadily at a constant rate $\dot{\mathcal{E}}$, where the dot denotes the derivative with respect to time, as in Section 1.1 of Vol. 1, so that $\mathcal{E} = \dot{\mathcal{E}}t$, and the source charges an initially uncharged capacitor C through a resistor R . Try a solution of the form $Q = C\dot{\mathcal{E}}(t - \tau) + Q_1e^{-t/\tau}$, with $\tau = RC$ and Q_1 a constant to be determined, and find Q . Show that at times much longer than τ the potential difference across the capacitor is the value the electromotive force had a time τ earlier.
6. While a capacitor charges in the circuit of Figure 38, find the time at which the potential differences across the resistor and the capacitor are equal. Show that the current in the circuit of Figure 40 reaches half its final value at the same multiple of the time constant L/R .

Magnetostatics

5.1 Magnetism at School

Magnetism at school begins with permanent magnets and compasses, and then connects magnetism with electricity through the forces on currents and on moving charges, and through the fields that currents produce. You have seen all of this at school, so this section states the results briefly, in the vector notation of the course. The two sections that follow take the subject further. Section 5.2 works out the forces and torques that a magnetic field exerts on circuits, and derives the fields of currents from the Biot–Savart law. Section 5.3 treats Ampère’s law and writes the laws of magnetostatics in differential form.

Magnets and Field Lines

A bar magnet has two poles, called north and south, and like poles repel while unlike poles attract. A magnet free to turn, such as a compass needle, comes to rest with its north pole pointing roughly towards the geographic north of the Earth, because the Earth itself behaves as a large magnet. The region around a magnet in which such forces act is described by the magnetic field $\mathbf{B}$, and the field can be drawn with field lines. A field line is a curve whose direction at each point is the direction in which the north pole of a small compass would point, and the field is strong where the lines are

close together. Outside a magnet the lines run from the north pole to the south pole, as in the left-hand part of Figure 41, and they never cross, since the field has only one direction at each point.

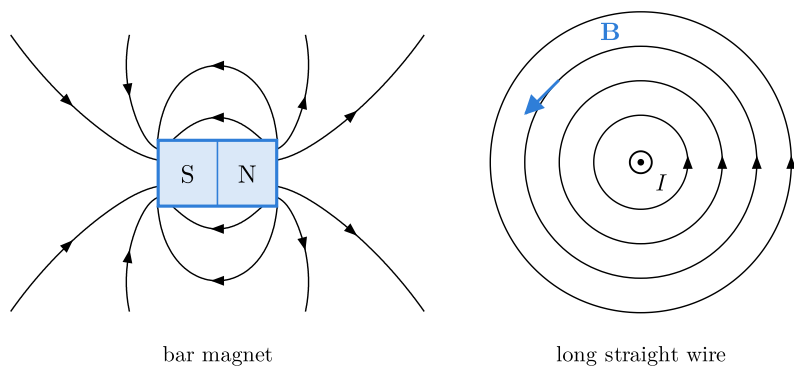


Figure 41. Left: the field lines outside a bar magnet, which run from its north pole N to its south pole S. A short solenoid carrying a current has field lines of the same pattern outside it. Right: the field lines of a long straight wire carrying a current I out of the page, which are circles centred on the wire and run anticlockwise when viewed with the current coming towards the viewer.

Oersted found in 1820 that a current deflects a compass needle, so that a current produces a magnetic field. The field lines of a long straight wire are circles centred on the wire, as in the right-hand part of Figure 41, and their direction is given by the right-hand grip rule: if the thumb of the right hand points along the current, the fingers curl in the direction of the field. A coil of wire carrying a current produces a field whose pattern outside the coil is the same as that of a bar magnet, and a coil wound on an iron core, which strengthens the field a great deal, is called an electromagnet. Electromagnets are used, for example, in electric bells, in relays, which are switches operated by an electromagnet, and in the cranes that lift scrap iron, where the field can be switched off by switching off the current.

The Magnetic Force on a Current and on a Moving Charge

A straight wire of length ℓ carrying a current I in a uniform field of strength B is observed to experience a force of magnitude

$$F = BI\ell \sin \theta, \quad (5.1)$$

where θ is the angle between the wire and the field. The force is perpendicular both to the wire and to the field, and its sense is given by Fleming's left-hand rule. Let $\boldsymbol{\ell}$ be a vector of length ℓ pointing along the wire in the direction of the current. The magnitude and the direction of the force are then both contained in the vector product

$$\mathbf{F} = I\boldsymbol{\ell} \times \mathbf{B}, \quad (5.2)$$

since the magnitude of a vector product is the product of the magnitudes and the sine of the angle between the vectors, and its direction is fixed by the right-hand rule, which gives the same sense as Fleming's rule. The unit of B is the tesla, defined so that a field of one tesla exerts a force of one newton on each metre of a wire that carries one ampere at right angles to the field.

A current is a flow of charge, and the force on the wire is the sum of the forces on the moving charges inside it. A charge q moving with velocity $\mathbf{v}$ in a magnetic field experiences the force $q\mathbf{v} \times \mathbf{B}$, whose magnitude is $qvB \sin \theta$, with θ now the angle between the velocity and the field. If there is an electric field as well, the total force on the charge is

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad (5.3)$$

which is called the Lorentz force. The magnetic part of this force is always perpendicular to the velocity, so it does no work on the charge and cannot change its speed.

Circular Motion in a Uniform Field

Let a particle of charge q and mass m move with speed v at right angles to a uniform field $\mathbf{B}$, as in Figure 42. The magnetic force has the constant magnitude qvB and is always perpendicular to the velocity, so the particle moves on a circle at constant speed, with the magnetic force providing the centripetal force. Setting

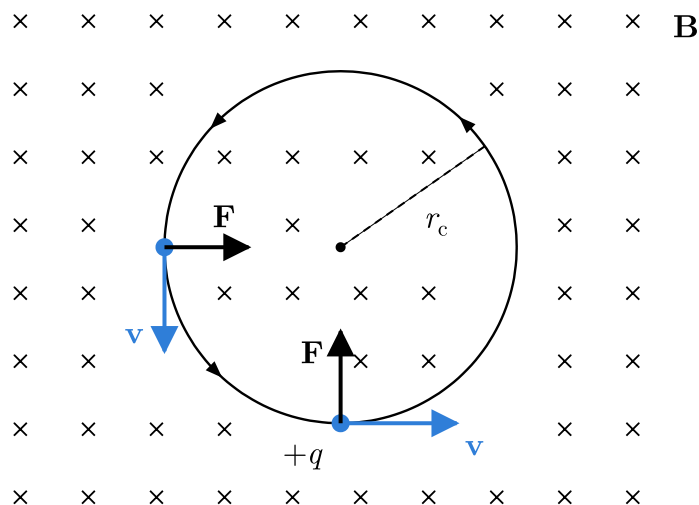


Figure 42. A positive charge q moving at right angles to a uniform field $\mathbf{B}$ that points into the page, shown by the crosses. The magnetic force $\mathbf{F}$ is perpendicular to the velocity $\mathbf{v}$ and points towards the centre of a circle of radius r_c , which the charge goes round anticlockwise.

$qvB = mv^2/r_c$ gives the radius of the circle,

$$r_c = \frac{mv}{qB}. \quad (5.4)$$

The time taken to go once round the circle is $T = 2\pi r_c/v = 2\pi m/qB$, which does not depend on the speed, since a faster charge goes round a proportionally larger circle. The corresponding angular frequency

$$\omega_c = \frac{qB}{m} \quad (5.5)$$

is called the cyclotron frequency, after the cyclotron, an accelerator in which charged particles circle in a magnetic field and are accelerated by an alternating electric field each time they cross a gap, which is possible because the period stays the same as they speed up. The same radius formula is the basis of the mass spectrometer, in which ions of the same charge and speed are bent into circles whose radii are proportional to their masses. The speed is often selected first by passing the ions through crossed electric and magnetic fields, in which only ions with $qE = qvB$, that is with

$v = E/B$, go straight through.

The Fields of a Wire and a Solenoid

The field of a long straight wire carrying a current I has the magnitude

$$B = \frac{\mu_0 I}{2\pi s} \quad (5.6)$$

at a distance s from the wire, where μ_0 is called the magnetic constant. A long solenoid, a coil of N turns wound evenly on a cylinder of length ℓ much greater than its radius, has $n = N/\ell$ turns per unit length, and when it carries a current I the field inside it is directed along the axis and has the magnitude

$$B = \mu_0 n I, \quad (5.7)$$

the same at every point inside, away from the ends. The field outside a long solenoid is very small, and its direction inside is given by the right-hand grip rule applied to the turns: if the fingers of the right hand curl in the direction of the current, the thumb points along the field.

Magnetic Flux

The magnetic flux through a flat surface of area A in a uniform field is

$$\Phi_B = BA \cos \theta, \quad (5.8)$$

where θ is the angle between the field and the normal to the surface. The flux is largest when the field crosses the surface at right angles, and zero when the field lies in the plane of the surface. Its unit is the weber, which is defined as one tesla multiplied by one square metre, and since the flux through a unit area perpendicular to the field is B , the field is sometimes called the magnetic flux density.

5.2 Magnetic Forces and the Biot–Savart Law

This section first works out the forces and torques that a given magnetic field exerts on currents and on moving charges, then derives the fields that currents produce, and finally uses those fields to find the force between two currents. The first half leads to the

magnetic dipole moment of a current loop, and the second rests on the Biot–Savart law.

The Force on a Current Element

In Section 4.2 the current density in a metal was written as $\mathbf{J} = -en_e\mathbf{v}_d$, where n_e is the number density of the conduction electrons and $\mathbf{v}_d$ their drift velocity. A small volume $d\mathcal{V}$ of the metal contains $n_e d\mathcal{V}$ conduction electrons, and the magnetic force on each is $-e\mathbf{v}_d \times \mathbf{B}$, so the total magnetic force on them is

$$d\mathbf{F} = -en_e d\mathcal{V} \mathbf{v}_d \times \mathbf{B} = \mathbf{J} \times \mathbf{B} d\mathcal{V}. \quad (5.9)$$

The electrons cannot leave the metal, so this force acts on the metal. For a thin wire of cross-sectional area A , a length $d\ell$ of the wire has volume $A d\ell$, and JA is the current I . Let $d\boldsymbol{\ell}$ be the vector of length $d\ell$ pointing along the wire in the direction of the current, which is the direction of $\mathbf{J}$. Then $\mathbf{J} d\mathcal{V} = I d\boldsymbol{\ell}$, and the force on this piece of the wire is

$$d\mathbf{F} = I d\boldsymbol{\ell} \times \mathbf{B}. \quad (5.10)$$

The product $I d\boldsymbol{\ell}$ is called a current element. For a straight wire in a uniform field, adding up the elements gives the school result $I\boldsymbol{\ell} \times \mathbf{B}$ of Section 5.1, and for a whole circuit C in any field the total force is the line integral

$$\mathbf{F} = I \oint_C d\boldsymbol{\ell} \times \mathbf{B}, \quad (5.11)$$

in which the circle on the integral sign shows that the path is closed.

In a uniform field, $\mathbf{B}$ can be taken outside the integral, which gives $\mathbf{F} = I \left(\oint_C d\boldsymbol{\ell} \right) \times \mathbf{B}$. The integral $\oint_C d\boldsymbol{\ell}$ is the sum of the small displacements taken in going once round the circuit, and since the path ends where it began, this sum is zero. The net force on a closed circuit carrying a steady current in a uniform field is therefore zero, for a circuit of any shape. You will see in Problem 1 that the same reasoning gives the force on a curved wire between two fixed points.

The Torque on a Current Loop

A zero net force does not mean a zero torque. The coil of an electric motor, for example, experiences a magnetic torque that turns it,

although the net magnetic force on it is zero. Let us work out the torque on a rectangular loop with sides of lengths ℓ_1 and ℓ_2 , carrying a current I in a uniform field $\mathbf{B}$ and free to turn about an axis through its centre that is parallel to the sides of length ℓ_1 and perpendicular to $\mathbf{B}$. Figure 43 shows the loop edge-on, looking along this axis, so that the sides of length ℓ_1 , labelled 1 and 2, are perpendicular to the page and the other two sides lie in planes parallel to it.

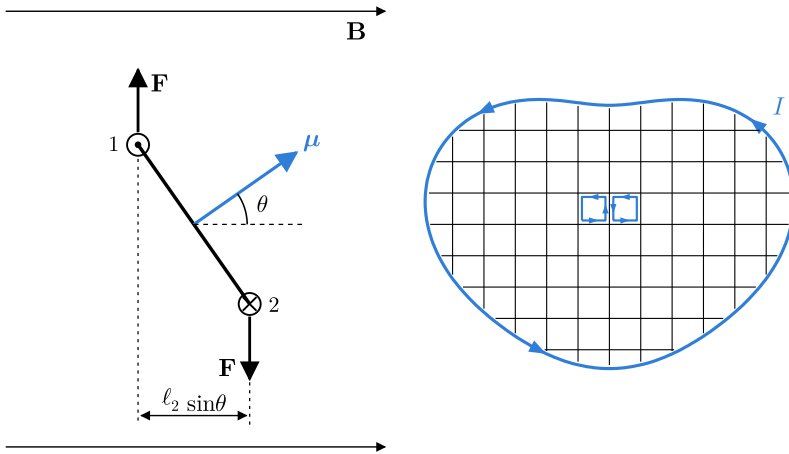


Figure 43. Left: a rectangular loop in a uniform field $\mathbf{B}$, seen edge-on. Side 1 carries the current out of the page and side 2 carries it into the page. The forces $\mathbf{F}$ on these sides are equal and opposite, and their lines of action are a distance $\ell_2 \sin \theta$ apart, where θ is the angle between the field and the normal to the loop, drawn as the arrow labelled $\boldsymbol{\mu}$. Right: a flat loop of any shape divided into small rectangles, each carrying the current I round the same way. On a side shared by two neighbouring rectangles the two currents flow in opposite directions.

Sides 1 and 2 are perpendicular to the field, so each experiences a force of magnitude $I\ell_1 B$, perpendicular to the field, and the vector product shows that the force on side 1 points up the page and the force on side 2 down it. The two forces cancel, but their lines of action are separated by the distance $\ell_2 \sin \theta$ shown in Figure 43, where θ is the angle between the field and the normal to the loop,

and so they exert a torque of magnitude

$$\Gamma = I\ell_1\ell_2 B \sin \theta = IAB \sin \theta, \quad (5.12)$$

where $A = \ell_1\ell_2$ is the area of the loop. The forces on the other two sides are equal and opposite, perpendicular to the page, and both act along the line through the centre of the loop parallel to the axis, so they exert no torque about any point and only stretch or compress the loop. Since the net force on the loop is zero, the torque is the same about every point, as shown in Section 3.2 of Vol. 1 of the Fermi Notes on Physics.

Let $\hat{\mathbf{n}}$ be the unit normal to the loop whose direction is given by the right-hand rule: if the fingers of the right hand curl round the loop in the direction of the current, the thumb points along $\hat{\mathbf{n}}$. The vector $\mathbf{a} = A\hat{\mathbf{n}}$ is called the vector area of the loop, and the vector

$$\boldsymbol{\mu} = I\mathbf{a} \quad (5.13)$$

is called the magnetic dipole moment of the loop, or simply its magnetic moment. Its unit is the ampere square metre. With this definition the torque on the loop is

$$\boldsymbol{\Gamma} = \boldsymbol{\mu} \times \mathbf{B}, \quad (5.14)$$

since $|\boldsymbol{\mu} \times \mathbf{B}| = IAB \sin \theta$, and the right-hand rule for the vector product gives a torque into the page in Figure 43, which turns the loop clockwise and so turns $\boldsymbol{\mu}$ towards $\mathbf{B}$. The torque vanishes when the moment is parallel or antiparallel to the field. A coil of N turns has N times the moment of a single turn, $\boldsymbol{\mu} = N I \mathbf{a}$. The formula has the same form as the torque $\mathbf{p} \times \mathbf{E}$ on an electric dipole found in Section 3.1, with $\boldsymbol{\mu}$ in place of $\mathbf{p}$ and $\mathbf{B}$ in place of $\mathbf{E}$.

The result holds for a flat loop of any shape. To see this, we cover the loop with a grid of small rectangles, as in the right-hand part of Figure 43, and imagine the current I flowing round each rectangle in the same sense as round the loop. On every side that two rectangles share, the two currents are equal and opposite and their forces cancel, and the currents left on the outer edges of the grid make up the loop itself, provided the rectangles are small enough to follow its outline. The torque on the loop is therefore the sum of the torques on the rectangles, $\sum I \mathbf{da} \times \mathbf{B}$, where $\mathbf{da}$ is the vector area of one rectangle. All the $\mathbf{da}$ point the same way, so their sum is the vector area $\mathbf{a}$ of the loop, and $\boldsymbol{\Gamma} = I\mathbf{a} \times \mathbf{B}$ again. A loop that does not lie

in a plane can be treated in the same way by covering any surface bounded by the loop with small patches, and its vector area is then the sum of the vector areas of the patches.

The Energy of a Dipole

To turn a loop from an angle θ to $\theta + d\theta$ against the magnetic torque, keeping its current fixed, we must do work $\mu B \sin \theta d\theta$ on it. This work is stored as potential energy, and choosing the energy to be zero when the moment is perpendicular to the field gives

$$U(\theta) = \int_{\pi/2}^{\theta} \mu B \sin \theta' d\theta' = -\mu B \cos \theta, \quad (5.15)$$

which can be rewritten as

$$U = -\boldsymbol{\mu} \cdot \mathbf{B}. \quad (5.16)$$

The energy is lowest when the moment points along the field, which is therefore the orientation of stable equilibrium, and highest when it points against the field, where the equilibrium is unstable. This is the energy that was quoted in Section 4.2 of Vol. 3, together with the moment of a current loop, to explain the Stern–Gerlach experiment and to relate the magnetic moment of an electron to its angular momentum.

In a field that varies from place to place, the energy of a dipole of fixed moment depends on where the dipole is. If U is treated as a potential energy, the force on the dipole is minus the gradient of U , as for any conservative force in Section 2.3 of Vol. 1, and therefore

$$\mathbf{F} = \nabla (\boldsymbol{\mu} \cdot \mathbf{B}), \quad (5.17)$$

where the gradient acts on $\mathbf{B}$ alone, with $\boldsymbol{\mu}$ held fixed. For a moment along z the z -component of this force is $F_z = \mu \partial B_z / \partial z$, which is the force used in Section 4.2 of Vol. 3. A dipole pointing along the field is pulled towards the region where the field is stronger, and a uniform field exerts no force on it, as we found above.

The force can also be seen directly from the forces on the elements of the loop. We take a small circular loop of radius b , with its moment along the axis of a field whose lines converge in the direction of the moment, as in Figure 44. At the rim the field has a component B_s along the outward radius, which is negative here, since the lines

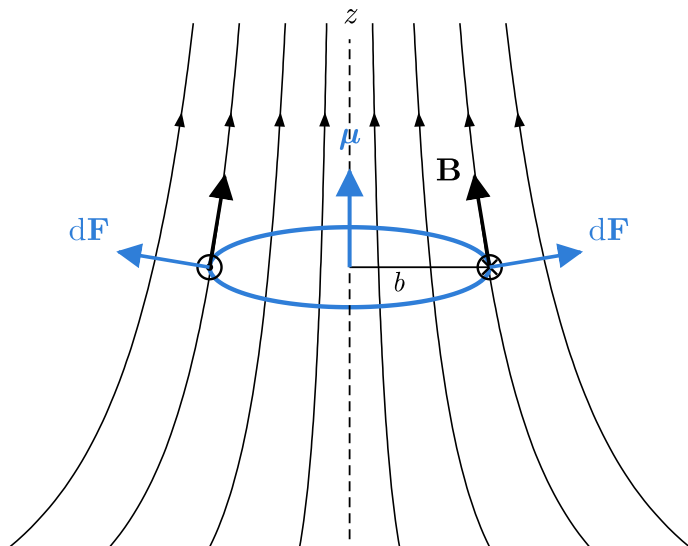


Figure 44. A small loop of radius b with its moment μ along the axis of a field whose lines converge upwards. At the rim the field $\mathbf{B}$ has an inward component, so the force $d\mathbf{F}$ on each element of the loop, which is perpendicular to the field, has an upward component. The current flows into the page on the right of the loop and out of it on the left.

converge. The force $I d\boldsymbol{\ell} \times \mathbf{B}$ on each element is perpendicular to the field, so it has an upward component, equal to $-IB_s d\ell$. Adding up the elements round the rim gives

$$F_z = -2\pi b I B_s, \quad (5.18)$$

which is positive, so the loop is pulled into the stronger field. You will check in Problem 3 that this agrees with $\mu \partial B_z / \partial z$ for a particular field of this kind.

Helical Motion of a Charged Particle

In Section 5.1 a charge moving at right angles to a uniform field went round a circle. A charge whose velocity has a component along the field moves on a helix, as we now show. Let the field be $B\hat{\mathbf{z}}$ and let the charge q have mass m . Newton's second law with the force

$q\mathbf{v} \times \mathbf{B}$ gives, component by component,

$$\frac{dv_x}{dt} = \omega_c v_y, \quad \frac{dv_y}{dt} = -\omega_c v_x, \quad \frac{dv_z}{dt} = 0, \quad (5.19)$$

with $\omega_c = qB/m$ the cyclotron frequency of Section 5.1. The component v_z along the field is therefore constant, and we call it $v_{\parallel}$. Differentiating the first equation with respect to time and substituting the second gives $d^2v_x/dt^2 = -\omega_c^2 v_x$, which is the equation of simple harmonic motion. If we choose the zero of time when the velocity perpendicular to the field points along x , and let $v_{\perp}$ be the magnitude of that perpendicular velocity, the solution is

$$v_x = v_{\perp} \cos \omega_c t, \quad v_y = -v_{\perp} \sin \omega_c t. \quad (5.20)$$

Integrating once more, with the origin placed on the axis of the motion, gives

$$\begin{aligned} x &= r_c \sin \omega_c t, & y &= r_c \cos \omega_c t, \\ z &= v_{\parallel} t, & r_c &= \frac{mv_{\perp}}{qB}. \end{aligned} \quad (5.21)$$

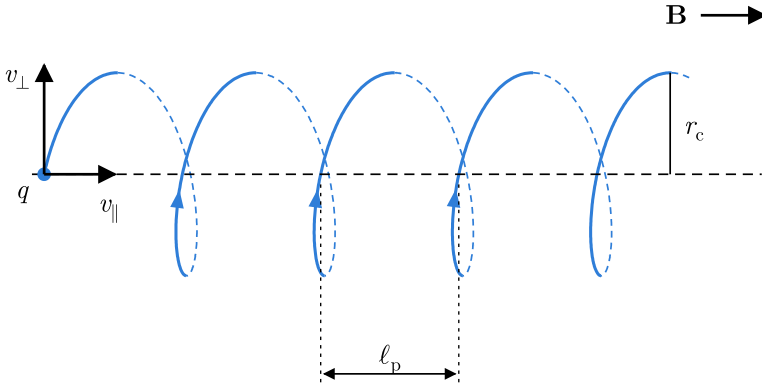


Figure 45. The helical path of a positive charge q in a uniform field $\mathbf{B}$ that points to the right. At the start the velocity has a component $v_{\parallel}$ along the field and a component $v_{\perp}$ across it. The path winds round a cylinder of radius r_c and advances a distance ℓ_p , the pitch, in each turn. The parts of the path behind the axis are dashed.

The projection of the motion onto the xy -plane is the circle of Section 5.1, now with the radius set by $v_{\perp}$, while the charge drifts

steadily along the field. The path is a helix, as in Figure 45, and in one period $2\pi/\omega_c$ the charge advances a distance

$$\ell_p = \frac{2\pi m v_{\parallel}}{qB}, \quad (5.22)$$

called the pitch of the helix. A positive charge goes round clockwise when viewed from a point towards which the field points, and a negative charge goes round anticlockwise. The helix explains, for example, why charged particles arriving from the Sun tend to spiral along the field lines of the Earth towards the polar regions, where they produce the aurorae.

The Biot–Savart Law

We now turn from the force that a field exerts on currents to the field that currents produce. In 1820 Biot and Savart measured the field of long wires carrying steady currents, and it is found by experiment that the field of any steady circuit is reproduced by adding up contributions from its current elements. The current element $I d\boldsymbol{\ell}'$ at the source point $\mathbf{r}'$ contributes to the field at the field point $\mathbf{r}$ the amount

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I d\boldsymbol{\ell}' \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3}, \quad (5.23)$$

and the field of a circuit C is the sum of these contributions,

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \oint_C \frac{d\boldsymbol{\ell}' \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3}. \quad (5.24)$$

This is called the Biot–Savart law. For currents spread through a volume, $I d\boldsymbol{\ell}'$ is replaced by $\mathbf{J}(\mathbf{r}') dV'$ and the line integral by a volume integral, and for a surface current $\mathbf{K}$ it is replaced by $\mathbf{K}(\mathbf{r}') da'$.

Like Coulomb's law of Section 1.2, the law gives a contribution that falls off as the inverse square of the distance, and the contributions of different elements add as vectors. Its source, however, is a vector, and the field of an element is perpendicular both to the element and to the line joining the element to the field point, so it circles round the line of the element. The law applies to steady currents, which do not change in time. The subject of such currents and their fields is called magnetostatics. By the continuity equation of

Section 4.2, a steady current has $\nabla \cdot \mathbf{J} = -\partial\rho/\partial t = 0$, so no charge builds up anywhere and the current flows round closed circuits. An isolated current element therefore cannot carry a steady current, and experiment tests only the integral round a closed circuit.

The Field of a Straight Segment

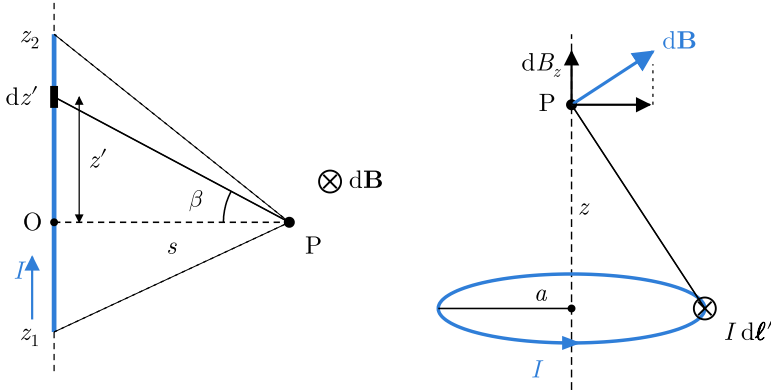


Figure 46. Left: a straight segment of wire from z_1 to z_2 carrying a current I , and a field point P at perpendicular distance s from the foot O of the perpendicular. The element dz' at height z' is seen from P at the angle β to PO , and its field $d\mathbf{B}$ at P points into the page. Right: a circular loop of radius a carrying a current I , and a point P on its axis at height z . The field $d\mathbf{B}$ of the element $I d\ell'$ at the right-hand end of the loop, whose current flows into the page, is perpendicular to the line joining the element to P , and dB_z is its component along the axis.

Let us look first at a straight segment of wire along the z -axis, from z_1 to z_2 , carrying a current I in the $+z$ direction, and a field point P at a perpendicular distance s from the axis, as in the left-hand part of Figure 46. We set up coordinates such that the foot O of the perpendicular from P is the origin. Let $\hat{\mathbf{s}}$ be the unit vector from O towards P , so that $\mathbf{r} = s\hat{\mathbf{s}}$, and let $\hat{\phi} = \hat{\mathbf{z}} \times \hat{\mathbf{s}}$ be the azimuthal unit vector of cylindrical coordinates about the wire, which are treated, with their unit vectors, in Section 3.4 of Vol. 4 of the Fermi Notes on Physics. The element at height z' has $d\ell' = dz' \hat{\mathbf{z}}$

and $\mathbf{r} - \mathbf{r}' = s\hat{\mathbf{s}} - z'\hat{\mathbf{z}}$, so

$$d\boldsymbol{\ell}' \times (\mathbf{r} - \mathbf{r}') = s dz' \hat{\boldsymbol{\phi}}, \quad |\mathbf{r} - \mathbf{r}'| = (s^2 + z'^2)^{1/2}, \quad (5.25)$$

since $\hat{\mathbf{z}} \times \hat{\mathbf{z}} = 0$. Every element gives a field along $\hat{\boldsymbol{\phi}}$, and

$$\mathbf{B} = \frac{\mu_0 I s}{4\pi} \hat{\boldsymbol{\phi}} \int_{z_1}^{z_2} \frac{dz'}{(s^2 + z'^2)^{3/2}}. \quad (5.26)$$

The integral is of the same kind as the one worked out for the charged segment in Section 1.2. The substitution $z' = s \tan \beta$, where β is the angle at P shown in Figure 46, gives $dz' = s \sec^2 \beta d\beta$ and $s^2 + z'^2 = s^2 \sec^2 \beta$, so the integrand becomes $\cos \beta d\beta / s^2$, and

$$\mathbf{B} = \frac{\mu_0 I}{4\pi s} (\sin \beta_2 - \sin \beta_1) \hat{\boldsymbol{\phi}}, \quad (5.27)$$

where β_1 and β_2 are the angles at which the two ends are seen, with $\sin \beta_i = z_i / (s^2 + z_i^2)^{1/2}$. For an infinite wire $\beta_1 \rightarrow -\pi/2$ and $\beta_2 \rightarrow \pi/2$, and the field becomes $\mu_0 I / 2\pi s$ along $\hat{\boldsymbol{\phi}}$, which is the school result of Section 5.1 with its direction given by the right-hand grip rule. An isolated segment cannot carry a steady current, but the result can be applied to each side of a circuit made of straight pieces, and you will use it in Problem 2 to find the field at the centre of a square loop.

The Field on the Axis of a Circular Loop

We turn next to a circular loop of radius a in the xy -plane, centred on the origin and carrying a current I anticlockwise when viewed from above, and a field point P on the axis at height z , as in the right-hand part of Figure 46. Every element of the loop is perpendicular to the line joining it to P, and every such line has the length $(a^2 + z^2)^{1/2}$, so each element $d\boldsymbol{\ell}'$ contributes a field of the same magnitude,

$$|d\mathbf{B}| = \frac{\mu_0}{4\pi} \frac{I d\ell'}{a^2 + z^2}. \quad (5.28)$$

The field $d\mathbf{B}$ is perpendicular to the line from the element to P and lies in the plane containing that line and the axis. The angle between $d\mathbf{B}$ and the axis is therefore equal to the angle between the line and the plane of the loop, whose cosine is $a / (a^2 + z^2)^{1/2}$. The components of $d\mathbf{B}$ perpendicular to the axis cancel in pairs,

since the elements at opposite ends of a diameter give equal and opposite components, and only the components along the axis add. The elements have a total length $2\pi a$, so

$$B_z = \frac{\mu_0 I}{4\pi} \frac{2\pi a}{a^2 + z^2} \frac{a}{(a^2 + z^2)^{1/2}} = \frac{\mu_0 I a^2}{2(a^2 + z^2)^{3/2}}. \quad (5.29)$$

At the centre of the loop this is $\mu_0 I/2a$. The field points along the moment $\boldsymbol{\mu}$ of the loop on both sides of it, and it is largest at the centre, as Figure 47 shows.

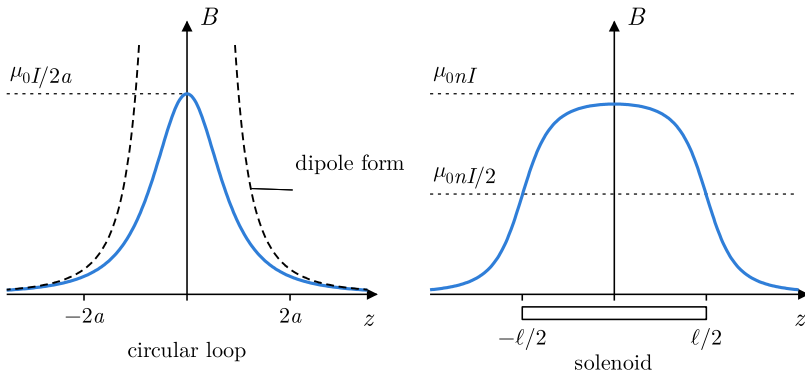


Figure 47. Left: the field on the axis of a circular loop of radius a , which is largest at the centre, where it equals $\mu_0 I/2a$. The dashed curves show the dipole form $\mu_0 \mu / 2\pi |z|^3$, which agrees with the exact field far from the loop. Right: the field on the axis of a solenoid whose length ℓ is several times its radius. The field is close to $\mu_0 n I$ inside, away from the ends, and close to $\mu_0 n I/2$ at each end.

Far from the loop, where $|z| \gg a$, the term a^2 in the denominator can be neglected, and with $\mu = I\pi a^2$ the field becomes

$$B_z \approx \frac{\mu_0 I a^2}{2|z|^3} = \frac{\mu_0 \mu}{2\pi |z|^3}. \quad (5.30)$$

The field far along the axis depends on the loop only through its magnetic moment, and a loop seen from a distance large compared with its size is therefore called a magnetic dipole. In Section 3.1 the field of an electric dipole on its axis was found to be $p/2\pi\epsilon_0|z|^3$ along $\mathbf{p}$, and the magnetic field of the loop is obtained from this

expression by replacing p/ϵ_0 with $\mu_0\mu$. The fields of the two dipoles have the same form in every direction at large distances, but the calculation for points off the axis is not treated in this course.

The Field on the Axis of a Solenoid

The result for the loop also gives the field on the axis of a solenoid of radius a and length ℓ , with n turns per unit length carrying a current I , lying along the z -axis from $-\ell/2$ to $\ell/2$. When the turns are closely wound, the slice of the solenoid between z' and $z' + dz'$ is a loop carrying the current $nI dz'$. By the formula for the field on the axis of a loop, with I replaced by $nI dz'$ and z by $z - z'$, this slice contributes to the field at the point z on the axis the amount

$$dB_z = \frac{\mu_0 n I a^2}{2} \frac{dz'}{[a^2 + (z' - z)^2]^{3/2}}. \quad (5.31)$$

Let $\xi = z' - z$ and substitute $\xi = a \tan \beta$, as for the segment. The integrand becomes $\cos \beta d\beta/a^2$, whose integral is $\sin \beta/a^2 = \xi/[a^2(a^2 + \xi^2)^{1/2}]$, and inserting the limits $\xi = -\ell/2 - z$ and $\xi = \ell/2 - z$ gives

$$B_z = \frac{\mu_0 n I}{2} \left[\frac{\ell/2 - z}{\sqrt{a^2 + (\ell/2 - z)^2}} + \frac{\ell/2 + z}{\sqrt{a^2 + (\ell/2 + z)^2}} \right]. \quad (5.32)$$

This field is plotted in the right-hand part of Figure 47. For a long solenoid, with $\ell \gg a$, each fraction in the brackets is close to one at points well inside, and $B_z \approx \mu_0 n I$, the school result of Section 5.1. At either end of a long solenoid one fraction is zero and the other is close to one, so the field there is half as large. This is expected, since two long solenoids placed end to end make a single longer one, and by superposition each contributes half of the field at the join.

The Force between Two Parallel Wires

Finally, let us combine the force that a magnetic field exerts on a current with the field that a current produces. Let two long parallel wires, a distance d apart, carry currents I_1 and I_2 in the same direction, as in Figure 48. The field of the first wire at the second has the magnitude $B_1 = \mu_0 I_1 / 2\pi d$ and is perpendicular both to the second wire and to the line joining the wires. The force

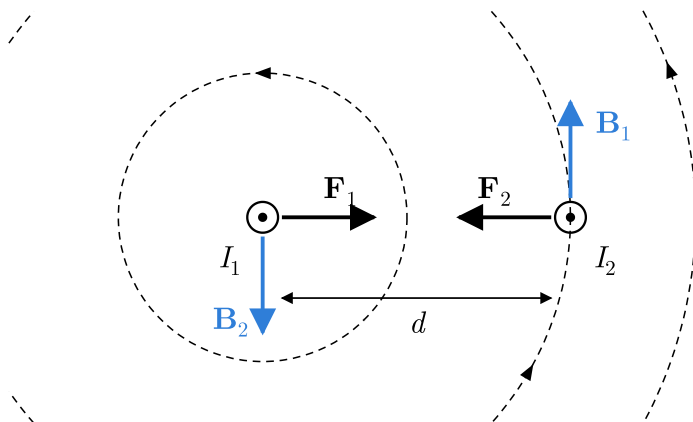


Figure 48. A pair of long parallel wires carrying currents I_1 and I_2 out of the page, a distance d apart, seen end-on. The dashed circles are field lines of the first wire. The field $\mathbf{B}_1$ of the first wire at the second, and the field $\mathbf{B}_2$ of the second wire at the first, give forces $\mathbf{F}_2$ and $\mathbf{F}_1$ that pull the wires together.

on a length ℓ of the second wire is $I_2\ell \times \mathbf{B}_1$, whose magnitude per unit length is

$$\frac{F}{\ell} = \frac{\mu_0 I_1 I_2}{2\pi d}, \quad (5.33)$$

and the vector product shows that it points towards the first wire. The same calculation with the wires exchanged gives an equal and opposite force on the first wire, so the forces obey Newton's third law. Parallel currents attract each other, and antiparallel currents repel. This force can be measured precisely with a current balance, in which the magnetic force between a fixed coil and a movable coil carrying the same current is balanced against a known weight, and until 2019 the force between two long parallel wires was used to define the ampere.

5.3 Ampère's Law and the Curl of the Magnetic Field

In Chapter 1 the electrostatic field was described in two ways, by Coulomb's law, which gives the field of each charge, and by Gauss's

law, which relates the flux of the field through a closed surface to the charge inside. We now do the same for the magnetic field of steady currents, showing first that the flux of $\mathbf{B}$ through any closed surface is zero, then introducing Ampère's law, which relates the circulation of $\mathbf{B}$ round a closed loop to the current through it and gives the field of a current with enough symmetry, and finally writing both laws in differential form with the help of the curl and Stokes' theorem. The curl and Stokes' theorem are treated in detail in Sections 3.1 and 3.3 of Vol. 4, and the account of them given here is brief.

The Flux of the Magnetic Field

The field lines of Figure 41 have no beginning and no end, since the lines of a magnet continue through the magnet itself and the lines of a wire are circles, whereas the lines of the electric field begin and end on charges. A magnet cut in two gives two magnets, and searches for an isolated magnetic pole, which is called a magnetic monopole, have so far found none.

The Biot–Savart law shows that the flux of $\mathbf{B}$ through a closed surface is zero. To see this, we consider the field of a single current element, and choose the z -axis along the element, with the origin at the element. By the calculation for the straight segment in Section 5.2, the field of the element is along $\hat{\phi}$, and its magnitude $B_\phi(s, z)$ depends only on the cylindrical coordinates s and z . The field has only the component B_ϕ , which does not depend on ϕ , so by the formula for the divergence in cylindrical coordinates worked out in Section 3.4 of Vol. 4, $\nabla \cdot \mathbf{B} = (1/s) \partial B_\phi / \partial \phi = 0$.

The field of any steady circuit is a sum of fields of this kind, each with an axis along its element, and the divergence of a sum is the sum of the divergences, so

$$\nabla \cdot \mathbf{B} = 0 \quad (5.34)$$

at every point away from the current elements. The result holds inside a conductor that carries a current as well, but the proof there is not treated in this course. By the divergence theorem of Section 1.3, the flux of $\mathbf{B}$ out of any closed surface S is then zero,

$$\oint_S \mathbf{B} \cdot d\mathbf{a} = 0, \quad (5.35)$$

which is sometimes called Gauss's law for magnetism. The differential form $\nabla \cdot \mathbf{B} = 0$ is the counterpart of $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ with no magnetic charge density on the right-hand side. It follows that the flux through a loop is the same for every surface bounded by the loop, since two such surfaces together form a closed surface.

Circulation and Ampère's Law

The line integral of a vector field round a closed path C is called the circulation of the field round C . Section 1.4 showed that the circulation of the electrostatic field round any closed path is zero, since its line integral between two points does not depend on the path.

We take first a long straight wire carrying a current I , and a circle of radius s centred on the wire in a plane perpendicular to it. The field has the constant magnitude $\mu_0 I/2\pi s$ on the circle and points along it, so, going round in the sense given by the right-hand grip rule,

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \frac{\mu_0 I}{2\pi s} 2\pi s = \mu_0 I, \quad (5.36)$$

which does not depend on the radius of the circle.

The same result holds for a loop of any shape that goes once round the wire, such as the one in the left-hand part of Figure 49. An element $d\mathbf{l}$ of the loop has a component along the radius from the wire, an azimuthal component of length $s d\phi$, and possibly a component along the wire. The field points along $\hat{\phi}$, so only the azimuthal component contributes, and

$$\mathbf{B} \cdot d\mathbf{l} = \frac{\mu_0 I}{2\pi s} s d\phi = \frac{\mu_0 I}{2\pi} d\phi. \quad (5.37)$$

The distance s has cancelled, and going once round the wire increases ϕ by 2π , so the circulation is again $\mu_0 I$. For a loop that does not go round the wire, as in the right-hand part of Figure 49, the angle ϕ increases along one part of the loop, from Q_2 to Q_1 , and decreases by the same amount along the other part, from Q_1 back to Q_2 , so the circulation is zero. For several long parallel wires the fields add, and the circulation is μ_0 times the total current through the loop, each current counted with a sign given by its direction.

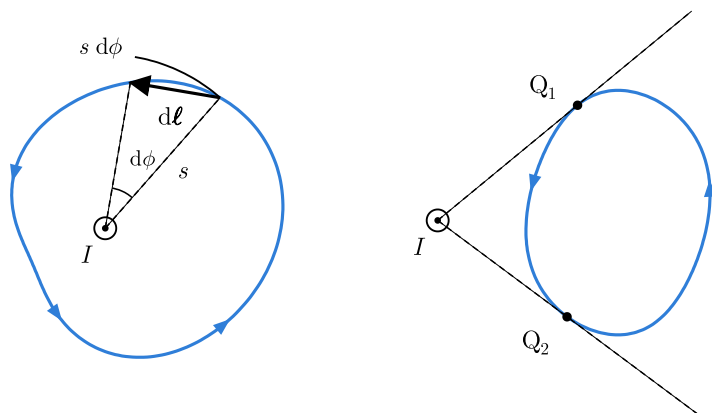


Figure 49. Loops round a long straight wire that carries a current I out of the page, seen end-on. Left: a loop that encloses the wire. The element $d\ell$ at distance s subtends the angle $d\phi$ at the wire, and only its azimuthal part, of length $s d\phi$, contributes to $\mathbf{B} \cdot d\ell$. Right: a loop that does not enclose the wire, which lies between the lines from the wire through its points Q_1 and Q_2 .

These results are examples of Ampère's law, which states that for any steady currents and any closed loop C ,

$$\oint_C \mathbf{B} \cdot d\ell = \mu_0 I_{\text{enc}}, \quad I_{\text{enc}} = \int_S \mathbf{J} \cdot d\mathbf{a}, \quad (5.38)$$

where S is any surface bounded by C , and the direction of $d\mathbf{a}$ is related to the sense in which C is traversed by the right-hand rule. The current I_{enc} is called the current enclosed by the loop. It is the same for every surface bounded by C , because a steady current has $\nabla \cdot \mathbf{J} = 0$, as noted in Section 5.2, so by the divergence theorem the flux of $\mathbf{J}$ out of the closed surface formed by two such surfaces is zero. We have derived the law for long straight wires, and the general derivation from the Biot–Savart law, for any distribution of steady currents, is not treated in this course.

Ampère's law gives the field when the symmetry of the currents fixes the direction of $\mathbf{B}$ and makes its magnitude constant along a suitable loop, so that B can be taken outside the integral. Such a loop is called an Amperian loop, and it plays the part that the

Gaussian surface played for Gauss's law in Section 1.3. The arrangements of current in the subsections that follow are examples in which this is the case, and we take them in turn.

The Long Solenoid

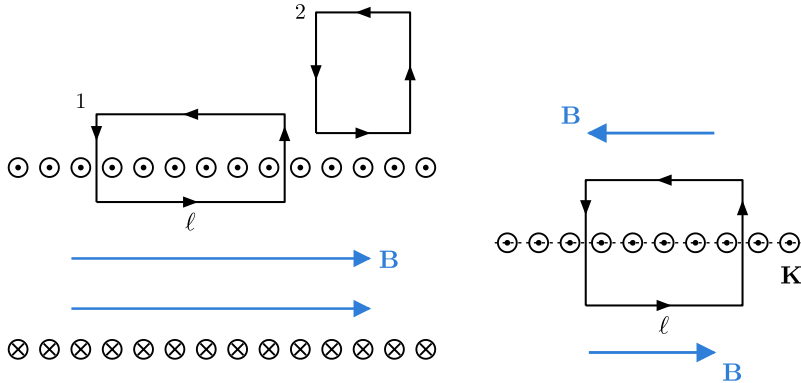


Figure 50. Left: a long solenoid cut along its axis, with the current coming out of the page in the upper row of turns and going into it in the lower row. Rectangle 1 straddles the upper row, and rectangle 2 lies wholly outside. Right: a plane sheet of surface current $\mathbf{K}$ coming out of the page, seen edge-on, with a rectangle that straddles the sheet. The field $\mathbf{B}$ points in opposite directions on the two sides. Each rectangle is traversed in the sense shown by the arrows.

As a first application, consider a long solenoid with n turns per unit length carrying a current I . When the turns are closely wound, the current is equivalent to a surface current of magnitude $K = nI$ flowing round the cylinder. By symmetry the field depends only on the distance s from the axis. It has no radial component B_s , since that would give a flux $2\pi s\ell B_s$ through the curved surface of a closed coaxial cylinder of radius s and length ℓ , while the fluxes through its two flat ends cancel, because the field depends only on s and so is the same on both ends, and the total flux out of the closed cylinder must be zero. It has no azimuthal component either, since by symmetry B_ϕ is constant on a circle coaxial with the solenoid, and Ampère's law then gives $2\pi s B_\phi = 0$, because the current flows round the cylinder and never crosses the flat disc bounded by the

circle. The field is therefore along the axis.

Rectangle 2 in Figure 50, with its long sides parallel to the axis at distances s_1 and s_2 from it, lies wholly outside the solenoid and encloses no current. Its short sides are perpendicular to the field and contribute nothing, so Ampère's law gives $[B_z(s_1) - B_z(s_2)]\ell = 0$, where ℓ is the length of the long sides. The field outside is therefore the same at all distances from the axis, and since it must vanish far from the solenoid, it is zero everywhere outside. Rectangle 1 has one long side inside and one outside, and it encloses $n\ell$ turns, each carrying the current I out of the page, so

$$B\ell = \mu_0 n\ell I, \quad B = \mu_0 nI. \quad (5.39)$$

The inner side may be at any distance from the axis, so the field is uniform inside, with the value found on the axis in Section 5.2.

The Toroid

Ampère's law applies in the same way to a toroid, a coil of N turns wound evenly on a ring whose axis is vertical, as in Figure 51. You will show in Problem 3 that the field of such a winding has no component along the radius from the axis and no component along the axis, so its lines are circles about the axis, and its magnitude depends only on s and on the height. A circle of radius s about the axis is then an Amperian loop. Circle 1, in the central hole, encloses no current, so the field there is zero. Circle 2, inside the ring, is crossed by the N turns on the inner edge, each carrying I upwards, so

$$2\pi s B = \mu_0 N I, \quad B = \frac{\mu_0 N I}{2\pi s} \quad (a < s < b). \quad (5.40)$$

Circle 3, outside the ring, is crossed by the N turns going up on the inner edge and the N turns going down on the outer edge, so the enclosed current is zero and the field outside is zero as well. The field of an ideal toroid is therefore confined to the ring. For a thin ring the number of turns per unit length is $n = N/2\pi s$, and the field is $\mu_0 nI$, as for a long solenoid bent into a circle.

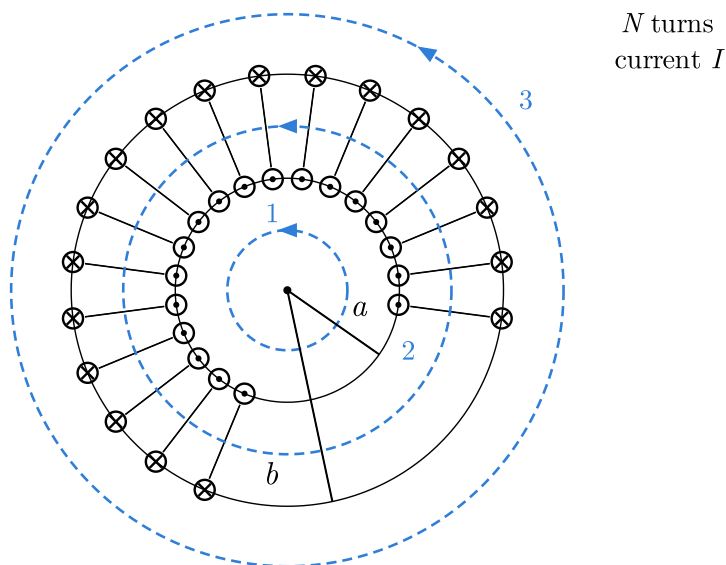


Figure 51. A toroidal winding of N turns carrying a current I , seen from above, with the current coming up through the page on the inner edge of the ring, of radius a , and going down on the outer edge, of radius b . Part of the winding is left out to show the radii. The dashed circles 1, 2, and 3 are Amperian loops in the central hole, inside the ring, and outside it.

A Plane Current Sheet

Finally, we apply Ampère's law to an infinite plane carrying a uniform surface current $\mathbf{K}$, seen edge-on in the right-hand part of Figure 50. The Biot–Savart law gives a field perpendicular to $\mathbf{K}$, since every current element is parallel to $\mathbf{K}$. The field also has no component perpendicular to the sheet, since two strips of current at equal distances on either side of the foot of the perpendicular from the field point give equal and opposite perpendicular components. Since the field of each strip circles round it, the field points in opposite directions on the two sides of the sheet, with the same magnitude B at equal distances. The rectangle in Figure 50 has its long sides, of length ℓ , parallel to the field on either side of the sheet, and it encloses the current $K\ell$. Each long side contributes

$B\ell$ to the circulation, so $2B\ell = \mu_0 K\ell$, and

$$B = \frac{\mu_0 K}{2}. \quad (5.41)$$

The long sides can be at any distance from the sheet, so the field is uniform on each side, and it can be written as $\mathbf{B} = \frac{1}{2}\mu_0 \mathbf{K} \times \hat{\mathbf{n}}$, where $\hat{\mathbf{n}}$ is the unit normal pointing from the sheet towards the field point. This has the same form as the field $\sigma/2\epsilon_0$ of a charged plane in Section 1.3, although the magnetic field is parallel to the sheet and the electric field perpendicular to it.

The Curl

In Section 1.3 the flux out of a small volume, divided by the volume, gave the divergence. In the same way, the circulation round a small loop, divided by its area, gives the component of the curl along the normal to the loop, as shown in Section 3.3 of Vol. 4: for a small flat loop of vector area $d\mathbf{a}$ the circulation is $(\nabla \times \mathbf{B}) \cdot d\mathbf{a}$, where the curl, defined in Section 3.1 of Vol. 4, has the Cartesian components

$$\nabla \times \mathbf{B} = \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z}, \frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x}, \frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right). \quad (5.42)$$

The curl measures how much a field circulates about a point. The velocity $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$ of a body rotating with angular velocity $\boldsymbol{\omega}$, found in Section 1.2 of Vol. 1, has curl $2\boldsymbol{\omega}$, as shown in Section 3.1 of Vol. 4, where the curl of the velocity of a flowing fluid is pictured with a small paddle wheel that turns where the curl is not zero. A field whose lines are circles may nevertheless have zero curl, and you will show in Problem 5 that the field $\mu_0 I/2\pi s$ of a straight wire, a multiple of the field treated in Problem 3 of Section 3.2 of Vol. 4, has zero curl everywhere outside the wire, because the weakening of the field with distance compensates for the curving of its lines.

Stokes' Theorem

Let S be a surface bounded by a closed loop C , with the directions of its surface elements related to the sense of C by the right-hand rule. Stokes' theorem, which is proved in Section 3.3 of Vol. 4, states that

$$\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{a} \quad (5.43)$$

for any vector field whose partial derivatives are continuous on S .

The statement quoted without proof in Section 2.3 of Vol. 1, that a force satisfying $\partial F_x/\partial y = \partial F_y/\partial x$ and the two similar equalities throughout a region with no holes in it is conservative there, is proved in Section 3.2 of Vol. 4, and it also follows from Stokes' theorem, since these equalities say that the curl of the force vanishes. The field of a wire shows why the condition on the region is needed, since its curl is zero everywhere outside the wire, while the circulation round a loop about the wire is $\mu_0 I$.

Ampère's Law in Differential Form

Ampère's law and Stokes' theorem together give, for any loop C and any surface S bounded by it,

$$\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{a} = \mu_0 \int_S \mathbf{J} \cdot d\mathbf{a}. \quad (5.44)$$

This holds for every surface, and in particular for a small flat patch of any orientation about any point, so the components of $\nabla \times \mathbf{B}$ and $\mu_0 \mathbf{J}$ along every direction are equal, and

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}, \quad (5.45)$$

which is Ampère's law in differential form. As a check, you will find in Problem 1 that the field inside a wire of radius a carrying a current I spread uniformly over its cross-section is $\mathbf{B} = (\mu_0 I/2\pi a^2)(-y, x, 0)$, which has the same form as the velocity $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$ above, with $\boldsymbol{\omega} = (\mu_0 I/2\pi a^2)\hat{\mathbf{z}}$, so its curl is $(\mu_0 I/\pi a^2)\hat{\mathbf{z}} = \mu_0 \mathbf{J}$. Outside the wire the curl is zero, as it must be where there is no current.

The Curl of the Electrostatic Field

The electrostatic field of Chapter 1 is the gradient of a potential, $\mathbf{E} = -\nabla V$, as shown in Section 1.4, and since the curl of a gradient is zero, by the identity proved in Section 3.1 of Vol. 4, $\nabla \times \mathbf{E} = 0$. Stokes' theorem gives the same conclusion from the vanishing circulation of the electrostatic field. The laws of electrostatics and

of magnetostatics can now be set side by side,

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0}, & \nabla \times \mathbf{E} &= 0, \\ \nabla \cdot \mathbf{B} &= 0, & \nabla \times \mathbf{B} &= \mu_0 \mathbf{J}.\end{aligned}\tag{5.46}$$

Problems

§5.1 Magnetism at School

1. A particle of mass m and charge q is accelerated from rest through a potential difference V and then enters a uniform field B at right angles to the field. Find the radius of its path and the time it takes to go once round.
2. Ions of charge q pass undeflected through a region with a uniform electric field E and a uniform magnetic field B_1 at right angles to each other and to the ions' velocity. They then enter a region with a uniform field B_2 alone, perpendicular to their velocity, and travel on a semicircle of radius r_c . Find the mass of the ions.
3. A horizontal straight rod of mass m and length ℓ hangs from two light vertical threads in a uniform vertical field B . When a current I flows through the rod, the threads swing out and come to rest at an angle to the vertical. Find the angle.
4. A long solenoid of radius a has n turns per unit length and carries a current I . Find the magnetic flux through a flat circular loop of radius a placed inside the solenoid with its normal at an angle θ to the axis, and through a loop of radius $b > a$ placed round the outside of the solenoid, coaxial with it.

§5.2 Magnetic Forces and the Biot–Savart Law

1. A wire carrying a current I runs along any path from a point P to a point Q in a uniform field $\mathbf{B}$. Show that the force on it is $I\boldsymbol{\ell}_{PQ} \times \mathbf{B}$, where $\boldsymbol{\ell}_{PQ}$ is the displacement from P to Q. Use this result to find the force on a semicircular wire of radius a lying in a plane perpendicular to the field.
2. A square loop of side ℓ carries a current I . Using the field of a straight segment, find the field at the centre of the square, and

compare it with the field at the centre of a circular loop whose circumference is the same as the perimeter of the square.

3. A field has the components $B_z = B_0(1 + \kappa z)$ and $B_s = -\frac{1}{2}B_0\kappa s$ in cylindrical coordinates, with no azimuthal component, where B_0 and κ are constants. A circular loop of radius b carrying a current I lies in the plane $z = z_0$, centred on the axis, with its moment along $+z$. Find the force on the loop by adding up the forces on its elements, as in Figure 44, and show that it equals $\mu \partial B_z / \partial z$.
4. A particle of mass m and charge q moves on a helix in a uniform field $\mathbf{B}$, with the speed $v_\perp$ across the field. Treating the circulating charge as a current loop, find the magnitude of its magnetic moment, and show that the moment points against the field for either sign of q .
5. A pair of circular loops of radius a share an axis and are a distance d apart, and they carry equal currents in the same sense. Show that the derivative of the field on the axis with respect to z vanishes at the point midway between the loops, and that the second derivative also vanishes there when $d = a$. A pair of loops with this spacing is called a pair of Helmholtz coils, and the field near its centre is very nearly uniform.
6. A spinning body has angular momentum $\mathbf{L}_s$ and a magnetic moment $\boldsymbol{\mu} = \eta \mathbf{L}_s$, where the constant η is the gyromagnetic ratio of Section 4.2 of Vol. 3, written γ there, and it is placed in a uniform field $\mathbf{B}$. Using $d\mathbf{L}_s/dt = \boldsymbol{\Gamma}$, show that the magnitude of $\mathbf{L}_s$ and its component along $\mathbf{B}$ are constant, and that $\mathbf{L}_s$ turns about the field with the angular frequency ηB .

§5.3 Ampère's Law and the Curl of the Magnetic Field

1. A coaxial cable consists of a long solid inner conductor of radius a , carrying a current I spread uniformly over its cross-section, and a thin outer cylindrical conductor of radius b , carrying the same current in the opposite direction. Find the magnetic field at all distances from the axis.
2. A pair of large parallel plane sheets carry uniform surface currents $\mathbf{K}$ and $-\mathbf{K}$. Find the field between the sheets and outside

them, and compare the result with the field of a long solenoid.

3. The current in each turn of a toroidal winding flows in a plane that contains the axis of the toroid, so every current element has components along $\hat{\mathbf{s}}$ and $\hat{\mathbf{z}}$ only. Let P be a field point in the xz -plane, and consider two current elements that are mirror images of each other in that plane. Using the Biot-Savart law, show that the sum of their fields at P is perpendicular to the xz -plane, and hence that the field of the toroid is azimuthal.
4. A toroid of N turns carrying a current I has a rectangular cross-section, with inner radius a , outer radius b , and height ℓ . Find the magnetic flux through one turn.
5. Show, using the curl in cylindrical coordinates worked out in Section 3.4 of Vol. 4, that the field $\mathbf{B} = (\mu_0 I / 2\pi s) \hat{\phi}$ of a straight wire has zero curl for $s > 0$. Explain, using Stokes' theorem, why its circulation round a circle about the wire is nevertheless not zero.
6. The field inside a long cylinder is $\mathbf{B} = \zeta s^2 \hat{\phi}$, where ζ is a constant and s is the distance from the axis. Find the current density from Ampère's law in differential form, and check the result by applying Ampère's law in integral form to a circle of radius s .
7. Verify Stokes' theorem for the field $\mathbf{B} = \zeta(-y, x, 0)$, where ζ is a constant, and a circle of radius b in the xy -plane centred on the origin, by working out both sides.

Magnetic Materials

6.1 Magnetisation and Bound Currents

Every material responds to a magnetic field in which it is placed by acquiring a magnetic moment, although for most materials the moment is small. This chapter describes that response, much as Chapter 3 described the response of matter to an electric field. In this section we first show that atoms carry magnetic dipole moments, then describe how an applied field lines those moments up, and finally show that a magnetised material produces the same magnetic field as a certain distribution of current, which is called the bound current.

Atoms as Magnetic Dipoles

An electron moving about the nucleus of an atom is a current. In the picture of the Bohr model it moves round a circular orbit and forms a small current loop, as shown in Figure 52, whose magnetic dipole moment, by Section 5.2, is the current multiplied by the vector area of the loop. Section 4.2 of Vol. 3 of the Fermi Notes on Physics used this result to show that an electron of charge $-e$ and mass m_e in a circular orbit has a magnetic moment equal to $-e/2m_e$ times its angular momentum. Let $\mathbf{L}_{\text{orb}}$ be the orbital angular momentum

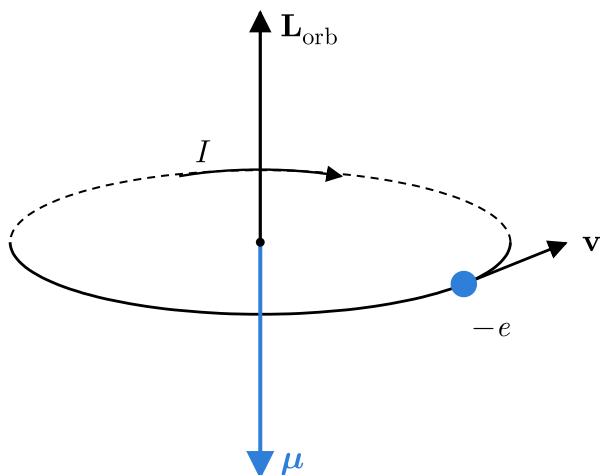


Figure 52. An electron moving round a circular orbit, drawn in perspective. Its orbital angular momentum $\mathbf{L}_{\text{orb}}$ points up, by the right-hand rule applied to its velocity $\mathbf{v}$. Since the charge of the electron is negative, the current I circulates in the opposite sense, and the magnetic moment $\boldsymbol{\mu}$ (accent) points down.

of the electron. As vectors,

$$\boldsymbol{\mu} = -\frac{e}{2m_e} \mathbf{L}_{\text{orb}}. \quad (6.1)$$

The minus sign appears because the conventional current flows in the opposite sense to the motion of the negatively charged electron. You will show in Problem 1 that the relation holds for any orbit that lies in a plane.

Section 4.1 of Vol. 3 showed that the component of an orbital angular momentum along any chosen axis is a whole multiple $m_l \hbar$ of $\hbar$, so the component of the orbital moment along that axis is $-m_l$ times the quantity

$$\mu_B = \frac{e\hbar}{2m_e}, \quad (6.2)$$

which is called the Bohr magneton, and which sets the scale of the

magnetic moments of atoms.

The electron also carries the spin $\mathbf{S}_e$ of Section 4.2 of Vol. 3, whose component along any axis is $\pm\frac{1}{2}\hbar$. The magnetic moment associated with the spin is very close to $-(e/m_e)\mathbf{S}_e$, so its component along any axis is very close to $\mp\mu_B$. Although the spin moment cannot be pictured in detail as the moment of a circulating charge, it produces a magnetic field, and responds to one, in the same way as a small current loop with the same moment. Nuclei have magnetic moments too, but they are smaller by a factor of roughly the ratio of the proton mass to the electron mass, and we neglect them.

The moment of an atom is the vector sum of the orbital and spin moments of its electrons, and that of a hydrogen atom in its ground state, which has no orbital angular momentum by Section 5.2 of Vol. 3, comes from the spin of its electron. In atoms with many electrons the moments often cancel in pairs, so that an atom of helium or neon has no permanent moment, while an atom of iron has a permanent moment of a few Bohr magnetons. The theory of atoms with many electrons, which decides how their moments combine, is not treated in this course.

Diamagnetic, Paramagnetic, and Ferromagnetic Materials

Materials are classified by the way in which they respond to an applied field, and the most common classes are the following:

1. A diamagnetic material, such as water, copper, or bismuth, is made of atoms or molecules with no permanent moment. An applied field alters the orbital motion of their electrons, and each acquires a small moment opposite to the field.
2. A paramagnetic material, such as liquid oxygen or copper sulphate, is made of atoms or molecules with permanent moments, whose directions thermal motion keeps changing at random. An applied field lines the moments up partially, and the material acquires a moment along the field.
3. A ferromagnetic material, such as iron, nickel, or cobalt, is one in which the moments of neighbouring atoms line up with one another, even without an applied field, over regions called domains. An applied field enlarges the domains whose moments lie along

it, and the moment the material acquires is far larger than that of a paramagnet and depends on the history of the sample, so that a piece of iron can stay magnetised after the applied field is removed, and it is then a permanent magnet.

Every material has a diamagnetic response, but where the atoms have permanent moments it is usually much weaker than the paramagnetic or ferromagnetic response. There are also other classes, such as antiferromagnets, in which the moments of neighbouring atoms line up in opposite directions. The microscopic theory of diamagnetism, paramagnetism, and ferromagnetism is not treated in this course, and from here on a material is described only by the magnetic moment that a given volume of it carries.

A diamagnet and a paramagnet can be told apart in a non-uniform field. By Section 5.2, the energy $-\boldsymbol{\mu} \cdot \mathbf{B}$ of a moment that points along the field is lowered where the field is stronger, and the force on it points towards the stronger field, while a moment that points against the field is pushed towards the weaker field. Liquid oxygen, which is paramagnetic, is therefore held between the poles of a strong electromagnet, while a diamagnetic sample, such as a piece of bismuth, the material in which Faraday studied the effect in 1845, is pushed out of the strong field.

Alignment of the Moments by a Field

We found in Section 5.2 that a dipole of moment $\boldsymbol{\mu}$ in a uniform field $\mathbf{B}$ experiences a torque and has an energy given by

$$\boldsymbol{\Gamma} = \boldsymbol{\mu} \times \mathbf{B}, \quad U = -\boldsymbol{\mu} \cdot \mathbf{B}. \quad (6.3)$$

We set up coordinates such that the field points along the z -axis, and let θ be the angle between the moment and the z -axis, as in spherical coordinates, so that $U = -\mu B \cos \theta$. As Figure 53 shows, the energy is least, $-\mu B$, when the moment points along the field, and greatest, $+\mu B$, when it points against the field, and the torque, of magnitude $\mu B \sin \theta$, acts to turn the moment towards the field.

A moment that carries angular momentum, such as that of an electron, responds to this torque by precessing about the field at a constant angle θ , as you showed for a spinning body in Problem 6 of Section 5.2, and as a spin does in Section 4.2 of Vol. 3. The angle

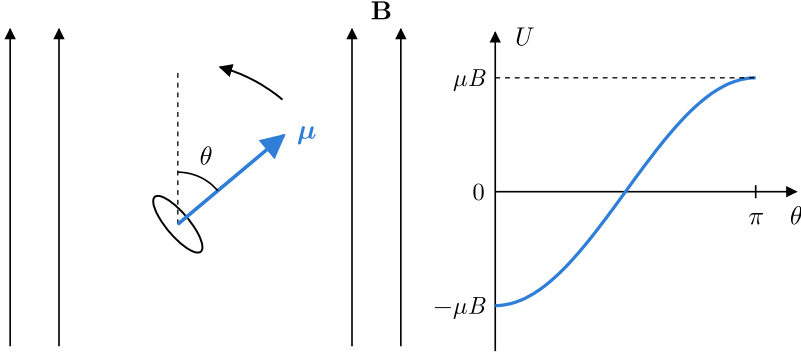


Figure 53. Left: a small current loop with magnetic moment μ at an angle θ to a uniform field $\mathbf{B}$. The torque $\mu \times \mathbf{B}$ acts to turn the moment towards the field, in the sense of the curved arrow. Right: the energy $U = -\mu B \cos \theta$ of the moment, which is least when the moment points along the field and greatest when it points against it.

between the moment and the field decreases only when the atom exchanges energy with its surroundings, for example in collisions, and these exchanges can also increase that angle, so the balance is set by the temperature. Let $\mathcal{T}$ be the absolute temperature, written as in Section 3.1 because the letter T is kept in these notes for a period, and let k_B be Boltzmann's constant. The magnetic energy μB is then to be compared with the thermal energy $k_B \mathcal{T}$. This competition is worked out in Section 5.3 of Vol. 2 for a solid containing N moments of spin one half, each with a component $\pm \mu_B$ along the field, whose total moment along the field was found there to be

$$N \mu_B \tanh \frac{\mu_B B}{k_B \mathcal{T}}. \quad (6.4)$$

When $\mu_B B \gg k_B \mathcal{T}$ every moment lies along the field. When $\mu_B B \ll k_B \mathcal{T}$, the approximation $\tanh y \approx y$ for small y gives the total moment $N \mu_B^2 B / k_B \mathcal{T}$, which is Curie's law. In the fields of laboratory magnets at room temperature, $\mu_B B$ is a small fraction of $k_B \mathcal{T}$, so the alignment is slight, and the moment of a paramagnet is proportional to the field and inversely proportional to the absolute temperature.

Magnetisation

The magnetic field inside a material varies greatly over distances comparable with the size of an atom, and in this chapter we are concerned only with its average over a region that is small compared with the sample but contains many atoms. Let $\Delta\mathcal{V}$ be a volume of this kind, and let the atoms in it have moments $\boldsymbol{\mu}_i$. The magnetic moment per unit volume,

$$\mathbf{M} = \frac{1}{\Delta\mathcal{V}} \sum_i \boldsymbol{\mu}_i, \quad (6.5)$$

is called the magnetisation of the material at that place, and it is measured in amperes per metre. If the material contains n_d dipoles per unit volume with mean moment $\langle \boldsymbol{\mu} \rangle$, then $\mathbf{M} = n_d \langle \boldsymbol{\mu} \rangle$. The magnetisation of a paramagnet points along the applied field and that of a diamagnet points against it, while a bar magnet has a roughly uniform magnetisation that points along its length from its south pole to its north pole.

The magnetisation is a dipole moment per unit volume, as the polarisation $\mathbf{P}$ of Section 3.1 is, and the small volume $d\mathcal{V}$ carries the magnetic moment $\mathbf{M} d\mathcal{V}$. We shall now show, by replacing each small volume by a current loop with the same moment, that the field of all these moments is the field of a certain distribution of current, which can then be found with the Biot–Savart law of Section 5.2 or with Ampère’s law of Section 5.3.

The Bound Surface Current

As a first case, consider a slab with a uniform magnetisation $\mathbf{M}$ along the z -axis, divided into small blocks whose edges Δx , Δy , and Δz lie along the three axes, as shown in Figure 54. Each block has the moment $M \Delta x \Delta y \Delta z$ along z , and we replace it by a current I that flows round the four sides of the block parallel to the z -axis, anticlockwise as seen from the positive z -axis. The vector area enclosed by this current is $\Delta x \Delta y \hat{\mathbf{z}}$, and equating the moment $I \Delta x \Delta y$ to the moment of the block gives

$$I = M \Delta z. \quad (6.6)$$

Each side of the block therefore carries a surface current of density $I/\Delta z = M$, which does not depend on the size of the block.

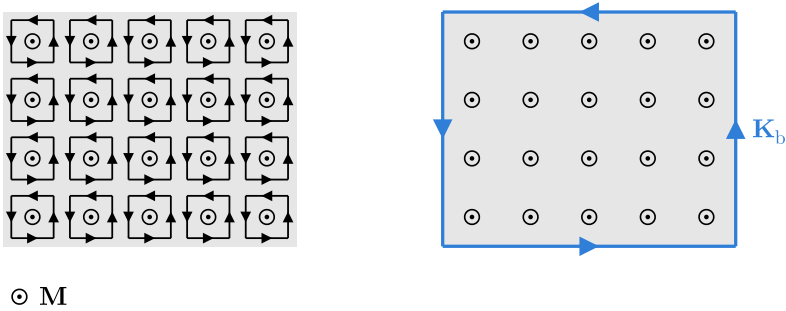


Figure 54. Left: a uniformly magnetised slab, seen from above, divided into small blocks, each of which is replaced by a current loop with the same moment. The magnetisation $\mathbf{M}$ points out of the page, and along each edge shared by two blocks the two loops carry equal currents in opposite directions. Right: the current that remains when these currents have cancelled, a bound surface current $\mathbf{K}_b$ (accent) that circulates round the edge of the slab.

On a face shared by two neighbouring blocks, the currents of the two blocks are equal and opposite, so they cancel, and inside the slab no current is left. The outer faces of the blocks at the edge of the slab have no neighbours, and their currents form a surface current of density M that circulates round the edge, as the right-hand part of Figure 54 shows. On the face whose outward unit normal is $\hat{\mathbf{n}} = \hat{\mathbf{x}}$, this current flows along $\hat{\mathbf{y}} = \hat{\mathbf{z}} \times \hat{\mathbf{x}}$. The top and bottom faces of the slab, whose normals are parallel to $\mathbf{M}$, carry no current, since the current of each block flows only on its side faces. The general result, that the component of the magnetisation along the normal to a surface gives no current there while the component along the surface gives a current at right angles to that component and to the normal, is contained in the formula

$$\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}, \quad (6.7)$$

where $\hat{\mathbf{n}}$ is the outward unit normal to the surface of the material. The quantity $\mathbf{K}_b$ is called the bound surface current density.

No charge is carried from one place to another by the bound current, and it cannot be led off through a wire to an ammeter, but its

magnetic field is that of an ordinary current of the same density.

The Bound Volume Current

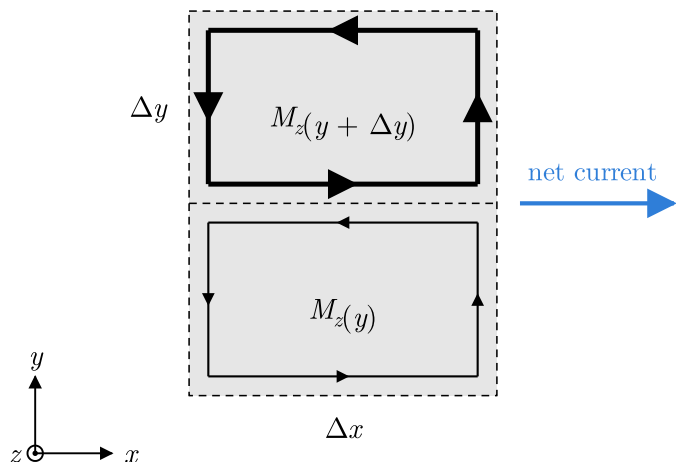


Figure 55. A pair of neighbouring blocks, seen from above, whose magnetisations along z differ, with $M_z(y + \Delta y)$ greater than $M_z(y)$, so that the upper loop carries the larger current, drawn heavier. Along the face the two blocks share, their currents flow in opposite directions and leave a net current (accent) along the x -axis.

If the magnetisation varies from place to place, the currents of neighbouring blocks no longer cancel. Suppose that M_z depends on y , and consider the two blocks of Figure 55, centred at y and at $y + \Delta y$, which share a face perpendicular to the y -axis. On that face the first block carries the current $M_z(y) \Delta z$ along $-x$ and the second carries $M_z(y + \Delta y) \Delta z$ along $+x$. By Taylor's theorem, to first order in Δy , the net current along $+x$ is

$$[M_z(y + \Delta y) - M_z(y)] \Delta z = \frac{\partial M_z}{\partial y} \Delta y \Delta z. \quad (6.8)$$

There is one such face in each area $\Delta y \Delta z$ of a plane perpendicular to the x -axis, so the current density along x is $\partial M_z / \partial y$.

A variation of M_y with z also gives a current along x . The blocks then carry currents $M_y \Delta y$ round their sides parallel to the y -axis, anticlockwise as seen from the positive y -axis, and on the face shared by blocks centred at z and at $z + \Delta z$ the first carries $M_y(z) \Delta y$ along $+x$ and the second carries $M_y(z + \Delta z) \Delta y$ along $-x$. A similar calculation gives the current density $-\partial M_y / \partial z$ along x , and a magnetisation along x contributes nothing along x , since the loops that replace it lie in planes perpendicular to the x -axis and their currents have no component along x . Adding the contributions of M_z and M_y ,

$$J_{b,x} = \frac{\partial M_z}{\partial y} - \frac{\partial M_y}{\partial z}, \quad (6.9)$$

and the other components follow by relabelling the axes cyclically. The right-hand side is the x -component of the curl of $\mathbf{M}$, as defined in Section 5.3 and in Section 3.1 of Vol. 4 of the Fermi Notes on Physics, so the current density inside the material, which is called the bound volume current density, is

$$\mathbf{J}_b = \nabla \times \mathbf{M}, \quad (6.10)$$

which vanishes for a uniform magnetisation, as we found for the slab.

At the surface of a real material the magnetisation falls to zero across a thin layer, and the surface current is the volume current of that layer. For a surface with outward normal $\hat{\mathbf{x}}$ and a magnetisation $M_z \hat{\mathbf{z}}$ parallel to it, the y -component of $\nabla \times \mathbf{M}$ is $-\partial M_z / \partial x$, and integrating it across the layer, from just inside, where the magnetisation is M_z , to just outside, where it is zero, gives

$$\int J_{b,y} dx = -(0 - M_z) = M_z. \quad (6.11)$$

This is a surface current of density M_z along $\hat{\mathbf{y}}$, which is $\mathbf{M} \times \hat{\mathbf{n}}$. Taken together, the bound volume and surface currents of a magnetised body produce the same magnetic field as the body.

A Uniformly Magnetised Cylinder

Let us find the bound currents of a cylinder of radius a and length ℓ , magnetised uniformly along its axis, which is a simple model of a bar magnet. We take the axis as the z -axis, as in Figure 56. The

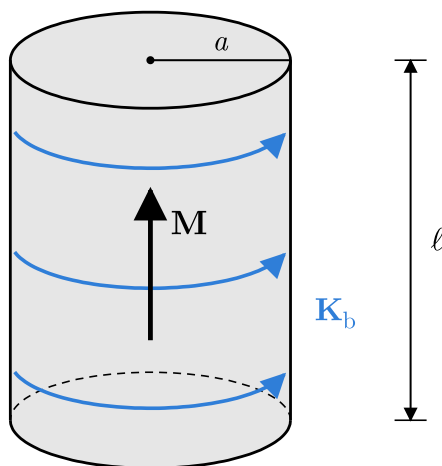


Figure 56. A cylinder of radius a and length ℓ with a uniform magnetisation $\mathbf{M}$ along its axis. The bound surface current $\mathbf{K}_b$ (accent) circulates round the curved surface, and no bound current flows on the flat ends or inside the cylinder.

magnetisation is uniform, so there is no bound volume current, and on the flat ends the normal is parallel to $\mathbf{M}$, so no surface current flows there. On the curved surface the outward normal is $\hat{\mathbf{s}}$, the unit vector pointing away from the axis, and

$$\mathbf{K}_b = M \hat{\mathbf{z}} \times \hat{\mathbf{s}} = M \hat{\phi}. \quad (6.12)$$

The cylinder therefore produces the same field as a solenoid of the same shape with $nI = M$, which is why the field lines of a bar magnet are so similar to those of a solenoid, a resemblance you may have noticed at school. If the cylinder is long, the field inside it, away from its ends, is the field $\mu_0 nI$ of a long solenoid found in Section 5.3, so that

$$\mathbf{B} = \mu_0 \mathbf{M}, \quad (6.13)$$

and the field outside it, away from its ends, is very small, and it goes to zero in the limit of an infinitely long cylinder.

For a cylinder of any length, the field on the axis is the field found

in Section 5.2 for a solenoid of the same radius and length, with nI replaced by M , since the strip of the curved surface between z' and $z' + dz'$ is a loop carrying the current $M dz'$ in place of $nI dz'$. Let the cylinder run from $z = -\ell/2$ to $z = \ell/2$, and let $z_{\pm} = \ell/2 \pm z$, which for a point inside the cylinder are its distances from the two ends. The field on the axis is then

$$B_z(z) = \frac{\mu_0 M}{2} \left[\frac{z_-}{\sqrt{a^2 + z_-^2}} + \frac{z_+}{\sqrt{a^2 + z_+^2}} \right]. \quad (6.14)$$

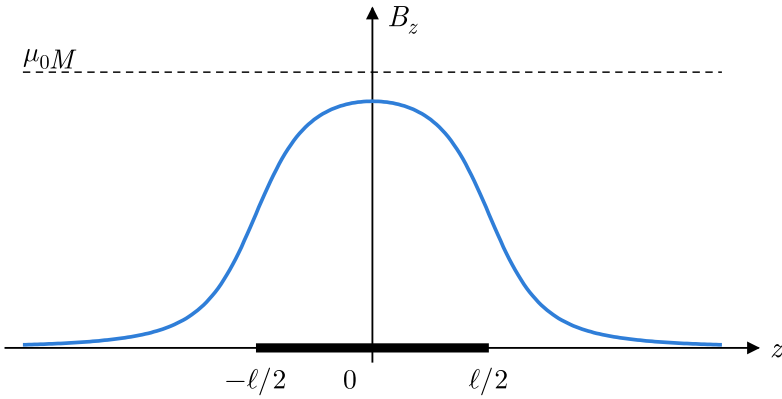


Figure 57. The field on the axis of a uniformly magnetised cylinder whose length is several times its radius, plotted against the position z along the axis, with the cylinder occupying the heavy segment from $-\ell/2$ to $\ell/2$. The dashed line marks the value $\mu_0 M$ inside a very long cylinder.

Figure 57 shows this field for a cylinder several times longer than its radius. At the centre, where $z_+ = z_- = \ell/2$, the field is

$$B_z(0) = \frac{\mu_0 M \ell}{\sqrt{\ell^2 + 4a^2}}, \quad (6.15)$$

which is less than $\mu_0 M$ and approaches it as ℓ/a grows. As was found in Section 5.2 for a long solenoid, the field at the centre of an end face of a long cylinder is half as large, $\frac{1}{2}\mu_0 M$. You will show in Problem 4 that far from the cylinder along its axis the field is that of a dipole whose moment is the magnetisation multiplied by the volume of the cylinder.

6.2 The Auxiliary Field and Linear Magnetic Materials

In the previous section we found that a magnetised material produces the same magnetic field as its bound currents. The bound currents depend on the magnetisation, however, and in a paramagnet or a diamagnet the magnetisation depends in turn on the field, so the field and the magnetisation have to be found together. To find them together, we first write Ampère's law in a form whose only source is the current that flows through conductors, which introduces a new field $\mathbf{H}$, then define the simplest class of magnetic materials, in which the magnetisation is proportional to that field, and finally compare $\mathbf{H}$ with the electric displacement $\mathbf{D}$ of Section 3.2.

Free and Bound Currents

The current in and around a magnetised material is of two kinds. The bound current, of density $\mathbf{J}_b = \nabla \times \mathbf{M}$, was the subject of Section 6.1. All other current is called free current, and its density is written $\mathbf{J}_f$. Free current is carried by charges that move through the material over large distances, such as the conduction electrons in a wire, and it is the current that a battery drives and an ammeter measures. In a solenoid wound on an iron core, for example, the current in the windings is free current, and the current on the surface of the magnetised core is bound current.

For steady currents, Ampère's law in the differential form of Section 5.3 holds with the total current density on the right,

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}_f + \mu_0 \mathbf{J}_b = \mu_0 \mathbf{J}_f + \mu_0 \nabla \times \mathbf{M}. \quad (6.16)$$

Each component of the curl is a difference of partial derivatives, so the curl of a sum of two fields is the sum of their curls, and the terms containing $\mathbf{B}$ and $\mathbf{M}$ can be collected on the left. Dividing by μ_0 gives

$$\nabla \times \left(\frac{\mathbf{B}}{\mu_0} - \mathbf{M} \right) = \mathbf{J}_f. \quad (6.17)$$

The Auxiliary Field

Let us define the field

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M}, \quad (6.18)$$

which is called the auxiliary field. Ampère's law then becomes

$$\nabla \times \mathbf{H} = \mathbf{J}_f. \quad (6.19)$$

The auxiliary field is measured in amperes per metre, the same unit as the magnetisation, and outside matter, where $\mathbf{M} = 0$, it is simply $\mathbf{B}/\mu_0$. Some books call $\mathbf{H}$ the magnetic field and $\mathbf{B}$ the magnetic flux density or the magnetic induction, while these notes keep the name magnetic field for $\mathbf{B}$.

The integral form follows from Stokes' theorem of Section 5.3. We integrate $\nabla \times \mathbf{H}$ over a surface bounded by a closed curve and turn the surface integral into a line integral round the curve, which gives

$$\oint \mathbf{H} \cdot d\boldsymbol{\ell} = I_{f, \text{enc}}, \quad (6.20)$$

where $I_{f, \text{enc}}$ is the free current through the surface. This relation is called Ampère's law in matter. Where the arrangement is symmetric enough, it gives $\mathbf{H}$ from the free currents alone, in the same way as Ampère's law gave $\mathbf{B}$ in Section 5.3, and $\mathbf{B}$ then follows once the magnetisation is known. A bound surface current does not appear on the right either, since we found in Section 6.1 that it is the bound volume current of a thin layer at the surface of the material, and that current is part of $\nabla \times \mathbf{M}$, which has been absorbed into $\mathbf{H}$.

Ampère's law in matter fixes the curl of $\mathbf{H}$ and says nothing about its divergence. Since $\nabla \cdot \mathbf{B} = 0$, as shown in Section 5.3, the divergence is

$$\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}, \quad (6.21)$$

which is not zero wherever the magnetisation has a divergence. At the ends of a bar magnet, for example, the magnetisation stops, and its divergence is not zero there. We return to this point at the end of the section.

Linear Magnetic Materials

In a paramagnet or a diamagnet the magnetisation is produced by the applied field, and in fields that are not too strong it is proportional to the field. It is conventional to express this proportionality in terms of $\mathbf{H}$. A material is called a linear magnetic material if

$$\mathbf{M} = \chi_m \mathbf{H}, \quad (6.22)$$

where the dimensionless constant χ_m is called the magnetic susceptibility of the material. Substituting this relation into the definition of $\mathbf{H}$ gives $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = \mu_0(1 + \chi_m)\mathbf{H}$, which is written

$$\mathbf{B} = \mu_r \mu_0 \mathbf{H}, \quad \mu_r = 1 + \chi_m. \quad (6.23)$$

The constant μ_r is called the relative permeability, and the product $\mu_r \mu_0$ is called the permeability of the material. The relation $\mathbf{M} = \chi_m \mathbf{H}$ describes an isotropic material, one whose response is the same for a field in any direction. The response of a liquid such as water, for example, is of this kind. A crystal whose response depends on the direction of the field relative to its axes needs a more general relation, which is not treated in this course.

For a paramagnet, χ_m is positive, and for a diamagnet it is negative. For a paramagnet and a diamagnet alike its magnitude is much smaller than one, so that μ_r is close to one and $\mathbf{B}$ differs from $\mu_0 \mathbf{H}$ only by the small amount $\mu_0 \mathbf{M}$. For a paramagnet with $\mu_B B \ll k_B \mathcal{T}$, we found in Section 6.1, from the result of Section 5.3 of Vol. 2, that the magnetisation is proportional to $B/\mathcal{T}$, and since B is very close to $\mu_0 H$ the susceptibility of such a paramagnet is inversely proportional to the absolute temperature. The linear relation fails for a paramagnet when $\mu_B B$ becomes comparable with $k_B \mathcal{T}$, since the moments then approach full alignment and the magnetisation stops growing in proportion to the field. This relation does not describe a ferromagnet, whose magnetisation depends on the history of the sample as well as on the present field, and the magnetisation curves of ferromagnets are not treated in this course.

The magnetisation of a weakly magnetic material could equally well have been written as proportional to $\mathbf{B}$, and when $|\chi_m| \ll 1$ the difference is negligible. The reason for preferring $\mathbf{H}$ has to do with what is controlled in the laboratory, and it is given at the end of this section.

In a homogeneous linear material, one in which χ_m has the same value everywhere, the bound volume current is fixed by the free current. The constant χ_m can be taken outside each partial derivative in the curl, so that

$$\mathbf{J}_b = \nabla \times \mathbf{M} = \chi_m \nabla \times \mathbf{H} = \chi_m \mathbf{J}_f. \quad (6.24)$$

Where no free current flows through such a material, there is no bound volume current, and the bound current flows only on the surface.

A Solenoid Filled with a Linear Material

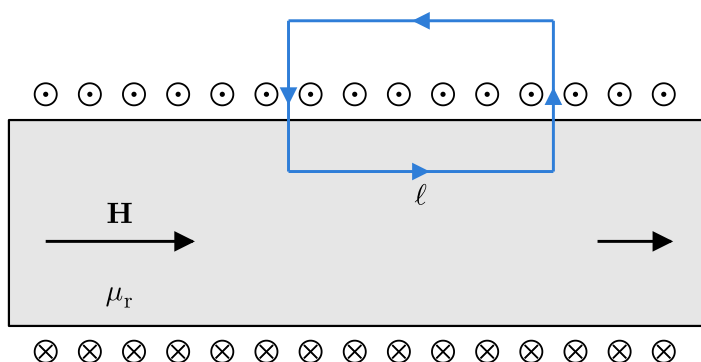


Figure 58. A long solenoid filled with a linear magnetic material of relative permeability μ_r , cut by a plane containing its axis. The windings carry current out of the page along the top and into the page along the bottom, and the field $\mathbf{H}$ inside points along the axis. The rectangular loop (accent) has one side of length ℓ inside the material, parallel to the axis, and the opposite side outside the solenoid.

The first arrangement we treat is a long solenoid with n turns per unit length carrying a current I , whose interior is filled with a linear material of relative permeability μ_r , as shown in Figure 58. The free current flows round the axis in the windings, and the bound current flows round the axis on the curved surface of the material, so every current in the arrangement flows round the axis on a cylinder

coaxial with it. The argument of Section 5.3 for a long solenoid applies to such an arrangement, and it shows that $\mathbf{B}$ points along the axis and is zero outside. The magnetisation points along the axis inside the material and is zero outside it, so $\mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}$ also points along the axis and is zero outside the solenoid.

We now apply Ampère's law in matter to the rectangular loop of Figure 58. Only the side inside the material contributes to the line integral, since $\mathbf{H}$ is zero on the outer side and perpendicular to the two short sides, and the free current through the loop is $n\ell I$. Therefore $H\ell = n\ell I$, and

$$H = nI, \quad B = \mu_r\mu_0 nI, \quad M = \chi_m nI. \quad (6.25)$$

The auxiliary field is the same as in an empty solenoid, since it is fixed by the free current alone, while the magnetic field is larger by the factor μ_r .

The result can be checked against the bound current. The magnetisation is uniform inside the material, so there is no bound volume current, and the bound surface current on the curved surface of the material has density $M = \chi_m nI$, by the result for the cylinder of Section 6.1. For $\chi_m > 0$ it circulates in the same sense as the current in the windings. The total current per unit length is then $nI + \chi_m nI$, and the field of a long solenoid carrying this current is $\mu_0(1 + \chi_m)nI$, in agreement with the value above. In a diamagnetic filling the bound current circulates against the windings, and the field is slightly smaller than in the empty solenoid.

The magnetic flux through each turn is $\Phi_B = \mu_r\mu_0 nIA$, where A is the area of the cross-section, so the flux is larger by the factor μ_r than for the empty solenoid. A core of iron, which is not a linear material, increases the flux by a very much larger factor, and this is one reason the electromagnets of Section 5.1 are wound on iron cores.

A Wire of Magnetisable Material

Let us now turn to a long straight wire of radius a , made of a linear material of relative permeability μ_r , which carries a free current I spread uniformly over its cross-section. A wire of aluminium, which is paramagnetic, is an example of such a wire. The free current flows

parallel to the axis, and its density depends only on the distance s from the axis. By the argument you used for the solid inner conductor of the coaxial cable in Problem 1 of Section 5.3, applied to the free and bound currents together, $\mathbf{B}$ circles the axis with a magnitude that depends only on s , and so do $\mathbf{M} = \chi_m \mathbf{B} / \mu_r \mu_0$ and $\mathbf{H}$. Ampère's law in matter, applied to a circle of radius s centred on the axis, gives $2\pi s H = I_{\text{f,enc}}$, and the enclosed free current is $I s^2 / a^2$ inside the wire and I outside it. Therefore

$$H = \frac{I s}{2\pi a^2} \quad (s < a), \quad H = \frac{I}{2\pi s} \quad (s > a). \quad (6.26)$$

Inside the wire $B = \mu_r \mu_0 H$, and outside it $B = \mu_0 H$, so

$$B = \frac{\mu_r \mu_0 I s}{2\pi a^2} \quad (s < a), \quad B = \frac{\mu_0 I}{2\pi s} \quad (s > a). \quad (6.27)$$

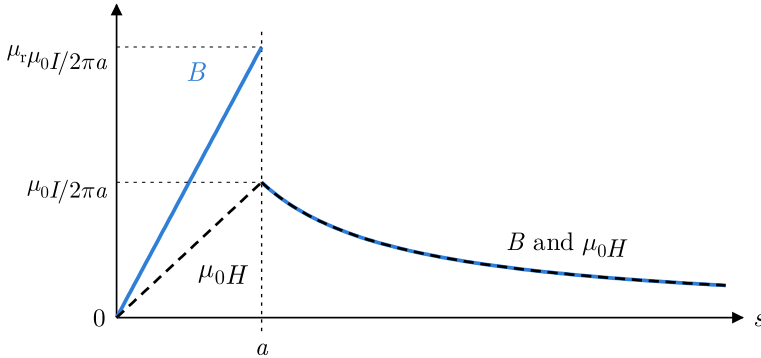


Figure 59. The magnetic field B (accent) and the product $\mu_0 H$ (dashed) for a long straight wire of radius a , made of a linear material with μ_r greater than one and carrying a current I , plotted against the distance s from the axis. Outside the wire the two curves coincide, and at the surface of the wire B jumps while H is continuous.

Figure 59 shows the two fields, drawn with a relative permeability far larger than that of any paramagnet so that the jump in B can be seen. The auxiliary field is continuous at the surface of the wire, while the magnetic field jumps there from $\mu_r \mu_0 I / 2\pi a$ just inside to $\mu_0 I / 2\pi a$ just outside. Outside the wire the magnetic field is the

same as for a wire of non-magnetic material.

The jump in B is produced by the bound currents, which we now find. The magnetisation inside the wire is $\mathbf{M} = \chi_m \mathbf{H}$, which circles the axis with magnitude $\chi_m I s / 2\pi a^2$. Let $\hat{\mathbf{z}}$ point along the free current, whose density is $I/\pi a^2$. The wire is a homogeneous linear material, so the bound volume current density is $\chi_m I/\pi a^2$ along $\hat{\mathbf{z}}$, and the total bound current it carries through the cross-section is $\chi_m I$. At the surface the outward normal is $\hat{\mathbf{s}}$, and since $\hat{\phi} \times \hat{\mathbf{s}} = -\hat{\mathbf{z}}$, the bound surface current density is

$$\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{s}} = -\frac{\chi_m I}{2\pi a} \hat{\mathbf{z}}, \quad (6.28)$$

which carries a total current $-\chi_m I$ round the circumference $2\pi a$. The bound volume and surface currents add up to zero, so a circle outside the wire encloses only the free current I , and Ampère's law of Section 5.3 gives $B = \mu_0 I / 2\pi s$ there. A circle of radius s inside the wire encloses the free current $I s^2 / a^2$ and the bound current $\chi_m I s^2 / a^2$, and Ampère's law gives $B = \mu_0 (1 + \chi_m) I s / 2\pi a^2$. The values of B inside and outside the wire agree with those found from $\mathbf{H}$. The surface current flows against the free current when $\chi_m > 0$, and it is this current that lowers the field from its value just inside the surface to its value just outside.

Comparing the Auxiliary Field with the Displacement

The auxiliary field was constructed in the same way as the electric displacement $\mathbf{D}$ of Section 3.2, and the two fields have the following features in common:

1. Each is defined so as to absorb the bound sources. The displacement $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$ absorbs the bound charge, and the auxiliary field $\mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}$ absorbs the bound current.
2. Each obeys a law in which only free sources appear. Gauss's law in matter, $\nabla \cdot \mathbf{D} = \rho_f$, and Ampère's law in matter, $\nabla \times \mathbf{H} = \mathbf{J}_f$, each give their field from the free sources alone when the arrangement is symmetric enough, as they did for the filled capacitor of Section 3.2 and the filled solenoid above.
3. In a linear material, each is proportional to the field it accompanies, with $\mathbf{D} = \epsilon_r \epsilon_0 \mathbf{E}$ and $\mathbf{B} = \mu_r \mu_0 \mathbf{H}$.

The polarisation is added in the definition of $\mathbf{D}$, while the magnetisation is subtracted in the definition of $\mathbf{H}$. The difference in sign comes from the bound sources themselves, since the bound charge density $-\nabla \cdot \mathbf{P}$ of Section 3.1 carries a minus sign and the bound current density $\nabla \times \mathbf{M}$ does not.

Another difference concerns what the free sources determine. Gauss's law in matter fixes the divergence of $\mathbf{D}$, and in the same way Ampère's law in matter fixes the curl of $\mathbf{H}$. The curl of $\mathbf{D}$ and the divergence of $\mathbf{H}$ are not fixed by the free sources, since $\nabla \times \mathbf{D} = \nabla \times \mathbf{P}$, using $\nabla \times \mathbf{E} = 0$ from Section 5.3, and $\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}$. Neither of these need vanish, so $\mathbf{H}$ is not determined by the free current alone, and it can be non-zero in an arrangement with no free current anywhere.

The uniformly magnetised cylinder of Section 6.1 shows this. It carries no free current. If the cylinder is long, the field inside it, away from its ends, is $\mu_0 \mathbf{M}$, so $\mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}$ is very small inside. Outside, $\mathbf{M}$ is zero and $\mathbf{B}$, away from the ends, is very small, so $\mathbf{H}$ is very small everywhere, and it vanishes in the limit of an infinitely long cylinder. For a cylinder of finite length, the field $B_z(0)$ at the centre, found in Section 6.1, gives

$$H_z(0) = \frac{B_z(0)}{\mu_0} - M = -M \left(1 - \frac{\ell}{\sqrt{\ell^2 + 4a^2}} \right), \quad (6.29)$$

which is not zero and points opposite to the magnetisation. The magnetisation of the finite cylinder stops at the flat ends, so $\nabla \cdot \mathbf{M}$ is not zero there, and this divergence is the source of the auxiliary field inside the cylinder. You will find in Problem 5 how $H_z(0)$ behaves for a long cylinder and for a thin disc.

In a region with no free current, the equations for $\mathbf{H}$, which are $\nabla \times \mathbf{H} = 0$ and $\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}$, have the same form as the equations for the electrostatic field in a region with no free charge, which are $\nabla \times \mathbf{E} = 0$ and $\nabla \cdot \mathbf{E} = -\nabla \cdot \mathbf{P}/\epsilon_0$, with $\mathbf{M}$ in place of $\mathbf{P}/\epsilon_0$. In this respect $\mathbf{H}$ resembles $\mathbf{E}$ more closely than it resembles $\mathbf{D}$, and the correspondence runs through the definitions as well, since $\mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}$ has the same form as $\mathbf{E} = \mathbf{D}/\epsilon_0 - \mathbf{P}/\epsilon_0$.

The fields $\mathbf{D}$ and $\mathbf{H}$ also differ in how they are related to what an experimenter controls. In a capacitor filled with a dielectric, the potential difference between the plates is usually fixed by a

battery, and the potential difference is the line integral of $\mathbf{E}$ between the plates, which is one reason the susceptibility of Section 3.2 is defined through $\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$. In an electromagnet, the current in the windings is fixed by a power supply, and this free current fixes $\mathbf{H}$ through Ampère's law in matter, which is one reason the magnetic susceptibility is defined through $\mathbf{M} = \chi_m \mathbf{H}$.

Problems

§6.1 Magnetisation and Bound Currents

1. An electron moves under a central force in an orbit of any shape that lies in a plane, returning to its starting point after a period T . Show that its magnetic moment is $\boldsymbol{\mu} = -(e/2m_e) \mathbf{L}_{\text{orb}}$, as for the circular orbit of Figure 52.
2. A long cylinder of radius a has a uniform magnetisation $\mathbf{M} = M\hat{\mathbf{x}}$, perpendicular to its axis, which lies along the z -axis. Find the bound currents, and show that the total bound current flowing along the cylinder is zero.
3. A slab of material occupies the region $0 < x < \ell$ and extends without limit in y and z . Its magnetisation is $\mathbf{M} = M_0(x/\ell) \hat{\mathbf{z}}$, where M_0 is a constant. Find the bound volume and surface currents, and show that the total bound current per unit length along z is zero. Using the field of a current sheet found in Section 5.3, show that $\mathbf{B} = \mu_0 \mathbf{M}$ inside the slab and $\mathbf{B} = 0$ outside it.
4. For the uniformly magnetised cylinder of Figure 56, show that at a distance z along the axis with $z \gg \ell$ and $z \gg a$ the field is $\mu_0 \mu / 2\pi z^3$, where $\mu = M\pi a^2 \ell$ is the total moment of the cylinder.
5. A sphere of radius a has a uniform magnetisation $\mathbf{M} = M\hat{\mathbf{z}}$. Find its bound currents. A thin spherical shell of the same radius, carrying a uniform surface charge density σ , rotates about the z -axis with angular velocity ω . Find the surface current density on the shell, and show that it is the same as the bound surface current density of the sphere, so that the shell and the sphere produce the same magnetic field, if $\sigma\omega a = M$.

§6.2 The Auxiliary Field and Linear Magnetic Materials

1. A coaxial cable consists of a long solid inner conductor of radius a , carrying a current I spread uniformly over its cross-section, and a thin outer conducting shell of radius b , carrying the same current in the opposite direction. Both conductors are non-magnetic, and the space between them is filled with a linear material of susceptibility χ_m . Find $\mathbf{H}$, $\mathbf{B}$, and $\mathbf{M}$ everywhere, and find the bound currents. Check that Ampère's law of Section 5.3, with the free and bound currents together, gives the same $\mathbf{B}$ between the conductors.
2. A toroid of N turns carrying a current I is wound on a ring of linear material of relative permeability μ_r , whose cross-section is a rectangle with inner radius a , outer radius b , and height ℓ . Find $\mathbf{H}$ and $\mathbf{B}$ inside the ring, and find the magnetic flux through the cross-section of the ring.
3. A long solenoid of radius b , with n turns per unit length carrying a current I , contains a coaxial rod of linear material of radius $a < b$ and susceptibility χ_m , with air, whose susceptibility may be neglected, in the space between the rod and the windings. Show that $H = nI$ everywhere inside the solenoid, find B in the rod and in the air, and find the magnetic flux through one turn.
4. A small sample of a diamagnetic material, of volume $\mathcal{V}$, mass m , and susceptibility $\chi_m < 0$ with $|\chi_m| \ll 1$, is placed on the vertical axis of a magnet whose field on the axis points along z and has magnitude $B_z(z)$, with z measured upwards. As found in Section 5.2, the force on a small dipole of moment $\mu_z \hat{\mathbf{z}}$ at a point on the axis of such a field is $\mu_z \partial B_z / \partial z$ along z . Show that the sample floats where $B_z \partial B_z / \partial z = \mu_0 mg / \chi_m \mathcal{V}$, and explain why it must be placed where the magnitude of the field decreases with height.
5. For the uniformly magnetised cylinder of Section 6.1, find the limits of $H_z(0)$ when $\ell \gg a$ and when $\ell \ll a$. Find $B_z(0)$ in the second limit, and describe the field and the auxiliary field at the centre of a thin disc magnetised along its axis.

Electromagnetic Induction and Alternating-Current Circuits

7.1 Induction and Alternating Current at School

A steady current produces a magnetic field, as Chapter 5 described, and the converse process, in which a changing magnetic field drives a current, is called electromagnetic induction. You will have met induction at school, together with the generators, transformers, and alternating currents that depend on it, and this section runs through those results in the notation of the course. Sections 7.2 and 7.3 take induction to university level, first deriving the induced electromotive force from the forces on the charges and writing Faraday's law as a statement about the electric field, and then treating inductance and the energy stored in a magnetic field. Sections 7.4 to 7.6 turn to alternating currents and to the circuits that they drive.

Magnetic Flux and Flux Linkage

Section 5.1 defined the magnetic flux through a surface. For a flat coil of area A in a uniform field $\mathbf{B}$, whose normal makes an angle θ with the field, the flux through each turn is

$$\Phi_B = BA \cos \theta, \quad (7.1)$$

measured in webers, with $1 \text{ Wb} = 1 \text{ T m}^2$. The flux is largest when the plane of the coil is perpendicular to the field, and zero when the field lies in the plane of the coil. For a coil of N turns, each threaded by the same flux, the quantity $N\Phi_B$ is called the flux linkage.

Faraday's Law and Lenz's Law

When the flux linkage of a coil changes, an electromotive force is induced in the coil. Faraday found by experiment that the size of this electromotive force is equal to the rate of change of the flux linkage, and Lenz's law gives its direction: the induced electromotive force drives a current whose magnetic field opposes the change in flux that produced it. Faraday's result and Lenz's law are combined in

$$\mathcal{E} = -\frac{d(N\Phi_B)}{dt}, \quad (7.2)$$

in which the minus sign expresses Lenz's law. The flux through a coil can be changed by moving a magnet towards it or away from it, by changing the current in a nearby electromagnet, by moving or rotating the coil in a field, or by changing the area of the coil. As an example of Lenz's law, when the north pole of a bar magnet is pushed towards a coil, the current induced in the coil makes the face of the coil nearest the magnet act as a north pole, which repels the approaching magnet. Work must therefore be done to push the magnet in, and this work is the source of the electrical energy that appears in the circuit.

The Motional Electromotive Force

A straight conductor of length ℓ , moving with speed v in a uniform field B , with its length, its velocity, and the field all perpendicular to one another, has an electromotive force

$$\mathcal{E} = B\ell v \quad (7.3)$$

induced between its ends. If the conductor slides along a pair of rails joined at one end, the area of the circuit grows at the rate ℓv , so the flux through the circuit grows at the rate $B\ell v$, and Faraday's law gives the same result. The direction of the induced current is given by Fleming's right-hand rule, or by Lenz's law, which requires

the magnetic force BIl on the current in the conductor to oppose its motion.

The Rotating Coil and the Generator

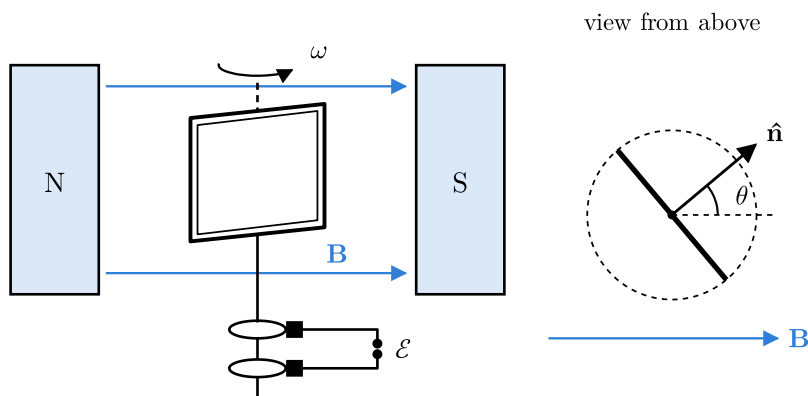


Figure 60. A coil rotating with angular velocity ω between the poles of a magnet, with its ends connected to the external circuit through slip rings and brushes. The view from above shows the angle θ between the normal $\hat{n}$ to the coil and the field B .

A coil of N turns and area A , rotating with angular velocity ω in a uniform field about an axis perpendicular to the field, is drawn in Figure 60. If the normal to the coil points along the field at $t = 0$, the angle between them at time t is $\theta = \omega t$, and the flux linkage is $NBA \cos \omega t$. Faraday's law then gives

$$\mathcal{E} = NBA\omega \sin \omega t, \quad (7.4)$$

an electromotive force that varies sinusoidally with time and has the peak value $\mathcal{E}_0 = NBA\omega$. The electromotive force is largest when the plane of the coil is parallel to the field, where the flux through the coil is zero but is changing most rapidly, and it is zero when the plane of the coil is perpendicular to the field, where the flux is largest and momentarily not changing. In an alternating-current generator the ends of the coil are connected to the external circuit through slip rings, on which carbon brushes press, so that

the connections are fixed and the output reverses its sign every half turn, as the electromotive force does. In a direct-current generator a split-ring commutator exchanges the connections every half turn, so that the output keeps the same sign. An electric motor is the same device operated in reverse, with a current supplied to the coil from outside, and its rotating coil has induced in it a back electromotive force that opposes the supply.

The Transformer

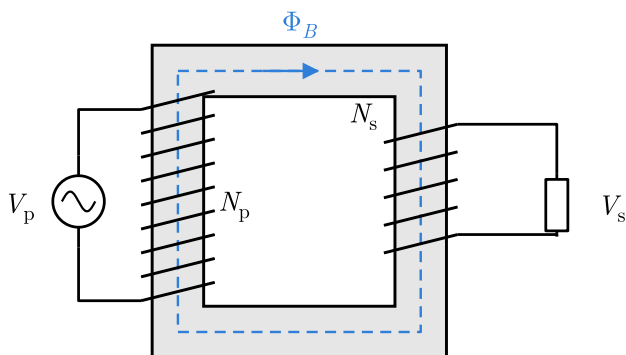


Figure 61. A transformer, with a primary coil of N_p turns connected to an alternating supply and a secondary coil of N_s turns connected to a load, both wound on a core of soft iron that guides the magnetic flux Φ_B (dashed) through both coils.

A transformer, drawn in Figure 61, has a primary coil of N_p turns and a secondary coil of N_s turns, wound on a common core of soft iron. The core guides the magnetic flux, so that, to a good approximation, the same flux passes through every turn of both coils. An alternating potential difference V_p applied to the primary coil produces an alternating flux in the core, and this flux induces the same electromotive force in each turn of both coils. The potential differences across the two coils are therefore in the ratio of their numbers of turns,

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}. \quad (7.5)$$

A transformer with $N_s > N_p$ is called a step-up transformer, and one with $N_s < N_p$ is called a step-down transformer. A steady current in the primary coil produces a steady flux, which induces nothing in the secondary coil, so a transformer works only with currents that change in time, such as alternating currents. An ideal transformer wastes no energy, so the power supplied to the primary coil is equal to the power taken from the secondary coil, $V_p I_p = V_s I_s$, and the currents are in the inverse ratio of the turns, $I_s/I_p = N_p/N_s$. A real transformer loses a small fraction of the power as heat in its coils and its core.

Transformers make it possible to transmit electrical power over long distances at a high voltage. For a given power, a higher voltage means a smaller current, and the power $I^2 R$ dissipated in the resistance R of the transmission cables is correspondingly smaller.

Alternating Current and Mains Electricity

A current that reverses its direction periodically is called an alternating current. The current delivered to a resistor by a generator of the kind described above is sinusoidal,

$$I = I_0 \sin \omega t, \quad (7.6)$$

where I_0 is the peak current, and the potential difference across the resistor is $V = V_0 \sin \omega t$, with $V_0 = I_0 R$. The frequency is $f = \omega/2\pi$ and the period is $T = 1/f$. Since the current is positive for half of each period and negative for the other half, its mean value is zero, and the size of an alternating current can be described instead by its root-mean-square value. For a sinusoid the root-mean-square values of the current and the potential difference are

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}}, \quad V_{\text{rms}} = \frac{V_0}{\sqrt{2}}. \quad (7.7)$$

The root-mean-square current is the steady current that would produce heat in a resistor at the same mean rate as the alternating current does, so the mean power $\langle \mathcal{P} \rangle$ delivered to a resistor is

$$\langle \mathcal{P} \rangle = I_{\text{rms}} V_{\text{rms}} = I_{\text{rms}}^2 R = \frac{V_{\text{rms}}^2}{R}, \quad (7.8)$$

which is half of the peak power $I_0 V_0$.

The mains supply is an alternating potential difference whose root-mean-square value and frequency are fixed by the supplier and differ from one country to another. The values marked on appliances, and displayed by most alternating-current meters, are root-mean-square values. A domestic circuit has a live wire, whose potential alternates; a neutral wire, which is kept close to the potential of the earth; and an earth wire, which is connected to any metal casing of an appliance, so that the casing cannot remain at a dangerous potential if a fault connects it to the live wire. A fuse or a circuit breaker in the live wire breaks the circuit if the current becomes too large.

7.2 Faraday's Law and Lenz's Law

The relation $\mathcal{E} = -d(N\Phi_B)/dt$ of Section 7.1, which is called the flux rule and for a single loop reads $\mathcal{E} = -d\Phi_B/dt$, covers two physically different situations: a conductor moving through a magnetic field that does not change in time, and a conductor at rest in a field that does. In this section we first derive the electromotive force of a moving conductor from the magnetic force on its charges, then show that a changing magnetic field is accompanied by an electric field, which gives Faraday's law in integral and in differential form, and finally show that the sign in the flux rule, which is Lenz's law, is required by the conservation of energy.

The Motional Electromotive Force

Section 4.2 defined the electromotive force round a circuit as the line integral of the force per unit charge that drives the current,

$$\mathcal{E} = \oint \frac{\mathbf{F}}{q} \cdot d\boldsymbol{\ell}. \quad (7.9)$$

As an example, consider the circuit drawn in Figure 62. The circuit consists of a pair of long parallel rails, a distance ℓ apart, lying along the x -direction in the xy -plane, joined at one end through a resistor R , and a conducting rod lies across them at a distance x from the resistor. A uniform field $\mathbf{B} = B\hat{\mathbf{z}}$ points out of the page, and the rod is pulled along the rails with velocity $\mathbf{v} = v\hat{\mathbf{x}}$. A charge q in the rod moves with the rod, and so experiences the magnetic force

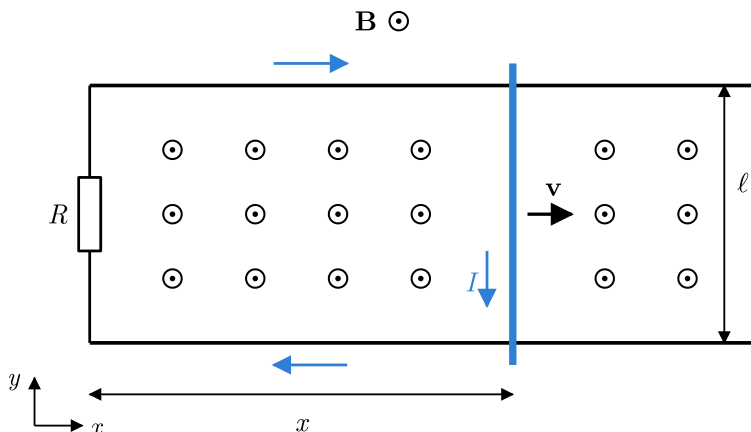


Figure 62. A conducting rod pulled with velocity $\mathbf{v}$ along a pair of rails a distance ℓ apart, which are joined at one end through a resistor R . The uniform field $\mathbf{B}$ points out of the page, and the induced current I circulates clockwise.

of Section 5.1,

$$q\mathbf{v} \times \mathbf{B} = qvB \hat{\mathbf{x}} \times \hat{\mathbf{z}} = -qvB \hat{\mathbf{y}}, \quad (7.10)$$

which is directed along the rod. The magnetic force that comes from the drift of the charges along the wires is perpendicular to the wires and contributes nothing to the line integral, so the electromotive force comes from the motion of the rod alone, where the force per unit charge has the constant magnitude vB along the length ℓ , and

$$\mathcal{E} = B\ell v, \quad (7.11)$$

which is the result quoted in Section 7.1. The force pushes positive charge along the rod in the $-\hat{\mathbf{y}}$ direction, so the current circulates clockwise as seen in Figure 62, and it has the value $I = B\ell v/R$ if R is the total resistance of the circuit.

The magnetic force on a moving charge is perpendicular to its velocity and does no work, and yet the circuit receives energy at the rate $\mathcal{E}I$. The resolution of this apparent contradiction is as follows. Once the current flows, a charge in the rod has a velocity $-v_d \hat{\mathbf{y}}$ along the rod as well as the velocity $v \hat{\mathbf{x}}$ of the rod, and the

magnetic force on it is

$$q(v\hat{\mathbf{x}} - v_d\hat{\mathbf{y}}) \times B\hat{\mathbf{z}} = -qvB\hat{\mathbf{y}} - qv_dB\hat{\mathbf{x}}. \quad (7.12)$$

The scalar product of this force with the total velocity is $-qv_dvB + qv_dvB = 0$, so the force does no work. Its first term does work on the charge at the rate qv_dvB as it pushes the charge along the rod, and its second term pushes the charge against the motion of the rod, so the agent pulling the rod must balance it, doing work at the same rate qv_dvB . The energy delivered to the circuit is therefore supplied by the agent, and the magnetic force only redirects it.

The Flux Rule

The flux of $\mathbf{B}$ through the circuit of Figure 62, with the normal along $\hat{\mathbf{z}}$, is $\Phi_B = B\ell x$, and its rate of change is $B\ell v$, the magnitude of the electromotive force found above. To compare the signs, we relate the direction of the line integral to the direction of the normal by the right-hand rule used for Stokes' theorem in Section 5.3: when the thumb of the right hand points along the normal, the fingers curl in the positive direction of circulation. With the normal along $\hat{\mathbf{z}}$, the positive direction is anticlockwise in Figure 62, which runs along the rod in the $+\hat{\mathbf{y}}$ direction, so the line integral of the force per unit charge is $-vB\ell$. Therefore

$$\mathcal{E} = -\frac{d\Phi_B}{dt} \quad (7.13)$$

for the sliding rod, in agreement with the flux rule.

To show that the flux rule holds for any loop of wire moving in a static field, we let each element $d\boldsymbol{\ell}$ of the loop move with a velocity $\mathbf{v}$ that may differ from one element to the next, so that the loop can change its shape as well as its position. In a time dt the element moves through $\mathbf{v}dt$ and sweeps out a small parallelogram, whose vector area is $(\mathbf{v}dt) \times d\boldsymbol{\ell}$, and the elements together sweep out the ribbon drawn in Figure 63. A surface spanning the loop at time t , the ribbon, and a surface spanning the loop at time $t + dt$ together enclose a volume, and Section 5.3 showed that the flux of $\mathbf{B}$ out of any closed surface is zero. The flux through the loop at $t + dt$ therefore differs from the flux at t by the flux through the ribbon,

$$d\Phi_B = \oint \mathbf{B} \cdot (\mathbf{v}dt \times d\boldsymbol{\ell}), \quad (7.14)$$

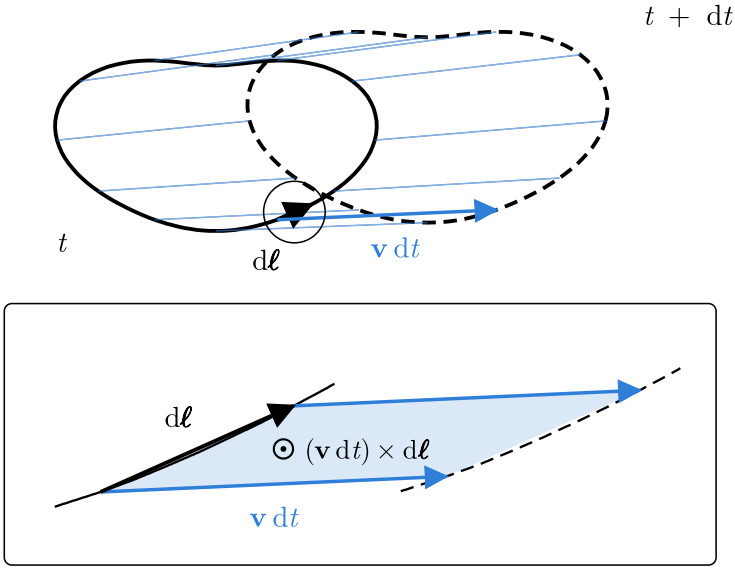


Figure 63. A loop of wire at time t (solid) and at time $t + dt$ (dashed). Each element $d\ell$ moves through $\mathbf{v} dt$, and together the elements sweep out a ribbon joining the two positions of the loop. The inset shows, at a larger scale, the parallelogram swept out by the circled element, and its normal $(\mathbf{v} dt) \times d\ell$, which points out of the page.

where the order of the factors in the vector product gives the ribbon the orientation for which the flux increases when the loop grows. This can be checked on the sliding rod, where $d\ell$ points along $+\hat{\mathbf{y}}$, $\mathbf{v} \times d\ell$ points along $+\hat{\mathbf{z}}$, and the integral is $B\ell v dt$.

To rewrite the integrand we use the identity $\mathbf{G} \cdot (\mathbf{X} \times \mathbf{Y}) = \mathbf{Y} \cdot (\mathbf{G} \times \mathbf{X})$, which says that a scalar triple product is unchanged when its three vectors are permuted cyclically, and which was used in Section 7.3 of Vol. 1 of the Fermi Notes on Physics. The identity holds because the scalar triple product is the determinant whose rows are the components of $\mathbf{G}$, $\mathbf{X}$, and $\mathbf{Y}$, and a cyclic permutation of the rows is made up of two exchanges of rows, each of which changes the sign of the determinant, as shown in Section 4.1 of Vol. 4 of the

Fermi Notes on Physics. Applying the identity, and then reversing the order of the vector product, gives $\mathbf{B} \cdot (\mathbf{v} \times d\boldsymbol{\ell}) = d\boldsymbol{\ell} \cdot (\mathbf{B} \times \mathbf{v}) = -(\mathbf{v} \times \mathbf{B}) \cdot d\boldsymbol{\ell}$, so that

$$\frac{d\Phi_B}{dt} = - \oint (\mathbf{v} \times \mathbf{B}) \cdot d\boldsymbol{\ell} = -\mathcal{E}. \quad (7.15)$$

The flux rule therefore holds for any loop of wire that moves or changes its shape in a static field, and since the flux of $\mathbf{B}$ through a closed surface vanishes, the flux may be computed through any surface bounded by the loop. The argument assumed that each element of the loop moves with the conducting material, as it does in a loop of wire and in the rod of Figure 62; where the path of the current moves through the material of a conductor, as it does in a metal disc that rotates between a contact at its rim and a contact at its axis, the electromotive force is best found by working out the force on the charges directly.

The Induced Electric Field

In one of Faraday's experiments of 1831, a magnet was moved towards a coil held at rest, and a current flowed in the coil, the same current as flows when the coil is moved towards a stationary magnet with the same relative velocity. For the stationary coil, however, the charges are at rest before the current starts, and the magnetic force $q\mathbf{v} \times \mathbf{B}$ on them is zero, so some other force must set them moving. A force on charges at rest is by definition an electric force, and we conclude that a magnetic field that changes in time is accompanied by an electric field. Experiment shows that the flux rule holds in this situation as well, with the electromotive force now given by the line integral of the electric field round the loop, so that for a loop at rest

$$\oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \int \mathbf{B} \cdot d\mathbf{a}, \quad (7.16)$$

where the surface integral is taken over any surface bounded by the loop, oriented by the right-hand rule. This relation is called Faraday's law. The electric field is present whether or not a wire lies along the loop, so Faraday's law holds for any closed path in space. The reason why moving the magnet and moving the coil give the same electromotive force lies in the relativity of electric and magnetic fields, which is not treated in this course.

The induced field differs in one respect from the electrostatic field of Chapter 1. Section 1.4 showed that the line integral of an electrostatic field round any closed path is zero, just as Section 2.3 of Vol. 1 showed that the work done round a closed path by a conservative force is zero, and this property allowed the potential to be defined there. The line integral of the induced field round a path enclosing a changing flux is not zero, so the induced field is not conservative and cannot be written as the gradient of a potential.

The Field of a Solenoid with a Changing Current

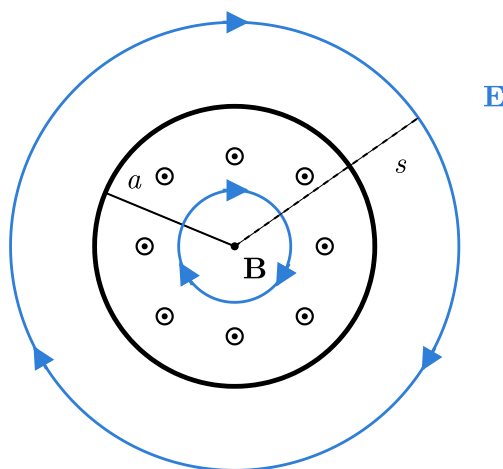


Figure 64. The cross-section of a long solenoid of radius a whose field $\mathbf{B}$, pointing out of the page, is increasing. The induced electric field $\mathbf{E}$ (accent) circles the axis clockwise, inside the solenoid and outside it, and s is the distance from the axis.

We next apply Faraday's law to a long solenoid of radius a with n turns per unit length, carrying a current I that increases slowly with time. Section 5.3 found the field inside the solenoid to be $\mu_0 n I \hat{\mathbf{z}}$ and the field outside to be zero, and provided the current changes slowly these expressions hold at each instant. The cross-section is drawn in Figure 64. By symmetry the magnitude of the induced field depends only on the distance s from the axis,

and a radial component is excluded by Gauss's law, since there is no charge to account for its flux through a coaxial cylinder, while a component along the axis is excluded by the symmetry of the solenoid under reflection in a plane perpendicular to its axis, which leaves the windings and their current unchanged but would reverse such a component. We therefore take the field lines to be circles about the axis, $\mathbf{E} = E_\phi(s)\hat{\phi}$, like the lines of $\mathbf{B}$ round a straight wire. Applying Faraday's law to a circle of radius $s < a$, taken anticlockwise as seen from the $+z$ direction so that its normal is $\hat{\mathbf{z}}$, gives

$$2\pi s E_\phi = -\pi s^2 \mu_0 n \frac{dI}{dt}, \quad (7.17)$$

so that inside the solenoid

$$E_\phi = -\frac{\mu_0 n s}{2} \frac{dI}{dt}. \quad (7.18)$$

A circle of radius $s > a$ encloses the whole flux $\pi a^2 \mu_0 n I$, and the same calculation gives, outside the solenoid,

$$E_\phi = -\frac{\mu_0 n a^2}{2s} \frac{dI}{dt}. \quad (7.19)$$

The calculation has the same structure as the application of Ampère's law to a wire in Section 5.3, with the circulation of the field round a circle fixed by what the circle encloses. The electric field outside the solenoid is not zero, although the magnetic field there is, because a circle of radius $s > a$ encloses the changing flux. For an increasing current the field lines circulate clockwise as seen from $+z$, opposite to the anticlockwise current in the windings, so a ring of wire round the solenoid would carry a current that opposes the increase of the flux, as Lenz's law requires.

Faraday's Law in Differential Form

Stokes' theorem, stated in Section 5.3, turns the line integral in Faraday's law into a surface integral of the curl, $\oint \mathbf{E} \cdot d\boldsymbol{\ell} = \int (\nabla \times \mathbf{E}) \cdot d\mathbf{a}$. The surface does not move, so the derivative with respect to time can be taken inside the surface integral, where it becomes a partial derivative, and Faraday's law becomes

$$\int \left(\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} \right) \cdot d\mathbf{a} = 0, \quad (7.20)$$

and since this holds for every surface, including very small surfaces of every orientation about any point, the integrand must vanish everywhere:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \quad (7.21)$$

This is Faraday's law in differential form. When the fields do not change in time it reduces to $\nabla \times \mathbf{E} = 0$, the result for electrostatic fields found in Section 5.3.

Let us check the differential form on the field inside the solenoid. In Cartesian components the unit vector $\hat{\phi}$ is $(-y/s, x/s, 0)$, so with the abbreviation $\dot{B} = \mu_0 n dI/dt$ for the rate of change of the magnetic field inside, the electric field there is $\mathbf{E} = -\frac{1}{2}\dot{B}(-y, x, 0)$, that is $E_x = \frac{1}{2}\dot{B}y$ and $E_y = -\frac{1}{2}\dot{B}x$. The curl of this field has only a z -component,

$$(\nabla \times \mathbf{E})_z = \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -\frac{\dot{B}}{2} - \frac{\dot{B}}{2} = -\dot{B}, \quad (7.22)$$

which is $-\partial B_z/\partial t$, as Faraday's law requires. You will show in Problem 2 that the field outside the solenoid has zero curl, in agreement with the vanishing of $\partial \mathbf{B}/\partial t$ there.

Lenz's Law and the Conservation of Energy

The minus sign in Faraday's law expresses Lenz's law, and the sliding rod of Figure 62 shows that this sign is required by the conservation of energy. The current $I = B\ell v/R$ in the rod experiences the force $I\boldsymbol{\ell} \times \mathbf{B}$ of Section 5.2, where the vector $\boldsymbol{\ell}$ has the length of the rod and points along the rod in the direction of the current, which is $-\hat{y}$. The force on the rod is therefore

$$\mathbf{F} = -I\ell B \hat{y} \times \hat{z} = -\frac{B^2 \ell^2 v}{R} \hat{x}, \quad (7.23)$$

which opposes the motion of the rod. An agent keeping the rod moving at constant velocity must push it with an equal and opposite force, doing work at the rate

$$\frac{B^2 \ell^2 v^2}{R} = I^2 R, \quad (7.24)$$

which is the rate at which heat is produced in the resistance of the circuit. With the opposite sign, the magnetic force on the rod

would point along its velocity, and a rod given a small push would accelerate without limit while heating the resistor, creating energy from nothing. In Problem 1 you will follow the energy of a rod that is released and allowed to slow down.

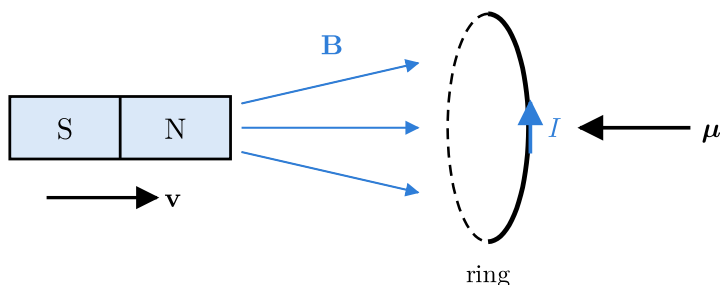


Figure 65. A bar magnet moving with velocity $\mathbf{v}$ towards a ring, with its north pole in front. The current I induced in the ring gives the ring a magnetic moment $\boldsymbol{\mu}$ pointing back towards the magnet, and the ring is repelled.

Let us now take a bar magnet moving towards a ring of wire with its north pole in front, as drawn in Figure 65. The flux through the ring, measured away from the magnet, increases, so the induced current gives the ring a magnetic moment $\boldsymbol{\mu}$ pointing back towards the magnet. Section 5.2 showed that a dipole in a non-uniform field experiences the force $\nabla(\boldsymbol{\mu} \cdot \mathbf{B})$, and here $\boldsymbol{\mu} \cdot \mathbf{B}$ is negative and larger in magnitude closer to the magnet, so the force pushes the ring away from the magnet, opposing the relative motion. In Elihu Thomson's jumping ring, an aluminium ring placed over the iron core of a coil is thrown upwards in the same way when the current in the coil is switched on.

The Rotating Coil

As a final example, we derive the electromotive force of the rotating coil of Section 7.1 from the force on its charges, for a rectangular coil of N turns, with sides of length ℓ_1 parallel to the axis of rotation

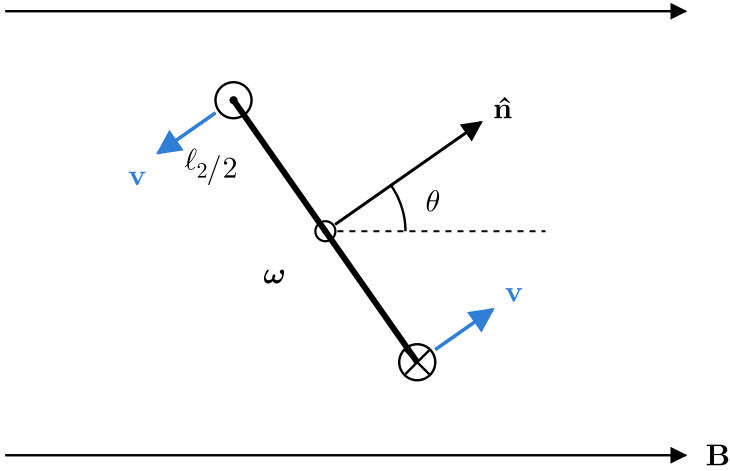


Figure 66. The rotating coil seen along its axis, which points out of the page. The sides parallel to the axis move with velocity $\mathbf{v}$ along $\pm\hat{\mathbf{n}}$, where $\hat{\mathbf{n}}$ is the normal to the coil at an angle θ to the field $\mathbf{B}$, and the force per unit charge on them points out of the page on one side and into it on the other.

and sides of length ℓ_2 perpendicular to it, rotating with angular velocity $\omega = \omega\hat{\mathbf{z}}$ about its central axis in a uniform field $\mathbf{B} = B\hat{\mathbf{x}}$. Figure 66 shows the coil seen along the axis, with its normal $\hat{\mathbf{n}} = (\cos\theta, \sin\theta, 0)$ at an angle $\theta = \omega t$ to the field. Each side of length ℓ_1 lies a distance $\ell_2/2$ from the axis and moves with speed $\omega\ell_2/2$ along $\pm\hat{\mathbf{n}}$. On the side whose velocity is $-\frac{1}{2}\omega\ell_2\hat{\mathbf{n}}$, the force per unit charge is

$$\mathbf{v} \times \mathbf{B} = -\frac{\omega\ell_2 B}{2} \hat{\mathbf{n}} \times \hat{\mathbf{x}} = \frac{\omega\ell_2 B}{2} \sin\theta \hat{\mathbf{z}}, \quad (7.25)$$

since $\hat{\mathbf{n}} \times \hat{\mathbf{x}} = -\sin\theta \hat{\mathbf{z}}$, and on the opposite side, which moves the other way, the force per unit charge is reversed. This force per unit charge is parallel to the axis, so it acts along the sides of length ℓ_1 , in the direction of travel round the turn on both of them, and across the sides of length ℓ_2 , which contribute nothing. Each turn therefore has an electromotive force $2 \times \ell_1 \times \frac{1}{2}\omega\ell_2 B \sin\theta$, and with

$A = \ell_1 \ell_2$ the coil has

$$\mathcal{E} = NBA\omega \sin \omega t, \quad (7.26)$$

which agrees with the flux rule applied to the flux linkage $NBA \cos \omega t$ in Section 7.1. You will show in Problem 4 that the mechanical power needed to turn the coil is equal to the electrical power that it delivers.

7.3 Inductance and Magnetic Energy

Section 4.1 stated the relation $\mathcal{E} = -L dI/dt$ for an inductor, and Section 4.3 used it to find how the current grows and decays in an RL circuit. Faraday's law, established in Section 7.2, now allows us to obtain this relation from the magnetic field of the circuit and to calculate the constant L for circuits of simple shape. The same argument applied to a pair of circuits gives their mutual inductance, which we take first. Following that, we calculate the energy stored in an inductor, and show that it can be regarded as stored in the magnetic field, with an energy density $B^2/2\mu_0$.

Mutual Inductance

Let us consider two separate circuits at rest, labelled 1 and 2. A current I_1 in circuit 1 produces a magnetic field $\mathbf{B}_1$, which the Biot-Savart law of Section 5.2 gives as an integral round circuit 1 with I_1 as a constant factor in front. The field at every point is therefore proportional to I_1 , and so is its flux through circuit 2. If circuit 2 is a coil of several turns, the relevant quantity is its flux linkage, the total flux through all of its turns, and we write

$$\Phi_{21} = L_{21} I_1, \quad (7.27)$$

where Φ_{21} is the flux linkage of circuit 2 due to the current in circuit 1. The constant L_{21} is called the mutual inductance of circuit 2 with respect to circuit 1, and it depends only on the shapes, sizes, and relative positions of the two circuits. If I_1 changes, Faraday's law gives an electromotive force in circuit 2,

$$\mathcal{E}_2 = -L_{21} \frac{dI_1}{dt}. \quad (7.28)$$

In the same way, a current I_2 in circuit 2 produces a flux linkage $\Phi_{12} = L_{12} I_2$ in circuit 1. The unit of inductance is the henry, with

$1 \text{ H} = 1 \text{ Wb A}^{-1}$, which is the same as 1 Vs A^{-1} . The primary and secondary coils of a transformer, described in Section 7.1, form a pair of circuits with a large mutual inductance, and the electromotive force in the secondary coil is of this kind.

It turns out that $L_{12} = L_{21}$ for any pair of circuits, so that a pair has a single mutual inductance. This is far from obvious, since the two flux linkages are computed from different fields through different surfaces. We first check the equality for a pair of coaxial solenoids, and then, later in this section, prove it in general from the conservation of energy.

Coaxial Solenoids

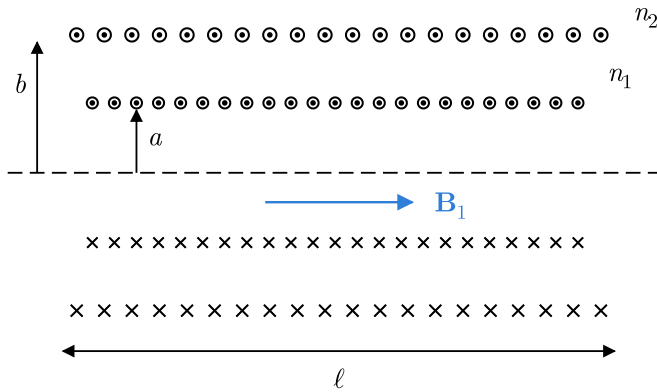


Figure 67. A pair of long coaxial solenoids of length ℓ , seen in section through the axis (dashed). The inner solenoid has radius a and n_1 turns per unit length, and the outer one has radius b and n_2 turns per unit length. The field $\mathbf{B}_1$ of the inner solenoid is confined within radius a .

For this check, consider the pair of long coaxial solenoids drawn in Figure 67. The inner solenoid has radius a and n_1 turns per unit length, the outer one has radius $b > a$ and n_2 turns per unit length, and both have a length ℓ that is large enough compared with b for the fields near the ends to be neglected. A current I_1 in the inner solenoid produces a field $\mu_0 n_1 I_1$ inside radius a and no

field outside it, by the result of Section 5.3. Each turn of the outer solenoid encloses the whole of this field, although the field fills only part of the area of the turn, so the flux through each outer turn is $\mu_0 n_1 I_1 \pi a^2$. The outer solenoid has $n_2 \ell$ turns, and therefore

$$L_{21} = \mu_0 n_1 n_2 \pi a^2 \ell. \quad (7.29)$$

A current I_2 in the outer solenoid produces a field $\mu_0 n_2 I_2$ everywhere inside radius b , including the interior of the inner solenoid. Each of the $n_1 \ell$ turns of the inner solenoid encloses only the part of this field that lies inside radius a , which is a flux $\mu_0 n_2 I_2 \pi a^2$, so

$$L_{12} = \mu_0 n_1 n_2 \pi a^2 \ell, \quad (7.30)$$

which is equal to L_{21} . The area πa^2 appears in the first calculation because the field of the inner solenoid is confined to radius a , and in the second because the turns of the inner solenoid enclose only that area.

Self-Inductance

A current in a single circuit also produces a flux through the circuit itself, and by the argument above the flux linkage of the circuit is proportional to its current. We write the flux linkage as LI , where the constant L is called the self-inductance, or simply the inductance, of the circuit. When the current changes, Faraday's law gives an electromotive force in the circuit itself,

$$\mathcal{E} = -L \frac{dI}{dt}, \quad (7.31)$$

which is the relation stated in Section 4.1. By Lenz's law this electromotive force opposes the change in the current, driving charge forwards when the current is falling and backwards when it is rising, and it is often called a back electromotive force. Every circuit has some self-inductance, since every current produces a field that threads the circuit carrying it, but in most circuits it can be neglected unless the current changes rapidly. A circuit component designed to have a large self-inductance, usually a coil of many turns, is called an inductor. We now work out the self-inductance for the following three examples.

The Inductance of a Solenoid

We start with a long solenoid of length ℓ and radius a , with $N = n\ell$ turns. Section 5.3 gives the field inside it as $B = \mu_0 nI$, so the flux through each turn is $\mu_0 nI\pi a^2$, and the flux linkage is $N\Phi_B = \mu_0 n^2 \pi a^2 \ell I$. The self-inductance is therefore

$$L = \mu_0 n^2 \pi a^2 \ell = \frac{\mu_0 N^2 A}{\ell}, \quad (7.32)$$

where $A = \pi a^2$ is the area of the cross-section. The inductance goes as the square of the number of turns, since doubling the number of turns on a given length doubles the field and also doubles the number of turns through which the field passes. If the solenoid is filled with a linear magnetic material, Section 6.2 showed that the field inside it becomes $\mu_r \mu_0 nI$, so the inductance is multiplied by μ_r , which is one reason why inductors are often wound on cores of magnetic material.

The Inductance of a Toroid

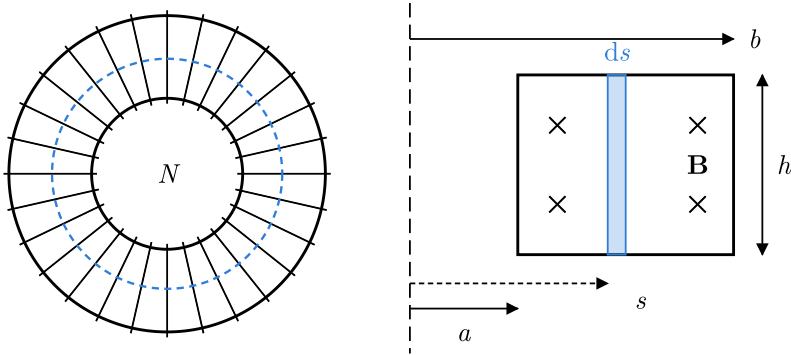


Figure 68. A toroid of N turns, seen from above (left), and its rectangular cross-section, of inner radius a , outer radius b , and height h , in a plane containing the axis (right). The field $\mathbf{B}$ crosses the section, and the strip of width ds at distance s from the axis (accent) carries the flux $Bh ds$. The dashed circle (accent) in the view from above is a line of $\mathbf{B}$ inside the ring.

For a second example, let us take a toroid of N turns wound on a ring of rectangular cross-section, with inner radius a , outer radius b , and height h , as drawn in Figure 68. Section 5.3 found from Ampère's law that the field inside the toroid circles the axis with magnitude $B = \mu_0 NI / 2\pi s$ at a distance s from the axis, and that the field outside the toroid is zero. The field is not uniform over the cross-section, so we slice the cross-section into strips of width ds and height h , parallel to the axis, over each of which the field is constant. The flux through one turn, which you found in Problem 4 of Section 5.3, where the height of the toroid was called ℓ , is

$$\Phi_B = \int_a^b \frac{\mu_0 NI}{2\pi s} h \, ds = \frac{\mu_0 NI h}{2\pi} \ln \frac{b}{a}, \quad (7.33)$$

and multiplying by the number of turns N and dividing by I gives

$$L = \frac{\mu_0 N^2 h}{2\pi} \ln \frac{b}{a}. \quad (7.34)$$

Let us check the limit of a thin toroid, with $b - a \ll a$. The Taylor expansion of the logarithm gives $\ln(b/a) = \ln[1 + (b - a)/a] \approx (b - a)/a$, and then $L \approx \mu_0 N^2 h(b - a) / 2\pi a$. This is the inductance of a solenoid with cross-sectional area $h(b - a)$ and length $2\pi a$, as it should be, since a thin toroid is a long solenoid bent round into a circle.

The Inductance of a Coaxial Cable

The third example is a coaxial cable made of an inner conducting tube of radius a and an outer conducting tube of radius b , with a current I flowing along the inner tube and back along the outer one, as drawn in Figure 69. We take the tubes to be thin, so that the current flows on their surfaces. Ampère's law, applied to circles about the axis as you did for a coaxial cable in Problem 1 of Section 5.3, gives a field of magnitude $\mu_0 I / 2\pi s$ circling the axis between the tubes. Inside the inner tube a circle encloses no current, and outside the outer tube it encloses a total current of zero, so the field vanishes in both regions. A length ℓ of the cable, closed at its two ends, forms a circuit that is threaded by the flux through the shaded rectangle of Figure 69, which lies in a plane containing the axis and runs from $s = a$ to $s = b$. Slicing this rectangle into strips

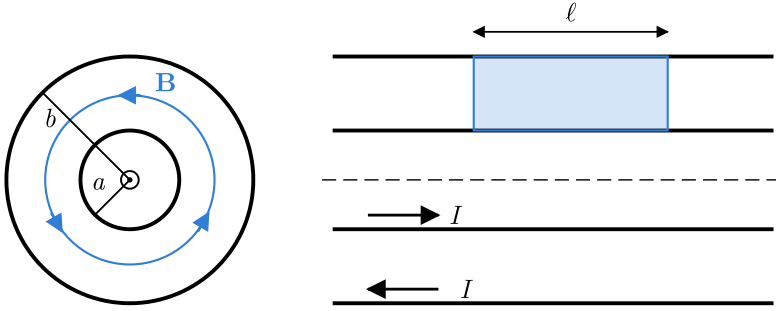


Figure 69. A coaxial cable made of an inner tube of radius a and an outer tube of radius b , seen end-on (left) with the circular lines of $\mathbf{B}$ between the tubes, and in section through the axis (right), with the current I flowing along the inner tube and back along the outer one. The flux through the shaded rectangle of length ℓ is the flux linking that length of cable.

of width ds and length ℓ , as for the toroid, gives

$$\Phi_B = \int_a^b \frac{\mu_0 I}{2\pi s} \ell ds = \frac{\mu_0 I \ell}{2\pi} \ln \frac{b}{a}, \quad (7.35)$$

so the inductance per unit length of the cable is

$$\frac{L}{\ell} = \frac{\mu_0}{2\pi} \ln \frac{b}{a}. \quad (7.36)$$

The Energy Stored in an Inductor

To set up a current I in an inductor, the source that drives the current must work against the back electromotive force. When the current has the value I' and is increasing at the rate dI'/dt , the back electromotive force has magnitude $L dI'/dt$, and the source does work on the charge passing through the inductor at the rate

$$\mathcal{P} = LI' \frac{dI'}{dt}. \quad (7.37)$$

Integrating over time as the current rises from zero to I gives the

total work done,

$$U = \int_0^I LI' dI' = \frac{1}{2}LI^2. \quad (7.38)$$

This energy is recovered when the current is reduced to zero, since the back electromotive force then drives the current forwards and returns the energy $\frac{1}{2}LI^2$ to the rest of the circuit. It is therefore stored in the inductor, as the energy $Q^2/2C$ found in Section 2.1 is stored in a capacitor. In the RL circuit of Section 4.3, whose current grows towards $\mathcal{E}/R$ with the time constant $\tau = L/R$, the energy supplied by the source is divided between heat produced in the resistor and the energy $\frac{1}{2}LI^2$ stored in the inductor, and when the source is removed the stored energy is converted into heat while the current decays. In Problem 2 you will check this division of the energy.

The Equality of the Mutual Inductances

The energy stored in two coupled circuits leads to a general proof that $L_{12} = L_{21}$. Let the two circuits have self-inductances L_1 and L_2 , and let their resistances be negligible, or let the heat produced in them be accounted for separately, so that the work done by the sources against the induced electromotive forces is the energy stored. Suppose that the currents are raised from zero to final values I_1 and I_2 in two stages.

In the first stage, I_1 is raised from zero while I_2 is held at zero. The source in circuit 1 does work $\frac{1}{2}L_1I_1^2$, and the source in circuit 2 does no work, since no current flows in circuit 2. In the second stage, I_1 is held fixed while I_2 is raised from zero. The source in circuit 2 does work $\frac{1}{2}L_2I_2^2$. The changing current in circuit 2 also induces an electromotive force $-L_{12}dI_2/dt$ in circuit 1, and to keep I_1 fixed the source in circuit 1 must work against it at the rate $I_1L_{12}dI_2/dt$, which gives a work $L_{12}I_1I_2$ over the second stage. The total work done is

$$U = \frac{1}{2}L_1I_1^2 + \frac{1}{2}L_2I_2^2 + L_{12}I_1I_2. \quad (7.39)$$

If the currents are raised in the other order, the same argument gives the same expression with L_{21} in place of L_{12} . The two orders end in the same state, with the same currents and the same fields,

and the energy stored in that state cannot depend on the order in which it was reached. If it did, we could raise the currents in the order that stores less energy and then lower them in the reverse of the other order, returning the circuits to their initial state with a net gain of energy, in contradiction with the conservation of energy. Therefore

$$L_{12} = L_{21} \quad (7.40)$$

for any pair of circuits, and the common value is called the mutual inductance of the pair.

The Energy Density of the Magnetic Field

Section 1.4 showed that the energy of a distribution of charge can be regarded as stored in its electric field, with an energy density $\frac{1}{2}\epsilon_0 E^2$. The energy of an inductor can be interpreted in a similar way. For a long solenoid, substituting $L = \mu_0 n^2 \pi a^2 \ell$ and $B = \mu_0 nI$ into $\frac{1}{2}LI^2$ gives

$$U = \frac{1}{2}\mu_0 n^2 I^2 \pi a^2 \ell = \frac{B^2}{2\mu_0} \pi a^2 \ell. \quad (7.41)$$

The field is uniform inside the solenoid and zero outside it, and $\pi a^2 \ell$ is the volume of the interior, so this energy is what we would find if the field carried an energy per unit volume

$$u = \frac{B^2}{2\mu_0}. \quad (7.42)$$

The toroid provides a check with a field that is not uniform. The part of the toroid between the radii s and $s + ds$ has volume $dV = 2\pi sh ds$, and integrating the energy density over the toroid gives

$$\int \frac{B^2}{2\mu_0} dV = \int_a^b \frac{1}{2\mu_0} \left(\frac{\mu_0 NI}{2\pi s} \right)^2 2\pi sh ds, \quad (7.43)$$

which is $(\mu_0 N^2 I^2 h / 4\pi) \ln(b/a)$, equal to $\frac{1}{2}LI^2$ with the inductance of the toroid found above. It can be shown that the energy of any system of steady currents is the integral of $B^2/2\mu_0$ over all space, but the proof uses the vector potential, which is not treated in this course. On this interpretation, the energy of an inductor is stored wherever its field is present, including the empty space inside the coil.

7.4 Alternating Current, Mean Power, and Rectification

The rotating coil of Section 7.2 produces an electromotive force that varies sinusoidally with time, and the mains supply described in Section 7.1 is of this kind. The subsections that follow describe first sinusoidal voltages and currents and the phase difference between them, then derive the mean power and the root-mean-square values quoted in Section 7.1, then describe how alternating voltages are measured with an oscilloscope, and finally show how an alternating supply is converted into a steady one by rectification and smoothing.

Sinusoidal Voltages and Currents

A sinusoidal potential difference can be written as

$$V(t) = V_0 \cos(\omega t + \psi), \quad (7.44)$$

where the amplitude V_0 is its peak value, ω is its angular frequency, and the angle ψ is called its phase constant, which fixes the value of V at $t = 0$. The frequency is $f = \omega/2\pi$, and the period is $T = 1/f = 2\pi/\omega$. Section 7.1 wrote alternating quantities with a sine in place of a cosine, and the two forms differ only in the phase constant, since $\sin \omega t = \cos(\omega t - \pi/2)$. A sinusoidal current is written in the same form, as $I(t) = I_0 \cos(\omega t + \psi')$ with a phase constant ψ' .

When two quantities oscillate at the same frequency, the difference between their phase constants is called their phase difference. If $V_a = V_{0a} \cos \omega t$ and $V_b = V_{0b} \cos(\omega t + \psi)$, with $0 < \psi < \pi$, then V_b reaches each of its maxima a time ψ/ω before V_a reaches the corresponding maximum. We say that V_b leads V_a by the phase ψ , or equivalently that V_a lags V_b by ψ . The current V/R in a resistor is in phase with the potential difference across it, since dividing by R changes only the amplitude.

A sum of two sinusoids of the same frequency, for any amplitudes and phase constants, is a third sinusoid of that frequency, as Section 1.2 of Vol. 4 shows. In particular, $V_1 \cos \omega t + V_2 \sin \omega t = V_0 \cos(\omega t - \psi)$, with $V_0 = \sqrt{V_1^2 + V_2^2}$ and $\tan \psi = V_2/V_1$, which follows from expanding $\cos(\omega t - \psi)$ with the identity for the cosine of a difference

of two angles.

Mean Power and Root-Mean-Square Values

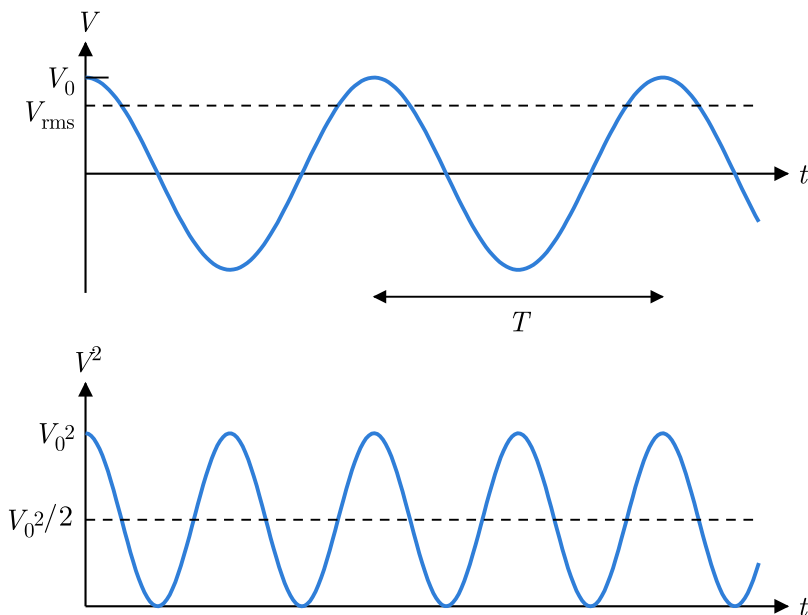


Figure 70. A sinusoidal potential difference $V = V_0 \cos \omega t$ (above), with its root-mean-square value V_{rms} marked, and its square (below), which oscillates at twice the frequency about its mean value $V_0^2/2$.

When a potential difference $V = V_0 \cos \omega t$ is applied across a resistor R , the power delivered to it is

$$\mathcal{P}(t) = \frac{V^2}{R} = \frac{V_0^2}{R} \cos^2 \omega t, \quad (7.45)$$

which oscillates between zero and V_0^2/R at twice the frequency of the supply, as the graph of V^2 in Figure 70 shows. Over a time long compared with a period, the energy delivered is fixed by the mean power $\langle \mathcal{P} \rangle$, the average of $\mathcal{P}$ over one period. The mean of a quantity over one period is its integral over the period divided by T . To find the mean of $\cos^2 \omega t$, we use the identity $\cos^2 \omega t = \frac{1}{2}(1 + \cos 2\omega t)$, which follows from the double-angle formula $\cos 2\omega t = 2 \cos^2 \omega t - 1$.

The integral of $\cos 2\omega t$ over a period is $[\sin 2\omega t/2\omega]_0^T = 0$, since $2\omega T = 4\pi$, and so

$$\langle \cos^2 \omega t \rangle = \frac{1}{T} \int_0^T \frac{1}{2} (1 + \cos 2\omega t) dt = \frac{1}{2}. \quad (7.46)$$

Similarly, the identity $\sin^2 \omega t = 1 - \cos^2 \omega t$ gives $\langle \sin^2 \omega t \rangle = \frac{1}{2}$, and the double-angle formula $\sin \omega t \cos \omega t = \frac{1}{2} \sin 2\omega t$ gives $\langle \sin \omega t \cos \omega t \rangle = 0$, since the integral of $\sin 2\omega t$ over a period also vanishes. The mean power delivered to the resistor is therefore

$$\langle \mathcal{P} \rangle = \frac{V_0^2}{2R}, \quad (7.47)$$

which is half of the peak power.

The root-mean-square value of a quantity that varies with time is the square root of the mean of its square, so that $V_{\text{rms}} = \sqrt{\langle V^2 \rangle}$. For a sinusoid, $\langle V^2 \rangle = V_0^2 \langle \cos^2 \omega t \rangle = V_0^2/2$, and the root-mean-square values of a sinusoidal potential difference and current are

$$V_{\text{rms}} = \frac{V_0}{\sqrt{2}}, \quad I_{\text{rms}} = \frac{I_0}{\sqrt{2}}, \quad (7.48)$$

which are the values quoted in Section 7.1. In terms of them, the mean power delivered to a resistor is

$$\langle \mathcal{P} \rangle = \frac{V_{\text{rms}}^2}{R} = I_{\text{rms}}^2 R = V_{\text{rms}} I_{\text{rms}}, \quad (7.49)$$

which has the same form as the power V^2/R delivered by a steady potential difference V . The reason for using root-mean-square values is that an alternating supply delivers the same mean power to a resistor as a steady supply whose potential difference equals the root-mean-square value of the alternating one, for any value of the resistance. The definition applies to any periodic waveform, and the factor $1/\sqrt{2}$ is particular to the sinusoid. A square wave that switches between $+V_0$ and $-V_0$, for example, has $V^2 = V_0^2$ at all times, so its root-mean-square value is V_0 . In Problem 1 you will find the root-mean-square value of a sawtooth waveform. The mean value of a sinusoid over a period is zero, since the integral of $\cos \omega t$ over a period vanishes like that of $\cos 2\omega t$, which is why its size can be described by the root-mean-square value instead.

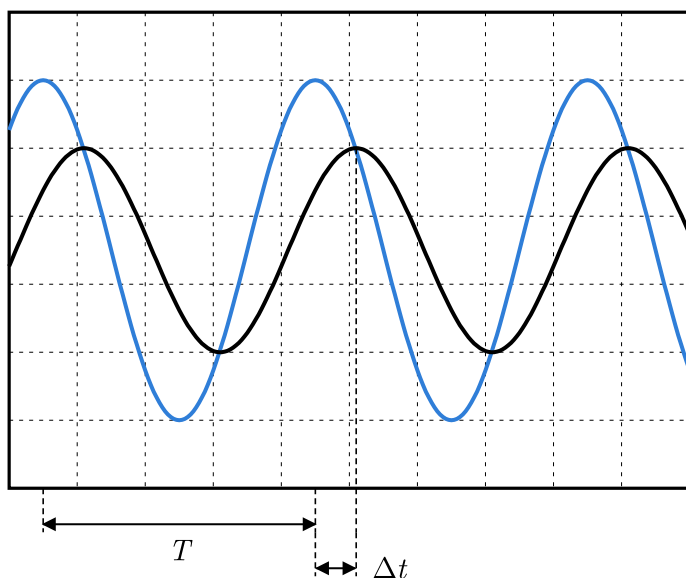


Figure 71. The screen of an oscilloscope showing two sinusoidal signals of the same frequency. The period T is the horizontal distance between successive peaks of one trace, and the trace of smaller amplitude reaches its peaks a time Δt after the other one.

The Oscilloscope Trace

An oscilloscope displays a potential difference as a function of time. In the older cathode-ray oscilloscope, a beam of electrons is deflected by the uniform electric fields between two pairs of parallel plates, as in Section 1.1, and strikes a fluorescent screen, which is coated with a material that glows where the electrons land, so that the beam makes a bright spot. A potential difference that rises steadily with time and then returns quickly to its starting value is applied to the plates that deflect the beam horizontally, and it sweeps the spot across the screen at constant speed, while the signal being measured is amplified and applied to the plates that deflect the beam vertically. A digital oscilloscope measures the signal at closely spaced instants and draws the same graph on a display. In either case the screen is ruled with a grid, as in Figure 71, and

two settings of the instrument convert distances on the grid into physical quantities. These are called the time base, which is the time represented by one horizontal division, and the vertical sensitivity, which is the potential difference represented by one vertical division.

The peak value of a sinusoidal signal is found by measuring the height of the trace from a trough to a peak, which is $2V_0$; this measurement does not depend on where the zero of the trace has been placed on the screen. The period is found from the horizontal distance between successive peaks, and the frequency is its reciprocal. Many oscilloscopes can display two signals together, and the phase difference between two signals of the same frequency is then found from the horizontal distance Δt between corresponding points of the two traces, such as their peaks. A delay Δt is the fraction $\Delta t/T$ of a period, and a whole period corresponds to a phase of 2π , so the phase difference is

$$\psi = \frac{2\pi \Delta t}{T}. \quad (7.50)$$

In Figure 71 the trace of smaller amplitude reaches its peaks a time Δt after the other trace, and so lags it by this phase. An oscilloscope also shows the shape of a waveform that is not sinusoidal, such as the output of the rectifiers treated next.

Half-Wave Rectification

Many devices need a steady potential difference, and converting an alternating supply into one whose sign does not change is called rectification. Rectification uses a diode, which is a circuit component that conducts current in one direction only. An ideal diode has zero resistance to a current in its forward direction, which is the direction of the arrow in its circuit symbol, and infinite resistance to a current in the reverse direction. A real semiconductor diode needs a small forward potential difference before it conducts appreciably, and we neglect this potential difference here, which is a valid approximation when the peak voltage of the supply is much larger.

The simplest rectifier, drawn at the upper left of Figure 72, is a single diode in series with the load resistor R and the supply

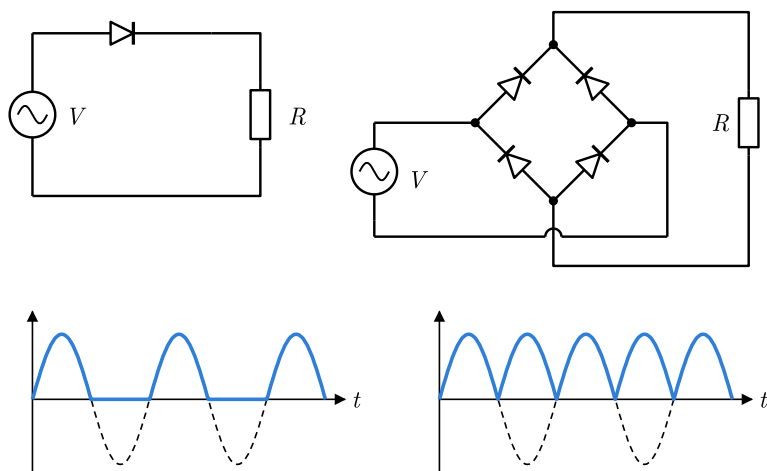


Figure 72. A half-wave rectifier (upper left), in which a single diode connects the supply V to the load R , and a bridge rectifier (upper right), in which four diodes connect the supply to the load. Below each circuit, the potential difference across the load (accent) is drawn with the supply voltage (dashed).

$V = V_0 \sin \omega t$. During the half-cycles in which V is positive the diode conducts, and the potential difference across the resistor is V . During the other half-cycles the diode does not conduct, and the potential difference across the resistor is zero. The output, drawn below the circuit, consists of positive half-cycles separated by gaps, and the arrangement is called a half-wave rectifier. The mean output over one period is

$$\langle V_{\text{out}} \rangle = \frac{1}{T} \int_0^{T/2} V_0 \sin \omega t \, dt = \frac{V_0}{\pi}, \quad (7.51)$$

since the integral is $(V_0/\omega)[- \cos \omega t]_0^{T/2} = 2V_0/\omega$, and $\omega T = 2\pi$. The square of the output is equal to $V_0^2 \sin^2 \omega t$ during the first half of the period and zero during the second half, and the integral of $\sin^2 \omega t$ over the first half-period is half of its integral over a whole period, since the graph of $\sin^2 \omega t$ repeats itself every half-period.

The mean of the square of the output is therefore half of $V_0^2/2$, and

$$\langle V_{\text{out}}^2 \rangle = \frac{V_0^2}{4}, \quad V_{\text{rms}} = \frac{V_0}{2}. \quad (7.52)$$

The mean power delivered to the load is $V_0^2/4R$, which is half of the mean power it would receive from the unrectified supply, because the supply delivers nothing during the half-cycles in which the diode blocks the current.

Full-Wave Rectification

A half-wave rectifier makes no use of the half-cycles of one sign. The arrangement drawn at the upper right of Figure 72, which is called a bridge rectifier, uses four diodes to pass both half-cycles to the load in the same direction. When the left terminal of the supply is positive, current flows from it through the upper-left diode to the top of the bridge, down through the load, back to the bottom of the bridge, and through the lower-right diode to the right terminal of the supply. When the right terminal is positive, current flows through the upper-right diode to the top of the bridge, down through the load in the same direction as before, and back through the lower-left diode to the left terminal. In each half-cycle the other pair of diodes has the potential difference across it in the reverse direction and carries no current. The potential difference across the load is therefore $|V_0 \sin \omega t|$, drawn below the bridge, and the arrangement is called a full-wave rectifier. You will show in Problem 3 that its mean output is $2V_0/\pi$ and that its root-mean-square output is $V_0/\sqrt{2}$, the same as that of the unrectified supply, which is to be expected since the square of the output is equal to the square of the supply at every instant.

Smoothing with a Capacitor

The output of a rectifier has a fixed sign, but its size varies with time. It can be smoothed by connecting a capacitor C across the load resistor R , as drawn for a half-wave rectifier at the left of Figure 73. The half-wave rectifier is treated first. As the supply rises towards its peak, the diode conducts and charges the capacitor, whose potential difference follows the supply up to V_0 . Once the supply has passed its peak and begins to fall, the side of the diode

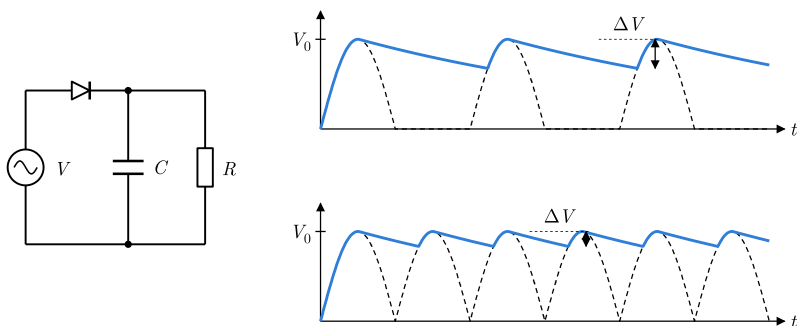


Figure 73. A half-wave rectifier with a smoothing capacitor C across the load R (left), together with the output of a half-wave rectifier (above right) and of a full-wave rectifier (below right) with such a capacitor, computed for the same time constant RC and drawn with the unsmoothed output (dashed). After the first peak the output falls slowly between peaks, and the size of this fall is the ripple ΔV .

connected to the capacitor is at the higher potential, so the diode stops conducting, and the capacitor discharges through the load. Section 4.3 showed that the potential difference across a capacitor discharging through a resistance R is

$$V_C = V_0 e^{-t'/\tau}, \quad \tau = RC, \quad (7.53)$$

where t' is the time since the discharge began. The discharge continues until the rising supply voltage on the next cycle is equal to the capacitor voltage, when the diode conducts again and recharges the capacitor to V_0 . The resulting output, computed and drawn in Figure 73, falls and rises by a small amount ΔV in every period, which is called the ripple.

To estimate the ripple, suppose that the time constant is long compared with the period, $RC \gg T$. The capacitor voltage then falls only slightly during one period, so the rising supply is equal to it shortly before the next peak, and the discharge lasts for nearly a whole period T . Expanding the exponential to first order, $e^{-T/\tau} \approx 1 - T/\tau$, gives

$$\Delta V \approx \frac{V_0 T}{RC} = \frac{V_0}{fRC}. \quad (7.54)$$

The same result follows from the charge on the capacitor. During the discharge the load current is close to V_0/R , so in a time T the capacitor loses a charge V_0T/R , and its potential difference falls by this charge divided by C . In a full-wave rectifier the capacitor is recharged twice in every period, and each discharge lasts only about $T/2$, so the ripple is halved,

$$\Delta V \approx \frac{V_0}{2fRC}, \quad (7.55)$$

as the graph at the lower right of Figure 73 shows. The ripple is reduced by a larger capacitance, and it grows when the load draws a larger current, which happens when R is smaller. The mean output is close to $V_0 - \frac{1}{2}\Delta V$, and it approaches the peak voltage of the supply as the ripple is reduced. The diode conducts only for a short time before each peak, and in that time it must replace all the charge that the load has drawn from the capacitor during the whole period, so while it conducts it carries a current much larger than the mean current in the load. In Problem 4 you will estimate the time for which it conducts. Where a steadier output is needed, further filtering is often added, or an electronic regulator, which is a circuit that holds its output at a fixed potential difference while its input and the current drawn by the load vary.

7.5 Phasors and Complex Impedance

In the circuits of Section 4.3, which contained a resistor with either a capacitor or an inductor, the current rose or fell exponentially. In a circuit containing both a capacitor and an inductor, energy can pass back and forth between the electric field of the capacitor and the magnetic field of the inductor, and the current oscillates. We begin by studying these oscillations, which obey the same equations as the mechanical oscillators of Sections 4.2 and 4.3 of Vol. 1, and then develop the method of complex amplitudes, or phasors, which turns the analysis of a circuit driven by a sinusoidal source into algebra. The following section applies the method to resonance and to the power delivered to a circuit.

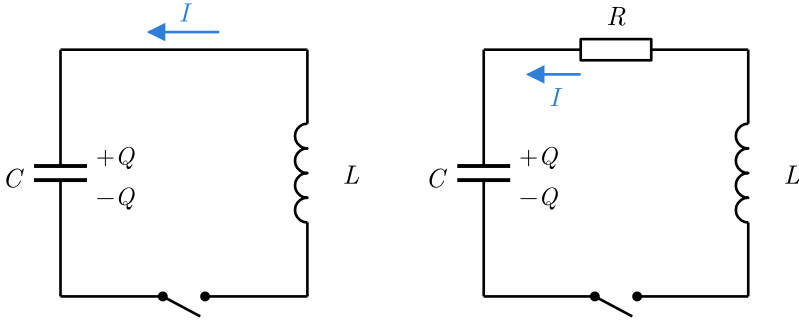


Figure 74. An LC circuit (left) and a series LCR circuit (right). The capacitor carries a charge Q on its upper plate, and the current I is taken as positive in the direction that carries charge onto that plate. Each circuit is completed by closing the switch at $t = 0$.

The LC Circuit

Let a capacitor C whose upper plate carries a charge Q_0 be connected at $t = 0$ across an inductor L by closing the switch in the left-hand circuit of Figure 74. The resistance of the circuit is taken to be negligible. Let Q be the charge on the upper plate at time t , and let I be the current round the circuit, taken as positive in the direction that carries charge onto the upper plate, so that $I = dQ/dt$. Going round the circuit in the direction of I , the potential falls by $L dI/dt$ across the inductor, as in Section 4.3, and by Q/C across the capacitor, from its upper plate to its lower plate. Kirchhoff's loop law requires the total change of potential round the circuit to be zero, so that

$$L \frac{dI}{dt} + \frac{Q}{C} = 0, \quad (7.56)$$

and substituting $I = dQ/dt$ gives

$$\frac{d^2Q}{dt^2} = -\omega_0^2 Q, \quad \omega_0 = \frac{1}{\sqrt{LC}}. \quad (7.57)$$

This is the equation of simple harmonic motion, whose solution was worked out in Section 4.2 of Vol. 1. With $Q = Q_0$ and $I = 0$ at

$t = 0$, the solution is

$$Q = Q_0 \cos \omega_0 t, \quad I = -\omega_0 Q_0 \sin \omega_0 t. \quad (7.58)$$

The angular frequency ω_0 is called the resonant angular frequency of the circuit, and the current, which can be written as $\omega_0 Q_0 \cos(\omega_0 t + \pi/2)$, is a quarter of a period ahead of the charge.

The energy of the circuit is the sum of the energy $Q^2/2C$ of the capacitor, found in Section 2.1, and the energy $\frac{1}{2}LI^2$ of the inductor, found in Section 7.3. Substituting the solution gives

$$\frac{Q^2}{2C} + \frac{1}{2}LI^2 = \frac{Q_0^2}{2C} \cos^2 \omega_0 t + \frac{1}{2}L\omega_0^2 Q_0^2 \sin^2 \omega_0 t, \quad (7.59)$$

and since $L\omega_0^2 = 1/C$, the total is $Q_0^2/2C$ at all times, while the energy passes from the capacitor to the inductor and back twice in every period.

The LC circuit is an electrical analogue of a mass m on a spring of stiffness κ , whose equation of motion $m d^2x/dt^2 = -\kappa x$ becomes the equation of the circuit when the displacement x is replaced by the charge Q , the velocity by the current I , the mass by the inductance L , and the stiffness by $1/C$. The kinetic energy $\frac{1}{2}mv^2$ then becomes the magnetic energy $\frac{1}{2}LI^2$, and the potential energy $\frac{1}{2}\kappa x^2$ becomes the electric energy $Q^2/2C$.

The Damped LCR Circuit

A real circuit has resistance, and the right-hand circuit of Figure 74 includes a resistor R in series with the capacitor and the inductor. Going round the circuit, the potential now also falls by IR across the resistor, and Kirchhoff's loop law gives

$$L \frac{d^2Q}{dt^2} + R \frac{dQ}{dt} + \frac{Q}{C} = 0. \quad (7.60)$$

This has the form of the equation of motion of the damped oscillator of Section 4.3 of Vol. 1, a mass m on a spring of stiffness κ subject to a resistive force proportional to its velocity, with the resistance R corresponding to the constant of proportionality between the resistive force and the velocity. The resistor removes energy from the circuit as the resistive force does from the oscillator; differentiating

the total energy with respect to time, with the product rule and then the circuit equation, gives

$$\frac{d}{dt} \left(\frac{Q^2}{2C} + \frac{1}{2} L I^2 \right) = I \left(\frac{Q}{C} + L \frac{dI}{dt} \right) = -I^2 R, \quad (7.61)$$

so the energy of the circuit decreases at the rate at which heat is produced in the resistor.

Dividing the circuit equation by L and letting $\gamma = R/L$ gives

$$\frac{d^2 Q}{dt^2} + \gamma \frac{dQ}{dt} + \omega_0^2 Q = 0. \quad (7.62)$$

This is the equation of the damped oscillator solved in Section 4.3 of Vol. 1 and in Section 5.2 of Vol. 4, whose damping constant corresponds here to $\gamma/2 = R/2L$. When $R < 2\sqrt{L/C}$, so that $\gamma < 2\omega_0$, the circuit is lightly damped, and the charge oscillates as $Q = Q_1 e^{-\gamma t/2} \cos(\omega_1 t + \psi)$ with $\omega_1 = \sqrt{\omega_0^2 - \gamma^2/4}$; when $R > 2\sqrt{L/C}$ it is heavily damped, and the charge decays without oscillating; and when $R = 2\sqrt{L/C}$ it is critically damped, with $Q = (Q_1 + Q_2 t)e^{-\gamma t/2}$, which returns to zero without oscillating and sooner than in a heavily damped circuit with the same inductance and capacitance.

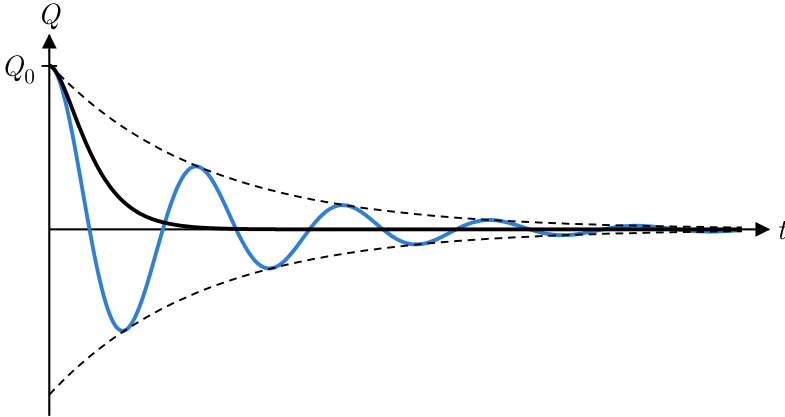


Figure 75. The charge on the capacitor of a series LCR circuit released with charge Q_0 and no current, for light damping (accent), with its decaying envelope (dashed), and for critical damping.

The discharge of a capacitor in the lightly damped and the critically damped cases is drawn in Figure 75. In the lightly damped case the energy of the circuit goes as the square of the amplitude, so averaged over a period it decays as $e^{-\gamma t}$, and $1/\gamma = L/R$, the time in which the energy falls by a factor e , is the time constant τ of the RL circuit of Section 4.3.

Complex Exponential Currents and Voltages

We now drive the circuit with a source whose potential difference varies sinusoidally, $V(t) = V_0 \cos \omega t$. The equations of a circuit of resistors, capacitors, and inductors are linear with real constant coefficients, and Section 4.2 of Vol. 1 showed that such equations can be solved with complex exponentials, taking the real part at the end, since the real part of a complex solution is itself a solution. The complex exponential and phasors are treated at greater length in Section 1.2 of Vol. 4. For the driven oscillator, Section 4.3 of Vol. 1 found that the motion settles, once the transient has died away, into an oscillation at the driving frequency. In the same way, once the solutions of the undriven circuit have decayed, every current and potential difference in a driven circuit oscillates with the angular frequency ω of the source, each with an amplitude and a phase constant to be found.

We therefore write the potential difference of the source as the real part of a complex exponential,

$$V(t) = \operatorname{Re} [\tilde{V} e^{i\omega t}] , \quad (7.63)$$

where the complex number $\tilde{V}$ is called the complex amplitude of V . Writing $\tilde{V}$ in polar form as $V_0 e^{i\psi}$ gives $V(t) = V_0 \cos(\omega t + \psi)$, so the modulus of the complex amplitude is the peak value and its argument is the phase constant; for the source above, $\tilde{V} = V_0$. Currents are written in the same way, $I(t) = \operatorname{Re} [\tilde{I} e^{i\omega t}]$. Differentiating $\tilde{I} e^{i\omega t}$ with respect to time multiplies it by $i\omega$, so once the common factor $e^{i\omega t}$ has been cancelled, the differential equations of the circuit become algebraic equations between complex amplitudes.

The method relies on linearity, since the real part of a sum is the sum of the real parts. It cannot be used for products, since the real part of a product of two complex numbers generally differs from

the product of their real parts, so a product such as the power VI has to be calculated from the real current and potential difference.

Phasors

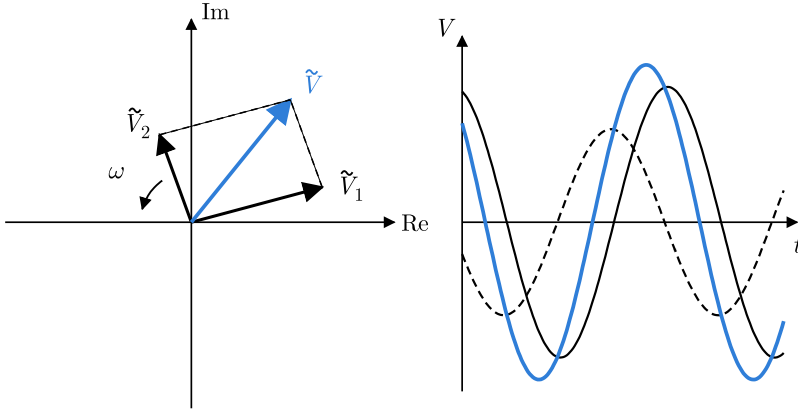


Figure 76. The phasors $\tilde{V}_1$ and $\tilde{V}_2$ and their sum $\tilde{V}$ (accent), which is the diagonal of the parallelogram that they form. The diagram rotates anticlockwise at the angular velocity ω . On the right are the corresponding sinusoids, the real parts of the rotating phasors, with the sinusoid of $\tilde{V}_1$ drawn solid, that of $\tilde{V}_2$ dashed, and their sum in the accent colour.

The complex number $\tilde{V}e^{i\omega t}$, drawn as an arrow from the origin in the complex plane, is called a phasor; it rotates anticlockwise with angular velocity ω , and its projection on the real axis is $V(t)$, as Section 1.2 of Vol. 4 shows. A diagram of such arrows drawn at $t = 0$, where they are the complex amplitudes themselves, is called a phasor diagram, and a phasor ahead of another in the anticlockwise direction leads it. The phasor of the sum of two sinusoids of the same frequency is the vector sum of their phasors, as drawn in Figure 76, which gives the result of Section 7.4 on the sum of two sinusoids as the diagonal of a parallelogram. In these notes the length of a phasor is the peak value, although some books use the root-mean-square value.

The Impedances of a Resistor, a Capacitor, and an Inductor

For a circuit component carrying a current I , let V be the potential difference across it, measured as the fall in potential in the direction of I . The ratio of the complex amplitudes of these two quantities,

$$Z = \frac{\tilde{V}}{\tilde{I}}, \quad (7.64)$$

is called the complex impedance of the circuit component. It is measured in ohms, and it depends on the angular frequency but not on time. Its modulus is the ratio of the peak potential difference to the peak current, and its argument is the phase by which the potential difference leads the current. We now find the impedances of the following three circuit components.

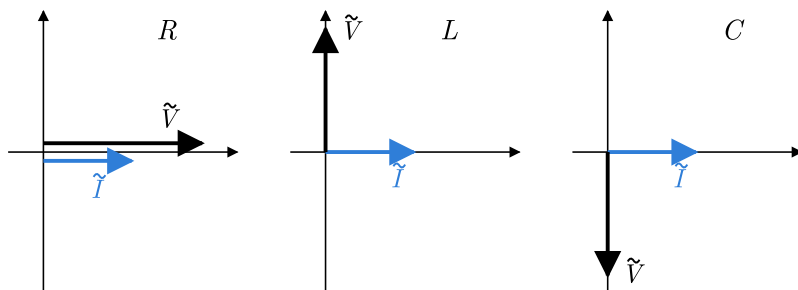


Figure 77. Phasor diagrams for a resistor, an inductor, and a capacitor, with the current phasor $\tilde{I}$ (accent) drawn along the real axis. The potential difference $\tilde{V}$ is in phase with the current for the resistor, leads it by a quarter of a period for the inductor, and lags it by a quarter of a period for the capacitor.

For a resistor, $V = IR$ at every instant, so $\tilde{V} = R\tilde{I}$ and

$$Z_R = R, \quad (7.65)$$

with the potential difference and the current in phase.

For an inductor, $V = L dI/dt$, and the derivative with respect to time becomes a factor $i\omega$, so $\tilde{V} = i\omega L\tilde{I}$ and

$$Z_L = i\omega L. \quad (7.66)$$

Since $i = e^{i\pi/2}$, the potential difference across an inductor leads the current by a quarter of a period. The modulus $X_L = \omega L$ is called the reactance of the inductor. The reactance increases with the angular frequency, because the back electromotive force is proportional to the rate of change of the current, which for a given amplitude is proportional to ω .

For a capacitor, let Q be the charge on the plate into which the current I flows. Then $V = Q/C$ and $I = dQ/dt = C dV/dt$, so $\tilde{I} = i\omega C \tilde{V}$ and

$$Z_C = \frac{1}{i\omega C} = -\frac{i}{\omega C}. \quad (7.67)$$

The current through a capacitor therefore leads the potential difference across it by a quarter of a period, and the reactance of the capacitor, $X_C = 1/\omega C$, decreases with the angular frequency. At low frequency the charge CV_0 needed to bring the capacitor to the peak potential difference is delivered over a long time, so the current is small, while at high frequency the same charge must be delivered in a short time, so the current is large. The phasors of the current and the potential difference for the three circuit components are drawn in Figure 77.

Kirchhoff's Laws with Complex Impedances

Kirchhoff's junction law states that the currents flowing into a junction add up to zero at every instant, so that

$$\sum_j \operatorname{Re} [\tilde{I}_j e^{i\omega t}] = \operatorname{Re} \left[\left(\sum_j \tilde{I}_j \right) e^{i\omega t} \right] = 0. \quad (7.68)$$

At $t = 0$ this says that the real part of $\sum_j \tilde{I}_j$ vanishes, and at $\omega t = \pi/2$, where $e^{i\omega t} = i$, it says that the imaginary part vanishes, so the complex amplitudes themselves satisfy the junction law, $\sum_j \tilde{I}_j = 0$. The loop law holds for complex amplitudes in the same way. With $\tilde{V} = Z\tilde{I}$ for each circuit component, the formulas for combining resistors in series and in parallel, which follow from Kirchhoff's laws and from the proportionality of V and I alone, carry over to impedances. Circuit components in series carry the same current, and their potential differences add, so their combined impedance is

$$Z = Z_1 + Z_2. \quad (7.69)$$

Circuit components in parallel have the same potential difference, and their currents add, so

$$\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2}. \quad (7.70)$$

Once the impedance $Z = |Z|e^{i\psi}$ of a network connected to a source $\tilde{V} = V_0$ is known, the current drawn from the source is $\tilde{I} = V_0/Z = (V_0/|Z|)e^{-i\psi}$. The current therefore has the peak value $V_0/|Z|$, and it lags the potential difference of the source by the phase ψ .

Simple Filters

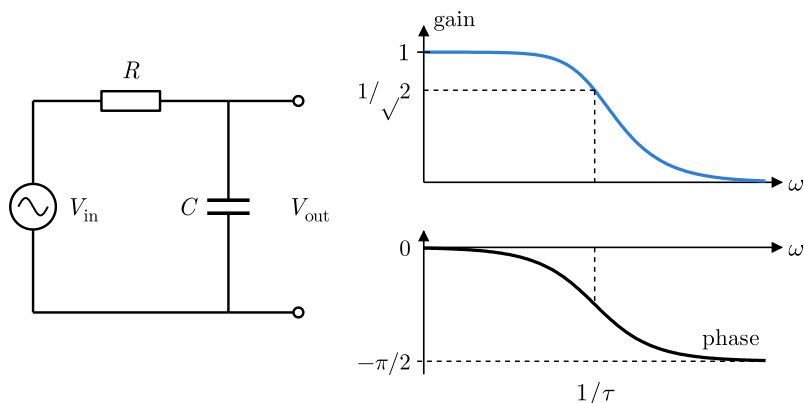


Figure 78. A low-pass filter made of a resistor R and a capacitor C (left), and its gain and the phase of its output relative to its input, plotted against the angular frequency on a logarithmic scale (right). At $\omega = 1/\tau$, where $\tau = RC$, the gain is $1/\sqrt{2}$ and the output lags by $\pi/4$.

As an example of the method, consider the circuit of Figure 78, in which a source with complex amplitude $\tilde{V}_{\text{in}}$ is connected across a resistor R and a capacitor C in series, and the output $\tilde{V}_{\text{out}}$ is taken across the capacitor. We suppose that the output is connected to an instrument that draws a negligible current, such as an oscilloscope, so that the resistor and the capacitor carry the same current. The current is $\tilde{V}_{\text{in}}/(Z_R + Z_C)$, and the output is Z_C times the current, so

$$\frac{\tilde{V}_{\text{out}}}{\tilde{V}_{\text{in}}} = \frac{Z_C}{Z_R + Z_C} = \frac{1}{1 + i\omega RC} = \frac{1}{1 + i\omega\tau}, \quad (7.71)$$

where $\tau = RC$ is the time constant of Section 4.3, and the second step multiplies the numerator and the denominator by $i\omega C$. The ratio of the output amplitude to the input amplitude is called the gain of the circuit, and it is the modulus of this expression,

$$\frac{|\tilde{V}_{\text{out}}|}{|\tilde{V}_{\text{in}}|} = \frac{1}{\sqrt{1 + \omega^2 \tau^2}}. \quad (7.72)$$

The argument of $1/(1 + i\omega\tau)$ is minus the argument of $1 + i\omega\tau$, so the output lags the input by the phase $\arctan \omega\tau$. The gain and the phase are drawn in Figure 78. When $\omega\tau \ll 1$ the gain is close to one, and the output follows the input, since the capacitor has time to charge fully in each cycle. When $\omega\tau \gg 1$ the gain goes as $1/\omega\tau$, and the output is small and lags the input by nearly a quarter of a period. The circuit therefore passes low frequencies and suppresses high ones, and it is called a low-pass filter. At $\omega = 1/\tau$ the gain is $1/\sqrt{2}$, so the mean power that the output would deliver to a given load is half its low-frequency value, and this angular frequency, often taken as the boundary between the two regions, is called the cut-off frequency. A low-pass filter can be used, for example, to remove rapidly varying noise from a slowly varying signal. You will show in Problem 3 that the same circuit components, with the output taken across the resistor, form a high-pass filter, which passes high frequencies and suppresses low ones.

7.6 Resonance and Power in Alternating-Current Circuits

A circuit that contains an inductor and a capacitor has a resonant angular frequency $\omega_0 = 1/\sqrt{LC}$, found in Section 7.5, at which its charge oscillates when the circuit is left to itself. When such a circuit is driven by a sinusoidal source, its response depends strongly on how close the angular frequency of the source is to ω_0 , as the response of the driven mechanical oscillator of Section 4.3 of Vol. 1 depends on how close the driving frequency is to the natural frequency. Here we work out this response with the complex impedances of Section 7.5, first for a series circuit and then for a parallel one, and finally express the mean power delivered to a circuit in terms of complex amplitudes.

The Driven Series LCR Circuit

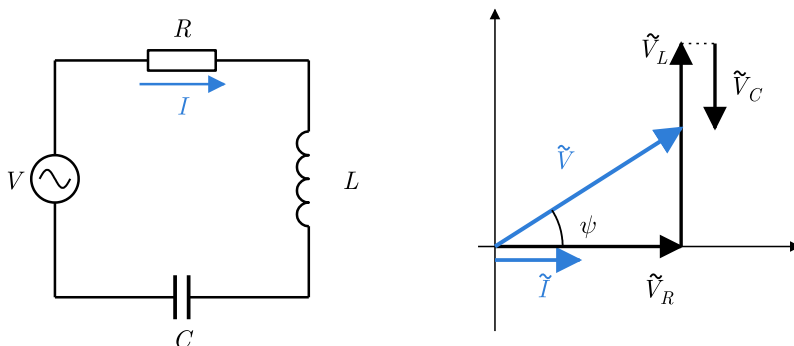


Figure 79. A series LCR circuit driven by a sinusoidal source V (left), and its phasor diagram (right) for $\omega L > 1/\omega C$. The current phasor $\tilde{I}$ (accent) lies along the real axis, the phasors $\tilde{V}_R$, $\tilde{V}_L$, and $\tilde{V}_C$ are added head to tail, and their sum $\tilde{V}$ (accent) leads the current by the angle ψ .

We now connect a resistor R , an inductor L , and a capacitor C in series with a source $V(t) = V_0 \cos \omega t$, as drawn in Figure 79. The three circuit components carry the same current, and their impedances add, so the circuit has the impedance

$$Z = R + i\omega L + \frac{1}{i\omega C} = R + i\left(\omega L - \frac{1}{\omega C}\right). \quad (7.73)$$

The modulus and the argument of this impedance are given by

$$|Z| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}, \quad \tan \psi = \frac{\omega L - 1/\omega C}{R}, \quad (7.74)$$

with ψ between $-\pi/2$ and $\pi/2$, since the real part of Z is positive. The complex amplitude of the current is $\tilde{I} = V_0/Z$, so the current is

$$I(t) = \frac{V_0}{|Z|} \cos(\omega t - \psi), \quad (7.75)$$

which has the peak value $I_0 = V_0/|Z|$ and lags the source by the phase ψ . When $\omega L > 1/\omega C$, the reactance of the inductor is the larger one, ψ is positive, and the current lags the source; when

$\omega L < 1/\omega C$, the reactance of the capacitor is the larger one, ψ is negative, and the current leads the source.

The phasor diagram of Figure 79 shows the same result geometrically. The current phasor $\tilde{I}$ is drawn along the real axis. The potential difference across the resistor, $\tilde{V}_R = R\tilde{I}$, lies along it; the potential difference across the inductor, $\tilde{V}_L = i\omega L\tilde{I}$, points a quarter turn ahead of it; and the potential difference across the capacitor, $\tilde{V}_C = \tilde{I}/i\omega C$, points a quarter turn behind it. By the loop law, the phasor $\tilde{V}$ of the source is the sum of these three phasors. Since $\tilde{V}_L$ and $\tilde{V}_C$ point in opposite directions, their sum has length $|\omega L - 1/\omega C|I_0$, and this sum and $\tilde{V}_R$ are the two shorter sides of a right-angled triangle whose hypotenuse is $\tilde{V}$. The length of the hypotenuse is $|Z|I_0$, and the angle between $\tilde{V}$ and $\tilde{I}$ is ψ .

This current is called the steady-state solution. The general solution adds to it the solutions of the undriven circuit found in Section 7.5, which in the lightly damped case decay as $e^{-\gamma t/2}$, so after a time long compared with $2/\gamma = 2L/R$ only the steady state remains.

Resonance

The peak current $I_0 = V_0/|Z|$ is largest when $|Z|$ is smallest, which happens when the reactances of the inductor and the capacitor are equal, $\omega L = 1/\omega C$, that is when

$$\omega = \omega_0 = \frac{1}{\sqrt{LC}}, \quad (7.76)$$

the resonant angular frequency of the undamped circuit. At this angular frequency $Z = R$, the peak current is V_0/R , and the current is in phase with the source. The large response of the circuit near ω_0 is called resonance. Away from resonance the current falls, towards zero at low frequencies, where the reactance of the capacitor becomes large, and towards zero at high frequencies, where the reactance of the inductor becomes large. The peak current and the phase ψ are plotted against ω in Figure 80.

The corresponding result for the mechanical oscillator in Section 4.3 of Vol. 1 differs in one respect, since there the amplitude of the displacement has its maximum slightly below the natural frequency. The current in the circuit corresponds to the velocity of the mechanical oscillator, and the amplitude of the velocity has

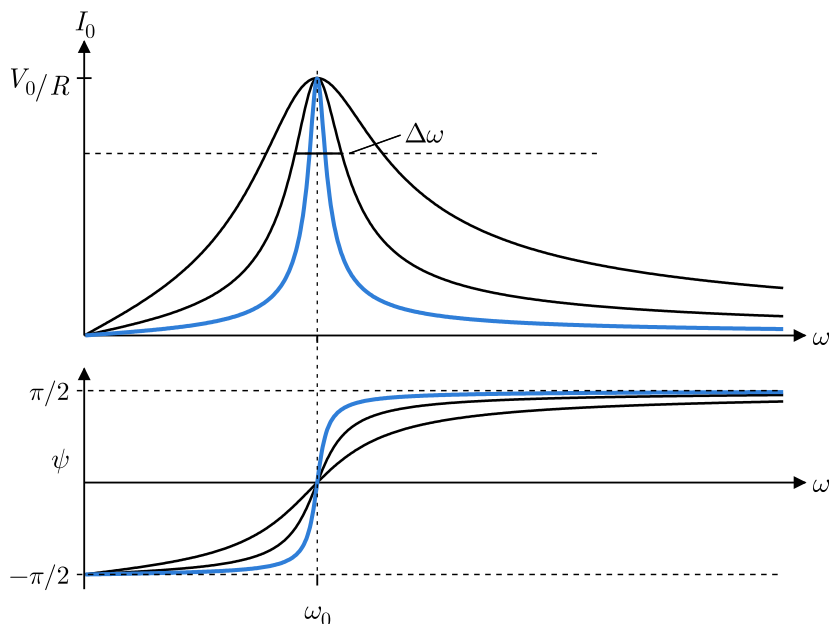


Figure 80. The peak current I_0 in a driven series LCR circuit (above) and the phase ψ by which it lags the source (below), plotted against the angular frequency of the source for three values of the quality factor Q , the sharpest curve (accent) having the largest Q . The horizontal line marks $1/\sqrt{2}$ of the peak current, and the width $\Delta\omega$ is marked on the middle curve.

its maximum exactly at the natural frequency, as you showed in Problem 3 of Section 4.3 of Vol. 1. The charge on the capacitor corresponds to the displacement, and you will show in Problem 1 that its amplitude has its maximum below ω_0 , as the displacement does.

At resonance, the potential differences across the inductor and the capacitor are equal in magnitude and opposite in phase, so they cancel in the loop law, and the source drives current through the resistance alone. Each of these potential differences separately has the amplitude $\omega_0 L I_0 = \omega_0 L V_0 / R$, which can be much larger than the amplitude V_0 of the source when $\omega_0 L \gg R$. Many radio receivers make use of resonance to select one station. The aerial

picks up small voltages at many frequencies, and a resonant circuit, whose capacitance is adjusted to bring ω_0 to the frequency of the chosen station, responds strongly to that station and only weakly to the others.

The Width of the Resonance and the Quality Factor

All of the mean power delivered to the series circuit is dissipated in the resistor, and by Section 7.4 this power is $\frac{1}{2}I_0^2R$, so that

$$\langle \mathcal{P} \rangle = \frac{V_0^2 R}{2[R^2 + (\omega L - 1/\omega C)^2]}. \quad (7.77)$$

The mean power has its maximum value $V_0^2/2R$ at ω_0 , and it falls to half of this value when $(\omega L - 1/\omega C)^2 = R^2$, that is when $\omega L - 1/\omega C = \pm R$. Multiplying these two conditions by ω/L turns them into the quadratic equations

$$\omega^2 \mp \gamma\omega - \omega_0^2 = 0, \quad (7.78)$$

with $\gamma = R/L$ as in Section 7.5, the upper sign belonging to $+R$. Each equation has one positive root, and the two positive roots are

$$\omega_{\pm} = \sqrt{\omega_0^2 + \frac{\gamma^2}{4}} \pm \frac{\gamma}{2}. \quad (7.79)$$

The full width of the resonance, measured between the two angular frequencies at which the mean power is half its maximum, is therefore

$$\Delta\omega = \omega_+ - \omega_- = \gamma = \frac{R}{L}, \quad (7.80)$$

a result that holds exactly, for any amount of damping. The sharpness of the resonance is measured by the dimensionless ratio

$$\mathcal{Q} = \frac{\omega_0}{\Delta\omega} = \frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}, \quad (7.81)$$

which is called the quality factor of the circuit, as it is for the mechanical oscillator of Section 4.3 of Vol. 1, whose resonance has a width equal to the damping coefficient of the resistive force divided by the mass, corresponding to R/L here. A circuit with a small resistance has a large quality factor and a narrow resonance, and the curves in Figure 80 show how the peak becomes narrower

as Q increases. In terms of the quality factor, the condition $\gamma < 2\omega_0$ of Section 7.5 for light damping is $Q > \frac{1}{2}$.

The quality factor also compares the energy stored in the circuit with the energy dissipated in it. At resonance the current is $I_0 \cos \omega_0 t$, and since $I = dQ/dt$, the charge on the capacitor is $(I_0/\omega_0) \sin \omega_0 t$. The stored energy is

$$\frac{1}{2} L I_0^2 \cos^2 \omega_0 t + \frac{I_0^2}{2\omega_0^2 C} \sin^2 \omega_0 t = \frac{1}{2} L I_0^2, \quad (7.82)$$

since $1/\omega_0^2 C = L$, and it is constant in time. The energy dissipated in one period is the mean power $\frac{1}{2} I_0^2 R$ times the period $2\pi/\omega_0$, so that

$$2\pi \times \frac{\text{energy stored}}{\text{energy dissipated per period}} = 2\pi \times \frac{\frac{1}{2} L I_0^2}{\pi I_0^2 R / \omega_0} = Q, \quad (7.83)$$

which is the relation found for the freely oscillating mechanical system in Section 4.3 of Vol. 1.

The Parallel Resonant Circuit

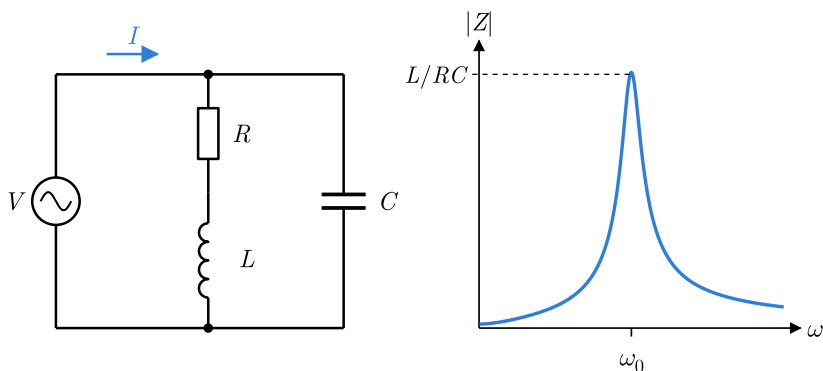


Figure 81. A parallel resonant circuit (left), in which a capacitor C is connected in parallel with an inductor L whose coil has resistance R , and the modulus of its impedance plotted against the angular frequency of the source (right). The impedance has a sharp maximum, close to L/RC , near the resonant angular frequency ω_0 .

In the circuit of Figure 81 the source is connected across a capacitor C in parallel with an inductor L , and the inductor has a resistance

R , which represents the resistance of the wire of its coil. Let us take first the case $R = 0$. The formula of Section 7.5 for circuit components in parallel gives

$$\frac{1}{Z} = \frac{1}{i\omega L} + i\omega C = \frac{1 - \omega^2 LC}{i\omega L}, \quad (7.84)$$

so that

$$Z = \frac{i\omega L}{1 - \omega^2/\omega_0^2}, \quad (7.85)$$

which becomes infinite at $\omega = \omega_0$. At resonance the source therefore supplies no current, although each branch carries a current of amplitude $V_0/\omega_0 L$. The currents in the two branches are equal in magnitude and opposite in phase, so their sum, which is the current drawn from the source, vanishes. A current circulates round the loop formed by the inductor and the capacitor, and energy passes back and forth between them, as in the LC circuit of Section 7.5.

When the resistance is included, the impedance of the branch containing the inductor is $R + i\omega L$, and

$$\frac{1}{Z} = \frac{1}{R + i\omega L} + i\omega C = \frac{R - i\omega L}{R^2 + \omega^2 L^2} + i\omega C, \quad (7.86)$$

where the first fraction has been multiplied above and below by $R - i\omega L$. The impedance is real, so that the current drawn from the source is in phase with the source, when the imaginary part of $1/Z$ vanishes, which requires

$$\omega C = \frac{\omega L}{R^2 + \omega^2 L^2}. \quad (7.87)$$

This condition gives $R^2 + \omega^2 L^2 = L/C$, or

$$\omega^2 = \omega_0^2 - \frac{R^2}{L^2} = \omega_0^2 \left(1 - \frac{1}{Q^2}\right), \quad (7.88)$$

where $Q = \omega_0 L/R$ is the quality factor that the inductor, its resistance, and the capacitor would have if they were connected in series. At this angular frequency the impedance is the reciprocal of the real part of $1/Z$,

$$Z = \frac{R^2 + \omega^2 L^2}{R} = \frac{L}{RC} = Q^2 R. \quad (7.89)$$

For $Q \gg 1$ this angular frequency is very close to ω_0 , the impedance there is much larger than R , and $|Z|$ has a sharp maximum close to

this angular frequency, as drawn in Figure 81. A parallel resonant circuit therefore has a large impedance near its resonant frequency and a small one elsewhere, which is the reverse of the behaviour of a series circuit. When such a circuit is connected, for example, in series with a resistor across a signal containing many frequencies, the potential difference across it is large for the frequencies near ω_0 and small for the others.

Mean Power in Terms of Complex Amplitudes

Let a circuit component or a network carry the current $I(t) = \text{Re}[\tilde{I}e^{i\omega t}]$, with the potential difference $V(t) = \text{Re}[\tilde{V}e^{i\omega t}]$ across it. The power delivered to it is the product VI , which, as noted in Section 7.5, cannot be found by multiplying the complex amplitudes. As you showed in Problem 7 of Section 1.2 of Vol. 4, the mean over a period of the product of two sinusoids of the same frequency with complex amplitudes $\tilde{V}$ and $\tilde{I}$ is $\frac{1}{2}\text{Re}(\tilde{V}\tilde{I}^*)$, so that the mean power is

$$\langle \mathcal{P} \rangle = \frac{1}{2}\text{Re}(\tilde{V}\tilde{I}^*) . \quad (7.90)$$

Writing $\tilde{V} = V_0e^{i\psi_V}$ and $\tilde{I} = I_0e^{i\psi_I}$ gives $\tilde{V}\tilde{I}^* = V_0I_0e^{i(\psi_V-\psi_I)}$. With $\psi = \psi_V - \psi_I$, the phase by which the potential difference leads the current, the mean power becomes

$$\langle \mathcal{P} \rangle = \frac{1}{2}V_0I_0 \cos \psi = V_{\text{rms}}I_{\text{rms}} \cos \psi . \quad (7.91)$$

For a circuit component of impedance Z , substituting $\tilde{V} = Z\tilde{I}$ gives $\langle \mathcal{P} \rangle = \frac{1}{2}|\tilde{I}|^2 \text{Re } Z$, so only the real part of an impedance takes up power. For an ideal inductor or an ideal capacitor, $\psi = \pm\pi/2$, and the mean power is zero; energy is stored in the circuit component during one quarter of each period and returned to the source during the next. For the series LCR circuit, $\text{Re } Z = R$, and the mean power is $\frac{1}{2}I_0^2R$, which is the expression used above.

The Power Factor

The factor $\cos \psi$ is called the power factor of the load. For a given mean power and a given supply voltage, a load with a small power factor draws a larger current than a purely resistive load does, since

$I_{\text{rms}} = \langle \mathcal{P} \rangle / V_{\text{rms}} \cos \psi$. The larger current produces more heat in the resistance of the cables that supply the load, without delivering any more power to the load itself, and for this reason electricity suppliers often require large consumers to keep their power factor close to one. Many industrial loads, such as electric motors, behave as a resistance in series with an inductance, and the current they draw lags the supply voltage.

The power factor of such a load can be corrected by connecting a capacitor in parallel with it, as follows. Let the load have the impedance $R + i\omega L$. The calculation made above for the parallel resonant circuit shows that the load and a capacitor C in parallel have a real impedance when

$$C = \frac{L}{R^2 + \omega^2 L^2} . \quad (7.92)$$

The capacitor takes no mean power, so the mean power delivered to the load is unchanged, while the combination has a real impedance, so the current drawn from the supply is in phase with the supply voltage and the power factor of the combination is one. The current drawn from the supply is therefore smaller than before by the factor $\cos \psi$ of the uncorrected load, as you will show in Problem 3. The current through the load itself is unchanged; the part of it that is a quarter of a period out of phase with the supply voltage now passes back and forth between the load and the capacitor, and no longer flows through the supply cables.

Problems

§7.1 Induction and Alternating Current at School

1. A flat coil of N turns and area A lies with its plane perpendicular to a uniform field B . The field is reduced steadily to zero in a time t_1 . Find the electromotive force induced in the coil. If the coil is part of a closed circuit of total resistance R , show that the charge that flows round the circuit is NBA/R , independent of t_1 .
2. A straight metal rod of length ℓ is perpendicular to a uniform field B , and it moves with speed v in a direction perpendicular to the rod that makes an angle θ with the field. Find the

electromotive force between the ends of the rod.

3. A coil of N turns and area A rotates f times per second in a uniform field B , about an axis perpendicular to the field. Find the peak value and the root-mean-square value of the electromotive force. Describe how the output changes when the rate of rotation is doubled.
4. An ideal transformer has N_p turns on its primary coil, which is connected to a supply of root-mean-square voltage V_p , and N_s turns on its secondary coil, which is connected to a resistor R . Find the root-mean-square currents in the secondary and primary coils, and the mean power taken from the supply.
5. A power station delivers a mean power $\langle \mathcal{P} \rangle$ at a root-mean-square voltage V_{rms} through transmission cables of total resistance R_{line} . Show that the mean power dissipated in the cables is $\langle \mathcal{P} \rangle^2 R_{\text{line}} / V_{\text{rms}}^2$, and find by what factor this loss changes if the voltage is first raised by a step-up transformer with turns ratio N_s/N_p .

§7.2 Faraday's Law and Lenz's Law

1. The rod of Figure 62 has mass m and slides without friction. At $t = 0$ it has velocity v_1 along the rails, and it is then left to itself. Find its velocity as a function of time, and show that the total heat produced in the resistor is equal to the initial kinetic energy of the rod.
2. Write the electric field outside the solenoid of this section, $E_\phi = -\mu_0 n a^2 (dI/dt)/2s$, in Cartesian components, and show that its curl is zero. Explain why its line integral round a circle of radius $s > a$ is nevertheless not zero, and why this does not contradict Stokes' theorem.
3. A long straight wire carries a steady current I . A square loop of side ℓ lies in a plane containing the wire, with two of its sides parallel to the wire, and moves directly away from the wire with speed v . When the nearer side is at a distance s from the wire, find the electromotive force in the loop, first from the flux rule and then from the magnetic force on the charges in the

sides parallel to the wire. Find also the direction of the induced current in the nearer side.

4. The rotating coil of this section is part of a closed circuit of total resistance R , and its self-inductance is negligible. Using the torque on a current loop from Section 5.2, show that the power needed to keep the coil turning at constant angular velocity is equal to $\mathcal{E}^2/R$ at every instant.
5. A circular ring of wire of radius b and resistance R is placed coaxially round the long solenoid of this section, with $b > a$, and the current in the solenoid increases at a constant rate. Find the current in the ring, and check that it is equal to the line integral of the induced electric field round the ring divided by R .

§7.3 Inductance and Magnetic Energy

1. Find the energy stored per unit length in the coaxial cable of this section by integrating the energy density of its magnetic field, and check that it is equal to $\frac{1}{2}(L/\ell)I^2$.
2. In the RL circuit of Section 4.3, a source of constant electromotive force $\mathcal{E}$ is connected at $t = 0$, and the current grows as $I = (\mathcal{E}/R)(1 - e^{-t/\tau})$ with $\tau = L/R$. Show that at every instant the power delivered by the source is equal to the rate at which heat is produced in the resistor plus the rate of increase of $\frac{1}{2}LI^2$. The current having reached $\mathcal{E}/R$, the source is removed and replaced by a wire. Find the total heat produced in the resistor as the current decays.
3. Coils with self-inductances L_1 and L_2 and mutual inductance L_{12} are connected in series, so that the same current flows through both. Show that the combination has inductance $L_1 + L_2 + 2L_{12}$ when the coils are connected so that their fluxes add, and $L_1 + L_2 - 2L_{12}$ when they are connected so that their fluxes oppose.
4. A long straight wire and a rectangular loop lie in the same plane, with two sides of the loop, of length ℓ , parallel to the wire at distances a and b from it. Treating the wire as part of a circuit whose other parts are far away, find the mutual inductance of the wire and the loop. If the current in the wire is $I_0 \cos \omega t$, find the electromotive force in the loop.

5. Coils of N_1 and N_2 turns are wound on a common core that confines the magnetic flux, so that every turn of either coil is threaded by the same flux, and this flux is $\Lambda(N_1 I_1 + N_2 I_2)$ when the coils carry currents I_1 and I_2 , where Λ is a constant fixed by the core. Show that $L_1 = \Lambda N_1^2$, $L_2 = \Lambda N_2^2$, and $L_{12} = L_{21} = \Lambda N_1 N_2$, so that $L_{12}^2 = L_1 L_2$. Show also that the electromotive forces induced in the two coils are in the ratio N_2/N_1 , which is the transformer relation of Section 7.1.

§7.4 Alternating Current, Mean Power, and Rectification

1. A sawtooth waveform rises steadily from $-V_0$ to $+V_0$ during each period and then drops instantaneously back to $-V_0$. Find its mean value and its root-mean-square value, and compare the mean power that it delivers to a resistor with that delivered by a sinusoid of the same peak value.
2. An oscilloscope has a vertical sensitivity V_{div} per division and a time base t_{div} per division. A sinusoidal trace measures η divisions from trough to peak, and its successive peaks are ξ divisions apart. Find the root-mean-square value and the frequency of the signal. A second trace, of the same frequency, reaches its peaks ζ divisions later than the first. Find the phase difference between the two signals, and say which of them leads.
3. Show that the output $|V_0 \sin \omega t|$ of a full-wave rectifier has mean value $2V_0/\pi$ and root-mean-square value $V_0/\sqrt{2}$, and find the mean power that it delivers to a load R .
4. In a smoothed half-wave rectifier with $RC \gg T$, the diode begins to conduct when the rising supply voltage $V_0 \cos \omega t$, with t measured from a peak, is equal to the capacitor voltage $V_0 - \Delta V$. Show that the diode conducts for a time of about $\sqrt{2\Delta V/V_0}/\omega$ before each peak, and hence that, while it conducts, the mean current through it is larger than the mean load current by a factor of about $2\pi/\sqrt{2\Delta V/V_0}$.
5. A smoothed rectifier supplies a load that draws a mean current I_{load} . Show that the ripple is approximately I_{load}/fC for a half-wave rectifier and $I_{\text{load}}/2fC$ for a full-wave rectifier, and explain why the ripple in these expressions does not depend on the re-

sistance of the load when the mean current is given.

§7.5 Phasors and Complex Impedance

1. At $t = 0$ the capacitor of an LC circuit is uncharged and the current is I_1 . Find $Q(t)$ and $I(t)$, and show that the largest charge on the capacitor is $I_1\sqrt{LC}$. Check this result by the conservation of energy.
2. A capacitor carrying a charge Q_0 is connected at $t = 0$ across an inductor L and a resistor R in series, with $R = 2\sqrt{L/C}$. Find $Q(t)$, and show that the charge never changes sign.
3. In the circuit of Figure 78 the output is taken across the resistor in place of the capacitor. Find the ratio $\tilde{V}_{\text{out}}/\tilde{V}_{\text{in}}$, the gain, and the phase of the output relative to the input, and describe their behaviour when $\omega\tau \ll 1$ and when $\omega\tau \gg 1$.
4. Find the impedance, its modulus, and its argument for a resistor R in series with an inductor L , and for a resistor R in parallel with a capacitor C .
5. Use phasors to write $V_0 \cos \omega t + V_0 \sin \omega t$ as a single sinusoid, and check the result with the identity used in Section 7.4.
6. An inductor L and a resistor R are connected in series across a source, and the output is taken across the resistor. Show that this circuit is a low-pass filter with the same gain as the filter of this section, with the time constant $\tau = L/R$.

§7.6 Resonance and Power in Alternating-Current Circuits

1. Show that in the driven series LCR circuit the amplitude of the potential difference across the capacitor is largest when $\omega^2 = \omega_0^2 - \gamma^2/2$, provided that $\gamma^2 < 2\omega_0^2$, and compare this result with the amplitude of the displacement of the driven oscillator in Section 4.3 of Vol. 1.
2. Show that at resonance the amplitudes of the potential differences across the inductor and the capacitor of a series LCR circuit are each QV_0 , and find the amplitude of the potential difference across the inductor and the capacitor taken together.

3. A load consisting of a resistance R in series with an inductance L is connected to a supply of root-mean-square voltage V_{rms} and angular frequency ω . Find its power factor, the mean power it takes, and the root-mean-square current drawn from the supply. Then find the root-mean-square current drawn from the supply when the capacitor of this section is connected in parallel with the load, and show that it is smaller by the factor $\cos \psi$ while the mean power is unchanged.
4. Show that the two angular frequencies at which the mean power in a series LCR circuit is half its maximum satisfy $\omega_+ \omega_- = \omega_0^2$, and that for $Q \gg 1$ they are approximately $\omega_0 \pm \omega_0/2Q$.
5. A source with complex amplitude $\tilde{V}$ has an internal impedance $Z_{\text{int}} = R_{\text{int}} + iX_{\text{int}}$, and it is connected to a load of impedance $R + iX$. Show that the mean power delivered to the load is $\frac{1}{2}|\tilde{V}|^2 R / [(R + R_{\text{int}})^2 + (X + X_{\text{int}})^2]$, and that this power is largest when $X = -X_{\text{int}}$ and $R = R_{\text{int}}$.
6. The parallel resonant circuit of this section, with $R = 0$, is driven by the source $V_0 \cos \omega_0 t$. Find the current in each branch, and show that the energy stored in the circuit is constant and equal to $\frac{1}{2}CV_0^2$.

Maxwell's Equations and Conservation Laws

8.1 The Displacement Current and Maxwell's Equations

In the previous chapters we found an equation for the divergence and an equation for the curl of the electric field, and a similar pair of equations for the magnetic field. In this section we first show that one of these equations, Ampère's law, cannot hold for fields that change in time, then add to it the term that Maxwell proposed in the early 1860s, and finally collect the four equations in their complete form and show that the conservation of charge follows from them. The section after this one writes the same equations for fields in matter.

The Field Equations So Far

Gauss's law in differential form, $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$, was derived in Section 1.3 from Coulomb's law. In Section 5.3 we found that the divergence of the magnetic field is zero and that its curl is $\mu_0 \mathbf{J}$, the second result being Ampère's law, which was stated for steady currents. The same section showed that the curl of the electrostatic field of Chapter 1 is zero, and in Section 7.2 Faraday's law replaced this

result, for fields that change in time, by $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$. Collected together, the four equations as they stand are

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0}, & \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \cdot \mathbf{B} &= 0, & \nabla \times \mathbf{B} &= \mu_0 \mathbf{J}.\end{aligned}\tag{8.1}$$

Of these equations only Faraday's law contains a time derivative. The other three were obtained from Coulomb's law and the Biot-Savart law, which describe static charges and steady currents, so their derivations do not tell us whether they hold when the charges and currents change. Faraday's law shows that the curl of $\mathbf{E}$ depends on how quickly $\mathbf{B}$ changes, and this raises the following question: does the curl of $\mathbf{B}$ depend in the same way on how quickly $\mathbf{E}$ changes?

Ampère's Law and a Charging Capacitor

Consider, for example, a parallel-plate capacitor that is being charged by a current I in a long straight wire, as in Figure 82. Let $\mathcal{L}$ be a loop that encircles the wire. In the integral form of Section 5.3, Ampère's law reads

$$\oint_{\mathcal{L}} \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}}, \tag{8.2}$$

in which I_{enc} is the current through a surface bounded by $\mathcal{L}$. As Section 5.3 states, the law holds for any surface bounded by the loop, so we are free to choose the surface.

The flat disc $\mathcal{S}_1$ in Figure 82 is pierced by the wire, and the current through it is I . The surface $\mathcal{S}_2$ is bounded by the same loop, but it bulges out and passes between the plates of the capacitor, where there is no wire, so the current through it is zero. The circulation on the left-hand side depends only on the loop and on the field along it, and yet Ampère's law gives $\mu_0 I$ for it with the first surface and zero with the second, so Ampère's law cannot be correct while the capacitor is charging. No such difficulty arose in Section 5.3, where the charges were static and the currents steady. Here the charge on each plate grows in magnitude, and so does the electric field between the plates.

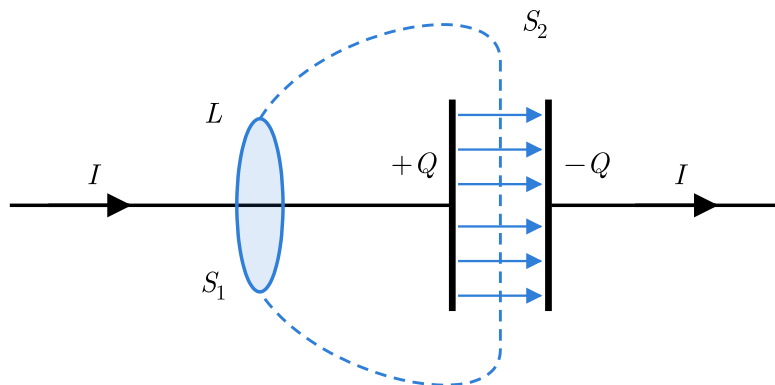


Figure 82. A parallel-plate capacitor being charged by a current I . The loop $\mathcal{L}$ (accent ellipse) encircles the wire, and two surfaces are bounded by it: the flat disc $\mathcal{S}_1$ (shaded), which the wire pierces, and the bag-shaped surface $\mathcal{S}_2$ (dashed), which passes between the plates, where there is an electric field (accent arrows) and no current.

The Divergence of a Curl

The same difficulty appears in the differential form of Ampère's law. To show this we need the identity, proved in Section 3.1 of Vol. 4 of the Fermi Notes on Physics, that for any vector field $\mathbf{G}$ whose second partial derivatives are continuous the divergence of the curl of $\mathbf{G}$ is zero,

$$\nabla \cdot (\nabla \times \mathbf{G}) = 0. \quad (8.3)$$

Taking the divergence of both sides of Ampère's law, $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$, and using this identity gives $\nabla \cdot \mathbf{J} = 0$. The continuity equation of Section 4.2, on the other hand, states that

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}, \quad (8.4)$$

which expresses the conservation of charge, since a current flowing out of a small volume carries charge away from it. The equation $\nabla \cdot \mathbf{J} = 0$ and the continuity equation agree only where the charge density does not change in time. At the surface of each plate of the charging capacitor, where the current in the wire ends, the charge

density changes in time and the divergence of $\mathbf{J}$ is not zero, so Ampère's law contradicts the conservation of charge there.

The Displacement Current

To make Ampère's law consistent with the continuity equation, we need to add to $\mathbf{J}$ a term whose divergence is $\partial\rho/\partial t$, and Gauss's law supplies one. Differentiating $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ with respect to time, and exchanging the order of the time derivative and the spatial derivatives in the divergence, gives

$$\frac{\partial\rho}{\partial t} = \epsilon_0 \nabla \cdot \frac{\partial\mathbf{E}}{\partial t}, \quad (8.5)$$

and substituting this into the continuity equation gives

$$\nabla \cdot \left(\mathbf{J} + \epsilon_0 \frac{\partial\mathbf{E}}{\partial t} \right) = 0. \quad (8.6)$$

The quantity in the bracket has zero divergence whether or not the currents are steady. Maxwell replaced $\mathbf{J}$ in Ampère's law by this quantity, which gives

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial\mathbf{E}}{\partial t}. \quad (8.7)$$

This equation is called the Ampère–Maxwell law, and the quantity $\mathbf{J}_d = \epsilon_0 \partial\mathbf{E}/\partial t$ is called the displacement current density. When the charges are static and the currents steady, the electric field does not change, the added term vanishes, and the equation reduces to Ampère's law, so the results of Chapter 5 are unaffected.

The name comes from Maxwell's picture of the vacuum as a medium whose parts are slightly displaced by an electric field, a picture that was later abandoned. In a vacuum the displacement current involves no motion of charge, and it appears in the equation for the curl of $\mathbf{B}$ in the same way as the current density, so a changing electric field is accompanied by a magnetic field just as a current is.

The integral form of the Ampère–Maxwell law follows from Stokes' theorem of Section 5.3, as that of Ampère's law in matter did in Section 6.2. For a loop $\mathcal{L}$ and a surface $\mathcal{S}$ bounded by it, both fixed

in space, integrating the Ampère–Maxwell law over $\mathcal{S}$ gives

$$\oint_{\mathcal{L}} \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}} + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}, \quad (8.8)$$

where $\Phi_E = \int_{\mathcal{S}} \mathbf{E} \cdot d\mathbf{a}$ is the flux of the electric field through $\mathcal{S}$, with $d\mathbf{a}$ related to the direction of the loop by the right-hand rule of Section 5.3, and the time derivative can be taken outside the integral because the surface does not move.

Let us apply this law to the capacitor of Figure 82, whose plates have area A and carry charges $\pm Q$. Section 2.1 showed that, when the effects at the edges are neglected, the field between the plates is uniform, with magnitude $Q/\epsilon_0 A$, and that the field outside is zero. The flux through $\mathcal{S}_2$ is therefore Q/ϵ_0 , and since $dQ/dt = I$, the right-hand side of the integral form for $\mathcal{S}_2$ is

$$\mu_0 \epsilon_0 \frac{d\Phi_E}{dt} = \mu_0 \frac{dQ}{dt} = \mu_0 I. \quad (8.9)$$

For the disc $\mathcal{S}_1$ of Figure 82, which lies where there is no electric field, the right-hand side is $\mu_0 I$ from the current alone. The surfaces $\mathcal{S}_1$ and $\mathcal{S}_2$ now give the same circulation, and you will show in Problem 5 that any two surfaces bounded by the same loop do so.

The Size of the Displacement Current

The displacement current is zero for the steady currents with which Ampère's law was established, and it is small in many circuits in which the currents change. To estimate its size, consider a conductor of conductivity σ_c in which the field oscillates as $\mathbf{E} = \mathbf{E}_0 \cos \omega t$. By Ohm's law in the form of Section 4.2, the conduction current density $\sigma_c \mathbf{E}$ has amplitude $\sigma_c E_0$, while the displacement current density $\epsilon_0 \partial \mathbf{E} / \partial t$ has amplitude $\epsilon_0 \omega E_0$. The ratio of the second amplitude to the first is

$$\frac{\epsilon_0 \omega}{\sigma_c} = \omega \frac{\epsilon_0}{\sigma_c}. \quad (8.10)$$

The time ϵ_0 / σ_c is the time constant with which, as Section 4.2 showed, a charge placed inside the conductor decays. For a metal this time is far shorter than the period of the alternating currents used in circuits, so inside a metal the displacement current is negligible. Between the plates of a capacitor the displacement current is

the only term, but the magnetic field it produces is weak unless the electric field changes quickly, and this field is difficult to measure directly.

The Magnetic Field between the Plates

We now work out the magnetic field inside a capacitor whose plates are discs of radius a , with a separation much smaller than a , charged by a current I in a straight wire that runs along the common axis of the plates, as in Figure 83.

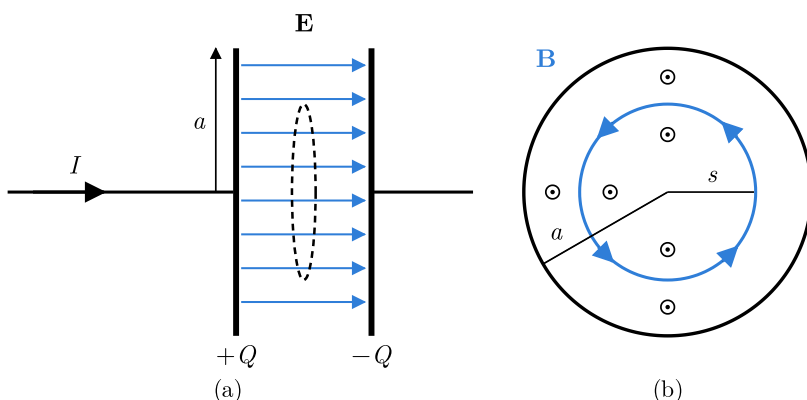


Figure 83. A capacitor with circular plates of radius a , charged by a current I . (a) Side view, with the uniform field $\mathbf{E}$ between the plates (accent arrows) and, seen edge-on, a circle of radius s (dashed) centred on the axis midway between the plates. (b) The same circle (accent) seen along the axis, with the current flowing towards the viewer; the dots mark the field $\mathbf{E}$, which points towards the viewer and grows as the plates charge, and the arrows mark the magnetic field $\mathbf{B}$ that circulates around the axis.

We look for the magnetic field in the plane midway between the plates, at a distance s from the axis, and suppose that the current changes slowly enough that at each instant the field between the plates is the uniform field $Q/\epsilon_0\pi a^2$ of Section 2.1. The arrangement is unchanged by a rotation about the axis, so in this plane the radial, azimuthal, and axial components of $\mathbf{B}$ depend only on s . A radial component of $\mathbf{B}$ would give a non-zero flux out of a closed cylinder

coaxial with the plates, which $\nabla \cdot \mathbf{B} = 0$ forbids, as you showed for the coaxial cable in Problem 1 of Section 5.3. A component along the axis, which by the symmetry could depend only on s , is excluded by the Ampère–Maxwell law applied to a narrow rectangle in a plane containing the axis, with its short sides parallel to the axis. No current and no flux of $\mathbf{E}$ pass through such a rectangle, so the axial component has the same value at the two distances from the axis at which the short sides lie, and since the field vanishes far from the capacitor, this value is zero. The field in the midplane is therefore azimuthal, and its lines are circles centred on the axis, on which its magnitude $B(s)$ is constant.

Let $\mathcal{L}$ now be the circle of radius s in the midplane, drawn in Figure 83, and let $\mathcal{S}$ be the flat disc that it bounds, which no conduction current crosses. For $s < a$ the flux through the disc is the uniform field multiplied by the area πs^2 , and the Ampère–Maxwell law gives

$$2\pi s B = \mu_0 \epsilon_0 \frac{d}{dt} \left(\frac{Q \pi s^2}{\epsilon_0 \pi a^2} \right) = \mu_0 I \frac{s^2}{a^2}, \quad (8.11)$$

so that

$$B = \frac{\mu_0 I s}{2\pi a^2}, \quad s < a. \quad (8.12)$$

For $s > a$, still in the midplane, the disc $\mathcal{S}$ contains the whole of the field between the plates, whose flux is Q/ϵ_0 , and the same calculation gives

$$B = \frac{\mu_0 I}{2\pi s}, \quad s > a, \quad (8.13)$$

which is the field at a distance s from a long straight wire carrying the current I , found in Section 5.2.

The field is drawn in Figure 84. Outside the plates it is the field that the wire would produce if it ran straight through the gap, and inside them it has the same form as the field inside a thick wire of radius a carrying a current I spread uniformly over its cross-section, which you found in Problem 1 of Section 5.3, and you will see the reason for this agreement in Problem 3. The direction of $\mathbf{B}$ is given by the right-hand rule, with the thumb along $\partial \mathbf{E} / \partial t$, which points in the direction of the current in the wire, as in Figure 83(b).

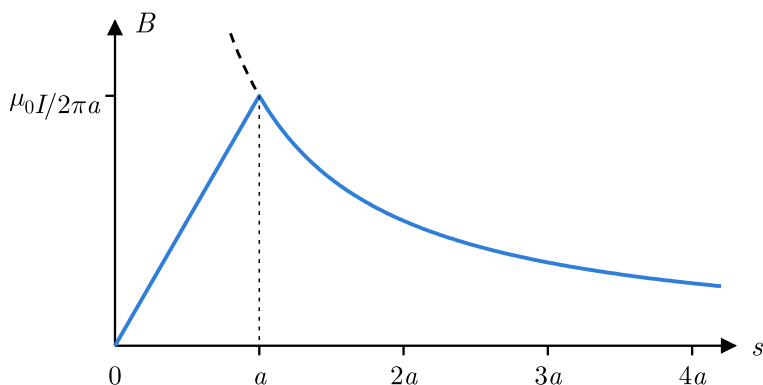


Figure 84. The magnitude of the magnetic field (accent) in the plane midway between the plates of the capacitor of Figure 83, as a function of the distance s from the axis. It grows in proportion to s inside the plates and goes as $1/s$ outside them, where it equals the field $\mu_0 I/2\pi s$ of the wire. The dashed curve continues $\mu_0 I/2\pi s$ a little way inside $s = a$ for comparison.

Maxwell's Equations

With the displacement current included, the equations for the divergence and the curl of the electric and magnetic fields are

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0}, & \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \cdot \mathbf{B} &= 0, & \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}, \end{aligned} \quad (8.14)$$

which are called Maxwell's equations. In this form every charge and every current is counted in ρ and $\mathbf{J}$, including those bound in the atoms and molecules of a material, and the only constants that appear are ϵ_0 and μ_0 , so the equations in this form are often called Maxwell's equations in vacuum. Together with the force $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ of Section 5.1, they describe how charges produce fields and how fields act on charges. For static charges and steady currents they reduce to the equations of Chapters 1 and 5. Maxwell's equations can be interpreted as follows:

1. Gauss's law states that lines of $\mathbf{E}$ begin on positive charges and end on negative ones.
2. The equation $\nabla \cdot \mathbf{B} = 0$, sometimes called Gauss's law for magnetism, states that there are no magnetic charges, so lines of $\mathbf{B}$ neither begin nor end.
3. Faraday's law states that a changing magnetic field is accompanied by a circulating electric field.
4. The Ampère–Maxwell law states that a current and a changing electric field are each accompanied by a circulating magnetic field.

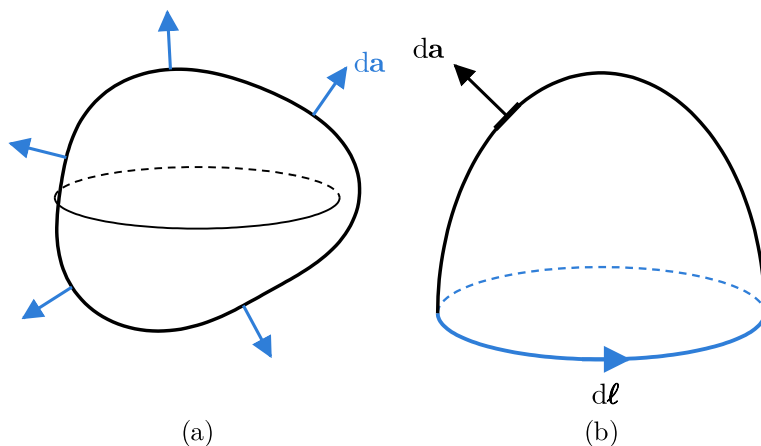


Figure 85. (a) A closed surface, whose surface elements $d\mathbf{a}$ (accent) point outwards. (b) An open surface bounded by a closed curve (accent), with the dashed part of the curve lying behind the surface. The curve is traversed in the direction of the line element $d\ell$, and the surface element $d\mathbf{a}$ points to the side given by the right-hand rule: with the fingers of the right hand curled along $d\ell$, the thumb points along $d\mathbf{a}$.

The integral forms follow from the divergence theorem of Section 1.3 and Stokes' theorem of Section 5.3, with the orientations of Figure 85. For a closed surface $\mathcal{S}$ that bounds a fixed volume $\mathcal{V}$, with the surface element $d\mathbf{a}$ pointing outwards, the two equations

for the divergence become

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0}, \quad \oint_S \mathbf{B} \cdot d\mathbf{a} = 0, \quad (8.15)$$

in which Q_{enc} is the charge inside $\mathcal{V}$. For a fixed open surface $\mathcal{S}$ bounded by a loop $\mathcal{L}$, with $d\mathbf{a}$ related to the direction of $d\boldsymbol{\ell}$ by the right-hand rule, the two equations for the curl become

$$\oint_{\mathcal{L}} \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi_B}{dt}, \quad (8.16)$$

which is Faraday's law in the form of Section 7.2, and the integral form of the Ampère–Maxwell law found above,

$$\oint_{\mathcal{L}} \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}} + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}, \quad (8.17)$$

in which Φ_B and Φ_E are the fluxes of the magnetic and electric fields through $\mathcal{S}$, and I_{enc} is the current through $\mathcal{S}$.

The Conservation of Charge

The continuity equation was used above as a requirement that the Ampère–Maxwell law had to meet, and it can now be derived from Maxwell's equations. When we take the divergence of both sides of the Ampère–Maxwell law, the left-hand side vanishes by the identity above, and exchanging the order of the derivatives on the right gives

$$0 = \mu_0 \nabla \cdot \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial}{\partial t} (\nabla \cdot \mathbf{E}). \quad (8.18)$$

Substituting ρ/ϵ_0 for $\nabla \cdot \mathbf{E}$ from Gauss's law and dividing by μ_0 gives

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0. \quad (8.19)$$

The conservation of charge is therefore a consequence of Maxwell's equations, and charges and currents that did not satisfy the continuity equation could not be the sources of any fields that obey them.

Integrating the continuity equation over a fixed volume $\mathcal{V}$ and applying the divergence theorem to the term in $\mathbf{J}$ gives

$$\frac{dQ_{\text{enc}}}{dt} = -\oint_S \mathbf{J} \cdot d\mathbf{a}, \quad (8.20)$$

where $\mathcal{S}$ is the closed surface that bounds $\mathcal{V}$, so the charge inside any volume changes only by the amount that the current carries through its surface. This statement is the local conservation of charge, of which the continuity equation of Section 4.2 is the differential form. The constancy of the total charge of an isolated system would by itself allow a charge to vanish at one point and reappear at the same instant at another, whereas the local statement requires the charge to travel between the points, crossing every surface that separates them.

The equations for the divergence are consistent with the equations for the curl in a similar way. The divergence of Faraday's law gives $\partial(\nabla \cdot \mathbf{B})/\partial t = 0$, so a magnetic field whose divergence is zero at one instant keeps a zero divergence, and for sources that satisfy the continuity equation the divergence of the Ampère–Maxwell law gives $\partial(\nabla \cdot \mathbf{E} - \rho/\epsilon_0)/\partial t = 0$, so Gauss's law, once it holds, continues to hold.

The fields also carry energy, with the densities $\frac{1}{2}\epsilon_0 E^2$ and $B^2/2\mu_0$ found in Sections 1.4 and 7.3. The local conservation of the energy and the momentum of the fields, which for energy is expressed by Poynting's theorem, is not treated in this course.

8.2 Maxwell's Equations in Matter and the Boundary Conditions

In Chapters 3 and 6 static fields in matter were described by bound charges and currents, and by the auxiliary fields $\mathbf{D}$ and $\mathbf{H}$, for which the equations $\nabla \cdot \mathbf{D} = \rho_f$ and $\nabla \times \mathbf{H} = \mathbf{J}_f$ contain the free charges and currents alone. We now extend those results to fields that change in time, finding first the additional current that flows inside a material when its polarisation changes, then writing Maxwell's equations in terms of $\mathbf{D}$ and $\mathbf{H}$, and finally using the integral forms of the equations to find how the fields change across the boundary between two materials, which explains how field lines bend at such a boundary.

Free and Bound Sources

The bound charges and currents of a material belong to its atoms and molecules and are described by the polarisation $\mathbf{P}$ and the magnetisation $\mathbf{M}$. Section 3.1 showed that a polarised material carries a bound charge density $\rho_b = -\nabla \cdot \mathbf{P}$ and, on its surface, a bound surface charge density $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$, where $\hat{\mathbf{n}}$ is the outward normal. Section 6.1 showed that a magnetised material carries a bound current density $\mathbf{J}_b = \nabla \times \mathbf{M}$ and a bound surface current density $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$. All other charges and currents, such as the charge placed on a capacitor plate or the current of the conduction electrons in a wire, are called free, with densities ρ_f and $\mathbf{J}_f$. For static fields the total densities are $\rho = \rho_f + \rho_b$ and $\mathbf{J} = \mathbf{J}_f + \mathbf{J}_b$.

The Polarisation Current

When the polarisation of a material changes, the bound charges move, and their motion is a current. To find it, we model each molecule as a dipole, as in Section 3.1, made of a negative charge $-q$ and a positive charge q displaced from it by $\boldsymbol{\xi}$, so that its dipole moment is $\mathbf{p} = q\boldsymbol{\xi}$. If there are n_d molecules per unit volume, all with the same displacement, the polarisation, which is the dipole moment per unit volume, is

$$\mathbf{P} = n_d q \boldsymbol{\xi}. \quad (8.21)$$

Suppose now that the applied field grows, so that $\boldsymbol{\xi}$ grows, and let the negative charges be fixed while the positive charges move with velocity $d\boldsymbol{\xi}/dt$, as in Figure 86. Let ΔA be the area of a small flat surface inside the material, perpendicular to $\boldsymbol{\xi}$. In a short time dt the positive charges that cross the surface are those that lie within a distance $|d\boldsymbol{\xi}/dt| dt$ of it on one side, and there are $n_d \Delta A |d\boldsymbol{\xi}/dt| dt$ of them. The charge that crosses the surface per unit area and per unit time is therefore $n_d q |d\boldsymbol{\xi}/dt|$, and the current density, which points along the velocity, is

$$\mathbf{J}_p = n_d q \frac{d\boldsymbol{\xi}}{dt} = \frac{\partial \mathbf{P}}{\partial t}. \quad (8.22)$$

This current density is called the polarisation current density. If the negative charges move as well, with velocity $\mathbf{v}_-$, while the positive charges move with velocity $\mathbf{v}_+$, they contribute $-n_d q \mathbf{v}_-$ to

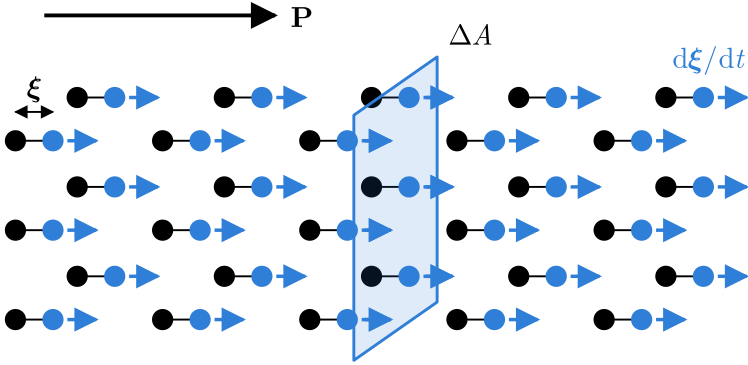


Figure 86. A model of a polarised material in which each molecule is a negative charge (solid) and a positive charge (accent) separated by ξ , drawn in two layers, one behind the other. While the polarisation $\mathbf{P}$ grows, the positive charges move with velocity $d\xi/dt$ (accent arrows), and some of them cross a small flat surface of area ΔA (lightly shaded) perpendicular to ξ .

the current density, and the total $n_d q(\mathbf{v}_+ - \mathbf{v}_-) = n_d q d\xi/dt$ is the same, since only the rate of change of the separation enters. Since $\mathbf{J}_p$ is expressed in terms of the polarisation alone, it can be checked against the continuity equation without reference to the model, and differentiating $\rho_b = -\nabla \cdot \mathbf{P}$ with respect to time gives

$$\frac{\partial \rho_b}{\partial t} = -\nabla \cdot \frac{\partial \mathbf{P}}{\partial t} = -\nabla \cdot \mathbf{J}_p, \quad (8.23)$$

so the bound charge in any small volume changes by the amount that the polarisation current carries in or out. The bound current $\mathbf{J}_b = \nabla \times \mathbf{M}$ of a magnetised material, on the other hand, circulates and changes no charge density, as you will show in Problem 1. When the fields change in time, the total current density is therefore

$$\mathbf{J} = \mathbf{J}_f + \mathbf{J}_b + \mathbf{J}_p. \quad (8.24)$$

Maxwell's Equations in Matter

We now substitute the free and bound sources into Maxwell's equations. Gauss's law becomes $\epsilon_0 \nabla \cdot \mathbf{E} = \rho_f - \nabla \cdot \mathbf{P}$, and with the displacement $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$ of Section 3.2 it takes the form $\nabla \cdot \mathbf{D} = \rho_f$ found there. The Ampère–Maxwell law, divided by μ_0 , becomes

$$\frac{1}{\mu_0} \nabla \times \mathbf{B} = \mathbf{J}_f + \nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (8.25)$$

Moving the curl of $\mathbf{M}$ to the left-hand side and collecting the time derivatives of $\epsilon_0 \mathbf{E}$ and $\mathbf{P}$ on the right gives

$$\nabla \times \left(\frac{\mathbf{B}}{\mu_0} - \mathbf{M} \right) = \mathbf{J}_f + \frac{\partial}{\partial t} (\epsilon_0 \mathbf{E} + \mathbf{P}), \quad (8.26)$$

and the quantity on the left is the curl of the auxiliary field $\mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}$ of Section 6.2. The other two equations, $\nabla \cdot \mathbf{B} = 0$ and Faraday's law, contain no charges or currents and are unchanged, so Maxwell's equations in matter are

$$\begin{aligned} \nabla \cdot \mathbf{D} &= \rho_f, & \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \cdot \mathbf{B} &= 0, & \nabla \times \mathbf{H} &= \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}. \end{aligned} \quad (8.27)$$

In the last equation, which is called the Ampère–Maxwell law in matter, the displacement current density and the polarisation current density have combined into $\partial \mathbf{D}/\partial t$, which some books call the displacement current density, while in these notes $\mathbf{J}_d$ keeps its meaning $\epsilon_0 \partial \mathbf{E}/\partial t$. For static fields the equation reduces to Ampère's law in matter, $\nabla \times \mathbf{H} = \mathbf{J}_f$, of Section 6.2. The integral forms follow as in Section 8.1. For a closed surface $\mathcal{S}$ that encloses a free charge $Q_{f,\text{enc}}$, Gauss's law in matter becomes

$$\oint_{\mathcal{S}} \mathbf{D} \cdot d\mathbf{a} = Q_{f,\text{enc}}, \quad (8.28)$$

and for an open surface $\mathcal{S}$ bounded by a loop $\mathcal{L}$, through which the free current is $I_{f,\text{enc}}$, the last equation becomes

$$\oint_{\mathcal{L}} \mathbf{H} \cdot d\boldsymbol{\ell} = I_{f,\text{enc}} + \frac{d}{dt} \int_{\mathcal{S}} \mathbf{D} \cdot d\mathbf{a}. \quad (8.29)$$

These equations are used together with relations that give $\mathbf{D}$ in terms of $\mathbf{E}$ and $\mathbf{H}$ in terms of $\mathbf{B}$, which are called constitutive

relations. For the linear materials of Sections 3.2 and 6.2 they are

$$\mathbf{D} = \epsilon_r \epsilon_0 \mathbf{E}, \quad \mathbf{B} = \mu_r \mu_0 \mathbf{H}. \quad (8.30)$$

These relations were established for static fields. The response of materials to rapidly changing fields, which the polarisation and the magnetisation need not follow, is not treated in this course, and the static values of ϵ_r and μ_r are used here.

A Charging Capacitor Filled with a Dielectric

To see how the polarisation current enters, consider the parallel-plate capacitor of Section 3.2, with plates of area A and the gap between them filled with a non-magnetic linear dielectric of relative permittivity ϵ_r , while a current I charges the plates to $\pm Q$. Section 3.2 showed that, when the effects at the edges are neglected, the fields between the plates have magnitudes $D = Q/A$ and $E = Q/\epsilon_r \epsilon_0 A$. The flux of $\partial \mathbf{D}/\partial t$ through a surface crossing the whole region between the plates is therefore I , and for a loop around the wire the integral form for $\mathbf{H}$ gives the circulation I with this surface and with a surface pierced by the wire, as for the empty capacitor of Section 8.1.

Inside the dielectric, the displacement current and the polarisation current carry different parts of the current. The displacement current density is $\epsilon_0 \partial E/\partial t = I/\epsilon_r A$, and the polarisation current density is the remainder,

$$\frac{\partial P}{\partial t} = \frac{\partial D}{\partial t} - \epsilon_0 \frac{\partial E}{\partial t} = \left(1 - \frac{1}{\epsilon_r}\right) \frac{I}{A}. \quad (8.31)$$

Of the current that crosses the region between the plates, the fraction $1/\epsilon_r$ is displacement current, and the rest is polarisation current, carried by the small shifts of the bound charges within the molecules of the dielectric, which change the bound charge on its faces. Since $\mathbf{M} = 0$, we have $\mathbf{H} = \mathbf{B}/\mu_0$, and for circular plates the magnetic field is the field found for the empty capacitor in Section 8.1.

The Boundary Conditions

At the boundary between two materials, which may carry a free surface charge or current, the fields can change abruptly, and the integral forms of Maxwell's equations determine how. For an interface between medium 1 and medium 2, let $\hat{\mathbf{n}}$ be the unit normal to the interface pointing from medium 1 into medium 2. Any field can be split at the interface into a normal component and a tangential part, which for $\mathbf{D}$ are

$$D_{\perp} = \mathbf{D} \cdot \hat{\mathbf{n}}, \quad \mathbf{D}_{\parallel} = \mathbf{D} - D_{\perp} \hat{\mathbf{n}}, \quad (8.32)$$

and similarly for the other fields, with a subscript 1 or 2 for the value just on the corresponding side of the interface.

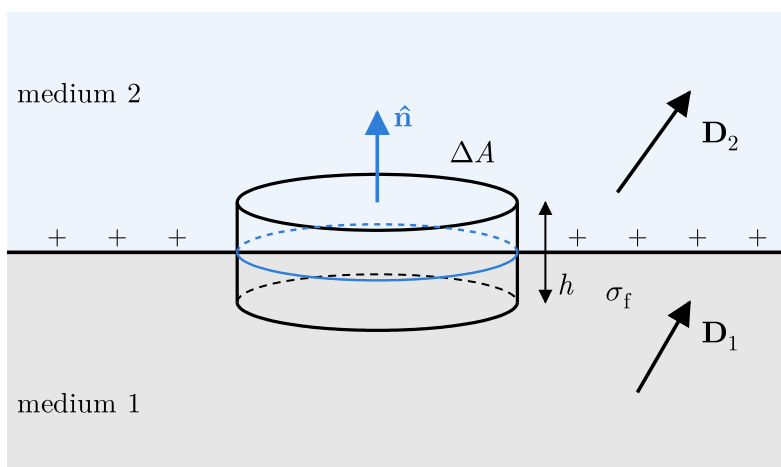


Figure 87. A pillbox of face area ΔA and height h straddling the interface between two media, with its axis along the unit normal $\hat{\mathbf{n}}$, which points from medium 1 into medium 2. The interface carries a free surface charge density σ_f , and the accent ellipse marks the part of the interface inside the pillbox, with its far half dashed, as for the lower rim. The arrows $\mathbf{D}_1$ and $\mathbf{D}_2$ show the displacement just below and just above the interface.

We apply Gauss's law in matter first to a small pillbox that straddles the interface, as in Figure 87, with a face of area ΔA in each medium, small enough that the fields are uniform over it, and with height h . In the integral form of Gauss's law in matter, the upper face contributes $\mathbf{D}_2 \cdot \hat{\mathbf{n}} \Delta A$ and the lower face $-\mathbf{D}_1 \cdot \hat{\mathbf{n}} \Delta A$, while

the contribution of the curved side, whose area is proportional to h , vanishes as h tends to zero, provided the fields are finite. The free charge inside is $\sigma_f \Delta A$, where σ_f is the free surface charge density, together with any free volume charge, whose amount is proportional to h and also vanishes in the limit. In the limit, therefore,

$$D_{2\perp} - D_{1\perp} = \sigma_f. \quad (8.33)$$

The same argument gives $B_{2\perp} = B_{1\perp}$ from $\oint \mathbf{B} \cdot d\mathbf{a} = 0$, and $\epsilon_0(E_{2\perp} - E_{1\perp}) = \sigma$ from the vacuum form of Gauss's law, where σ is the total surface charge density, free and bound. For a conductor in electrostatic equilibrium, with no field inside, the last condition gives the field σ/ϵ_0 just outside found in Section 2.1.

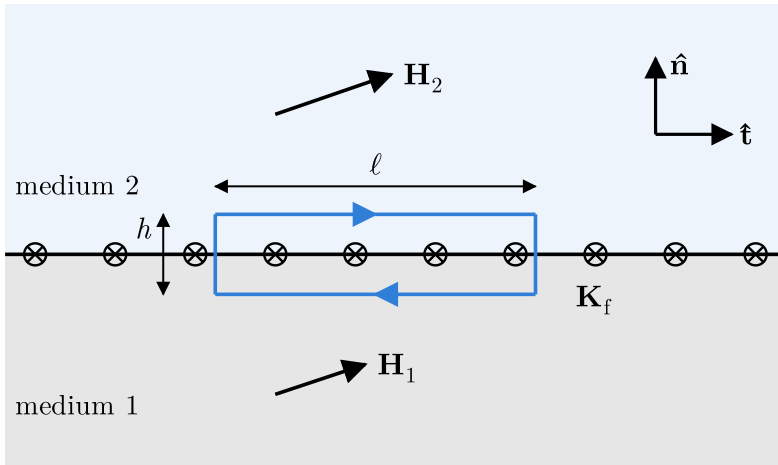


Figure 88. A narrow rectangular loop (accent) of length ℓ and width h straddling the interface, with its long sides parallel to a unit vector $\hat{\mathbf{t}}$ that lies in the interface. The long side in medium 2 is traversed along $\hat{\mathbf{t}}$ and the long side in medium 1 against it. The interface carries a free surface current density $\mathbf{K}_f$, drawn here flowing into the page. The unit normal $\hat{\mathbf{n}}$ points from medium 1 into medium 2, and the arrows $\mathbf{H}_1$ and $\mathbf{H}_2$ show the auxiliary field on the two sides of the interface.

Next, we take a narrow rectangular loop that straddles the interface, as in Figure 88, with long sides of length ℓ parallel to a unit vector $\hat{\mathbf{t}}$ in the interface and short sides of length h . The long side in medium 2 is traversed along $\hat{\mathbf{t}}$, and by the right-hand rule

the surface bounded by the loop has the unit normal $\hat{\mathbf{n}} \times \hat{\mathbf{t}}$, which points into the page in Figure 88. In Faraday's law the long sides contribute $(\mathbf{E}_2 - \mathbf{E}_1) \cdot \hat{\mathbf{t}} \ell$, while the short sides, and the rate of change of the flux of $\mathbf{B}$ through the area ℓh , contribute amounts that vanish as h tends to zero. Since $(\mathbf{E}_2 - \mathbf{E}_1) \cdot \hat{\mathbf{t}} = 0$ for every direction $\hat{\mathbf{t}}$ in the interface, the tangential part of $\mathbf{E}$ is continuous,

$$\mathbf{E}_{2\parallel} = \mathbf{E}_{1\parallel}. \quad (8.34)$$

For the same loop, the Ampère–Maxwell law in matter gives the circulation $(\mathbf{H}_2 - \mathbf{H}_1) \cdot \hat{\mathbf{t}} \ell$ in the limit, and the term in $\mathbf{D}$ vanishes in the same way as the term in $\mathbf{B}$ did, but the free current through the loop need not vanish. A free surface current density $\mathbf{K}_f$, which by Section 4.2 is the current per unit length across a line perpendicular to it, carries the current $\mathbf{K}_f \cdot (\hat{\mathbf{n}} \times \hat{\mathbf{t}}) \ell$ through the loop, while any free volume current contributes an amount proportional to ℓh . Dividing by ℓ gives

$$(\mathbf{H}_2 - \mathbf{H}_1) \cdot \hat{\mathbf{t}} = \mathbf{K}_f \cdot (\hat{\mathbf{n}} \times \hat{\mathbf{t}}). \quad (8.35)$$

To rewrite the right-hand side, we use the cyclic property of the scalar triple product, $\mathbf{G} \cdot (\mathbf{X} \times \mathbf{Y}) = \mathbf{Y} \cdot (\mathbf{G} \times \mathbf{X})$ for any three vectors, which was proved in Section 7.2. With $\mathbf{G} = \mathbf{K}_f$, $\mathbf{X} = \hat{\mathbf{n}}$, and $\mathbf{Y} = \hat{\mathbf{t}}$, the right-hand side becomes $(\mathbf{K}_f \times \hat{\mathbf{n}}) \cdot \hat{\mathbf{t}}$. The vector $\mathbf{K}_f \times \hat{\mathbf{n}}$ is perpendicular to $\hat{\mathbf{n}}$ and so lies in the interface, and since the equation holds for every direction $\hat{\mathbf{t}}$ in the interface,

$$\mathbf{H}_{2\parallel} - \mathbf{H}_{1\parallel} = \mathbf{K}_f \times \hat{\mathbf{n}}. \quad (8.36)$$

In vacuum the same argument gives $\mathbf{B}_{2\parallel} - \mathbf{B}_{1\parallel} = \mu_0 \mathbf{K} \times \hat{\mathbf{n}}$, where $\mathbf{K}$ is the total surface current density, which agrees with the jump $\mu_0 K$ in the field of the infinite current sheet of Section 5.3.

The conditions that the fields satisfy at an interface, which are called the boundary conditions, are therefore

$$D_{2\perp} - D_{1\perp} = \sigma_f, \quad B_{2\perp} - B_{1\perp} = 0, \quad (8.37)$$

for the normal components, and

$$\mathbf{E}_{2\parallel} - \mathbf{E}_{1\parallel} = 0, \quad \mathbf{H}_{2\parallel} - \mathbf{H}_{1\parallel} = \mathbf{K}_f \times \hat{\mathbf{n}}, \quad (8.38)$$

for the tangential parts. The time derivatives dropped out because they are proportional to the area of the loop, so the conditions hold for fields that change in time. At the boundary between two

linear materials that carries no free charge and no free current, the constitutive relations turn these conditions into

$$\epsilon_{r1}E_{1\perp} = \epsilon_{r2}E_{2\perp}, \quad \frac{\mathbf{B}_{1\parallel}}{\mu_{r1}} = \frac{\mathbf{B}_{2\parallel}}{\mu_{r2}}, \quad (8.39)$$

together with the continuity of $\mathbf{E}_{\parallel}$ and of $B_{\perp}$, where ϵ_{r1} and μ_{r1} belong to medium 1 and ϵ_{r2} and μ_{r2} to medium 2.

The Bending of Field Lines

Let β_1 and β_2 be the angles that the electric field makes with the normal on the two sides of the interface, so that $\tan \beta_1 = E_{1\parallel}/E_{1\perp}$ and $\tan \beta_2 = E_{2\parallel}/E_{2\perp}$, where $E_{\parallel}$ is the magnitude of $\mathbf{E}_{\parallel}$. The magnitudes of the tangential parts are equal on the two sides of the interface, and the normal components are in the inverse ratio of the relative permittivities, so

$$\frac{\tan \beta_2}{\tan \beta_1} = \frac{E_{1\perp}}{E_{2\perp}} = \frac{\epsilon_{r2}}{\epsilon_{r1}}. \quad (8.40)$$

A line of $\mathbf{E}$ that passes from empty space into a dielectric with $\epsilon_r > 1$ therefore bends away from the normal. The same calculation for the magnetic field gives $\tan \beta_2 / \tan \beta_1 = \mu_{r2} / \mu_{r1}$ for the angles that $\mathbf{B}$ makes with the normal. Where there is no free surface charge, the normal components of $\mathbf{D}$ and of $\mathbf{B}$ are continuous, and so are the lines of $\mathbf{D}$ and of $\mathbf{B}$, while the normal component of $\mathbf{E}$ changes because the interface carries a bound surface charge, on which some lines of $\mathbf{E}$ end or begin.

A Dielectric Slab in a Uniform Field

As an example of the use of the boundary conditions, consider a slab of linear dielectric of relative permittivity ϵ_r , with parallel faces that extend indefinitely, in a uniform applied field $\mathbf{E}_a$ that makes an angle β_0 with the normal $\hat{\mathbf{n}}$ to the faces, as in Figure 89(a). We look for a solution with a uniform polarisation, for which $-\nabla \cdot \mathbf{P} = 0$ and the bound charge lies on the faces, with density $P_{\perp} = \mathbf{P} \cdot \hat{\mathbf{n}}$ on the face whose outward normal is $\hat{\mathbf{n}}$ and $-P_{\perp}$ on the other face.

Each face is an infinite plane of charge, which by Section 1.3 produces a field of magnitude $|P_{\perp}|/2\epsilon_0$ on either side. As for the plates of the capacitor of Section 2.1, the fields of the faces cancel outside

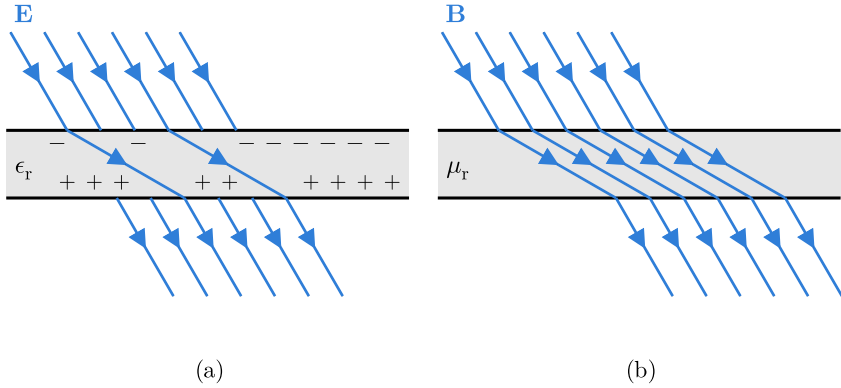


Figure 89. Field lines crossing a slab of linear material whose faces are horizontal, in a uniform applied field that makes an angle with the normal to the faces. (a) Lines of $\mathbf{E}$ crossing a dielectric with $\epsilon_r > 1$. Some of the lines end on the negative bound charge of the upper face and new lines begin on the positive bound charge of the lower face, so that the field inside is weaker and makes a larger angle with the normal. (b) Lines of $\mathbf{B}$ crossing a magnetic material with $\mu_r > 1$. The lines are continuous, and inside the slab they are closer together and make a larger angle with the normal.

the slab and add inside it, so the field outside is $\mathbf{E}_a$ and the field inside is

$$\mathbf{E} = \mathbf{E}_a - \frac{P_\perp}{\epsilon_0} \hat{\mathbf{n}}. \quad (8.41)$$

The tangential part is unchanged, $\mathbf{E}_\parallel = \mathbf{E}_{a\parallel}$. For the normal component we use $\mathbf{P} = \chi_e \epsilon_0 \mathbf{E}$ from Section 3.2, which gives $E_\perp = E_{a\perp} - \chi_e E_\perp$, and solving for $E_\perp$ gives

$$E_\perp = \frac{E_{a\perp}}{1 + \chi_e} = \frac{E_{a\perp}}{\epsilon_r}. \quad (8.42)$$

The polarisation $\chi_e \epsilon_0 \mathbf{E}$ is then uniform, as assumed, and the results agree with the boundary conditions, since the tangential part of $\mathbf{E}$ is continuous and $\epsilon_r E_\perp = E_{a\perp}$ expresses the continuity of $D_\perp$. The angle β that the field inside makes with the normal satisfies

$$\tan \beta = \frac{E_\parallel}{E_\perp} = \epsilon_r \tan \beta_0, \quad (8.43)$$

and the field inside is weaker than the applied field. In Figure 89(a), some lines of $\mathbf{E}$ end on the negative bound charge of the face at which the field enters, whose outward normal points against $\mathbf{P}$, and new lines begin on the positive bound charge of the other face. We can check the result in limiting cases. For an applied field normal to the faces, the field inside is $\mathbf{E}_a/\epsilon_r$, as in the filled capacitor of Section 3.2 with its free charge held fixed, and for an applied field parallel to the faces there is no bound charge and the field inside is $\mathbf{E}_a$.

The lines of $\mathbf{B}$ bend in the same way at the faces of a slab of linear magnetic material, where $\tan \beta = \mu_r \tan \beta_0$, as you will show in Problem 3 from the bound surface currents on the faces. The lines of $\mathbf{B}$ continue across the faces, and for $\mu_r > 1$ they make a larger angle with the normal inside the slab, where they lie closer together, as Figure 89(b) shows. For the paramagnetic and diamagnetic materials named in Section 6.1, whose susceptibilities are very small compared with one, the bending is slight, while for a material with a relative permeability much larger than one the lines inside run nearly parallel to the faces, and you will show in Problem 6 that a line of $\mathbf{B}$ that meets the surface of such a material from inside, at an angle to the normal that is not close to a right angle, leaves nearly along the normal. The lines in the slab, which run nearly parallel to the faces, meet the face through which they leave at an angle to the normal that is close to a right angle, and since the formula you will find in Problem 6 divides $\tan \beta = \mu_r \tan \beta_0$ by μ_r , they leave the slab at the angle β_0 at which they entered it.

Problems

§8.1 The Displacement Current and Maxwell's Equations

1. A parallel-plate capacitor with plates of area A and separation d is connected to a source whose potential difference is $V_0 \cos \omega t$. Neglecting the effects at the edges of the plates, find the displacement current between the plates, and show that it is equal to the current in the wires.
2. A conductor of conductivity σ_c carries a current driven by a field $\mathbf{E}_0 \cos \omega t$. Using the amplitudes of the conduction current

density and of the displacement current density found in the text, find the angular frequency at which they are equal. Find also the phase difference between the two current densities.

3. In Problem 1 of Section 5.3 you found the magnetic field inside a long straight wire of radius a that carries a current I spread uniformly over its cross-section. Compare this field with the field between the plates of the capacitor of Figure 83, and explain the agreement in terms of the displacement current density.
4. The capacitor of Figure 83, with plates of radius a and capacitance C , carries a charge Q_0 at $t = 0$, when it begins to discharge through a resistor R . Using the result of Section 4.3 for the discharge, find the magnitude of the magnetic field in the midplane at a distance $s < a$ from the axis as a function of time, and say how its direction compares with the direction of the field while the capacitor is being charged.
5. Let $\mathcal{S}_1$ and $\mathcal{S}_2$ be two surfaces bounded by the same loop $\mathcal{L}$, with their surface elements related to the direction of $\mathcal{L}$ by the right-hand rule, and suppose that $\mathcal{S}_2$ lies on the side of $\mathcal{S}_1$ towards which the surface elements of $\mathcal{S}_1$ point, so that together the two surfaces enclose a volume. Using the integral forms of Gauss's law and of the conservation of charge, show that $I_{\text{enc}} + \epsilon_0 d\Phi_E/dt$ has the same value for the two surfaces.
6. A small radioactive source at the origin emits charged particles equally in all directions, so that its charge changes in time and a current flows radially outwards. Show that $\mathbf{J} + \epsilon_0 \partial \mathbf{E} / \partial t$ is zero at every point outside the source, and explain why this result is consistent with the absence of any magnetic field, which the spherical symmetry of the arrangement requires.

§8.2 Maxwell's Equations in Matter and the Boundary Conditions

1. Show that $\nabla \cdot \mathbf{J}_p = -\partial \rho_b / \partial t$ and that $\nabla \cdot \mathbf{J}_b = 0$. Explain why the bound current of a magnetised material does not change the charge density anywhere, while the polarisation current does.
2. The region between the plates of a parallel-plate capacitor of area A is filled with two slabs parallel to the plates, of thicknesses

- d_1 and d_2 and relative permittivities ϵ_{r1} and ϵ_{r2} . The plate in contact with the first slab carries free charge Q and the other plate $-Q$. Neglecting the effects at the edges, use the boundary conditions to find the field in each slab, the capacitance, and the bound surface charge density on the interface between the slabs.
3. A slab of linear magnetic material of relative permeability μ_r , with parallel faces that extend indefinitely, is placed in a uniform applied magnetic field $\mathbf{B}_a$ that makes an angle β_0 with the normal $\hat{\mathbf{n}}$ to the faces, as in Figure 89(b). Assume that the magnetisation $\mathbf{M}$ inside is uniform. Using the bound surface currents on the two faces and the field of an infinite current sheet from Section 5.3, show that the field inside is $\mathbf{B}_a + \mu_0 \mathbf{M}_{\parallel}$, and hence that the normal component of $\mathbf{B}$ is unchanged, that $\mathbf{B}_{\parallel} = \mu_r \mathbf{B}_{a\parallel}$, and that the angle inside satisfies $\tan \beta = \mu_r \tan \beta_0$.
 4. A conductor in electrostatic equilibrium, carrying a free surface charge density σ_f , is surrounded by a linear dielectric of relative permittivity ϵ_r . Use the boundary conditions to find the field just outside the conductor, and find the bound surface charge density on the dielectric where it touches the conductor.
 5. The region between the plates of a parallel-plate capacitor of area A and separation d is filled with a material of relative permittivity ϵ_r and conductivity σ_c , and the plates are connected to a source of potential difference $V_0 \cos \omega t$. Neglecting the effects at the edges, find the current in the wires. Show that the integral form of $\nabla \times \mathbf{H} = \mathbf{J}_f + \partial \mathbf{D} / \partial t$, applied to a loop around the capacitor with a surface that passes between the plates, gives the same current, and show that the capacitor behaves as a resistor and a capacitor connected in parallel.
 6. A line of $\mathbf{B}$ inside a linear magnetic material of relative permeability μ_r meets the flat surface of the material at an angle β to the normal and passes into empty space, where it makes an angle β' with the normal. Find $\tan \beta'$, and show that when μ_r is much larger than one the line leaves nearly along the normal unless β is close to a right angle.

Electromagnetic Waves and Their Propagation

9.1 Plane Electromagnetic Waves

In this chapter we show that the Maxwell equations of Section 8.1 have solutions in which electric and magnetic fields travel through empty space as waves. This section first reviews electromagnetic waves as they are taught at school, then derives a wave equation for the fields from Maxwell's equations, and finally finds the plane-wave solutions and the relation between their electric and magnetic fields. The section that follows treats the polarisation of the waves and the energy and momentum that they carry.

Electromagnetic Waves at School

Electromagnetic waves appear in school physics as a family of waves that travel through a vacuum, with no medium to carry them, all at the same speed c . The members of the family differ in their frequency f , and in order of increasing frequency they are called

1. radio waves,
2. microwaves,
3. infrared radiation,

4. visible light,
5. ultraviolet radiation,
6. X-rays, and
7. gamma rays.

Together they make up the electromagnetic spectrum, in which neighbouring ranges merge into one another with no sharp boundaries.

At school the distance between neighbouring crests of a wave is called the wavelength and is written λ . In this course, λ is the line charge density of Section 1.2, so we describe the spacing of the crests by the wave number k , defined so that the wavelength is $2\pi/k$, just as the angular frequency $\omega = 2\pi f$ is defined so that the period is $2\pi/\omega$. In this notation the school relation between the speed, the frequency, and the wavelength is

$$c = f \times \frac{2\pi}{k} = \frac{\omega}{k}. \quad (9.1)$$

Electromagnetic waves are transverse, which means that their electric and magnetic fields oscillate at right angles to the direction of travel; the two fields are also at right angles to each other. Light whose electric field oscillates along a single fixed direction is said to be plane polarised, or linearly polarised, while the light from the Sun or from a filament lamp is unpolarised, which means that the direction of its electric field changes rapidly and at random. A polarising filter, such as the sheet of plastic in a pair of polarising sunglasses, transmits the light whose electric field lies along one direction in the sheet, called its transmission axis, and a second filter whose transmission axis is at right angles to that of the first blocks the light completely.

The intensity of a wave is the power that it carries across unit area of a surface held at right angles to its direction of travel, and for light spreading out equally in all directions from a small source it goes as $1/r^2$.

Maxwell predicted in 1865 that light is an electromagnetic wave, and between 1886 and 1888 Hertz produced and detected radio

waves with electric circuits. How oscillating charges produce electromagnetic waves, the subject called radiation, is not treated in this course.

Maxwell's Equations in Empty Space

In a region of space that contains no charge and no current, $\rho = 0$ and $\mathbf{J} = 0$, and the Maxwell equations of Section 8.1 become a pair of equations for the divergences of the fields,

$$\nabla \cdot \mathbf{E} = 0, \quad \nabla \cdot \mathbf{B} = 0, \quad (9.2)$$

and a pair for their curls,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (9.3)$$

By Faraday's law of Section 7.2, a magnetic field that changes in time is accompanied by an electric field, and through the displacement current of Section 8.1 an electric field that changes in time is accompanied by a magnetic field. This coupling suggests that a disturbance of the fields might sustain itself and travel through space with no charges or currents along its path. To find out whether it can, we eliminate $\mathbf{B}$ from the curl equations, which requires an identity for the curl of a curl.

The Curl of a Curl

The identity, which you proved in Problem 9 of Section 3.1 of Vol. 4 of the Fermi Notes on Physics, reads

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}, \quad (9.4)$$

where $\nabla^2 \mathbf{E}$ is the vector whose Cartesian components are $\nabla^2 E_x$, $\nabla^2 E_y$, and $\nabla^2 E_z$, and $\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$ is the Laplacian of Section 1.4. It holds for any vector field whose components can be differentiated twice.

The Wave Equation

We take the curl of both sides of Faraday's law. The curl involves derivatives with respect to position, and since the order of partial differentiations makes no difference, the curl of $\partial \mathbf{B}/\partial t$ is the derivative with respect to time of $\nabla \times \mathbf{B}$. Using the Ampère–Maxwell law

in empty space for $\nabla \times \mathbf{B}$ gives

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial}{\partial t}(\nabla \times \mathbf{B}) = -\mu_0\epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}. \quad (9.5)$$

By the identity above, the left-hand side is $\nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$, and in empty space $\nabla \cdot \mathbf{E} = 0$, so the left-hand side is $-\nabla^2 \mathbf{E}$. Therefore

$$\nabla^2 \mathbf{E} = \mu_0\epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}, \quad (9.6)$$

and you will show in Problem 1 that the curl of the Ampère–Maxwell law, together with Faraday’s law, gives the same equation for the magnetic field,

$$\nabla^2 \mathbf{B} = \mu_0\epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}. \quad (9.7)$$

An equation of this form is called the wave equation in three dimensions, and each Cartesian component of $\mathbf{E}$ and of $\mathbf{B}$ satisfies it.

The constant $\mu_0\epsilon_0$ has the units of an inverse speed squared, as you will check in Problem 5, so the relation

$$c = \frac{1}{\sqrt{\mu_0\epsilon_0}} \quad (9.8)$$

defines a speed, and the next subsection explains the choice of the symbol c .

We consider first a field that depends only on z and t , for which the wave equation for the component E_x becomes

$$\frac{\partial^2 E_x}{\partial z^2} = \frac{1}{c^2} \frac{\partial^2 E_x}{\partial t^2}. \quad (9.9)$$

This equation is the one-dimensional wave equation, whose general solution is found in Section 7.1 of Vol. 4 by a change of variables: it is $E_x = w_1(z - ct) + w_2(z + ct)$, where w_1 and w_2 are any functions that can be differentiated twice.

The value of $w_1(z - ct)$ at the point $z + ct$ and time t is $w_1(z)$, the value at the point z at time zero, so this disturbance moves a distance ct in the positive z -direction in the time t without changing its shape, as Figure 90 shows, and in the same way $w_2(z + ct)$ travels at the speed c in the negative z -direction.

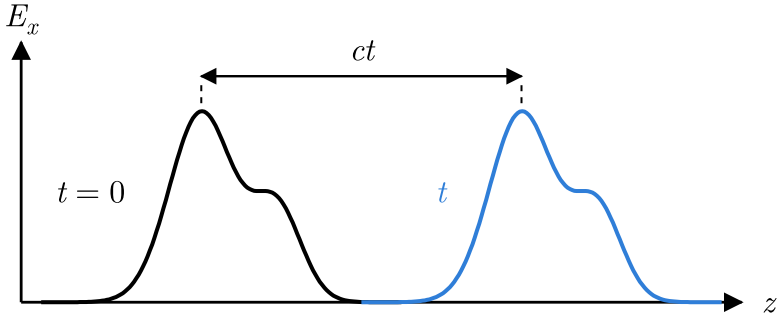


Figure 90. A disturbance $E_x = w_1(z - ct)$ at $t = 0$ (solid) and at a later time t (accent). The disturbance has the same shape at both instants, and it has moved a distance ct along the z -axis.

The Speed of the Waves

The constant ϵ_0 is found from the force between two charges at rest and μ_0 from the force between two steady currents, and neither measurement involves light. Weber and Kohlrausch compared the two kinds of measurement in 1856, and when Maxwell used their result to work out $1/\sqrt{\mu_0\epsilon_0}$, he found that it agreed, within the accuracy of the measurements, with the speed of light that Fizeau had measured directly in 1849. From this agreement Maxwell concluded that light is an electromagnetic wave travelling at the speed $c = 1/\sqrt{\mu_0\epsilon_0}$.

Plane Waves

A field that depends on position only through z has the same value at every point of each plane $z = \text{constant}$. A wave of this kind is called a plane wave, and the planes on which the field is uniform are called its wave fronts. A plane wave fills all of space and is an idealisation, but it describes well the light from a distant star over the aperture of a telescope.

A plane wave can travel in any direction. Let $\hat{\mathbf{k}}$ be a unit vector in the direction of travel. The component of the position vector

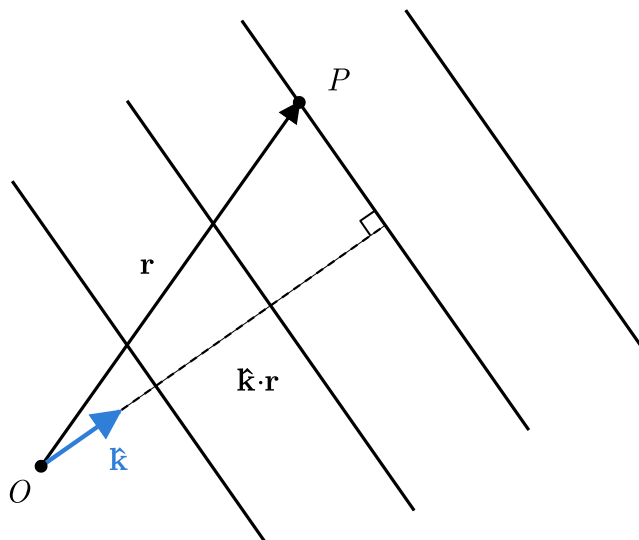


Figure 91. Planes perpendicular to a unit vector $\hat{\mathbf{k}}$, seen edge-on. Every point P of one plane has a position vector $\mathbf{r}$ with the same component $\hat{\mathbf{k}} \cdot \mathbf{r}$ along $\hat{\mathbf{k}}$, which is the distance of the plane from the origin O .

$\mathbf{r}$ along $\hat{\mathbf{k}}$ is $\hat{\mathbf{k}} \cdot \mathbf{r}$, and the points with the same value of $\hat{\mathbf{k}} \cdot \mathbf{r}$ lie on a plane perpendicular to $\hat{\mathbf{k}}$, as Figure 91 shows. A plane wave travelling in the direction of $\hat{\mathbf{k}}$ therefore has the form

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{w}(\hat{\mathbf{k}} \cdot \mathbf{r} - ct), \quad (9.10)$$

in which $\mathbf{w}$ is a vector function of a single variable. With $\xi = \hat{k}_x x + \hat{k}_y y + \hat{k}_z z - ct$, the chain rule gives $\partial^2 \mathbf{E} / \partial x^2 = \hat{k}_x^2 \mathbf{w}''$, with similar results for y and z , so that $\nabla^2 \mathbf{E} = (\hat{k}_x^2 + \hat{k}_y^2 + \hat{k}_z^2) \mathbf{w}'' = \mathbf{w}''$, since $\hat{\mathbf{k}}$ is a unit vector. The second derivative with respect to time is $c^2 \mathbf{w}''$, and the wave equation is satisfied.

The Transverse Character of the Waves

A solution of the wave equation need not satisfy Maxwell's equations, since the wave equation was obtained from them by differentiating, and differentiating an equation loses information. To find which plane waves are allowed, we return to Maxwell's equations.

Let the wave travel along z , so that $\mathbf{E}$ and $\mathbf{B}$ depend only on z and t and their derivatives with respect to x and y vanish. Gauss's law in empty space then gives

$$\nabla \cdot \mathbf{E} = \frac{\partial E_z}{\partial z} = 0, \quad (9.11)$$

and the z -component of the Ampère–Maxwell law gives

$$\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} = \mu_0 \epsilon_0 \frac{\partial E_z}{\partial t} = 0. \quad (9.12)$$

The component E_z is therefore the same at every z and at every instant. A uniform, constant field carries no disturbance from one place to another and forms no part of the wave, so we set $E_z = 0$. The same argument, with $\nabla \cdot \mathbf{B} = 0$ and the z -component of Faraday's law, shows that B_z is uniform and constant, and we set $B_z = 0$ as well. The electric and magnetic fields of a plane wave therefore lie in the plane perpendicular to the direction of travel, which is what is meant by saying that electromagnetic waves are transverse. You will show in Problem 3 that a field along the direction of travel, which satisfies the wave equation, fails to satisfy Gauss's law.

The Relation between the Electric and Magnetic Fields

Let the wave travel in the positive z -direction, with $E_x = w_x(z - ct)$ and $E_y = w_y(z - ct)$. Since $E_z = 0$ and nothing depends on x or y , the x -component of $\nabla \times \mathbf{E}$ is $-\partial E_y/\partial z$ and its y -component is $\partial E_x/\partial z$, and the x - and y -components of Faraday's law read

$$\frac{\partial B_x}{\partial t} = \frac{\partial E_y}{\partial z}, \quad \frac{\partial B_y}{\partial t} = -\frac{\partial E_x}{\partial z}. \quad (9.13)$$

The derivative of $w_x(z - ct)$ with respect to z is w'_x , while its derivative with respect to time is $-cw'_x$, so that $\partial E_x/\partial z = -(1/c)\partial E_x/\partial t$, and the same holds for E_y . The equations above therefore become $\partial B_y/\partial t = (1/c)\partial E_x/\partial t$ and $\partial B_x/\partial t = -(1/c)\partial E_y/\partial t$. Integrating with respect to time, and leaving out the constants of integration, which are fields that do not change in time and do not belong to the wave, gives

$$B_x = -\frac{E_y}{c}, \quad B_y = \frac{E_x}{c}. \quad (9.14)$$

The vector product $\hat{\mathbf{z}} \times (E_x \hat{\mathbf{x}} + E_y \hat{\mathbf{y}})$ is $E_x \hat{\mathbf{y}} - E_y \hat{\mathbf{x}}$, so these two relations can be written together as

$$\mathbf{B} = \frac{1}{c} \hat{\mathbf{z}} \times \mathbf{E}. \quad (9.15)$$

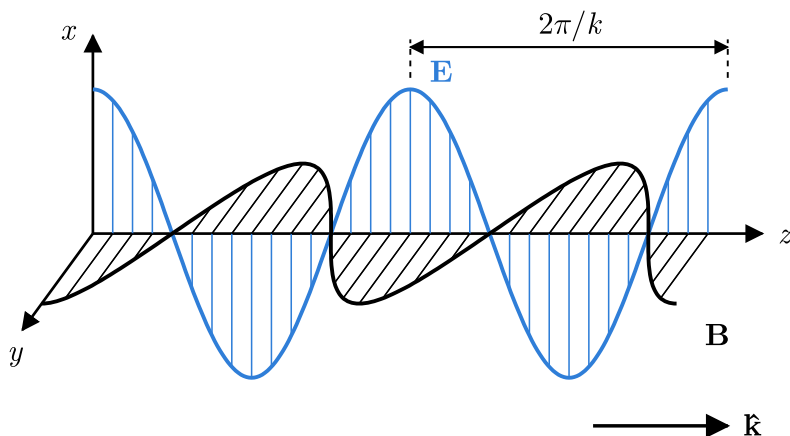


Figure 92. A snapshot of a sinusoidal plane wave travelling along the z -axis, in the direction of the unit vector $\hat{\mathbf{k}}$, with $\mathbf{E}$ (accent) along the x -axis and $\mathbf{B}$ (solid) along the y -axis, which is drawn obliquely and points out of the page. The two fields are in phase, and the crests of each are a distance $2\pi/k$ apart.

The relation between the fields has the following consequences, which are shown for a sinusoidal wave in Figure 92:

1. the magnetic field is perpendicular both to the electric field and to the direction of travel;
2. the magnitude of $\mathbf{B}$ is $|\mathbf{E}|/c$ at every point and at every instant, so the two fields rise and fall together; and
3. the vector $\mathbf{E} \times \mathbf{B}$ points in the direction of travel, since $\hat{\mathbf{z}} \times \mathbf{E} = E_x \hat{\mathbf{y}} - E_y \hat{\mathbf{x}}$ and $\mathbf{E} \times (E_x \hat{\mathbf{y}} - E_y \hat{\mathbf{x}}) = E_x^2 \hat{\mathbf{x}} \times \hat{\mathbf{y}} - E_y^2 \hat{\mathbf{y}} \times \hat{\mathbf{x}} = (E_x^2 + E_y^2) \hat{\mathbf{z}}$.

Repeating the calculation with the z -axis chosen along the direction

of travel $\hat{\mathbf{k}}$ of any plane wave gives $\mathbf{B} = (1/c) \hat{\mathbf{k}} \times \mathbf{E}$.

Sinusoidal Plane Waves

The light of a laser beam can be described, to a good approximation, by a plane wave whose fields vary sinusoidally, such as

$$\mathbf{E} = E_0 \hat{\mathbf{x}} \cos(kz - \omega t + \psi), \quad (9.16)$$

in which the amplitude E_0 , the wave number k , and the angular frequency ω are positive constants and ψ is a phase angle. Since $kz - \omega t = k(z - \omega t/k)$, this field is a function of $z - ct$ only if $\omega = ck$, which is the relation $c = \omega/k$ quoted from school at the start of the section. By the relation between the fields found above, the magnetic field of the wave is

$$\mathbf{B} = \frac{E_0}{c} \hat{\mathbf{y}} \cos(kz - \omega t + \psi). \quad (9.17)$$

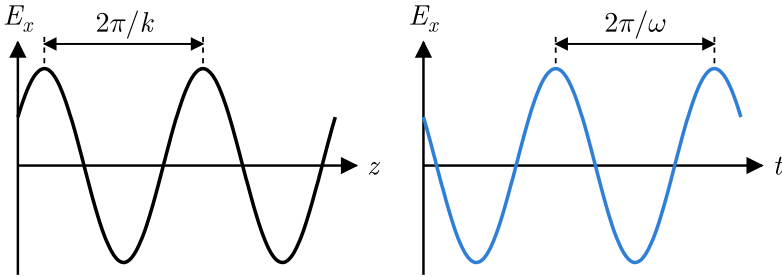


Figure 93. The field $E_x = E_0 \cos(kz - \omega t + \psi)$ of a sinusoidal plane wave, drawn against z at a fixed instant (left) and against t at a fixed point (right). Neighbouring crests are a distance $2\pi/k$ apart along the z -axis, and at a fixed point they pass at intervals of $2\pi/\omega$.

At a fixed instant the field repeats itself whenever z increases by $2\pi/k$, and at a fixed point it repeats itself whenever t increases by the period $T = 2\pi/\omega$, as Figure 93 shows. A sinusoidal plane wave travelling in the direction of a unit vector $\hat{\mathbf{k}}$ has the form

$$\mathbf{E} = \mathbf{E}_0 \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \psi), \quad \mathbf{k} = k\hat{\mathbf{k}}, \quad (9.18)$$

where the vector $\mathbf{k}$ is called the wave vector and $\mathbf{E}_0$ is a constant

vector. The transverse character of the wave requires $\mathbf{k} \cdot \mathbf{E}_0 = 0$, and since $\hat{\mathbf{k}}/c = \mathbf{k}/\omega$, the magnetic field is $\mathbf{B} = (1/\omega) \mathbf{k} \times \mathbf{E}$.

Complex Notation

In Section 7.5 the currents and voltages of a circuit were written as the real parts of complex exponentials, as the displacement of an oscillator was in Section 4.2 of Vol. 1 of the Fermi Notes on Physics and in the treatment of phasors in Section 1.2 of Vol. 4, and the same method simplifies calculations with sinusoidal waves. We write the field as

$$\mathbf{E} = \text{Re} \left[\tilde{\mathbf{E}}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \right], \quad (9.19)$$

where the complex vector amplitude $\tilde{\mathbf{E}}_0$ has components that are complex numbers, each with a modulus and a phase. For the wave of the previous subsection, $\tilde{\mathbf{E}}_0 = E_0 e^{i\psi} \hat{\mathbf{x}}$. The exponent is written with $-\omega t$, as for the plane waves of Section 2.1 of Vol. 3, while Section 7.5 wrote the currents and voltages of circuits with $e^{i\omega t}$. The two conventions give the same real field, since a complex number and its complex conjugate have the same real part, and a complex amplitude in one convention is the complex conjugate of the amplitude in the other. Maxwell's equations are linear and their coefficients are real, so by the argument of Section 4.2 of Vol. 1 the real part of a complex solution is a real solution. We may therefore work with the complex field $\tilde{\mathbf{E}} = \tilde{\mathbf{E}}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$ and the corresponding complex magnetic field $\tilde{\mathbf{B}} = \tilde{\mathbf{B}}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$, and take the real parts at the end.

Differentiating the complex field gives $\partial \tilde{\mathbf{E}} / \partial t = -i\omega \tilde{\mathbf{E}}$ and $\partial \tilde{\mathbf{E}} / \partial x = ik_x \tilde{\mathbf{E}}$, with similar results for y and z . The x -component of the curl is then $ik_y \tilde{E}_z - ik_z \tilde{E}_y$, which is i times the x -component of $\mathbf{k} \times \tilde{\mathbf{E}}$, and the same holds for the other components, so that

$$\nabla \cdot \tilde{\mathbf{E}} = i\mathbf{k} \cdot \tilde{\mathbf{E}}, \quad \nabla \times \tilde{\mathbf{E}} = i\mathbf{k} \times \tilde{\mathbf{E}}. \quad (9.20)$$

Substituting the complex fields into Maxwell's equations in empty space and cancelling the common exponential and the common factor i gives the divergence equations

$$\mathbf{k} \cdot \tilde{\mathbf{E}}_0 = 0, \quad \mathbf{k} \cdot \tilde{\mathbf{B}}_0 = 0, \quad (9.21)$$

and the curl equations

$$\mathbf{k} \times \tilde{\mathbf{E}}_0 = \omega \tilde{\mathbf{B}}_0, \quad \mathbf{k} \times \tilde{\mathbf{B}}_0 = -\frac{\omega}{c^2} \tilde{\mathbf{E}}_0. \quad (9.22)$$

The divergence equations express the transverse character of the wave, and the first curl equation is the relation $\mathbf{B} = (1/\omega) \mathbf{k} \times \mathbf{E}$ found above. With the z -axis along $\mathbf{k}$, this relation gives the amplitude $\tilde{\mathbf{B}}_0 = (k/\omega) \hat{\mathbf{z}} \times \tilde{\mathbf{E}}_0$, and substituting this amplitude into the second curl equation gives $(k^2/\omega) \hat{\mathbf{z}} \times (\hat{\mathbf{z}} \times \tilde{\mathbf{E}}_0) = -(\omega/c^2) \tilde{\mathbf{E}}_0$. The divergence equation makes $\tilde{E}_{0z}$ zero, so $\hat{\mathbf{z}} \times \tilde{\mathbf{E}}_0 = \tilde{E}_{0x} \hat{\mathbf{y}} - \tilde{E}_{0y} \hat{\mathbf{x}}$, and taking the vector product of $\hat{\mathbf{z}}$ with this result gives $-\tilde{E}_{0x} \hat{\mathbf{x}} - \tilde{E}_{0y} \hat{\mathbf{y}} = -\tilde{\mathbf{E}}_0$. Multiplying both sides of the substituted equation by ω then gives $-k^2 \tilde{\mathbf{E}}_0 = -(\omega^2/c^2) \tilde{\mathbf{E}}_0$, so a wave with a non-zero amplitude requires $\omega = ck$.

9.2 Polarisation, Energy Flux, and Radiation Pressure

The electric field of a plane wave lies in the plane perpendicular to the direction of travel, and Maxwell's equations leave its direction within that plane free. We first describe how this direction can behave, then work out the energy that a plane wave carries and the rate at which it carries it, and finally find the momentum of the wave and the pressure that it exerts on a surface. These results are obtained for plane waves only; the general theorem relating the flow of energy to the fields, called Poynting's theorem, is not treated in this course.

Linear Polarisation

Let a sinusoidal plane wave travel along the z -axis, so that its electric field lies in the xy -plane. Applied to a wave, the word polarisation describes how the direction of the electric field behaves, and this meaning of the word is unrelated to the polarisation $\mathbf{P}$ of a material, the dipole moment per unit volume of Section 3.1. If the field has the form

$$\mathbf{E} = \mathbf{E}_0 \cos(kz - \omega t + \psi), \quad (9.23)$$

with $\mathbf{E}_0$ a constant real vector in the xy -plane, then at every point the field oscillates back and forth along the fixed direction of $\mathbf{E}_0$,

and the wave is said to be linearly polarised along that direction. The magnetic field, $(1/c)\hat{\mathbf{z}} \times \mathbf{E}$, oscillates along a fixed direction perpendicular to $\mathbf{E}_0$. The wave of Figure 92 is linearly polarised along the x -axis, and light that has passed through a polarising filter is linearly polarised along the transmission axis of the filter.

Circular and Elliptical Polarisation

Since Maxwell's equations are linear, two plane waves of the same frequency travelling along the z -axis can be added, and their sum is again a solution. Let the x - and y -components of the field have amplitudes E_1 and E_2 and differ in phase by ψ :

$$E_x = E_1 \cos(kz - \omega t), \quad E_y = E_2 \cos(kz - \omega t + \psi). \quad (9.24)$$

At a fixed point, which we take at $z = 0$, these become $E_x = E_1 \cos \omega t$ and $E_y = E_2 \cos(\omega t - \psi)$, since the cosine is an even function. As time passes, the tip of the vector $\mathbf{E}$ traces out a curve in the xy -plane, and the curve repeats itself every period.

If $\psi = 0$ or $\psi = \pi$, the ratio E_y/E_x is the constant $\pm E_2/E_1$, and the tip of the field moves back and forth along a straight line, so the wave is linearly polarised. If instead $E_1 = E_2 = E_0$ and $\psi = \pi/2$, the components at $z = 0$ are

$$E_x = E_0 \cos \omega t, \quad E_y = E_0 \sin \omega t. \quad (9.25)$$

The magnitude of the field is then E_0 at every instant, and its direction turns at the constant angular velocity ω , so the tip of the field moves round a circle and the wave is said to be circularly polarised. At $t = 0$ the field points along the x -axis, and a quarter of a period later it points along the y -axis, so an observer who faces the oncoming wave, with the x -axis pointing to the right and the y -axis upwards, sees the field rotate anticlockwise. With $\psi = -\pi/2$ the field rotates clockwise. The names left-circular and right-circular are attached to these two senses in opposite ways in different parts of physics, so these notes state the sense of rotation explicitly.

For other values of the amplitudes and of the phase difference, the curve is found by eliminating the time. Let $\mathcal{X} = E_x/E_1 = \cos \omega t$ and $\mathcal{Y} = E_y/E_2 = \cos(\omega t - \psi)$. Expanding the second cosine with the formula for the cosine of a difference gives $\mathcal{Y} =$

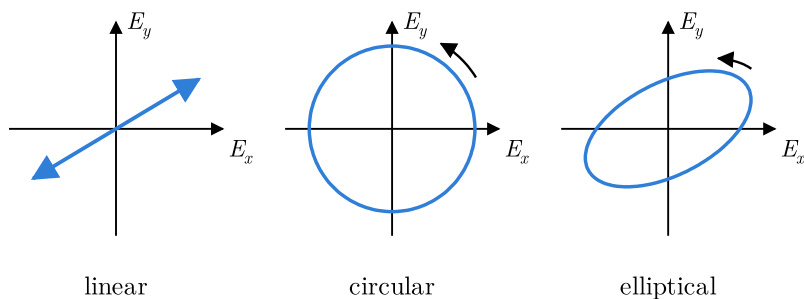


Figure 94. The path traced by the tip of the electric field at a fixed point during one period, for a linearly polarised wave (left), a circularly polarised wave (centre), and an elliptically polarised wave (right). In the centre and on the right the phase difference ψ lies between 0 and π , and the arrows show that the field rotates anticlockwise as seen by an observer facing the oncoming wave.

$\mathcal{X} \cos \psi + \sin \omega t \sin \psi$, so that $\mathcal{Y} - \mathcal{X} \cos \psi = \sin \omega t \sin \psi$. Squaring this equation and replacing $\sin^2 \omega t$ by $1 - \mathcal{X}^2$ gives

$$(\mathcal{Y} - \mathcal{X} \cos \psi)^2 = (1 - \mathcal{X}^2) \sin^2 \psi, \quad (9.26)$$

and multiplying out the square and using $\cos^2 \psi + \sin^2 \psi = 1$ gives

$$\mathcal{X}^2 - 2\mathcal{X}\mathcal{Y} \cos \psi + \mathcal{Y}^2 = \sin^2 \psi. \quad (9.27)$$

The left-hand side can be rewritten as $(\mathcal{X} - \mathcal{Y} \cos \psi)^2 + \mathcal{Y}^2 \sin^2 \psi$, so in the variables $\mathcal{X} - \mathcal{Y} \cos \psi$ and $\mathcal{Y} \sin \psi$ the curve is a circle of radius $|\sin \psi|$. Once both sides are divided by $\sin^2 \psi$, the equation is also the quadratic form $x_1^2 + 2\gamma x_1 x_2 + x_2^2 = 1$ of Section 4.3 of Vol. 4, with $x_1 = \mathcal{X}/|\sin \psi|$, $x_2 = \mathcal{Y}/|\sin \psi|$, and $\gamma = -\cos \psi$, whose eigenvalues $1 \pm \gamma$ are both positive when $\sin \psi \neq 0$, so the tip of the field moves round an ellipse centred on the origin, with its principal axes along the lines $\mathcal{Y} = \pm \mathcal{X}$, and the wave is said to be elliptically polarised. The axes of the ellipse are tilted with respect to the x - and y -axes unless $\psi = \pm \pi/2$, and linear and circular polarisation are limiting cases of elliptical polarisation, as Figure 94 shows. You will show in Problem 2 that the field rotates anticlockwise, for the observer facing the oncoming wave, when $0 < \psi < \pi$, and clockwise when $-\pi < \psi < 0$.

The light of the Sun or of a filament lamp is emitted by a very large number of atoms acting independently, so its polarisation changes at random, on a time scale far shorter than the response time of ordinary detectors, and the light is unpolarised.

The Energy Density of a Plane Wave

Section 1.4 found the energy density $\frac{1}{2}\epsilon_0 E^2$ of an electrostatic field, and Section 7.3 found the energy density $B^2/2\mu_0$ of the magnetic field of an inductor. These expressions hold for fields that change in time as well; the general argument for this is part of Poynting's theorem, which is not treated in this course. The energy density of an electromagnetic field in empty space is therefore

$$u = \frac{1}{2}\epsilon_0 E^2 + \frac{B^2}{2\mu_0}. \quad (9.28)$$

For a plane wave $|\mathbf{B}| = |\mathbf{E}|/c$, and since $c^2 = 1/\mu_0\epsilon_0$, the magnetic term is

$$\frac{B^2}{2\mu_0} = \frac{E^2}{2\mu_0 c^2} = \frac{1}{2}\epsilon_0 E^2. \quad (9.29)$$

The electric and magnetic fields of a plane wave therefore carry equal amounts of energy, and the energy density is $u = \epsilon_0 E^2$. For the linearly polarised wave $\mathbf{E} = E_0 \hat{\mathbf{x}} \cos(kz - \omega t)$, the energy density at a fixed point is $u = \epsilon_0 E_0^2 \cos^2(kz - \omega t)$, which oscillates between zero and $\epsilon_0 E_0^2$ twice in every period. For light the quantity usually measured is the average over a period, which we write with angle brackets. The average of $\cos^2$ over a period is one half, as Section 7.4 found by integration when it worked out the mean power in a resistor, and therefore

$$\langle u \rangle = \frac{1}{2}\epsilon_0 E_0^2. \quad (9.30)$$

For the circularly polarised wave of the previous subsection, $E^2 = E_x^2 + E_y^2 = E_0^2$ at every instant, and the energy density $\epsilon_0 E_0^2$ does not change in time.

Energy Flux and Intensity

To find the rate at which a plane wave carries energy, we consider a wave travelling along the z -axis and a surface of area A perpendicular

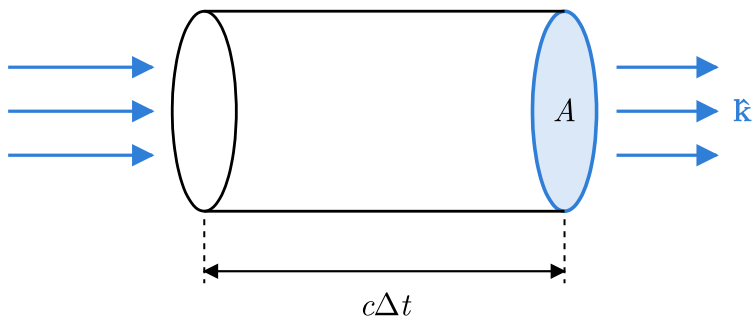


Figure 95. A plane wave travelling in the direction of $\hat{\mathbf{k}}$ crosses a surface of area A (accent) perpendicular to its direction of travel. The energy that crosses the surface in a time Δt is the energy that lay, at the start of that time, in the cylinder of length $c\Delta t$ behind the surface.

lar to the axis. The pattern of the fields moves along the axis at the speed c , so the energy that crosses the surface during a short time Δt is the energy that lay, at the start of that time, in the cylinder of volume $Ac\Delta t$ behind the surface drawn in Figure 95. Provided u changes little over the length $c\Delta t$, this energy is $uAc\Delta t$, and the energy crossing unit area in unit time is uc . To express this energy flow in terms of the fields, let

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}. \quad (9.31)$$

With $\mathbf{B} = (1/c) \hat{\mathbf{z}} \times \mathbf{E}$, the vector product $\mathbf{E} \times (\hat{\mathbf{z}} \times \mathbf{E})$ is $E^2 \hat{\mathbf{z}}$, as we found in Section 9.1, and since $1/\mu_0 c = \epsilon_0 c$,

$$\mathbf{S} = \frac{E^2}{\mu_0 c} \hat{\mathbf{z}} = \epsilon_0 c E^2 \hat{\mathbf{z}} = uc \hat{\mathbf{z}}. \quad (9.32)$$

For a plane wave, therefore, $\mathbf{S}$ points in the direction of travel, and its magnitude is the energy that crosses unit area perpendicular to that direction in unit time. The vector $\mathbf{S}$ is called the Poynting vector, after Poynting, who introduced it in 1884 to describe the flow of energy in general fields.

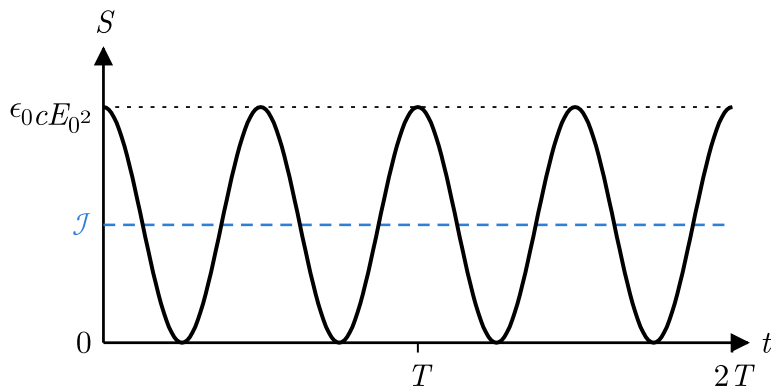


Figure 96. The magnitude S of the Poynting vector at a fixed point, for a linearly polarised sinusoidal wave of amplitude E_0 . It goes as $\cos^2 \omega t$ and oscillates between zero and $\epsilon_0 c E_0^2$, and its average over a period (accent) is the intensity $\mathcal{I}$.

The average of S over a period is called the intensity of the wave, and it is the power per unit area quoted at school. For a plane wave,

$$\mathcal{I} = \langle S \rangle = c \langle u \rangle, \quad (9.33)$$

and for the linearly polarised wave of amplitude E_0 , whose Poynting vector is drawn in Figure 96, the intensity is $\mathcal{I} = \frac{1}{2} \epsilon_0 c E_0^2$. The intensity goes as the square of the amplitude.

Since S is a product of fields, it has to be worked out from the real fields, because the real part of a product is not in general the product of the real parts. Its average over a period can nevertheless be found from the complex amplitudes, as the mean power of Section 7.6 was: for two quantities that are the real parts of $\tilde{X}e^{-i\omega t}$ and $\tilde{Y}e^{-i\omega t}$, the average of their product over a period is $\frac{1}{2} \text{Re}(\tilde{X}\tilde{Y}^*)$, a result you also found in Problem 7 of Section 1.2 of Vol. 4. Section 7.6 and that problem wrote the exponent as $e^{i\omega t}$, and the result is the same in either convention, since the amplitudes of the two conventions are complex conjugates of each other and $\text{Re}(\tilde{X}^*\tilde{Y}) = \text{Re}(\tilde{X}\tilde{Y}^*)$. For a wave travelling along z with complex amplitude $\tilde{\mathbf{E}}_0$, the average of $E^2 = E_x^2 + E_y^2$ is $\frac{1}{2}(|\tilde{E}_{0x}|^2 + |\tilde{E}_{0y}|^2)$,

and the intensity is

$$\mathcal{J} = \frac{1}{2}\epsilon_0 c \left(|\tilde{E}_{0x}|^2 + |\tilde{E}_{0y}|^2 \right). \quad (9.34)$$

For the elliptically polarised wave above, $|\tilde{E}_{0x}| = E_1$ and $|\tilde{E}_{0y}| = E_2$, so $\mathcal{J} = \frac{1}{2}\epsilon_0 c (E_1^2 + E_2^2)$ for every phase difference ψ , which is the sum of the intensities of the two linearly polarised components.

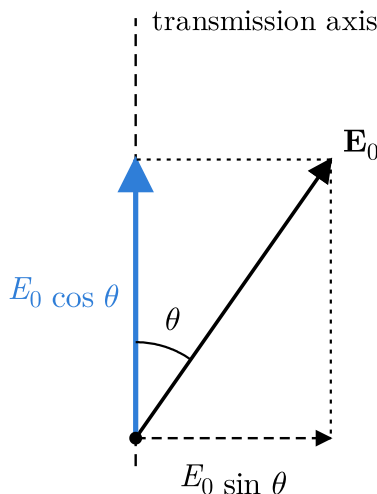


Figure 97. A linearly polarised wave falls on a polarising filter with its field $\mathbf{E}_0$ at an angle θ to the transmission axis. The filter transmits the component $E_0 \cos \theta$ along the axis (accent) and absorbs the component $E_0 \sin \theta$ perpendicular to it.

Let us now consider an ideal polarising filter, which transmits the component of the electric field along its transmission axis and absorbs the perpendicular component. A linearly polarised wave of intensity $\mathcal{J}_0$ and amplitude E_0 falls on the filter with its field at an angle θ to the transmission axis, as in Figure 97. The transmitted wave has amplitude $E_0 \cos \theta$, and since the intensity goes as the square of the amplitude, the transmitted intensity is

$$\mathcal{J} = \mathcal{J}_0 \cos^2 \theta. \quad (9.35)$$

This relation is called Malus's law, after Malus, who found it by experiment in 1809.

The Momentum Carried by the Wave

To find the momentum that a wave carries, we follow the forces that it exerts on the charges to which it gives its energy. Let a plane wave travelling along the z -axis fall on a material that contains charges free to move, such as the conduction electrons of a thin metal film. A charge q moving with velocity $\mathbf{v}$ in the fields of the wave feels the force $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ of Section 5.1. The magnetic force is perpendicular to $\mathbf{v}$ and does no work, so the wave gives energy to the charge at the rate $q\mathbf{E} \cdot \mathbf{v}$ at which the electric force does work. Since $E_z = 0$, the z -component of the force comes from the magnetic term alone, and with $B_x = -E_y/c$ and $B_y = E_x/c$ it is

$$F_z = q(v_x B_y - v_y B_x) = \frac{q}{c} (v_x E_x + v_y E_y) = \frac{q \mathbf{E} \cdot \mathbf{v}}{c}, \quad (9.36)$$

where the last step uses $E_z = 0$ once more. The force along the direction of travel is therefore the rate at which the wave gives energy to the charge, divided by c , and adding the forces on all the charges and integrating over time shows that the momentum given to the material along the direction of travel is the energy that the material absorbs, divided by c . The argument takes the fields at the charges to be those of the incident wave, and a full treatment, which includes the fields produced by the moving charges themselves, belongs to the theory of radiation, which is not treated in this course.

By the conservation of momentum, this momentum was carried by the wave, so a plane wave carries momentum along its direction of travel equal to its energy divided by c . The momentum per unit volume of the wave is therefore $\mathbf{g} = (u/c) \hat{\mathbf{z}}$, which, since $\mathbf{S} = uc \hat{\mathbf{z}}$, can be written as

$$\mathbf{g} = \frac{\mathbf{S}}{c^2} = \epsilon_0 \mathbf{E} \times \mathbf{B}. \quad (9.37)$$

The same ratio of momentum to energy belongs to a single photon, which by Section 1.2 of Vol. 3 carries energy hf and momentum hf/c , where h is Planck's constant.

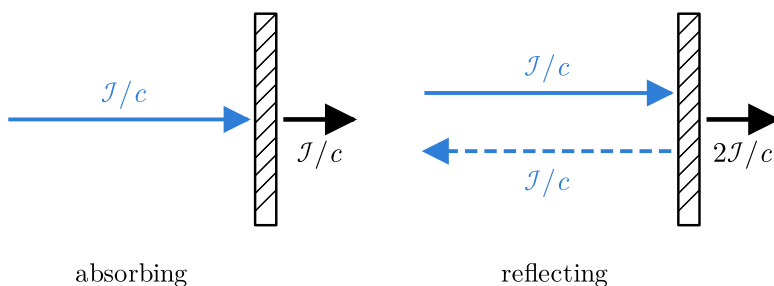


Figure 98. A plane wave of intensity $\mathcal{I}$ falling normally on a surface. The wave carries momentum $\mathcal{I}/c$ across unit area in unit time. An absorbing surface (left) takes up this momentum and feels a force $\mathcal{I}/c$ per unit area, while a perfectly reflecting surface (right) reverses the momentum and feels a force $2\mathcal{I}/c$ per unit area.

Radiation Pressure

Let a plane wave of intensity $\mathcal{I}$ fall at normal incidence on a surface that absorbs it completely, such as a plate coated with soot. During a time Δt , long compared with the period, an area A of the surface absorbs the energy $\mathcal{I}A\Delta t$, and by the result above it receives the momentum $\mathcal{I}A\Delta t/c$. The average force on the area is this momentum divided by Δt , and the force per unit area, which is called the radiation pressure, is

$$p_{\text{rad}} = \frac{\mathcal{I}}{c} = \langle u \rangle. \quad (9.38)$$

A perfect mirror absorbs no energy, so the reflected wave has the same intensity as the incident one and carries the same momentum in the opposite direction. The change in the momentum of the radiation is then twice as large, and the radiation pressure on a perfect mirror is $p_{\text{rad}} = 2\mathcal{I}/c$. The absorbing surface and the mirror are compared in Figure 98, and you will treat a surface that reflects part of the wave in Problem 6. The radiation pressure of sunlight is very small compared with the pressure of the atmosphere, but it was measured at the start of the twentieth century by Lebedev and by Nichols and Hull, with light falling on thin vanes hung from a

fine fibre in an evacuated vessel, so that the push of the light turned the vanes and twisted the fibre through a small angle that could be measured.

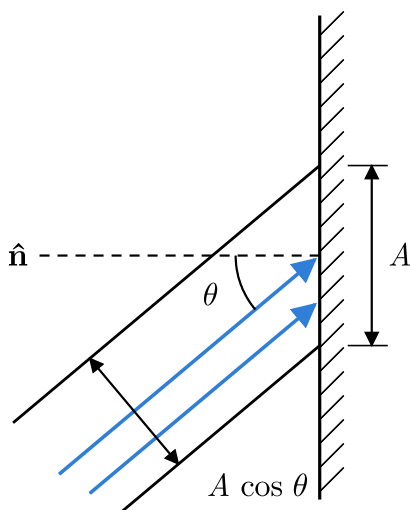


Figure 99. A beam of radiation falls on an absorbing wall at an angle θ to the normal $\hat{\mathbf{n}}$. A beam whose cross-section perpendicular to its direction of travel has area $A \cos \theta$ covers an area A of the wall.

Suppose now that the wave falls on the absorbing surface at an angle θ to the normal. As Figure 99 shows, the part of the beam that falls on an area A of the surface has a cross-section $A \cos \theta$ perpendicular to its direction of travel, so the energy absorbed per unit area of the surface in unit time is $\mathcal{J} \cos \theta$. The momentum given to the surface points along the direction of travel of the wave, and its component along the normal is a fraction $\cos \theta$ of it. The force per unit area along the normal is therefore

$$p_{\text{rad}} = \frac{\mathcal{J}}{c} \cos^2 \theta, \quad (9.39)$$

while the component of the momentum along the surface gives a force parallel to the surface.

The Pressure of Isotropic Radiation

Radiation that travels equally in all directions, with its energy shared evenly among them, is called isotropic; an example is the blackbody radiation in a cavity. Section 5.6 of Vol. 2 found, from the way the free energy of blackbody radiation depends on the volume, that isotropic radiation of energy density u exerts a pressure $u/3$ on the walls of its container. The results for plane waves give the same value. We treat the radiation as plane waves whose directions are spread evenly, so that the waves whose directions lie within a small solid angle $d\Omega$, in the sense of Section 1.3, have energy density $u d\Omega/4\pi$ and an intensity c times as large. By the result above, those that travel at an angle θ to the normal of an absorbing wall exert on it a pressure $(u d\Omega/4\pi) \cos^2 \theta$.

Let the wall be black, so that it absorbs all the radiation that falls on it and, in equilibrium, emits radiation of the same kind. The waves that fall on the wall have $0 \leq \theta < \pi/2$, with θ measured from the normal pointing into the wall. With the element of solid angle $d\Omega = \sin \theta d\theta d\phi$ of Section 4.1 of Vol. 3, the pressure that these waves exert is

$$p_{\text{in}} = \frac{u}{4\pi} \int_0^{2\pi} d\phi \int_0^{\pi/2} \cos^2 \theta \sin \theta d\theta. \quad (9.40)$$

Since $d(\cos \theta)/d\theta = -\sin \theta$, the chain rule shows that the integrand $\cos^2 \theta \sin \theta$ is the derivative of $-\frac{1}{3} \cos^3 \theta$, so the integral over θ is $[-\frac{1}{3} \cos^3 \theta]_0^{\pi/2} = 1/3$, and therefore

$$p_{\text{in}} = \frac{u}{4\pi} \times 2\pi \times \frac{1}{3} = \frac{u}{6}. \quad (9.41)$$

The radiation that the wall emits carries momentum away at the same rate, and as for the mirror the recoil of the wall adds an equal pressure, so the total pressure is $u/3$, in agreement with Section 5.6 of Vol. 2. A perfectly reflecting wall gives the same total, since each wave that falls on it is replaced by a reflected wave of equal intensity. A plane wave falling normally on an absorbing surface exerts the pressure $\langle u \rangle$, three times the pressure of isotropic radiation of the same energy density, and the factor of a third is the average of $\cos^2 \theta$ over all directions.

Problems

§9.1 Plane Electromagnetic Waves

1. Take the curl of the Ampère–Maxwell law in empty space, and use Faraday’s law and the identity for the curl of a curl to show that $\nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \partial^2 \mathbf{B} / \partial t^2$.
2. Consider the electric field $\mathbf{E} = E_0 \hat{\mathbf{x}} \sin kz \cos \omega t$ in empty space, with $\omega = ck$. Show that it satisfies the wave equation, find the magnetic field from Faraday’s law, and check that the two fields satisfy the Ampère–Maxwell law. Show that $\mathbf{E}$ is the sum of two sinusoidal waves travelling in opposite directions, and that $\mathbf{E}$ vanishes at every instant on the planes $z = j\pi/k$, where j is an integer.
3. Show that the field $\mathbf{E} = E_0 \hat{\mathbf{z}} \cos(kz - \omega t)$, with $\omega = ck$, satisfies the wave equation but does not satisfy Gauss’s law in empty space.
4. A plane wave in empty space has the electric field $\mathbf{E} = E_0 \hat{\mathbf{z}} \cos[k(x+y)/\sqrt{2} - \omega t]$. Find the direction in which it travels, the relation between ω and k , and its magnetic field.
5. Use Coulomb’s law and the force per unit length $\mu_0 I_1 I_2 / 2\pi d$ between two long parallel wires a distance d apart, carrying currents I_1 and I_2 , found in Section 5.2, to find the units of ϵ_0 and of μ_0 , and show that $1/\sqrt{\mu_0 \epsilon_0}$ has the units of a speed.

§9.2 Polarisation, Energy Flux, and Radiation Pressure

1. Unpolarised light of intensity $\mathcal{J}_0$ falls on an ideal polarising filter. Taking the direction of the field of the incident light to be distributed evenly over all angles in the plane of the filter, show that the transmitted intensity is $\mathcal{J}_0/2$. The transmitted light then passes through a second filter, whose transmission axis is at an angle θ to that of the first, and then through a third, whose transmission axis is at right angles to that of the first. Find the intensity of the light that emerges, and the angle θ for which it is largest.
2. Let β be the angle that the field $\mathbf{E}$ of the elliptically polarised wave of the text makes with the x -axis at $z = 0$, so that $\tan \beta =$

E_y/E_x . Show that $E^2 d\beta/dt = E_x dE_y/dt - E_y dE_x/dt$, and that for $E_x = E_1 \cos \omega t$ and $E_y = E_2 \cos(\omega t - \psi)$ this equals $\omega E_1 E_2 \sin \psi$. Hence show that the field rotates anticlockwise, for an observer facing the oncoming wave, when $0 < \psi < \pi$.

3. Show that the linearly polarised wave $\mathbf{E} = E_0 \hat{\mathbf{x}} \cos(kz - \omega t)$ is the sum of two circularly polarised waves of amplitude $E_0/2$ whose fields rotate in opposite senses. Show that the intensity of the linearly polarised wave is the sum of the intensities of the two circularly polarised waves.
4. A laser beam of power $\mathcal{P}$ has a circular cross-section of radius a , with the intensity uniform across it, and is linearly polarised. Find the intensity of the beam and the amplitudes of its electric and magnetic fields. The beam falls normally on a small body of mass m at rest in empty space, which absorbs it completely. Find the force on the body, and the speed of the body after it has absorbed the energy U from the beam, assuming that the speed stays small compared with c .
5. A flat, perfectly reflecting sail, a thin sheet of the kind used to drive a spacecraft by the pressure of sunlight, has area A and mass m and faces the Sun at a distance r from it. The Sun has mass m_{S} and radiates a total power $\mathcal{P}_{\text{S}}$ equally in all directions. Show that the ratio of the force of the radiation on the sail to the gravitational force on it does not depend on r , and find the largest mass per unit area of the sail for which the force of the radiation can balance the gravitational force.
6. A surface reflects a fraction $\mathcal{R}$ of the energy of a plane wave that falls on it normally and absorbs the rest. Show that the radiation pressure is $(1 + \mathcal{R})\mathcal{I}/c$. Show also that a perfect mirror on which a plane wave falls at an angle θ to the normal feels the pressure $(2\mathcal{I}/c) \cos^2 \theta$, given that the angle of reflection equals the angle of incidence.

Solutions

Chapter 1. Electrostatic Fields and Potential

§1.1 Electricity at School

1. The fields of the two charges point in opposite directions between them, so the field is zero at the distance x from q_1 at which $q_1/x^2 = q_2/(\ell - x)^2$. Taking the positive square root of both sides gives $\sqrt{q_1}(\ell - x) = \sqrt{q_2}x$, so $x = \ell\sqrt{q_1}/(\sqrt{q_1} + \sqrt{q_2})$.
2. The work done on the electron is eV , so $\frac{1}{2}m_e v^2 = eV$ and $v = \sqrt{2eV/m_e}$. The kinetic energy eV is e times V volts, which is V electronvolts by the definition of the electronvolt.
3. The acceleration towards one plate is $qE/m = qV/md$, and the particle spends a time ℓ/v between the plates. The deflection is $\frac{1}{2}(qV/md)(\ell/v)^2 = qV\ell^2/2mdv^2$. The velocity component perpendicular to the plates is then $(qV/md)(\ell/v)$, and dividing it by the parallel component v shows that the tangent of the angle between the final velocity and the plates is $qV\ell/mdv^2$.
4. The electric force qV/d balances the weight mg , so the charge has magnitude mgd/V . The electric force on the drop points upwards, and since the drop is negative, the field points downwards, from the upper plate to the lower one. The upper plate is therefore at the higher potential.

§1.2 Coulomb's Law and the Electric Field

1. Let the semicircle lie in the half-plane $x > 0$, symmetric about the x -axis and with its centre at the origin, and let ϕ run from $-\pi/2$ to $\pi/2$. The line density is $\lambda = Q/\pi a$, and the element at angle ϕ carries a charge $\lambda a d\phi$ at a distance a . By symmetry the y -components cancel, and the field points along $-\hat{\mathbf{x}}$ with magnitude $(\lambda/4\pi\epsilon_0 a) \int_{-\pi/2}^{\pi/2} \cos \phi d\phi = \lambda/2\pi\epsilon_0 a = Q/2\pi^2\epsilon_0 a^2$.
2. The element at z' is a distance $z_0 + z'$ from the field point, and its field there points along $-\hat{\mathbf{z}}$ when $\lambda > 0$. The field is $-\hat{\mathbf{z}}(\lambda/4\pi\epsilon_0) \int_0^\ell dz'/(z_0 + z')^2 = -\hat{\mathbf{z}}(\lambda/4\pi\epsilon_0)[1/z_0 - 1/(z_0 + \ell)] = -\hat{\mathbf{z}}\lambda\ell/4\pi\epsilon_0 z_0(z_0 + \ell)$. For $z_0 \gg \ell$ this becomes $-\hat{\mathbf{z}}Q/4\pi\epsilon_0 z_0^2$, with $Q = \lambda\ell$.
3. Each side lies at the perpendicular distance $s = (z^2 + \ell^2/4)^{1/2}$ from the field point, which is opposite its midpoint, so $\sin \beta_2 = -\sin \beta_1 = (\ell/2)/(s^2 + \ell^2/4)^{1/2}$ and the field of one side has magnitude $(\lambda/4\pi\epsilon_0 s) \ell/(s^2 + \ell^2/4)^{1/2}$, directed away from the side, in the plane through the axis perpendicular to the side. Its component along the axis is this magnitude times z/s , and the other components cancel between opposite sides. Adding the four sides gives $E_z = \lambda\ell z/[\pi\epsilon_0(z^2 + \ell^2/4)(z^2 + \ell^2/2)^{1/2}]$. For $z \gg \ell$ this becomes $\lambda\ell/\pi\epsilon_0 z^2 = 4\lambda\ell/4\pi\epsilon_0 z^2$, the field of a point charge equal to the total charge $4\lambda\ell$.
4. By the quotient rule, the derivative of $z/(a^2 + z^2)^{3/2}$ is $[(a^2 + z^2) - 3z^2]/(a^2 + z^2)^{5/2}$, which vanishes at $z = a/\sqrt{2}$. There $a^2 + z^2 = 3a^2/2$, and the field is $Q(a/\sqrt{2})/[4\pi\epsilon_0(3a^2/2)^{3/2}] = Q/6\sqrt{3}\pi\epsilon_0 a^2$.
5. The rings now run from $s' = a$ to $s' = b$, and the integral of the text gives $E_z = (\sigma z/2\epsilon_0)[(a^2 + z^2)^{-1/2} - (b^2 + z^2)^{-1/2}]$ for $z > 0$. As $a \rightarrow 0$ the first term becomes $1/z$, and the field of a disc of radius b is recovered.

§1.3 Electric Flux and Gauss's Law

1. The six faces are equivalent and together receive the total flux q/ϵ_0 , so each receives $q/6\epsilon_0$. With the charge at a corner, the cube is one of eight equal cubes meeting at that corner, which

together enclose the charge, so each receives a flux $q/8\epsilon_0$ through its outer faces. The three faces that meet at the corner contain the charge in their planes, so the field lies along them and they receive no flux. The flux through the whole surface is $q/8\epsilon_0$, and by symmetry each of the other three faces receives $q/24\epsilon_0$. Gauss's law does not apply directly here, since the charge lies on the surface, and the answer rests on the division into eight cubes.

2. A Gaussian cylinder of radius s and length ℓ encloses $\rho\pi s^2\ell$ for $s < a$ and $\rho\pi a^2\ell$ for $s > a$, and its flux is $E2\pi s\ell$. The field is $\rho s/2\epsilon_0$ inside and $\rho a^2/2\epsilon_0 s$ outside, pointing away from the axis. Inside, $\mathbf{E} = (\rho/2\epsilon_0)(x, y, 0)$, whose divergence is $(\rho/2\epsilon_0)(1+1) = \rho/\epsilon_0$.
3. By symmetry the field points along $\pm\hat{\mathbf{x}}$, away from the plane $x = 0$, with the same magnitude at x and $-x$. A pillbox with end faces of area A at $\pm x$ has flux $2EA$ and encloses $2\rho Ax$ for $x < \ell$ and $2\rho A\ell$ for $x > \ell$. The field is $\rho x/\epsilon_0$ along $\hat{\mathbf{x}}$ inside the slab and has magnitude $\rho\ell/\epsilon_0$ outside, pointing away from the slab.
4. A Gaussian sphere of radius r encloses no charge for $r < a$, Q_1 for $a < r < b$, and $Q_1 + Q_2$ for $r > b$. The field is therefore zero inside the inner shell, $Q_1\hat{\mathbf{r}}/4\pi\epsilon_0 r^2$ between the shells, and $(Q_1 + Q_2)\hat{\mathbf{r}}/4\pi\epsilon_0 r^2$ outside. The field outside vanishes when $Q_2 = -Q_1$.
5. The flux through a sphere of radius r is $4\pi r^2 E_0 r^2/a^2$, so $Q_{\text{enc}} = 4\pi\epsilon_0 E_0 r^4/a^2$. The charge in the shell between r and $r + dr$ is $\rho 4\pi r^2 dr = dQ_{\text{enc}} = 16\pi\epsilon_0 E_0 r^3 dr/a^2$, so $\rho = 4\epsilon_0 E_0 r/a^2$. For the second method, write $\mathbf{E} = (E_0/a^2)r\mathbf{r}$, so that $E_x = (E_0/a^2)rx$. By the product rule, $\partial(rx)/\partial x = r + x^2/r$, and adding the terms for y and z gives $3r + r = 4r$. Then $\nabla \cdot \mathbf{E} = 4E_0 r/a^2$, and $\rho = 4\epsilon_0 E_0 r/a^2$ as before.
6. The closed surface made of the hemisphere and the flat disc encloses no charge, so the total flux is zero. The field crosses the disc along its axis, giving a flux of magnitude $\pi a^2 E$ in through the disc, so the flux out through the curved surface is $\pi a^2 E$.

§1.4 The Electrostatic Potential and Field Energy

1. By the chain rule the derivative is $(1 + z'/\sqrt{z'^2 + s^2})/(z' + \sqrt{z'^2 + s^2})$, and writing the numerator over the common denominator $\sqrt{z'^2 + s^2}$ reduces it to $1/\sqrt{z'^2 + s^2}$. The potential is $V = (\lambda/4\pi\epsilon_0) \int_{-\ell}^{\ell} dz'/\sqrt{z'^2 + s^2} = (\lambda/2\pi\epsilon_0) \ln[(\ell + \sqrt{\ell^2 + s^2})/s]$, where the two limits have been combined using $(\sqrt{\ell^2 + s^2} + \ell)(\sqrt{\ell^2 + s^2} - \ell) = s^2$. Differentiating, $E_s = -\partial V/\partial s = \lambda\ell/(2\pi\epsilon_0 s\sqrt{\ell^2 + s^2})$. In Section 1.2 the point opposite the midpoint has $\sin\beta_2 = -\sin\beta_1 = \ell/\sqrt{\ell^2 + s^2}$, which gives the same field.
2. By Section 1.3 the field is zero inside and $Q\hat{\mathbf{r}}/4\pi\epsilon_0 r^2$ outside, so $V = Q/4\pi\epsilon_0 r$ outside and $V = Q/4\pi\epsilon_0 a$ everywhere inside, since the potential is continuous and has zero gradient there. The charge sits where $V = Q/4\pi\epsilon_0 a$, so $\frac{1}{2} \int \sigma V da = Q^2/8\pi\epsilon_0 a$. The field energy lies entirely outside and was found in the text to be $Q^2/8\pi\epsilon_0 a$.
3. There are six pairs, of which four are sides at distance ℓ and two are diagonals at distance $\sqrt{2}\ell$. The work is $(q^2/4\pi\epsilon_0\ell)(4 + 2/\sqrt{2}) = (4 + \sqrt{2})q^2/4\pi\epsilon_0\ell$.
4. The field inside is $\rho s/2\epsilon_0$ away from the axis, so $V(s) = -\int_0^s \rho s' ds'/2\epsilon_0 = -\rho s^2/4\epsilon_0 = -\rho(x^2 + y^2)/4\epsilon_0$. Its Laplacian is $-\rho(2 + 2)/4\epsilon_0 = -\rho/\epsilon_0$, as Poisson's equation requires.
5. For $\mathbf{E}_1$, $\partial E_x/\partial y = \kappa = \partial E_y/\partial x$, and the other two equalities hold because $E_z = 0$ and E_x and E_y do not depend on z . The potential is $V = -\kappa xy$, since $-\nabla(-\kappa xy) = \kappa(y, x, 0)$, and the charge density is $\rho = -\epsilon_0 \nabla^2 V = 0$. For $\mathbf{E}_2$, $\partial E_x/\partial y = \kappa$ while $\partial E_y/\partial x = -\kappa$, so it fails the test. A similar pair of forces appeared in Problem 1 of Section 2.3 of Vol. 1.
6. When the sphere has been built up to radius r , it carries the charge $Q(r) = Qr^3/a^3$ and its surface is at the potential $Q(r)/4\pi\epsilon_0 r$. Bringing in the next shell, of charge $dq = 3Qr^2 dr/a^3$, takes work $Q(r) dq/4\pi\epsilon_0 r$. Integrating from 0 to a gives $(3Q^2/4\pi\epsilon_0 a^6) \int_0^a r^4 dr = 3Q^2/20\pi\epsilon_0 a$.
7. The work is q times the potential at the centre, $qV(0) = 3qQ/8\pi\epsilon_0 a$, which is one and a half times the work needed to

bring the charge to the surface.

Chapter 2. Conductors and Capacitance

§2.1 Conductors and Capacitance

1. The field outside an isolated sphere carrying a charge q_a is that of a point charge at its centre, by Gauss's law applied to a concentric sphere as in Section 1.3, so the potential at a distance r from its centre is $q_a/4\pi\epsilon_0 r$, the potential of a point charge found in Section 1.4, with the potential zero at infinity, and its surface is at the potential $q_a/4\pi\epsilon_0 a$. The wire makes the spheres into one conductor at a single potential, so $q_a/a = q_b/b$, and with $q_a + q_b = Q$ this gives $q_a = Qa/(a + b)$ and $q_b = Qb/(a + b)$. The surface densities are $\sigma_a = q_a/4\pi a^2$ and $\sigma_b = q_b/4\pi b^2$, so $\sigma_a/\sigma_b = (a/b)(b^2/a^2) = b/a$. The field just outside is σ/ϵ_0 , so the smaller sphere has the larger field at its surface.
2. A closed surface in the material that surrounds the cavity has no flux through it, so it encloses no net charge, and the wall of the cavity carries $-q$. The conductor has net charge Q , so its outer surface carries $Q + q$. Rotating the arrangement about the centre leaves it unchanged, so the charge on the wall of the cavity and the charge on the outer surface are both spread uniformly. Gauss's law applied to spheres centred on the charge then gives a field $q\hat{\mathbf{r}}/4\pi\epsilon_0 r^2$ for $r < a$, zero for $a < r < b$, and $(Q + q)\hat{\mathbf{r}}/4\pi\epsilon_0 r^2$ for $r > b$. For $r > b$ the field is that of a point charge $Q + q$ at the centre, so the potential of a point charge found in Section 1.4 gives the potential of the conductor, $(Q + q)/4\pi\epsilon_0 b$.
3. As in Figure 14, the field between the plates induces charges on the faces of the slab, and the field inside the slab is zero. Let x_1 and x_2 be the gaps between the slab and the two plates, so that $x_1 + x_2 = d - h$. The slab is an isolated uncharged conductor, like the one in the dashed box of Figure 20(b), so the arrangement consists of two parallel-plate capacitors in series, of capacitances $\epsilon_0 A/x_1$ and $\epsilon_0 A/x_2$. Then $1/C = x_1/\epsilon_0 A + x_2/\epsilon_0 A = (d - h)/\epsilon_0 A$, which depends on x_1 and x_2 only through their sum.
4. The field between the conductors has magnitude $\lambda/2\pi\epsilon_0 s$, so the energy density is $u = \lambda^2/8\pi^2\epsilon_0 s^2$. The thin cylindrical shell

between the radii s and $s + ds$, of length ℓ , has volume $2\pi s\ell ds$, so the energy is $U = \int_a^b (\lambda^2 \ell / 4\pi\epsilon_0) ds/s = (\lambda^2 \ell / 4\pi\epsilon_0) \ln(b/a)$. With $C = 2\pi\epsilon_0 \ell / \ln(b/a)$ and $Q = \lambda\ell$, the energy $Q^2/2C = \lambda^2 \ell^2 \ln(b/a) / 4\pi\epsilon_0 \ell = (\lambda^2 \ell / 4\pi\epsilon_0) \ln(b/a)$ is the same expression.

5. With separation x , the charge is $Q = \epsilon_0 AV/x$ and the stored energy is $U = \epsilon_0 AV^2/2x$. The change in energy is $\Delta U = \frac{1}{2}\epsilon_0 AV^2(1/d_2 - 1/d_1)$, and the change in charge is $\Delta Q = \epsilon_0 AV(1/d_2 - 1/d_1)$, both negative since $d_2 > d_1$. The battery does work $V\Delta Q = \epsilon_0 AV^2(1/d_2 - 1/d_1)$ in moving the charge ΔQ onto the positive plate, and this work is negative, so energy is delivered to the battery. The attractive force at separation x has magnitude $Q^2/2\epsilon_0 A = \epsilon_0 AV^2/2x^2$, so the agent does work $\int_{d_1}^{d_2} \epsilon_0 AV^2 dx/2x^2 = \frac{1}{2}\epsilon_0 AV^2(1/d_1 - 1/d_2)$. Adding the work done by the agent and the work done by the battery gives $\frac{1}{2}\epsilon_0 AV^2(1/d_2 - 1/d_1) = \Delta U$, so the energy is conserved. With V fixed, $-dU/dx = +\epsilon_0 AV^2/2x^2$ would describe a repulsion. This derivative has the wrong sign because it leaves out the energy delivered to the battery, which is twice the decrease in U .
6. The charge $C_1 V_i$ is shared between the two capacitors, which reach a common potential difference V , so $C_1 V_i = (C_1 + C_2)V$ and $V = C_1 V_i / (C_1 + C_2)$. The initial energy is $\frac{1}{2}C_1 V_i^2$, and the final energy is $\frac{1}{2}(C_1 + C_2)V^2 = \frac{1}{2}C_1^2 V_i^2 / (C_1 + C_2)$, so the ratio of the final to the initial energy is $C_1 / (C_1 + C_2)$, and the energy falls by the fraction $C_2 / (C_1 + C_2)$. The missing energy is dissipated in the connecting wires while the charge flows.

Chapter 3. Electric Fields in Matter

§3.1 Dipoles and Polarisation

1. Moving the origin by $\mathbf{c}$ replaces each $\mathbf{r}_i$ by $\mathbf{r}_i - \mathbf{c}$, and the dipole moment becomes $\sum_i q_i \mathbf{r}_i - \mathbf{c} \sum_i q_i$, whose second term is zero because the total charge is zero. For the given charges the charge at the origin contributes nothing, and $\mathbf{p} = qa\hat{\mathbf{x}} + qa\hat{\mathbf{y}} = qa(\hat{\mathbf{x}} + \hat{\mathbf{y}})$.
2. The condition is $dr/r d\theta = E_r/E_\theta = 2\cos\theta/\sin\theta$. Multiplying by $d\theta$ and integrating, with the derivative of $\ln\sin\theta$ being $\cos\theta/\sin\theta$, gives $\ln r = 2\ln\sin\theta$ plus a constant, so $r =$

$r_0 \sin^2 \theta$. The gradient of $\cos \theta / r^2$ has components $-2 \cos \theta / r^3$ along $\hat{\mathbf{r}}$ and $-\sin \theta / r^3$ along $\hat{\boldsymbol{\theta}}$, which are proportional to E_r and E_θ , since $\cos \theta / r^2$ is a constant multiple of V . A step along a field line has components $(dr, r d\theta)$ proportional to $(2 \cos \theta, \sin \theta)$, and a step along an equipotential surface has zero scalar product with the gradient, so its components are proportional to $(\sin \theta, -2 \cos \theta)$. The scalar product of these two steps is $2 \cos \theta \sin \theta - 2 \sin \theta \cos \theta = 0$.

3. Seen from the dipole, the charge lies at the displacement $-\mathbf{r}\hat{\mathbf{r}}$, which is on the axis of the dipole on the side opposite to $\mathbf{p}$. The field of the dipole there is $[3(\mathbf{p} \cdot (-\hat{\mathbf{r}}))(-\hat{\mathbf{r}}) - \mathbf{p}]/4\pi\epsilon_0 r^3 = (3p - p)\hat{\mathbf{r}}/4\pi\epsilon_0 r^3 = 2p\hat{\mathbf{r}}/4\pi\epsilon_0 r^3$. The force on the charge is Q times this field, $2pQ\hat{\mathbf{r}}/4\pi\epsilon_0 r^3$, which is equal and opposite to the force on the dipole, as Newton's third law requires.
4. The field of the charge at the atom is $\mathbf{E} = Q\hat{\mathbf{r}}/4\pi\epsilon_0 r^2$, so the induced moment is $\mathbf{p} = \alpha Q\hat{\mathbf{r}}/4\pi\epsilon_0 r^2$, which lies along the radial direction. As in the example of the text, $\mathbf{p} \cdot \nabla$ is then $p \partial/\partial r$, and $\mathbf{F} = p(\partial E_r/\partial r)\hat{\mathbf{r}} = (\alpha Q/4\pi\epsilon_0 r^2)(-2Q/4\pi\epsilon_0 r^3)\hat{\mathbf{r}} = -2\alpha Q^2\hat{\mathbf{r}}/(4\pi\epsilon_0)^2 r^5$. The force depends on Q^2 , so it points towards the charge for either sign of Q , since the induced moment always lines up with the field.
5. The factors of $d\phi$ integrate to 2π in the numerator and in the denominator of the mean, and cancel. With the substitution $\eta = \cos \theta$, the denominator is $\int_{-1}^1 e^{\xi\eta} d\eta = 2 \sinh \xi/\xi$, and the numerator is $p \int_{-1}^1 \eta e^{\xi\eta} d\eta$. Integrating by parts, with $e^{\xi\eta}$ the factor integrated, gives $\int_{-1}^1 \eta e^{\xi\eta} d\eta = [\eta e^{\xi\eta}/\xi]_{-1}^1 - (1/\xi) \int_{-1}^1 e^{\xi\eta} d\eta = 2 \cosh \xi/\xi - 2 \sinh \xi/\xi^2$. Dividing gives $\langle p_E \rangle = p(\coth \xi - 1/\xi)$. For small ξ , the expansions $\cosh \xi \approx 1 + \xi^2/2$ and $\sinh \xi \approx \xi(1 + \xi^2/6)$ give $\coth \xi \approx (1/\xi)(1 + \xi^2/2)(1 - \xi^2/6) \approx 1/\xi + \xi/3$, so $\langle p_E \rangle \approx p\xi/3 = p^2 E/3k_B \mathcal{T}$.
6. The outward normal of a sphere is $\hat{\mathbf{r}}$, so $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{r}} = P \cos \theta$. Gauss's law applied to a sphere of radius r inside a ball of charge density ρ gives $E \cdot 4\pi r^2 = \rho \cdot \frac{4}{3}\pi r^3/\epsilon_0$, so the field inside the ball is $\rho(\mathbf{r} - \mathbf{r}_c)/3\epsilon_0$, where $\mathbf{r}_c$ is the position of its centre. With the positive ball centred at $\ell/2$ and the negative ball at $-\ell/2$, the field where the balls overlap is $\rho(\mathbf{r} - \ell/2)/3\epsilon_0 - \rho(\mathbf{r} + \ell/2)/3\epsilon_0 =$

$-\rho\ell/3\epsilon_0 = -\mathbf{P}/3\epsilon_0$. Outside, each ball produces the field of a point charge $\pm\frac{4}{3}\pi a^3\rho$ at its centre, by the shell theorem of Section 1.3, so the pair produces the field of a dipole of moment $\frac{4}{3}\pi a^3\rho\ell = \frac{4}{3}\pi a^3\mathbf{P}$, which becomes a point dipole as ℓ goes to zero with $\mathbf{P}$ fixed.

7. The x -component of the gradient is $\partial/\partial x'$ of $[(x-x')^2+(y-y')^2+(z-z')^2]^{-1/2}$, which by the chain rule is $(x-x')/|\mathbf{r}-\mathbf{r}'|^3$, and the other components follow in the same way. The potential of the dipoles in a small volume $d\mathcal{V}'$ is that of a point dipole $\mathbf{P} d\mathcal{V}'$, and adding these potentials gives the stated integral. With $g = 1/|\mathbf{r}-\mathbf{r}'|$ the integrand is $\mathbf{P} \cdot \nabla' g = \nabla' \cdot (g\mathbf{P}) - g \nabla' \cdot \mathbf{P}$. The divergence theorem turns the integral of the first term into $\oint g \mathbf{P} \cdot d\mathbf{a}'$ over the surface of the body, so that $4\pi\epsilon_0 V(\mathbf{r}) = \oint (\mathbf{P} \cdot \hat{\mathbf{n}}) d\mathbf{a}'/|\mathbf{r}-\mathbf{r}'| + \int (-\nabla' \cdot \mathbf{P}) d\mathcal{V}'/|\mathbf{r}-\mathbf{r}'|$, which is the potential of the charge densities $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$ and $\rho_b = -\nabla' \cdot \mathbf{P}$.

§3.2 The Electric Displacement and Linear Dielectrics

1. By the argument of the text, a pillbox with one face inside the positive plate shows that $D = \sigma_f = Q/A$ everywhere in the gap, in the slab and in the empty parts alike. The field is σ_f/ϵ_0 in the empty parts, whose total thickness is $d-w$, and $\sigma_f/\epsilon_r\epsilon_0$ in the slab, so $V = \sigma_f[(d-w) + w/\epsilon_r]/\epsilon_0$, and $C = Q/V = \epsilon_0 A/(d-w + w/\epsilon_r)$. Only the total thickness of the empty parts enters, so the position of the slab does not matter. The polarisation of the slab is $(\chi_e/\epsilon_r)\sigma_f$, so the bound surface charge density is $-(\chi_e/\epsilon_r)Q/A$ on the face nearer the positive plate and $+(\chi_e/\epsilon_r)Q/A$ on the other face.
2. Each plate is an equipotential surface, by Section 2.1, so the field has magnitude V/d in both halves. The displacement in the half with relative permittivity ϵ_{r1} is $\epsilon_{r1}\epsilon_0 V/d$, and a pillbox at the positive plate shows that this equals the free surface charge density there; similarly the other half has $\epsilon_{r2}\epsilon_0 V/d$. The total charge is $(A/2)(\epsilon_{r1} + \epsilon_{r2})\epsilon_0 V/d$, and dividing by V gives the stated capacitance, which is that of two capacitors of area $A/2$ connected in parallel, as in Section 2.1.
3. The charge on the positive plate is $\epsilon_0 AV/d$ before the slab is inserted and $\epsilon_r\epsilon_0 AV/d$ after, so the charge that flows is $(\epsilon_r -$

- 1) $\epsilon_0 AV/d = \chi_e \epsilon_0 AV/d$, and the free charge added per unit area is $\chi_e \epsilon_0 V/d$. The field in the slab is V/d , so its polarisation is $\epsilon_0 \chi_e V/d$, and the bound surface charge density on the face next to the positive plate is $-\epsilon_0 \chi_e V/d$, equal in magnitude to the free charge added per unit area.
4. We apply Gauss's law in matter to a cylinder of radius s and length h coaxial with the cable, with $a < s < b$. By symmetry $\mathbf{D}$ points along $\hat{\mathbf{s}}$ and has the same magnitude over the curved surface, and no displacement crosses the flat ends, so $D \cdot 2\pi sh = \lambda h$. Therefore $\mathbf{D} = \lambda \hat{\mathbf{s}}/2\pi s$ and $\mathbf{E} = \lambda \hat{\mathbf{s}}/2\pi \epsilon_r \epsilon_0 s$. The potential difference is $\int_a^b E ds = (\lambda/2\pi \epsilon_r \epsilon_0) \ln(b/a)$, and dividing λ by it gives the capacitance per unit length. The polarisation is $\mathbf{P} = \chi_e \lambda \hat{\mathbf{s}}/2\pi \epsilon_r s$, and on the inner surface of the dielectric $\hat{\mathbf{n}} = -\hat{\mathbf{s}}$, so $\sigma_b = -\chi_e \lambda/2\pi \epsilon_r a$. Multiplying by the circumference $2\pi a$ gives the bound charge per unit length, $-\chi_e \lambda/\epsilon_r$.
5. A sphere of radius $r < a$ encloses the free charge $\frac{4}{3}\pi r^3 \rho_f$, so $\mathbf{D} = \rho_f \mathbf{r}/3$ and $\mathbf{E} = \rho_f \mathbf{r}/3\epsilon_r \epsilon_0$ inside. Outside, the enclosed free charge is $\frac{4}{3}\pi a^3 \rho_f$, so $\mathbf{D} = \rho_f a^3 \hat{\mathbf{r}}/3r^2$ and $\mathbf{E} = \mathbf{D}/\epsilon_0$. The relation in the text gives $\rho_b = -(\chi_e/\epsilon_r)\rho_f$. Directly, $\mathbf{P} = (\chi_e/\epsilon_r)\rho_f \mathbf{r}/3$, and since $\nabla \cdot \mathbf{r} = \partial x/\partial x + \partial y/\partial y + \partial z/\partial z = 3$, we get $-\nabla \cdot \mathbf{P} = -(\chi_e/\epsilon_r)\rho_f$, as before. At the surface $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{r}} = \chi_e \rho_f a/3\epsilon_r$. The total bound charge is $-(\chi_e/\epsilon_r)\rho_f \cdot \frac{4}{3}\pi a^3 + (\chi_e \rho_f a/3\epsilon_r) \cdot 4\pi a^2 = 0$.
6. Integrating the field inwards from infinity, as in Section 1.4, the part of the path outside the shell gives $q/4\pi \epsilon_0 b$, and the part inside the dielectric gives $(q/4\pi \epsilon_r \epsilon_0)(1/r - 1/b)$, so $V(r) = (q/4\pi \epsilon_0)[1/\epsilon_r r + (\epsilon_r - 1)/\epsilon_r b]$. For the check, at a point with $a < r < b$ the charge q gives $q/4\pi \epsilon_0 r$. The bound charge $-\chi_e q/\epsilon_r$ on the inner surface lies inside the sphere through the point, and by the shell theorem of Section 1.3 it gives $-\chi_e q/4\pi \epsilon_r \epsilon_0 r$. The bound charge $\chi_e q/\epsilon_r$ on the outer surface produces no field inside that surface, by the same theorem, so its potential there is constant and equal to its value at the surface, $\chi_e q/4\pi \epsilon_r \epsilon_0 b$. Adding these, and using $1 - \chi_e/\epsilon_r = 1/\epsilon_r$ and $\chi_e = \epsilon_r - 1$, gives the same result.

Chapter 4. Electric Current and Circuits

§4.1 Circuits at School

1. Each series pair has resistance $2R$, and two resistances $2R$ in parallel give R . When the fifth resistor is added, the two mid-points are at the same potential, since each lies halfway along identical pairs, so no current flows through the fifth resistor and the combined resistance is still R .
2. The current is $\mathcal{E}/(R + R_{\text{int}})$, so the power in the load is $\mathcal{P} = \mathcal{E}^2 R / (R + R_{\text{int}})^2$. Differentiating with respect to R with the quotient rule gives $d\mathcal{P}/dR = \mathcal{E}^2 (R_{\text{int}} - R) / (R + R_{\text{int}})^3$, which is zero at $R = R_{\text{int}}$, positive below it, and negative above it. The greatest power is $\mathcal{E}^2 / 4R_{\text{int}}$.
3. The loop equations give $I_1 = (\mathcal{E}_1 - I_3 R_3) / R_1$ and $I_2 = (\mathcal{E}_2 - I_3 R_3) / R_2$. Adding these two expressions and using $I_1 + I_2 = I_3$ gives $\mathcal{E}_1 / R_1 + \mathcal{E}_2 / R_2 = I_3 (1 + R_3 / R_1 + R_3 / R_2)$. Multiplying both sides by $R_1 R_2$ and dividing by the bracket gives the stated result.
4. The load and R_2 in parallel have resistance $R_2 R_L / (R_2 + R_L)$, and the divider formula with this resistance in place of R_2 gives $V_{\text{out}} = V_{\text{in}} R_2 R_L / (R_1 R_2 + R_1 R_L + R_2 R_L)$. Dividing the numerator and the denominator by R_L and letting R_2 / R_L go to zero leaves $V_{\text{in}} R_2 / (R_1 + R_2)$.
5. The stored energy is $Q^2 / 2C$, which goes as $e^{-2t/\tau}$, and this falls to half its initial value when $2t/\tau = \ln 2$.
6. In the inductor $dI/dt = I_f / t_f$, so the electromotive force has magnitude LI_f / t_f .

§4.2 Current Density, Conductivity, and Ohm's Law

1. A current I flowing radially crosses every cylinder of radius s between a and b , so $J = I / 2\pi s \ell$ and $E = I / 2\pi \sigma_c s \ell$. Integrating the field from a to b gives $V = (I / 2\pi \sigma_c \ell) \ln(b/a)$, so $R = \ln(b/a) / 2\pi \sigma_c \ell$. Section 2.1 found $C = 2\pi \epsilon_0 \ell / \ln(b/a)$ for the coaxial cylinders, and the product of these expressions for R and C is ϵ_0 / σ_c . For the concentric spheres Section 2.1 found

$C = 4\pi\epsilon_0/(1/a - 1/b)$, and multiplying by the resistance of the shell again gives ϵ_0/σ_c .

2. Let us take a closed surface that surrounds the positive conductor and lies in the material. The current leaving the conductor is $I = \oint \mathbf{J} \cdot d\mathbf{a} = \sigma_c \oint \mathbf{E} \cdot d\mathbf{a}$, and by Gauss's law the last integral is Q/ϵ_0 for the assumed field. So $I = \sigma_c Q/\epsilon_0$, and $R = V/I = \epsilon_0 V/(\sigma_c Q) = \epsilon_0/(\sigma_c C)$, since $C = Q/V$. The two arrangements of Problem 1 are cases of this result.
3. Let x be the distance from the narrow end, so that the radius is $a + (b - a)x/\ell$. A slice of thickness dx has resistance $dx/(\sigma_c \pi[a + (b - a)x/\ell]^2)$, and the slices are in series, so their resistances add. Substituting the radius $s = a + (b - a)x/\ell$ of the slice, with $dx = \ell ds/(b - a)$, gives $R = [\ell/\pi\sigma_c(b - a)] \int_a^b s^{-2} ds = [\ell/\pi\sigma_c(b - a)](1/a - 1/b) = \ell/(\pi\sigma_c ab)$. When $a = b$ this is $\ell/(\sigma_c \pi a^2)$, the resistance of a uniform wire of cross-sectional area πa^2 .
4. The charge at distance s moves with speed ωs around the axis, so $K = \sigma\omega s$, directed around the axis. The current crossing a radius is $\int_0^a K ds = \frac{1}{2}\sigma\omega a^2$. In one revolution, which takes the time $2\pi/\omega$, all the charge $\sigma\pi a^2$ crosses the radius once, which gives the current $\sigma\pi a^2\omega/2\pi$, the same value.
5. The equation is that of a body falling from rest with linear drag, and the solution from Section 1.3 of Vol. 1, with the substitutions given in the text, is $\mathbf{v}_d = -(e\tau_c \mathbf{E}/m_e)(1 - e^{-t/\tau_c})$. At $t = \tau_c$ the bracket is $1 - 1/e$. In the wire $I = JA = n_e e v_d A$ in magnitude, so $v_d = I/(n_e e A)$.
6. From the text, $J = I/4\pi r^2$ and $E = I/4\pi\sigma_c r^2$, so $\mathbf{J} \cdot \mathbf{E} = I^2/(16\pi^2\sigma_c r^4)$. The volume between the spheres of radii r and $r + dr$ is $4\pi r^2 dr$, and integrating gives $(I^2/4\pi\sigma_c) \int_a^b r^{-2} dr = (I^2/4\pi\sigma_c)(1/a - 1/b) = I^2 R$.
7. By Section 3.2, the field in the dielectric is $E = Q/(\epsilon_r \epsilon_0 A)$, directed from the positive plate to the negative one. The current through the dielectric is $I = \sigma_c EA = \sigma_c Q/(\epsilon_r \epsilon_0)$, and it carries charge off the positive plate, so $dQ/dt = -(\sigma_c/\epsilon_r \epsilon_0)Q$. Solving this equation with $Q = Q_0$ at $t = 0$ gives the stated result, in

which A and d do not appear.

§4.3 Transients in RC and RL Circuits

1. In a step in which the electromotive force is raised from V_a to V_b , the charge $C(V_b - V_a)$ passes through the source, which does the work $CV_b(V_b - V_a)$, while the stored energy rises by $\frac{1}{2}C(V_b^2 - V_a^2)$. The difference, which is the heat produced, is $\frac{1}{2}C(V_b - V_a)^2$. With two steps of $\mathcal{E}/2$ the total heat is $2 \times \frac{1}{8}C\mathcal{E}^2 = \frac{1}{4}C\mathcal{E}^2$, and with N steps of $\mathcal{E}/N$ it is $N \times C\mathcal{E}^2/2N^2 = C\mathcal{E}^2/2N$.
2. Let Q_a and Q_b be the charges on the first and second capacitors. Charge is conserved, so $Q_a + Q_b = Q_0$, and the current is $I = dQ_b/dt$. The loop law gives $Q_a/C = IR + Q_b/C$, so $R dQ_b/dt = (Q_0 - 2Q_b)/C$. This is the charging equation with R replaced by $R/2$, $C\mathcal{E}$ replaced by $Q_0/2$, and the time constant $RC/2$, so $Q_b = \frac{1}{2}Q_0(1 - e^{-2t/RC})$ and $Q_a = \frac{1}{2}Q_0(1 + e^{-2t/RC})$. The initial energy is $Q_0^2/2C$, and the final energy is $2 \times (Q_0/2)^2/2C = Q_0^2/4C$. The current is $I = (Q_0/RC)e^{-2t/RC}$, and $\int_0^\infty I^2 R dt = (Q_0^2/RC^2)(RC/4) = Q_0^2/4C$, which is the energy lost.
3. During the decay $\int_{t_1}^\infty I^2 R dt = I_1^2 R \int_{t_1}^\infty e^{-2(t-t_1)/\tau} dt = I_1^2 R \tau / 2 = \frac{1}{2} L I_1^2$. During the growth, integrating the energy equation from $t = 0$, when $I = 0$, to the time at which $I = I_1$ gives $\int (\mathcal{E}I - I^2 R) dt = \frac{1}{2} L I_1^2$.
4. With the switch closed for a long time the current is steady, so there is no potential difference across the inductor and none across R_2 . No current then flows through R_2 , and the current through R_1 and the inductor is $\mathcal{E}/R_1$. When the switch is opened, the current through the inductor cannot change suddenly, and its only path is now through R_2 , so the current $\mathcal{E}/R_1$ flows through R_2 and the potential difference across it is $\mathcal{E}R_2/R_1$. The loop containing L and R_2 gives $L dI/dt + IR_2 = 0$, so the current and the potential difference decay as $e^{-R_2 t/L}$. When no resistor is connected across the inductor, the open switch itself acts as a resistance that is very large, so the potential difference across its contacts just after they separate is very large, and it can be large enough to make the air between the contacts conduct, which is seen as a spark.

5. The loop law gives $R dQ/dt + Q/C = \dot{\mathcal{E}} t$. Substituting the trial solution, the terms in Q_1 give $-RQ_1 e^{-t/\tau}/\tau + Q_1 e^{-t/\tau}/C = 0$, since $\tau = RC$, and the other terms give $RC\dot{\mathcal{E}} + \dot{\mathcal{E}}(t - \tau) = \dot{\mathcal{E}} t$, so the trial solution satisfies the equation for any Q_1 . The condition $Q = 0$ at $t = 0$ gives $Q_1 = C\dot{\mathcal{E}}\tau$, so $Q = C\dot{\mathcal{E}}(t - \tau + \tau e^{-t/\tau})$. When $t \gg \tau$ the exponential is negligible and $Q/C = \dot{\mathcal{E}}(t - \tau)$, which is the electromotive force at the time $t - \tau$.
6. The potential difference across the resistor is $IR = \mathcal{E}e^{-t/\tau}$ and across the capacitor is $\mathcal{E}(1 - e^{-t/\tau})$, and they are equal when $e^{-t/\tau} = \frac{1}{2}$, that is, at $t = \tau \ln 2$. In the RL circuit $I = (\mathcal{E}/R)(1 - e^{-t/\tau})$, which is half its final value when $e^{-t/\tau} = \frac{1}{2}$, again at $t = \tau \ln 2$ with $\tau = L/R$.

Chapter 5. Magnetostatics

§5.1 Magnetism at School

1. The speed after acceleration is given by $qV = \frac{1}{2}mv^2$, so $v = \sqrt{2qV/m}$. The radius is $r_c = mv/qB = \sqrt{2mV/q}/B$, and the time to go once round is $2\pi m/qB$, which does not depend on V .
2. In the crossed fields the electric and magnetic forces balance, $qE = qvB_1$, so $v = E/B_1$. In the second region $r_c = mv/qB_2$, so $m = qB_1B_2r_c/E$.
3. The magnetic force on the rod is horizontal, with magnitude $BI\ell$, since the rod is at right angles to the field. The tension in the threads balances the weight mg and this horizontal force, so the threads make an angle θ with the vertical given by $\tan \theta = BI\ell/mg$.
4. Inside the solenoid the field is uniform, with magnitude $\mu_0 nI$, so the flux through the inner loop is $\mu_0 nI\pi a^2 \cos \theta$. The field outside a long solenoid is very small, so the flux through the outer loop is only the flux through the part of the loop inside the solenoid, which is $\mu_0 nI\pi a^2$, the same as for a loop of radius a perpendicular to the axis.

§5.2 Magnetic Forces and the Biot–Savart Law

1. The field is uniform, so the force is $I \left(\int_P^Q d\boldsymbol{\ell} \right) \times \mathbf{B}$, and the sum of the small displacements from P to Q is the displacement $\boldsymbol{\ell}_{PQ}$ along any path. For the semicircle, $\boldsymbol{\ell}_{PQ}$ is a diameter, of length $2a$, perpendicular to the field, so the force has magnitude $2IaB$ and is perpendicular to both the diameter and the field.
2. Each side is seen from the centre at perpendicular distance $s = \ell/2$, and its ends are at $z = \pm\ell/2$, so $\sin\beta_2 - \sin\beta_1 = 2(\ell/2)/(\ell/\sqrt{2}) = \sqrt{2}$. Each side gives $\mu_0 I \sqrt{2}/2\pi\ell$, and the four sides give fields in the same direction, so the total is $2\sqrt{2}\mu_0 I/\pi\ell$. A circle with circumference 4ℓ has radius $2\ell/\pi$, and the field at its centre is $\mu_0 I/2a = \pi\mu_0 I/4\ell$. The ratio of the field of the square to that of the circle is $8\sqrt{2}/\pi^2$, which is slightly greater than one.
3. The element at azimuth ϕ is $d\boldsymbol{\ell} = b d\phi \hat{\boldsymbol{\phi}}$, and $\hat{\boldsymbol{\phi}} \times \hat{\mathbf{s}} = -\hat{\mathbf{z}}$ and $\hat{\boldsymbol{\phi}} \times \hat{\mathbf{z}} = \hat{\mathbf{s}}$. The z -component of $I d\boldsymbol{\ell} \times \mathbf{B}$ is therefore $-Ib d\phi B_s$, and the components along $\hat{\mathbf{s}}$ cancel round the loop. At the rim $B_s = -\frac{1}{2}B_0\kappa b$, so $F_z = -2\pi b I(-\frac{1}{2}B_0\kappa b) = \pi b^2 I B_0\kappa$. With $\mu = I\pi b^2$ and $\partial B_z/\partial z = B_0\kappa$, this is $\mu \partial B_z/\partial z$.
4. The charge goes round once in the time $2\pi m/|q|B$, so the current is $|q|^2 B/2\pi m$. The radius is $mv_\perp/|q|B$, so the area is $\pi m^2 v_\perp^2/q^2 B^2$, and the moment has magnitude $mv_\perp^2/2B$. With $\mathbf{B}$ along z , the solution in the text shows that a positive charge goes round clockwise when viewed from above, so by the right-hand rule its moment points along $-z$. A negative charge goes round anticlockwise, but the current it carries flows in the opposite direction to its motion, so its moment also points along $-z$.
5. Let the loops be at $z = \pm d/2$. The field on the axis is proportional to $w(z - d/2) + w(z + d/2)$, where $w(\xi) = (a^2 + \xi^2)^{-3/2}$. By the chain rule, $w'(\xi) = -3\xi(a^2 + \xi^2)^{-5/2}$, which is an odd function of ξ , so the first derivative of the sum vanishes at $z = 0$. Differentiating once more with the product rule gives $w''(\xi) = 3(4\xi^2 - a^2)(a^2 + \xi^2)^{-7/2}$, which is even, so the second derivative at $z = 0$ is $2w''(d/2) = 6(d^2 - a^2)(a^2 + d^2/4)^{-7/2}$, and this vanishes when $d = a$.

6. The torque is $\boldsymbol{\mu} \times \mathbf{B} = \eta \mathbf{L}_s \times \mathbf{B}$, so $d\mathbf{L}_s/dt = \eta \mathbf{L}_s \times \mathbf{B}$. Taking the scalar product with $\mathbf{L}_s$ gives $\frac{1}{2} d(L_s^2)/dt = 0$, and taking it with $\mathbf{B}$ gives $d(\mathbf{L}_s \cdot \mathbf{B})/dt = 0$, since a vector product is perpendicular to both its factors. With $\mathbf{B} = B\hat{\mathbf{z}}$ the components obey $dL_{sx}/dt = \eta BL_{sy}$ and $dL_{sy}/dt = -\eta BL_{sx}$, which are the equations for v_x and v_y in the helical motion with ω_c replaced by ηB . The perpendicular part of $\mathbf{L}_s$ therefore turns about the field at the angular frequency ηB , which is the Larmor frequency of Section 4.2 of Vol. 3.

§5.3 Ampère's Law and the Curl of the Magnetic Field

1. For $s < a$ the Amperian circle encloses the current Is^2/a^2 , so $B = \mu_0 Is/2\pi a^2$. For $a < s < b$ it encloses I , so $B = \mu_0 I/2\pi s$. For $s > b$ it encloses $I - I = 0$, so $B = 0$. In each region the field is azimuthal, since the current elements are axial, and a radial component would give a non-zero flux through the curved surface of a closed coaxial cylinder, while the fluxes through its two flat ends cancel because the field depends only on s .
2. Each sheet gives a uniform field of magnitude $\mu_0 K/2$ on either side, and the directions are given by $\frac{1}{2}\mu_0 \mathbf{K} \times \hat{\mathbf{n}}$. Between the sheets the unit normals from the two sheets are opposite, and so are the currents, so the two fields point the same way and add to $\mu_0 K$. Outside, the normals from the two sheets point the same way while the currents are opposite, so the fields cancel. The arrangement resembles a long solenoid flattened out, with $K = nI$ and a field $\mu_0 nI$ inside and zero outside.
3. Let P be at $(x_P, 0, z_P)$, let the first element be at (x, y, z) with components $(d\ell_x, d\ell_y, d\ell_z)$, and let its mirror image be at $(x, -y, z)$. Reflection in the xz -plane reverses the y -component of $\hat{\mathbf{s}}$ and leaves $\hat{\mathbf{z}}$ unchanged, so the second element has components $(d\ell_x, -d\ell_y, d\ell_z)$. The separations are $(x_P - x, -y, z_P - z)$ and $(x_P - x, y, z_P - z)$, which have the same length. Working out the vector products shows that their x -components are $d\ell_y(z_P - z) + d\ell_z y$ and $-d\ell_y(z_P - z) - d\ell_z y$, their z -components are $-d\ell_x y - d\ell_y(x_P - x)$ and $d\ell_x y + d\ell_y(x_P - x)$, and their y -components are both $d\ell_z(x_P - x) - d\ell_x(z_P - z)$. The x - and z -components cancel, so the sum points along $\hat{\mathbf{y}}$, which at P is

the azimuthal direction. The whole winding can be divided into such pairs, so the field at P is azimuthal, and since P can be any point, the field of the toroid is azimuthal everywhere.

4. Inside the ring $B = \mu_0 NI/2\pi s$, so the flux through a strip of the cross-section of height ℓ and width ds is $\mu_0 NI\ell ds/2\pi s$. Integrating from a to b gives $\Phi_B = (\mu_0 NI\ell/2\pi) \ln(b/a)$.
5. The field has only the component $B_\phi = \mu_0 I/2\pi s$, which depends on s alone, so by the curl in cylindrical coordinates of Section 3.4 of Vol. 4 the only component of the curl that can differ from zero is $(\nabla \times \mathbf{B})_z = (1/s) d(sB_\phi)/ds$, and this is zero, since $sB_\phi = \mu_0 I/2\pi$ is constant. Stokes' theorem needs a surface bounded by the circle on which the field has continuous derivatives, and every such surface is crossed by the wire, where the field is not of this form and the curl is $\mu_0 \mathbf{J}$. The circulation, $\mu_0 I$, comes from the current in the wire.
6. The field has only the component $B_\phi = \zeta s^2$, which depends on s alone, so by the curl in cylindrical coordinates of Section 3.4 of Vol. 4 the z -component of the curl is $(1/s) d(sB_\phi)/ds = (1/s) d(\zeta s^3)/ds = 3\zeta s$. The other components vanish, so $\mathbf{J} = (3\zeta s/\mu_0)\hat{\mathbf{z}}$. The current through a circle of radius s is $\int_0^s (3\zeta s'/\mu_0) 2\pi s' ds' = 2\pi\zeta s^3/\mu_0$, and μ_0 times this equals the circulation, which is the circumference $2\pi s$ times the field ζs^2 .
7. On the circle the field has magnitude ζb and points along the circle, so the circulation is the circumference $2\pi b$ times ζb , which is $2\pi\zeta b^2$. The curl is $2\zeta\hat{\mathbf{z}}$, as for the velocity of a rotating body, and its flux through the disc of radius b is $2\zeta\pi b^2$, which is the same.

Chapter 6. Magnetic Materials

§6.1 Magnetisation and Bound Currents

1. The electron passes any point of the orbit once in each period, so it is a current $-e/T$, counted positive in the direction of its motion. As the electron moves through $d\mathbf{r}$, the line from the centre of force to the electron sweeps out a thin triangle of vector area $\frac{1}{2}\mathbf{r} \times d\mathbf{r}$, so the vector area is swept out at the

rate $\frac{1}{2}\mathbf{r} \times \mathbf{v} = \mathbf{L}_{\text{orb}}/2m_e$, which is the areal speed of Kepler's second law in Section 5.4 of Vol. 1. A central force leaves $\mathbf{L}_{\text{orb}}$ constant, so in one period the vector area of the orbit is $\mathbf{L}_{\text{orb}}T/2m_e$. The moment is the current multiplied by the vector area, $(-e/T)(\mathbf{L}_{\text{orb}}T/2m_e) = -(e/2m_e)\mathbf{L}_{\text{orb}}$.

2. The magnetisation is uniform, so $\mathbf{J}_b = 0$. On the curved surface $\hat{\mathbf{n}} = \hat{\mathbf{s}} = \cos \phi \hat{\mathbf{x}} + \sin \phi \hat{\mathbf{y}}$, and $\hat{\mathbf{x}} \times \hat{\mathbf{s}} = \sin \phi \hat{\mathbf{z}}$, so $\mathbf{K}_b = M \sin \phi \hat{\mathbf{z}}$. The current flows along $+z$ on the side of the cylinder with $y > 0$ and along $-z$ on the side with $y < 0$. The total current along the cylinder is $\int_0^{2\pi} M \sin \phi a d\phi = 0$.
3. Only M_z is non-zero, and it depends only on x , so the only non-zero component of the curl is $J_{b,y} = \partial M_x / \partial z - \partial M_z / \partial x = -M_0 / \ell$. On the face $x = \ell$, where $\hat{\mathbf{n}} = \hat{\mathbf{x}}$, the surface current is $\mathbf{K}_b = M_0 \hat{\mathbf{z}} \times \hat{\mathbf{x}} = M_0 \hat{\mathbf{y}}$, and on the face $x = 0$ the magnetisation is zero, so no surface current flows there. Per unit length along z , the volume current carries $(-M_0 / \ell) \ell = -M_0$ along y and the surface current carries $+M_0$, so the total is zero. A sheet at x' carrying a current K per unit length along $+y$ produces the field $-\frac{1}{2}\mu_0 K \hat{\mathbf{z}}$ for $x > x'$ and $+\frac{1}{2}\mu_0 K \hat{\mathbf{z}}$ for $x < x'$. The slab is a stack of such sheets, and for a point outside it every sheet lies on the same side, so the fields add to $\mp \frac{1}{2}\mu_0$ times the total current, which is zero. For a point at x inside the slab, the sheets with $x' < x$ carry the current $-M_0 x / \ell$ and give $+\frac{1}{2}\mu_0 M_0 x / \ell$, while the sheets with $x' > x$, together with the surface current, carry $-M_0(\ell - x) / \ell + M_0 = M_0 x / \ell$ and give another $+\frac{1}{2}\mu_0 M_0 x / \ell$. The total is $B_z = \mu_0 M_0 x / \ell = \mu_0 M_z$.
4. Let $h(w) = w / \sqrt{a^2 + w^2}$, an odd function, so that $B_z = \frac{1}{2}\mu_0 M [h(z + \ell/2) - h(z - \ell/2)]$. Since $\ell \ll z$, Taylor's theorem gives $h(z + \ell/2) - h(z - \ell/2) \approx \ell h'(z)$, and $h'(z) = a^2 / (a^2 + z^2)^{3/2} \approx a^2 / z^3$ since $a \ll z$. Therefore $B_z \approx \mu_0 M a^2 \ell / 2z^3 = \mu_0 (M \pi a^2 \ell) / 2\pi z^3$, which is the field far along the axis of a dipole of moment $\mu = M \pi a^2 \ell$, found in Section 5.2.
5. The magnetisation is uniform, so $\mathbf{J}_b = 0$. On the surface $\hat{\mathbf{n}} = \hat{\mathbf{r}} = \sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}}$, so $\hat{\mathbf{z}} \times \hat{\mathbf{r}} = \sin \theta (\cos \phi \hat{\mathbf{y}} - \sin \phi \hat{\mathbf{x}}) = \sin \theta \hat{\boldsymbol{\phi}}$, and $\mathbf{K}_b = M \sin \theta \hat{\boldsymbol{\phi}}$. On the rotating shell, a point at $\mathbf{r}$ moves with the velocity $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$ of Section 1.2 of Vol. 1, which is $\omega a \sin \theta \hat{\boldsymbol{\phi}}$ for $\boldsymbol{\omega} = \omega \hat{\mathbf{z}}$. A moving surface charge

carries the surface current density $\sigma \mathbf{v} = \sigma \omega a \sin \theta \hat{\phi}$, which is the same as $\mathbf{K}_b$ when $\sigma \omega a = M$. The two arrangements then have the same currents everywhere, and the Biot–Savart law gives them the same field.

§6.2 The Auxiliary Field and Linear Magnetic Materials

1. By symmetry $\mathbf{H}$ circles the axis, and Ampère's law in matter round a circle of radius s gives $H = Is/2\pi a^2$ for $s < a$, $H = I/2\pi s$ for $a < s < b$, and $H = 0$ for $s > b$, where the enclosed free current is zero. The field is $B = \mu_0 H$ in the conductors and outside, and $B = (1 + \chi_m)\mu_0 I/2\pi s$ between the conductors, where $M = \chi_m I/2\pi s$, directed round the axis. The magnetisation has the same form as the field of a straight wire, whose curl you showed to vanish for $s > 0$ in Problem 5 of Section 5.3, so $\mathbf{J}_b = 0$, as expected in a homogeneous linear material with no free current. On the inner surface of the material the outward normal is $-\hat{\mathbf{s}}$, so $\mathbf{K}_b = M\hat{\phi} \times (-\hat{\mathbf{s}}) = (\chi_m I/2\pi a)\hat{\mathbf{z}}$, carrying a total $\chi_m I$ in the direction of the current in the inner conductor. On the outer surface the normal is $\hat{\mathbf{s}}$, and $\mathbf{K}_b = -(\chi_m I/2\pi b)\hat{\mathbf{z}}$, carrying a total $-\chi_m I$. A circle of radius s between the conductors encloses $I + \chi_m I$, so Ampère's law gives $B = \mu_0(1 + \chi_m)I/2\pi s$, as before.
2. By symmetry $\mathbf{H}$ circles the axis of the toroid, and a circle of radius s with $a < s < b$ inside the ring encloses the free current NI , so $H = NI/2\pi s$ and $B = \mu_r \mu_0 NI/2\pi s$. A circle outside the ring encloses no net free current, and $H = 0$ there. The flux through the cross-section is $\int_a^b B \ell ds = (\mu_r \mu_0 NI \ell / 2\pi) \ln(b/a)$, which is μ_r times the flux you found for the empty toroid in Problem 4 of Section 5.3.
3. All currents, the windings and the bound surface current of the rod, flow round the axis on coaxial cylinders, so, as for the filled solenoid, $\mathbf{B}$ and $\mathbf{H}$ point along the axis and vanish outside. A rectangular loop with one side of length ℓ inside the solenoid, at any distance from the axis, and the other side outside, encloses the free current $n\ell I$, so $H = nI$ in the rod and in the air alike. Then $B = (1 + \chi_m)\mu_0 nI$ in the rod and $B = \mu_0 nI$ in the air, and the flux through one turn is $\mu_0 nI [\pi b^2 + \chi_m \pi a^2]$.

4. The field produced by the sample itself is of order $\mu_0 M$, which is a fraction of order $|\chi_m|$ of the applied field, and to first order in χ_m we may take $\mathbf{H} = \mathbf{B}/\mu_0$ with $\mathbf{B}$ the applied field, and $M_z = \chi_m B_z/\mu_0$. The sample has the moment $\mu_z = M_z \mathcal{V} = \chi_m \mathcal{V} B_z/\mu_0$, and the upward force on it is $(\chi_m \mathcal{V}/\mu_0) B_z \partial B_z/\partial z$. Setting this equal to the weight mg gives the stated condition. Since $\chi_m < 0$ and $mg > 0$, the product $B_z \partial B_z/\partial z = \frac{1}{2} \partial(B_z^2)/\partial z$ must be negative, so B_z^2 must decrease upwards. Above the top of a vertical magnet, for example, the magnitude of the field decreases with height.
5. When $\ell \gg a$, the ratio $\ell/\sqrt{\ell^2 + 4a^2}$ approaches one and $H_z(0) \rightarrow 0$, as found for the long cylinder. When $\ell \ll a$, the ratio is approximately $\ell/2a$, which is small, so $H_z(0) \rightarrow -M$, and $B_z(0) = \mu_0 M \ell/\sqrt{\ell^2 + 4a^2} \approx \mu_0 M \ell/2a$, which is much smaller than $\mu_0 M$. At the centre of a thin disc the magnetic field is therefore small, and the auxiliary field is close to $-\mathbf{M}$, pointing opposite to the magnetisation with the same magnitude.

Chapter 7. Electromagnetic Induction and Alternating-Current Circuits

§7.1 Induction and Alternating Current at School

1. The flux linkage falls steadily from NBA to zero in the time t_1 , so the electromotive force has magnitude NBA/t_1 . The current is NBA/Rt_1 , and it flows for the time t_1 , so the charge that flows is NBA/R .
2. The rod sweeps out a flat area at the rate ℓv . The field is perpendicular to the rod and makes an angle θ with the velocity, so its component perpendicular to the swept area is $B \sin \theta$. The flux through the swept area therefore grows at the rate $B \ell v \sin \theta$, and this is the electromotive force.
3. The angular velocity is $\omega = 2\pi f$, so the peak electromotive force is $2\pi f NBA$ and its root-mean-square value is $\sqrt{2} \pi f NBA$. Doubling the rate of rotation doubles the peak value and also doubles the frequency of the output.
4. The secondary voltage is $V_p N_s/N_p$, so the secondary current is

$I_s = V_p N_s / N_p R$. The primary current is $I_s N_s / N_p = V_p N_s^2 / N_p^2 R$, and the mean power is $V_p I_p = V_p^2 N_s^2 / N_p^2 R$. The supply therefore behaves as if it were connected to a resistance $R(N_p/N_s)^2$.

5. The root-mean-square current in the cables is $\langle \mathcal{P} \rangle / V_{\text{rms}}$, and the mean power dissipated in them is this current squared times R_{line} , which is $\langle \mathcal{P} \rangle^2 R_{\text{line}} / V_{\text{rms}}^2$. The transformer multiplies V_{rms} by N_s/N_p , so the loss is divided by $(N_s/N_p)^2$.

§7.2 Faraday's Law and Lenz's Law

1. The force on the rod is $-B^2 \ell^2 v / R$ along the rails, so $m dv/dt = -B^2 \ell^2 v / R$, whose solution is $v = v_1 e^{-t/t_m}$ with $t_m = mR/B^2 \ell^2$. The rate of heating is $I^2 R = B^2 \ell^2 v^2 / R = (B^2 \ell^2 v_1^2 / R) e^{-2t/t_m}$, and integrating from $t = 0$ to infinity gives $(B^2 \ell^2 v_1^2 / R)(t_m/2) = \frac{1}{2} m v_1^2$.
2. With $\hat{\phi} = (-y, x, 0)/s$ and the abbreviation $\dot{B} = \mu_0 n dI/dt$, the field is $E_x = \frac{1}{2} \dot{B} a^2 y / (x^2 + y^2)$ and $E_y = -\frac{1}{2} \dot{B} a^2 x / (x^2 + y^2)$. This field is $-\frac{1}{2} \dot{B} a^2$ times $(-y, x, 0)/s^2$, whose curl you showed to vanish for $s > 0$ in Problem 5 of Section 5.3. The line integral round the circle is $2\pi s E_\phi = -\pi a^2 \dot{B}$. Stokes' theorem relates it to the curl over a surface bounded by the circle, and every such surface passes through the interior of the solenoid, where the curl is $-\dot{B} \hat{\mathbf{z}}$; the integral of the curl over the disc of radius a is $-\pi a^2 \dot{B}$, in agreement.
3. The field of the wire is $\mu_0 I / 2\pi s'$ at distance s' (Section 5.3), perpendicular to the plane of the loop. Slicing the loop into strips parallel to the wire gives the flux $\Phi_B = \int_s^{s+\ell} (\mu_0 I / 2\pi s') \ell ds' = (\mu_0 I \ell / 2\pi) \ln[(s + \ell)/s]$. Differentiating with $ds/dt = v$ gives $d\Phi_B/dt = (\mu_0 I \ell v / 2\pi) [1/(s + \ell) - 1/s]$, so the electromotive force has magnitude $\mu_0 I \ell^2 v / [2\pi s(s + \ell)]$. From the force, the nearer side has an electromotive force $\ell v B = \mu_0 I \ell v / 2\pi s$ and the farther side one of $\mu_0 I \ell v / [2\pi(s + \ell)]$, and these act in opposite directions round the loop, since the force per unit charge $\mathbf{v} \times \mathbf{B}$ has the same direction in space on both sides; their difference is the same result. Taking the wire along $\hat{\mathbf{z}}$ and the loop at positive x in the xz -plane, the field at the loop is along $\hat{\mathbf{y}}$ and $\mathbf{v} \times \mathbf{B}$ is along $\hat{\mathbf{z}}$, and since the nearer side, where the field is larger, has the larger

electromotive force, the current in the nearer side flows in the same direction as the current in the wire.

4. The current is $I = \mathcal{E}/R$, and the coil has magnetic moment $NI A$ along its normal, so by Section 5.2 the torque on it has magnitude $NIAB \sin \omega t$, and by Lenz's law this torque opposes the rotation. The power needed to turn the coil against it is $NIAB\omega \sin \omega t = I\mathcal{E} = \mathcal{E}^2/R$.
5. The flux through the ring is the flux inside the solenoid, $\pi a^2 \mu_0 n I$, since the field outside the solenoid is zero. The electromotive force has magnitude $\pi a^2 \mu_0 n dI/dt$, and the current is this divided by R , independent of b , circulating in the opposite direction to the current in the windings. The induced field at $s = b$ has magnitude $\mu_0 n a^2 (dI/dt)/2b$, and multiplying by the circumference $2\pi b$ gives the same electromotive force.

§7.3 Inductance and Magnetic Energy

1. The energy density between the tubes is $(\mu_0 I/2\pi s)^2/2\mu_0 = \mu_0 I^2/8\pi^2 s^2$, and a cylindrical shell of radius s , thickness ds , and unit length has volume $2\pi s ds$. The energy per unit length is $\int_a^b (\mu_0 I^2/4\pi s) ds = (\mu_0 I^2/4\pi) \ln(b/a)$, which is $\frac{1}{2}(L/\ell)I^2$ with $L/\ell = (\mu_0/2\pi) \ln(b/a)$.
2. Kirchhoff's loop law for the circuit is $\mathcal{E} = IR + L dI/dt$, and multiplying it by I gives $\mathcal{E}I = I^2 R + d(\frac{1}{2}LI^2)/dt$, using the chain rule. After the source is removed, the current is $I = (\mathcal{E}/R)e^{-t/\tau}$, with t now measured from the removal, and the heat produced is $\int_0^\infty (\mathcal{E}/R)^2 R e^{-2t/\tau} dt = (\mathcal{E}/R)^2 R \tau/2 = \frac{1}{2}L(\mathcal{E}/R)^2$, which is the energy that was stored in the inductor.
3. With the fluxes adding, the flux linkage of coil 1 is $L_1 I + L_{12} I$ and that of coil 2 is $L_2 I + L_{12} I$, using $L_{21} = L_{12}$. The total flux linkage of the combination is $(L_1 + L_2 + 2L_{12})I$. With the connections to one coil reversed, the mutual contributions change sign and the total is $(L_1 + L_2 - 2L_{12})I$.
4. The field of the wire is $\mu_0 I/2\pi s$, and slicing the loop into strips parallel to the wire, as you did in Problem 3 of Section 7.2, gives the flux $(\mu_0 I \ell/2\pi) \ln(b/a)$, so the mutual inductance is

$(\mu_0 \ell / 2\pi) \ln(b/a)$. The electromotive force is minus this times $dI/dt = -\omega I_0 \sin \omega t$, which is $(\mu_0 \ell \omega I_0 / 2\pi) \ln(b/a) \sin \omega t$.

5. The flux linkage of coil 1 is $N_1 \Lambda(N_1 I_1 + N_2 I_2) = \Lambda N_1^2 I_1 + \Lambda N_1 N_2 I_2$, so $L_1 = \Lambda N_1^2$ and $L_{12} = \Lambda N_1 N_2$. The same argument for coil 2 gives $L_2 = \Lambda N_2^2$ and $L_{21} = \Lambda N_1 N_2$. Then $L_1 L_2 = \Lambda^2 N_1^2 N_2^2 = L_{12}^2$. Since every turn is threaded by the same flux, the electromotive forces are $-N_1 d\Phi_B/dt$ and $-N_2 d\Phi_B/dt$, whose ratio is N_2/N_1 .

§7.4 Alternating Current, Mean Power, and Rectification

- During one period, from $t = 0$ to $t = T$, the waveform is $V = V_0(2t/T - 1)$. Its mean is zero, since it is odd about $t = T/2$. By the chain rule, $(T/6)(2t/T - 1)^3$ has the derivative $(2t/T - 1)^2$, which gives $\langle V^2 \rangle = (1/T) \int_0^T V^2 dt = (V_0^2/T) [(T/6)(2t/T - 1)^3]_0^T = V_0^2/3$, so $V_{\text{rms}} = V_0/\sqrt{3}$. The mean power is $V_0^2/3R$, which is two-thirds of the mean power $V_0^2/2R$ of the sinusoid.
- The peak value is $V_0 = \frac{1}{2}\eta V_{\text{div}}$, so $V_{\text{rms}} = \eta V_{\text{div}}/2\sqrt{2}$. The period is $T = \xi t_{\text{div}}$, so $f = 1/\xi t_{\text{div}}$. The delay is ζt_{div} , so the phase difference is $2\pi\zeta t_{\text{div}}/\xi t_{\text{div}} = 2\pi\zeta/\xi$, and the first signal leads the second.
- The output repeats every half-period, so its mean over a period is its mean over the first half-period, $(2/T) \int_0^{T/2} V_0 \sin \omega t dt = (2/T)(2V_0/\omega) = 2V_0/\pi$, using the integral found in the text. The square of the output is $V_0^2 \sin^2 \omega t$ at every instant, so its mean is $V_0^2/2$ and the root-mean-square value is $V_0/\sqrt{2}$. The mean power is $V_0^2/2R$, the same as for the unrectified supply.
- Near the peak, the Taylor expansion of the cosine gives $\cos \omega t \approx 1 - \frac{1}{2}\omega^2 t^2$, and setting $V_0(1 - \frac{1}{2}\omega^2 t^2) = V_0 - \Delta V$ gives $|t| = \sqrt{2\Delta V/V_0}/\omega$. During each period the load draws a charge of about $(V_0/R)T$, and the diode must deliver this charge in the conduction time, so its mean current while conducting is about $(V_0/R)T\omega/\sqrt{2\Delta V/V_0}$. Dividing by the mean load current V_0/R , and using $\omega T = 2\pi$, gives the factor $2\pi/\sqrt{2\Delta V/V_0}$.
- During a discharge of duration T the capacitor loses a charge of about $I_{\text{load}}T$, and its potential difference falls by $I_{\text{load}}T/C =$

I_{load}/fC ; for a full-wave rectifier the discharge lasts about $T/2$, which halves the ripple. The ripple depends on the charge drawn from the capacitor between recharges, which is fixed by the mean current and the time between recharges, so the resistance enters only through the current it draws.

§7.5 Phasors and Complex Impedance

1. The solution of the equation of simple harmonic motion with $Q = 0$ and $dQ/dt = I_1$ at $t = 0$ is $Q = (I_1/\omega_0) \sin \omega_0 t$, and differentiating gives $I = I_1 \cos \omega_0 t$. The largest charge is $I_1/\omega_0 = I_1 \sqrt{LC}$. The initial energy $\frac{1}{2} L I_1^2$ is all in the capacitor when the charge is largest, and setting it equal to $Q^2/2C$ gives $Q = I_1 \sqrt{LC}$.
2. With $R = 2\sqrt{L/C}$ the circuit is critically damped, $\gamma = 2\omega_0$, and $Q = (Q_1 + Q_2 t)e^{-\gamma t/2}$. The initial charge gives $Q_1 = Q_0$, and the initial current, $dQ/dt = Q_2 - \frac{1}{2}\gamma Q_0 = 0$, gives $Q_2 = \frac{1}{2}\gamma Q_0$. Therefore $Q = Q_0(1 + \frac{1}{2}\gamma t)e^{-\gamma t/2}$, which is positive for all $t \geq 0$.
3. The ratio is $Z_R/(Z_R + Z_C) = i\omega\tau/(1 + i\omega\tau)$, obtained by multiplying the numerator and the denominator by $i\omega C$. The gain is $\omega\tau/\sqrt{1 + \omega^2\tau^2}$, and the output leads the input by $\pi/2 - \arctan \omega\tau$, since the argument of i is $\pi/2$. When $\omega\tau \ll 1$ the gain is about $\omega\tau$ and the output leads by nearly $\pi/2$; when $\omega\tau \gg 1$ the gain is close to one and the output is nearly in phase with the input. The circuit passes high frequencies.
4. In series, $Z = R + i\omega L$, with modulus $\sqrt{R^2 + \omega^2 L^2}$ and argument $\arctan(\omega L/R)$. In parallel, $1/Z = 1/R + i\omega C$, so $Z = R/(1 + i\omega RC)$, with modulus $R/\sqrt{1 + \omega^2 R^2 C^2}$ and argument $-\arctan \omega RC$.
5. The phasor of $V_0 \cos \omega t$ is V_0 , and since $V_0 \sin \omega t = V_0 \cos(\omega t - \pi/2)$, its phasor is $V_0 e^{-i\pi/2} = -iV_0$. The sum is $V_0(1 - i) = \sqrt{2}V_0 e^{-i\pi/4}$, so the sum of the sinusoids is $\sqrt{2}V_0 \cos(\omega t - \pi/4)$. The identity of Section 7.4 with $V_1 = V_2 = V_0$ gives the amplitude $\sqrt{2}V_0$ and $\tan \psi = 1$, so $\psi = \pi/4$, in agreement.
6. The ratio is $Z_R/(Z_R + Z_L) = R/(R + i\omega L) = 1/(1 + i\omega L/R)$, which is the ratio for the RC filter with τ replaced by L/R . Its gain is $1/\sqrt{1 + \omega^2 \tau^2}$, which is close to one at low frequencies and

small at high frequencies.

§7.6 Resonance and Power in Alternating-Current Circuits

1. The complex amplitude of the potential difference across the capacitor is $\tilde{I}/i\omega C = V_0/(i\omega CZ)$, and $i\omega CZ = 1 - \omega^2 LC + i\omega RC$, so its modulus is $(V_0/LC)/\sqrt{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2}$, after dividing $i\omega CZ$ by LC . This is largest when the quantity under the square root is smallest, and setting its derivative with respect to ω^2 to zero gives $-2(\omega_0^2 - \omega^2) + \gamma^2 = 0$, so $\omega^2 = \omega_0^2 - \gamma^2/2$, which is positive provided $\gamma^2 < 2\omega_0^2$. With the damping constant of Vol. 1 equal to $\gamma/2$, this is the result $\omega_0^2 - 2(\gamma/2)^2$ found there for the displacement, as expected, since the charge corresponds to the displacement.
2. At resonance $I_0 = V_0/R$, and the amplitudes across the inductor and the capacitor are $\omega_0 LI_0$ and $I_0/\omega_0 C$, which are equal since $\omega_0^2 = 1/LC$, and each is $\omega_0 LV_0/R = QV_0$. The two potential differences are opposite in phase and cancel, so the potential difference across the inductor and the capacitor taken together is zero.
3. The impedance is $R + i\omega L$, so $\cos\psi = R/\sqrt{R^2 + \omega^2 L^2}$, the root-mean-square current is $V_{\text{rms}}/\sqrt{R^2 + \omega^2 L^2}$, and the mean power is $V_{\text{rms}} I_{\text{rms}} \cos\psi = V_{\text{rms}}^2 R/(R^2 + \omega^2 L^2)$. With the capacitor, the real part of $1/Z$ is unchanged at $R/(R^2 + \omega^2 L^2)$ and the imaginary part is zero, so the current drawn from the supply is $V_{\text{rms}} R/(R^2 + \omega^2 L^2)$. This is the previous current times $R/\sqrt{R^2 + \omega^2 L^2} = \cos\psi$, and the mean power $V_{\text{rms}}^2 R/(R^2 + \omega^2 L^2)$ is the same as before.
4. From the expressions for $\omega_{\pm}$, their product is $(\omega_0^2 + \gamma^2/4) - \gamma^2/4 = \omega_0^2$, using $(p+q)(p-q) = p^2 - q^2$. For $Q \gg 1$, $\gamma = \omega_0/Q$ is much smaller than ω_0 , the square root is close to ω_0 , and $\omega_{\pm} \approx \omega_0 \pm \gamma/2 = \omega_0 \pm \omega_0/2Q$.
5. The current is $\tilde{I} = \tilde{V}/(Z_{\text{int}} + R + iX)$, and the mean power in the load is $\frac{1}{2}|\tilde{I}|^2 R$, which is the stated expression. For any R , the denominator is smallest when $X = -X_{\text{int}}$, and the power is then $\frac{1}{2}|\tilde{V}|^2 R/(R + R_{\text{int}})^2$. Differentiating with respect to R by the

quotient rule gives a numerator proportional to $(R + R_{\text{int}}) - 2R$, which vanishes at $R = R_{\text{int}}$, where the power is largest.

6. The capacitor current is $C \, dV/dt = -\omega_0 C V_0 \sin \omega_0 t$. The inductor current satisfies $L \, dI_L/dt = V_0 \cos \omega_0 t$, so $I_L = (V_0/\omega_0 L) \sin \omega_0 t$, which is $\omega_0 C V_0 \sin \omega_0 t$ since $1/\omega_0 L = \omega_0 C$; the two currents cancel. The stored energy is $\frac{1}{2} C V_0^2 \cos^2 \omega_0 t + \frac{1}{2} L (V_0/\omega_0 L)^2 \sin^2 \omega_0 t$, and since $1/\omega_0^2 L = C$ the second term is $\frac{1}{2} C V_0^2 \sin^2 \omega_0 t$, so the total is $\frac{1}{2} C V_0^2$.

Chapter 8. Maxwell's Equations and Conservation Laws

§8.1 The Displacement Current and Maxwell's Equations

1. The field between the plates has magnitude V/d , where $V = V_0 \cos \omega t$ is the potential difference, so the flux through a surface that crosses the region between the plates is AV/d , and the displacement current is $\epsilon_0 \, d\Phi_E/dt = (\epsilon_0 A/d) \, dV/dt = -(\epsilon_0 A/d) \omega V_0 \sin \omega t$. The capacitance is $C = \epsilon_0 A/d$ by Section 2.1, and the current in the wires is $dQ/dt = C \, dV/dt$, which is the same.
2. The text found the amplitudes $\sigma_c E_0$ of the conduction current density and $\epsilon_0 \omega E_0$ of the displacement current density, and these amplitudes are equal when $\omega = \sigma_c/\epsilon_0$, the inverse of the decay time of Section 4.2. The conduction current density is $\sigma_c \mathbf{E}_0 \cos \omega t$ and the displacement current density is $-\epsilon_0 \omega \mathbf{E}_0 \sin \omega t$, and since $-\sin \omega t = \cos(\omega t + \pi/2)$, the displacement current density leads the conduction current density by a phase of $\pi/2$.
3. As you showed in Problem 1 of Section 5.3, the field inside the wire is $B = \mu_0 I s/2\pi a^2$, which is the field found between the plates. Between the plates the displacement current density $\epsilon_0 \, \partial E/\partial t$ is $I/\pi a^2$, spread uniformly over the plates in the same way as the current density in the wire, so a circle of radius $s < a$ encloses the fraction s^2/a^2 of the total in both cases, and the circulation is the same.
4. By Section 4.3 the charge is $Q = Q_0 e^{-t/RC}$, so the current flow-

ing onto the positive plate is $dQ/dt = -(Q_0/RC)e^{-t/RC}$. Replacing I by this rate in the calculation of the text gives a field of magnitude $B = \mu_0 s Q_0 e^{-t/RC} / (2\pi a^2 RC)$. The electric field is now decreasing, so $\partial \mathbf{E} / \partial t$ points against $\mathbf{E}$, and $\mathbf{B}$ circulates in the opposite direction to the field during charging.

5. The closed surface formed by $\mathcal{S}_1$ and $\mathcal{S}_2$ has outward surface elements equal to those of $\mathcal{S}_2$ on $\mathcal{S}_2$ and opposite to those of $\mathcal{S}_1$ on $\mathcal{S}_1$. Let I_1 and I_2 be the currents through the two surfaces, Φ_1 and Φ_2 the fluxes of $\mathbf{E}$ through them, and Q the charge enclosed between them. The net current out of the closed surface is $I_2 - I_1$, so the conservation of charge gives $I_2 - I_1 = -dQ/dt$, while Gauss's law gives $\Phi_2 - \Phi_1 = Q/\epsilon_0$. Therefore $(I_2 + \epsilon_0 d\Phi_2/dt) - (I_1 + \epsilon_0 d\Phi_1/dt) = -dQ/dt + dQ/dt = 0$.
6. Let $Q(r, t)$ be the charge inside a sphere of radius r centred on the source. By symmetry and Gauss's law, $\mathbf{E} = Q(r, t) \hat{\mathbf{r}} / 4\pi\epsilon_0 r^2$, and by symmetry the current density is radial, so the current out of the sphere is $4\pi r^2 J_r$. The conservation of charge gives $4\pi r^2 J_r = -\partial Q / \partial t$, while $\epsilon_0 \partial E_r / \partial t = (\partial Q / \partial t) / 4\pi r^2$, and the current term and the displacement term cancel. The Ampère–Maxwell law then requires $\nabla \times \mathbf{B} = 0$ everywhere outside the source. By the spherical symmetry, a magnetic field could only be radial, since no other direction at a point is distinguished by the arrangement, and a radial field $B(r)\hat{\mathbf{r}}$ with $B \neq 0$ would have a nonzero flux through a sphere centred on the source, which $\nabla \cdot \mathbf{B} = 0$ forbids. The field is therefore zero, in agreement with the Ampère–Maxwell law, whereas Ampère's law without the displacement current would have required a nonzero curl.

§8.2 Maxwell's Equations in Matter and the Boundary Conditions

1. Taking the divergence of $\mathbf{J}_p = \partial \mathbf{P} / \partial t$ and exchanging the order of the derivatives gives $\nabla \cdot \mathbf{J}_p = \partial(\nabla \cdot \mathbf{P}) / \partial t = -\partial \rho_b / \partial t$. The divergence of $\mathbf{J}_b = \nabla \times \mathbf{M}$ is zero by the identity of Section 8.1. By the continuity equation, a current density with zero divergence carries as much charge out of every small volume as it carries in, so the bound current $\mathbf{J}_b$ circulates and leaves the charge density unchanged, while the polarisation current changes

the bound charge density wherever the divergence of $\partial \mathbf{P}/\partial t$ is not zero.

2. By Section 3.2, $D = Q/A$ in the first slab. The interface carries no free charge, so $D_\perp$ is continuous and $D = Q/A$ in the second slab as well. The fields are $E_1 = Q/\epsilon_{r1}\epsilon_0 A$ and $E_2 = Q/\epsilon_{r2}\epsilon_0 A$, both pointing from the positive plate to the negative one. The potential difference is $E_1 d_1 + E_2 d_2$, so the capacitance is $C = \epsilon_0 A/(d_1/\epsilon_{r1} + d_2/\epsilon_{r2})$. With $\hat{\mathbf{n}}$ pointing from the first slab into the second, the total surface charge density on the interface is $\epsilon_0(E_2 - E_1)$, and since the interface carries no free charge, all of it is bound, with density $(Q/A)(1/\epsilon_{r2} - 1/\epsilon_{r1})$. As a check, the face of the first slab carries $P_1 = (1 - 1/\epsilon_{r1})Q/A$, and the face of the second slab carries $-P_2 = -(1 - 1/\epsilon_{r2})Q/A$, whose sum is the same.
3. The bound surface current density is $\mathbf{M} \times \hat{\mathbf{n}}$ on the face whose outward normal is $\hat{\mathbf{n}}$ and $-\mathbf{M} \times \hat{\mathbf{n}}$ on the other face, and only $\mathbf{M}_\parallel$ contributes to it, since $\hat{\mathbf{n}} \times \hat{\mathbf{n}} = 0$. By Section 5.3, each sheet produces a field of magnitude $\mu_0 |\mathbf{M}_\parallel|/2$, parallel to the faces and in opposite directions on its two sides. For two sheets carrying opposite currents, the fields cancel outside the slab and add inside it, as for a solenoid, to give $\mu_0 \mathbf{M}_\parallel$. For example, with $\hat{\mathbf{n}} = \hat{\mathbf{z}}$ and $\mathbf{M}_\parallel = M\hat{\mathbf{x}}$, the face with outward normal $\hat{\mathbf{z}}$ carries the current density $-M\hat{\mathbf{y}}$ and the other face carries $M\hat{\mathbf{y}}$, and each sheet produces the field $\frac{1}{2}\mu_0 M\hat{\mathbf{x}}$ in the slab between them. The field inside is therefore $\mathbf{B} = \mathbf{B}_a + \mu_0 \mathbf{M}_\parallel$, so $B_\perp = B_{a\perp}$. With $\mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}$, the tangential part is $\mathbf{H}_\parallel = \mathbf{B}_{a\parallel}/\mu_0$, and $\mathbf{M}_\parallel = \chi_m \mathbf{H}_\parallel$ gives $\mathbf{B}_\parallel = \mathbf{B}_{a\parallel} + \chi_m \mathbf{B}_{a\parallel} = \mu_r \mathbf{B}_{a\parallel}$. The magnetisation $\chi_m \mathbf{H}$ is uniform, as assumed, and the angle satisfies $\tan \beta = |\mathbf{B}_\parallel|/B_\perp = \mu_r \tan \beta_0$.
4. Inside the conductor $\mathbf{E}$ and $\mathbf{D}$ are zero. Taking medium 1 to be the conductor, the continuity of $\mathbf{E}_\parallel$ shows that the field just outside is normal to the surface, and the condition on $D_\perp$ gives $D_\perp = \sigma_f$, so the field just outside is $\sigma_f \hat{\mathbf{n}}/\epsilon_r \epsilon_0$. The total surface charge density is $\epsilon_0 E_\perp = \sigma_f/\epsilon_r$, so the bound surface charge density is $\sigma_f/\epsilon_r - \sigma_f = -(1 - 1/\epsilon_r)\sigma_f$, which partly cancels the free charge, as the bound charge around the point charge of Section 3.2 does.

5. The field between the plates is V/d , where $V = V_0 \cos \omega t$, so the free current density in the material is $\sigma_c V/d$ and the displacement is $D = \epsilon_r \epsilon_0 V/d$. The free charge on the positive plate is $Q = DA$, and by the conservation of charge the current I in the wire supplies both the conduction current $\sigma_c AV/d$ that leaves the plate through the material and the rate of growth $dQ/dt = \epsilon_r \epsilon_0 A(dV/dt)/d$ of its charge, so $I = V_0(A/d)(\sigma_c \cos \omega t - \epsilon_r \epsilon_0 \omega \sin \omega t)$. The integral form with a surface that crosses the whole region between the plates gives $\oint \mathbf{H} \cdot d\mathbf{l} = A(\sigma_c V/d + \epsilon_r \epsilon_0 dV/dt/d)$, which is the same current, and with a surface pierced by the wire it gives I directly. This current is $V/R + C dV/dt$ with $R = d/\sigma_c A$, the resistance of Section 4.2 for a block of length d and cross-section A , and $C = \epsilon_r \epsilon_0 A/d$, which are the currents in a resistor and a capacitor connected in parallel across the source.
6. Let the material be medium 1 and empty space medium 2. The normal component of $\mathbf{B}$ is continuous, and the tangential part of $\mathbf{B}$ is divided by μ_r on passing into empty space, so $\tan \beta' = \tan \beta / \mu_r$. When μ_r is much larger than one, $\tan \beta'$ is small, and the line leaves nearly along the normal, unless $\tan \beta$ is itself comparable with μ_r , which requires β to be close to a right angle.

Chapter 9. Electromagnetic Waves and Their Propagation

§9.1 Plane Electromagnetic Waves

1. The curl of the Ampère–Maxwell law is $\nabla \times (\nabla \times \mathbf{B}) = \mu_0 \epsilon_0 \partial(\nabla \times \mathbf{E})/\partial t$, since the curl and the derivative with respect to time can be taken in either order. By Faraday's law the right-hand side is $-\mu_0 \epsilon_0 \partial^2 \mathbf{B}/\partial t^2$. By the identity, the left-hand side is $\nabla(\nabla \cdot \mathbf{B}) - \nabla^2 \mathbf{B} = -\nabla^2 \mathbf{B}$, since $\nabla \cdot \mathbf{B} = 0$. Equating the two sides gives the result.
2. We have $\partial^2 E_x / \partial z^2 = -k^2 E_x$ and $\partial^2 E_x / \partial t^2 = -\omega^2 E_x$, so the wave equation holds when $\omega^2 = c^2 k^2$. Since $E_y = E_z = 0$ and E_x depends only on z and t , Faraday's law gives $\partial B_x / \partial t = 0$ and $\partial B_y / \partial t = -\partial E_x / \partial z = -k E_0 \cos kz \cos \omega t$. Integrating with respect to time gives

$B_y = -(kE_0/\omega) \cos kz \sin \omega t = -(E_0/c) \cos kz \sin \omega t$. The x -component of $\nabla \times \mathbf{B}$ is $-\partial B_y/\partial z = -(kE_0/c) \sin kz \sin \omega t$, and $\mu_0 \epsilon_0 \partial E_x/\partial t = -(\omega E_0/c^2) \sin kz \sin \omega t$, which is the same since $\omega/c^2 = k/c$; the other components of the Ampère–Maxwell law are zero on both sides. By the identity for the product of a sine and a cosine, $\sin kz \cos \omega t = \frac{1}{2}[\sin(kz - \omega t) + \sin(kz + \omega t)]$, so the field is $\frac{1}{2}E_0 \hat{\mathbf{x}}[\sin(kz - \omega t) + \sin(kz + \omega t)]$, a wave travelling towards positive z added to one travelling towards negative z . On the planes $z = j\pi/k$ the factor $\sin kz$ is zero, so $\mathbf{E}$ vanishes there at every instant. A wave of this kind is called a standing wave, and a standing wave with nodes of $\mathbf{E}$ at the walls is the kind of mode counted for a cavity with conducting walls in Section 5.6 of Vol. 2.

3. The field is a function of $z - ct$, so it satisfies the wave equation, since by the general solution quoted in the subsection on the wave equation every function of $z - ct$ that can be differentiated twice does. Its divergence is $\partial E_z/\partial z = -kE_0 \sin(kz - \omega t)$, which is not zero, so Gauss's law in empty space, $\nabla \cdot \mathbf{E} = 0$, is violated.
4. The phase is $\mathbf{k} \cdot \mathbf{r} - \omega t$ with $\mathbf{k} = (k/\sqrt{2})(\hat{\mathbf{x}} + \hat{\mathbf{y}})$, whose magnitude is k , so the wave travels in the direction $\hat{\mathbf{k}} = (\hat{\mathbf{x}} + \hat{\mathbf{y}})/\sqrt{2}$, and $\omega = ck$. The field is along $\hat{\mathbf{z}}$, which is perpendicular to $\hat{\mathbf{k}}$, as it must be. Since $\hat{\mathbf{x}} \times \hat{\mathbf{z}} = -\hat{\mathbf{y}}$ and $\hat{\mathbf{y}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}}$, the magnetic field is $\mathbf{B} = (1/c) \hat{\mathbf{k}} \times \mathbf{E} = (E_0/\sqrt{2}c)(\hat{\mathbf{x}} - \hat{\mathbf{y}}) \cos[k(x + y)/\sqrt{2} - \omega t]$.
5. From Coulomb's law, $\epsilon_0 = q_1 q_2 / (4\pi F r^2)$ for charges q_1 and q_2 a distance r apart with a force F between them, so ϵ_0 is measured in $\text{C}^2 \text{N}^{-1} \text{m}^{-2}$. From the force per unit length between the wires, $\mu_0 = 2\pi d (F/\ell) / (I_1 I_2)$, so μ_0 is measured in N A^{-2} , and since an ampere is defined as one coulomb per second, this is $\text{N s}^2 \text{C}^{-2}$. The product $\mu_0 \epsilon_0$ is therefore measured in $\text{s}^2 \text{m}^{-2}$, and $1/\sqrt{\mu_0 \epsilon_0}$ in m s^{-1} .

§9.2 Polarisation, Energy Flux, and Radiation Pressure

1. The transmitted intensity is $\mathcal{I}_0 \cos^2 \theta'$ for a field at angle θ' to the transmission axis, and its average over angles spread evenly between 0 and 2π is $\mathcal{I}_0/2$, since the average of $\cos^2 \theta'$ over a whole number of periods is one half, as in Section 7.4. The light leaving the first filter is linearly polarised along its axis, so by Malus's

law the second filter transmits $(\mathcal{I}_0/2)\cos^2\theta$. The third axis is at the angle $\pi/2 - \theta$ to the second, and the emerging intensity is $(\mathcal{I}_0/2)\cos^2\theta\sin^2\theta = (\mathcal{I}_0/8)\sin^2 2\theta$, using $\sin 2\theta = 2\sin\theta\cos\theta$. It is largest, equal to $\mathcal{I}_0/8$, when $\theta = \pi/4$. Without the middle filter no light emerges.

2. Differentiating $\tan\beta = E_y/E_x$ with respect to time, using the quotient rule on the right, gives $\sec^2\beta\,d\beta/dt = (E_x\,dE_y/dt - E_y\,dE_x/dt)/E_x^2$, and since $\sec^2\beta = E^2/E_x^2$, this gives the first result. For the given components, $E_x\,dE_y/dt - E_y\,dE_x/dt = \omega E_1 E_2 [\sin\omega t \cos(\omega t - \psi) - \cos\omega t \sin(\omega t - \psi)]$, and by the formula for the sine of a difference the bracket is $\sin[\omega t - (\omega t - \psi)] = \sin\psi$. For $0 < \psi < \pi$ the angle β therefore increases at every instant, and the field turns from the x -axis towards the y -axis, which is anticlockwise for the observer described in the text.
3. Let $\zeta = kz - \omega t$. The waves $\frac{1}{2}E_0[\hat{\mathbf{x}}\cos\zeta + \hat{\mathbf{y}}\cos(\zeta + \pi/2)]$ and $\frac{1}{2}E_0[\hat{\mathbf{x}}\cos\zeta + \hat{\mathbf{y}}\cos(\zeta - \pi/2)]$ are circularly polarised, with $\psi = \pi/2$ and $\psi = -\pi/2$, so their fields rotate in opposite senses. Their sum is $E_0\hat{\mathbf{x}}\cos\zeta + \frac{1}{2}E_0\hat{\mathbf{y}}(-\sin\zeta + \sin\zeta) = E_0\hat{\mathbf{x}}\cos\zeta$. By the result $\mathcal{I} = \frac{1}{2}\epsilon_0 c(E_1^2 + E_2^2)$, which the text found for the elliptically polarised wave and which holds for every phase difference ψ , each circularly polarised wave has intensity $\frac{1}{2}\epsilon_0 c[(E_0/2)^2 + (E_0/2)^2] = \frac{1}{4}\epsilon_0 cE_0^2$, and the sum of the two is $\frac{1}{2}\epsilon_0 cE_0^2$, which is the intensity of the linearly polarised wave.
4. The intensity is $\mathcal{I} = \mathcal{P}/\pi a^2$. From $\mathcal{I} = \frac{1}{2}\epsilon_0 cE_0^2$, the amplitude of the electric field is $E_0 = \sqrt{2\mathcal{P}/\pi a^2\epsilon_0 c}$, and the amplitude of the magnetic field is E_0/c . The force is the radiation pressure times the area, $(\mathcal{I}/c)\pi a^2 = \mathcal{P}/c$. The momentum given to the body is U/c , so its speed is U/mc .
5. At the distance r the intensity is $\mathcal{P}_S/4\pi r^2$, and the force on the sail is $2JA/c = \mathcal{P}_S A/2\pi r^2 c$. The gravitational force is $Gm_S m/r^2$, where G is the gravitational constant. The radiation force and the gravitational force each go as $1/r^2$, and their ratio is $\mathcal{P}_S A/(2\pi c G m_S m)$, which does not depend on r . The forces balance when $m/A = \mathcal{P}_S/(2\pi c G m_S)$, and a sail with a smaller mass per unit area is pushed away from the Sun.
6. In unit time, unit area of the surface receives momentum $\mathcal{I}/c$

from the incident wave. The reflected wave has intensity $\mathcal{R}\mathcal{I}$ and carries momentum $\mathcal{R}\mathcal{I}/c$ away in the opposite direction, so the total momentum given to the surface is $(1 + \mathcal{R})\mathcal{I}/c$, which is the pressure. For the mirror at oblique incidence, the energy arriving per unit area of the mirror in unit time is $\mathcal{I} \cos \theta$, with momentum $\mathcal{I} \cos \theta/c$ whose component along the normal is $(\mathcal{I}/c) \cos^2 \theta$. The reflected wave has the same intensity and makes the same angle with the normal on the other side, so the component of its momentum along the normal is reversed, while the component along the surface is unchanged. The momentum given to the mirror is therefore along the normal and equal to $(2\mathcal{I}/c) \cos^2 \theta$ per unit area in unit time.